Risk Bounds for Quantile Temporal-Spatial Analysis

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Abstract

This paper investigates risk bounds for quantile temporal-spatial analysis, focusing on estimation of a nonparametric quantile regression function from temporal-spatial data. We develop a general theory for quantile temporal-spatial estimators over a broad class of candidate functions under explicit assumptions. We consider both convex and nonconvex function classes. For convex classes, we apply the theory to trend filtering, high-dimensional quantile linear regression, and quantile K-NN fused lasso. For nonconvex classes, we apply the theory to quantile neural networks.

Whereas existing results for temporal-spatial estimators often assume sub-Gaussian errors, our risk bounds require no moment assumptions on the error variables and remain applicable to heavy-tailed errors. For trend filtering, our bound agrees with the minimax error rate for nonquantile estimators. We also derive risk bounds for the other models and provide algorithms for computing the estimators. Numerical experiments with simulated temporal-spatial data under normal and heavy-tailed noise illustrate the performance of the proposed estimators.

Keywords: Total Variation, nonparametric quantile regression, spatial-temporal analysis

1 Introduction

Technological advancements have significantly improved data collection and analysis techniques, resulting in an abundance of data with both temporal and spatial dimensions across various disciplines (Wen et al. (2019), Briz-Redón and Serrano-Aroca (2020), Liu et al. (2020), Tian et al. (2021), Jayasiri et al. (2022)). This proliferation of complex datasets has necessitated the development and adoption of advanced temporal-spatial models designed to address the unique challenges and harness the full potential of temporally and spatially diverse data.

Current leading results for temporal-spatial estimators typically assume that error distributions are sub-Gaussian. In contrast, quantile regression, known for its robustness and versatility across different sectors (Coad et al. (2006); Sanderson et al. (2006); Perlich et al. (2007); Benoit and Van den Poel (2009); Wasko and Sharma (2014); Pata and Schindler (2015); Belloni et al. (2023); Zhang et al. (2023)), often requires fewer presuppositions about error distributions. This approach diverges from conventional models that rely heavily on such assumptions. We extend the quantile regression approach to incorporate a thorough theory of risk bounds for an extensive array of quantile temporal-spatial functions, doing so with minimal assumptions about error variables. This analysis broadens the applicability of quantile regression for temporal-spatial data to both normal and heavy-tailed error distributions.

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This paper concentrates on nonparametric quantile regression within a temporal-spatial framework. We assume that our observations $\{(x_{ij}, y_{ij})\}_{i=1,j=1}^{n,m_i} \subseteq [0,1]^d \times \mathbb{R}$, where $\{x_{ij}\}_{i=1,j=1}^{n,m_i}$ represent the random locations of our temporal-spatial data $\{y_{ij}\}_{i=1,j=1}^{n,m_i}$. Here, $\left\{\delta_i : [0,1]^d \to \mathbb{R}\right\}_{i=1}^{n}$ signifies the spatial noise, while $\{e_{ij}\}_{i=1,j=1}^{n,m_i} \subseteq \mathbb{R}$ corresponds to the measurement errors.

We assume that the observations satisfy

$$y_{ij} = f^*(x_{ij}) + \delta_i(x_{ij}) + e_{ij}, \qquad i = 1, \dots, n, \quad j = 1, \dots, m_i. \tag{1}$$

Given a quantile level $\tau \in (0,1)$, let $f^* : [0,1]^d \to \mathbb{R}$ denote the deterministic target function. Our goal is to estimate $f^*(x_{ij})$, the conditional τ-quantile of y_{ij}, given by

$$f^*(x_{ij}) \;=\; F^{-1}_{y_{ij}|x_{ij}}(\tau), \quad \text{for } i = 1, \dots, n.$$

Here, $F_{y_{ij}|x_{ij}}$ is the conditional cumulative distribution function of y_{ij} given x_{ij}. Equivalently,

$$\mathbb{P}\left(y_{ij} \,\le\, f^*(x_{ij})|x_{ij}\right) \;=\; \mathbb{P}\left(\delta_i(x_{ij}) \,+\, e_{ij} \,\le\, 0|x_{ij}\right) \;=\; \tau.$$

1.1 Notation

For $s \in \mathbb{N}$, write $[s] = \{1, \dots, s\}$. Let Q be a distribution supported on $[0,1]^d$, and let

$$m = \left(\frac{1}{n}\sum_{i=1}^{n}\frac{1}{m_i}\right)^{-1}$$

denote the harmonic mean of the group sizes. For any measurable $f : [0,1]^d \to \mathbb{R}$ and $1 \le p < \infty$, define

$$\|f\|_p = \left(\int_{[0,1]^d} |f(x)|^p \, dQ(x)\right)^{1/p}, \qquad \|f\|_\infty = \sup_{x \in [0,1]^d} |f(x)|.$$

Let $L_p(Q) = \{f : [0,1]^d \to \mathbb{R} : \|f\|_p < \infty\}$. For $f, g : [0,1]^d \to \mathbb{R}$, define

$$\langle f, g\rangle_2 = \int_{[0,1]^d} f(x)g(x)\, dQ(x)$$

and the weighted empirical inner product and norm by

$$\langle f, g\rangle_{nm} = \frac{1}{n}\sum_{i=1}^{n}\frac{1}{m_i}\sum_{j=1}^{m_i} f(x_{ij})g(x_{ij}), \qquad \|f\|_{nm}^2 = \langle f, f\rangle_{nm}.$$

Given a set of elements $\{f(x_{ij})\}_{i=1,j=1}^{n,m_i} \subseteq \mathbb{R}$, we denote it as a vector $f \in \mathbb{R}^{\sum_{i=1}^n m_i}$. Let $\hat{f} \in \mathbb{R}^{\sum_{i=1}^n m_i}$ be an estimator of the set of elements $\{f(x_{ij})\}_{i=1,j=1}^{n,m_i} \subseteq \mathbb{R}$.

For any $A \subseteq \mathbb{R}^d$ consider $1_A(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{otherwise} \end{cases}$. For $s \in \mathbb{N}$, we set $1_s = (1, \dots, 1)^\top \in \mathbb{R}^s$. For two positive sequences $\{a_n\}_{n\in\mathbb{N}}$ and $\{b_n\}_{n\in\mathbb{N}}$, we write $a_n = O(b_n)$ or $a_n \lesssim b_n$ if $a_n \le C b_n$ with some constant $C > 0$ that does not depend on n, and $a_n = \Theta(b_n)$ or $a_n \asymp b_n$ if $a_n = O(b_n)$ and $b_n = O(a_n)$. Let X_n be a sequence of random $\mathbb{R}$-valued variables. For a deterministic or random $\mathbb{R}$-valued sequence a_n, we write $X_n = O_P(a_n)$ if $\lim_{M\to\infty}\limsup_{n\to\infty}\mathbb{P}(|X_n| > M a_n) \,=\, 0$. We also write $X_n = o_P(a_n)$ if $\limsup_{n\to\infty}\mathbb{P}(|X_n| > M a_n) \,=\, 0$ for all $M > 0$.

1.2 Summary of Results

Let $\rho_\tau(c) = \max\{\tau c, (\tau - 1)c\}$ be the usual check function for quantile regression. Let $\{(x_{i1}, \ldots, x_{im_i})\}_{i=1}^n$ and $\{(y_{i1}, \ldots, y_{im_i})\}_{i=1}^n$ be sequences of independent random variables. For a given quantile level $\tau \in (0, 1)$, let $f^*(x_{ij})$ be the conditional τ-quantile of y_{ij}. We study the optimization problem

$$\hat{f} = \underset{f \in \mathcal{F}}{\arg\min} \sum_{i=1}^n \frac{1}{n} \sum_{j=1}^{m_i} \frac{1}{m_i} \rho_\tau \left(y_{ij} - f(x_{ij})\right), \tag{2}$$

where $\mathcal{F}$ denotes the function class our estimator lives in. The optimization problem given in Equation (2) can be viewed as a constrained version of quantile regression for temporal-spatial data. Our investigation utilizes two distinct loss functions. The estimation loss, as delineated in Equation (2), employs the quantile loss. In contrast, our theoretical guarantees are articulated in the context of L_2 loss. In addition, we also introduce the Huber-type loss $\Delta^2(f)$ in (63), which, modulo constants, corresponds to a Huber loss, as delineated in Huber (1992). The primary rationale for focusing on the Huber loss stems from its emergence as a natural lower bound to the quantile population loss, and it is smooth and used in our derivation of theorems.

Our framework supposes that the sequences $\{(x_{i1}, \ldots, x_{im_i})\}_{i=1}^n$ and $\{e_{ij}\}_{i=1,j=1}^{n,m_i}$ and $\{\delta_i\}_{i=1}^n \subset L_2([0, 1])$ have elements that are identically distributed and β-mixing with coefficients having exponential decay, see the introduction below and Assumption 1 for the explicit details. We have temporal-dependent data across i. At every location $i \in \{1, \ldots, n\}$, we also assume that data have spatial dependence across j, which means $\delta_i(x_{ij})$ and $\delta_i(x_{ik})$ are dependent for $k \neq j$. The measurement errors $\{e_{ij}\}_{i=1,j=1}^{n,m_i}$ may be heavy-tailed, without moment assumptions.

We give a rigorous examination of beta-mixing block-independent instances and employ sophisticated techniques to deal with the data dependence. One of our major technical procedures to deal with dependent data is to split our data indices $[1 \ldots n]$ into even $\{\mathcal{J}_{e,l}\}_{l=1}^n$ and odd blocks $\{\mathcal{J}_{o,l}\}_{l=1}^n$, see Appendix H for the motivation and details. In particular, for any $L \in \{1, \ldots, n\}$, define

$$\begin{aligned} &\mathcal{J}_{e,l} = \{i \in \mathcal{I}_s \text{ for } s \mod 2 = 0 \text{ and } i \mod L = l\} \text{ and} \\ &\mathcal{J}_{o,l} = \{i \in \mathcal{I}_s \text{ for } s \mod 2 = 1 \text{ and } i \mod L = l\}, 1 \leq s \leq L. \end{aligned} \tag{3}$$

where $\mathcal{I}_s = \begin{cases} \{(s-1)L, sL\} & \text{when } 1 \leq s \leq K, \\ \{(KL, n]\} & \text{when } s = K + 1. \end{cases}$, $K = \lfloor n/L \rfloor$. So we have $\bigcup_{l=1}^L \mathcal{J}_{e,l} \cup \bigcup_{l=1}^L \mathcal{J}_{o,l} = \{1, \ldots, n\}$. Our analysis shows that if L is of order $O(\log n)$, the quantile estimator is consistent under exponentially decaying β-mixing.

We also have a sequence T_n tending to infinity that characterizes the tail probability of our spatial noise function δ_i and our measurement error e_{ij}, we show a very general set of distributions that could satisfy $\log(T_n) = o(n)$. See Assumption 4 for more details.

To estimate the optimal function f^*, the function class $\mathcal{F}$ in (2) is considered both convex and non-convex. For $\mathcal{F}$ that is characterized as a convex set, we also allow f^* is contained in $\mathcal{F}$, we give the convergence rate for convex $\mathcal{F}$ in Theorem 1. In particular, with the convex case of $\mathcal{F}$, under the above conditions, we show that the estimator $\hat{f}$ in (2) satisfies

$$\left\| \hat{f} - f^* \right\|_2^2 = O_p \left\{ b^2(B) \left(A + \frac{s \cdot \text{poly}(\log n) \log(T_n)}{n} \right) \right\},$$

where $b(B) = \max\left\{B^{1/2}, 1\right\}$ is a constant that depends on B, and c_0 satisfies (84).

The rate has a term A equal to local Rademacher complexity terms that depend on the complexity of the function class $\mathcal{F}$ within a neighborhood around the f^*, which gives a general result in the framework of empirical process theory. Moreover, it includes a term $s \cdot \text{poly}(\log(n)) \log(T_n) n^{-1}$, where s is contingent on the distributions of the sequence $\{(x_{i1}, y_{i1}, \ldots, x_{im_i}, y_{im_i})\}_{i=1}^n$ with $s < d$. For convex set of $\mathcal{F}$, our analysis is grounded in a very general function class. This class incorporates a wide array of statistical models that are of keen interest to various communities. Notable examples include Trend Filtering Guntuboyina et al. (2020); Tibshirani (2014), High-dimensional Linear Regression Fu and Knight (2000); Belloni and Chernozhukov (2011), and KNN-fused Lasso Ferradans et al. (2014); Madrid Padilla et al. (2020), among others.

For a nonconvex class $\mathcal{F}$, we also relax the requirement that f^* belong to $\mathcal{F}$. We first introduce the projection of f^* in $\mathcal{F}$ as $f_{\text{Proj}} := \inf_{f \in \mathcal{F}(L,r)} \|f - f^*\|_\infty$. Then we define ϕ_{nm} as an upper bound of error between the projection of f^* in $\mathcal{F}$ and itself, $\|f_{\text{Proj}} - f^*\|_\infty \leq \phi_{nm}$. We give the convergence rate for non-convex $\mathcal{F}$ in Theorem 7, satisfying

$$\left\|\hat{f} - f^*\right\|_2^2 = O_p\left\{b^2(B)\left(A_1 + \phi_{nm}^2 + \frac{s \cdot \text{poly}(\log n)\log(T_n)}{n}\right)\right\}.$$

The term A_1 is the local Rademacher complexity term associated with the function class $\text{Star}(\mathcal{F}, f^*)$. In the nonconvex case, we use $\text{Star}(\mathcal{F}, f^*)$, defined in (26), in place of $\mathcal{F}$. This broader class contains the estimators of interest and is used to derive the quantile convergence rate.

The theorem includes applications of neural network, with its rate in Theorem 10.

We propose a general theory for a quantile temporal-spatial estimator in (2) within a broad function class, extending current results beyond subGaussian error assumptions. This framework broadens the scope of quantile regression, accommodating both normal and heavy-tailed error distributions in temporal-spatial data analysis.

Our theoretical framework requires the function class to be bounded in the ℓ_∞ norm in order to obtain sharp convergence rates. This condition is readily met by continuous functions defined on bounded domains. In this context, we consider data points x originating from the unit hypercube $[0,1]^d$. Within the realm of nonparametric statistics, it is a common practice to assume that the covariates $x \in \mathbb{R}^d$ are uniformly distributed over $[0,1]^d$. This assumption is supported by the literature, for example, see Definition 3.4 in Györfi et al. (2002). Their model is a specific instance within the broader class of models we investigate, leading to sharp convergence rates. The boundedness condition also embodies the fundamental nonparametric analysis principle that favors localized over global complexity measures. These localized complexity measures are crucial for attaining optimal rates in nonparametric estimation tasks Koltchinskii (2001, 2006); Bartlett et al. (2005); Mendelson (2002); Bartlett and Mendelson (2002). Localization's importance in empirical process theory is underscored in Alexander (1987) and Geer (2000), and adopting this established technique allows us to further develop the theory for our specific problem.

The main proof first relates the Huber-type loss in (63) to localized complexity. We treat the estimator in (2) as an M-estimator, or empirical risk minimizer, following the framework of Mammen and Van De Geer (1997). The resulting Huber-type loss is controlled by localized Rademacher widths, as developed from Lemma 21 through Theorem 53.

To account for spatial dependence, we decompose the relevant risk into three terms and control them separately in Lemmas 26–33. We then apply symmetrization and contraction results from empirical process theory to express these controls in terms of localized Rademacher widths under the Huber-type metric.

Finally, we compare the localized Rademacher widths under the Huber-type and ℓ_2 metrics; this comparison is detailed in Lemmas 35–37. Together, these steps yield rates that do not require moment assumptions on the errors.

This theory is applied to three distinct models: trend filtering, high-dimensional Quantile Linear Regression, and Quantile K-NN fused Lasso. Traditional approaches for temporal-spatial-type estimators often rely on the assumption of sub-Gaussian error distributions. In contrast, our derived risk bounds are robust, making minimal assumptions on the error variables. Specifically, our framework eliminates the need for moment assumptions and is applicable even in scenarios characterized by heavy-tailed errors.

The convergence rate for trend filtering has been explored in Guntuboyina et al. (2020); Tibshirani (2014); Wang et al. (2015); Padilla et al. (2023) under the assumption of Gaussian errors and extended to heavy-tailed error distributions in Madrid Padilla and Chatterjee (2022); Zhang et al. (2023). In our analysis, trend filtering is considered within a subspace of our function class where functions satisfy $TV(f^{(k-1)}) \leq V$, with TV denoting the total variation operator on the $k-1$th derivative of f, and V acting as a tuning parameter. Given the presence of heavy-tailed error distributions and temporal-spatial dependence, the convergence rate is derived as $(nm)^{-\frac{2k}{2k+1}} + n^{-1}$, which is in line with the rate for non-quantile estimators presented in Theorem 1 of Padilla et al. (2023), up to logarithmic factors. This rate is shown to be minimax optimal for estimating signals within the function class with $V = O(1)$, adhering to the canonical scaling condition $V = O(1)$ commonly used in the literature for functions within a total variation ball of radius V, as discussed in Madrid Padilla and Chatterjee (2022) and Sadhanala et al. (2016). Our work advances the field by introducing and validating minor assumptions, demonstrating that the constrained quantile trend filtering estimator maintains robustness against heavy-tailed error distributions for temporal-spatial dependent data without the need for moment conditions.

Addressing the challenge of nonparametric quantile temporal-spatial data estimation for dimensions $d > 1$, we introduce a quantile temporal-spatial data adaptation of the K-Nearest Neighbor fused lasso estimator (K-NN-FL), inspired by the methodologies of Elmoataz et al. (2008), Ferradans et al. (2014), and Madrid Padilla and Chatterjee (2022). The K-NN-FL estimator operates through a two-step procedure. Initially, a K-nearest neighbor graph G_K is constructed with an edge set E_K, connecting two pairs (i_1, j_1) and (i_2, j_2) with an edge if $x_{i_1 j_1}$ ranks among the K-nearest neighbors of $x_{i_2 j_2}$, or vice versa. Subsequently, the graph fused lasso estimator is applied to G_K. Notably, our implementation of K-NN-FL diverges from the one in Madrid Padilla et al. (2020) by employing a distinct loss function tailored to temporal-spatial data, and from Padilla et al. (2023) by adapting the loss function to accommodate heavy-tailed error distributions. We demonstrate that the K-NN-FL estimator achieves a convergence rate of order $(nm)^{-1/d} + n^{-1}$, with respect to the Huber type loss, omitting logarithmic factors and under the condition that $d = o(n)$. This rate is valid provided that the true function f^* exhibits bounded variation and adheres to a property extending piecewise Lipschitz continuity. Hence, our findings significantly broaden the scope of Theorem 3 in Padilla et al. (2023), applying it within the quantile regression framework for temporal-spatial data analysis.

Our theoretical framework also finds application in high-dimensional regression. In contexts of weak sparsity and fixed design, we demonstrate that ℓ_1-constrained quantile regression is capable of consistently estimating the vector of regression coefficients without necessitating a restricted eigenvalue condition, diverging from the requirements outlined in Belloni and Chernozhukov (2011) and Fan et al. (2014). Further, we delve into high-dimensional quantile regression, specifically examining the constrained ℓ_1-QR estimator as initially defined by Knight and in Fu and Knight (2000), and later explored in Belloni and Chernozhukov (2011). The ℓ_1-QR, leveraging the quantile version of the lasso estimator from Tibshirani (1996), serves as a robust mechanism for variable selection and prediction in scenarios with high-dimensional covariates. We advance the discourse by extending the analysis presented in Theorem 5 of Madrid Padilla and Chatterjee (2022) to the temporal-spatial setting, introducing a nuanced dependence on the data and achieving a convergence rate of $(nm)^{-1/2} + n^{-1}s$.

Relevant theory for deep networks includes expressive-power results (Raghu et al., 2017), statistical risk bounds (Barron, 1989), stochastic-gradient methods (Reddi et al., 2019), energy landscapes (Choromanska et al., 2015), and dynamical-systems interpretations (Kumpati et al., 1990).

Convergence rates for neural-network regression estimators have been studied by Schmidt-Hieber (2020); Barron (1991); Oono and Suzuki (2019); Suzuki (2018); Kohler and Langer (2019). We consider a network with $\mathcal{L}$ hidden layers, r neurons per layer, and activation function σ. The neural-network estimator achieves a convergence rate of order $\max\{B,1\}\cdot \mathrm{poly}(\mathcal{L})\mathrm{poly}(r)\cdot(nm)^{-1} + \mathrm{poly}(\phi_{nm}) + \log(T_n)s\cdot n^{-1}$, up to logarithmic factors. Our analysis derives this rate for quantile temporal-spatial data with a neural-network function class.

Additionally, this paper presents algorithms specifically developed to compute the estimator efficiently, thus increasing its practical utility. An alternating direction method of multipliers (ADMM) tailored to our objective function is developed, with further details on ADMM algorithms available in Boyd et al. (2011). Our simulation studies encompass various scenarios, employing data points evenly spaced within $[0,1]^d$ or sampled from a uniform distribution over $[0,1]^d$. Unlike the assumptions of sub-Gaussian e_{ij} and $\delta_i(x)$ required by Padilla et al. (2023), our approach does not rely on such constraints, thereby accommodating potentially heavy-tailed distributions. We introduce noise stemming from either normal or heavy-tailed distributions. The experimental results affirm the convergence of the quantile temporal-spatial estimator with increasing n and m, in line with the theoretical framework proposed. Furthermore, our findings indicate superior performance of the estimator over trend filtering models, particularly with heavy-tailed distributions, and an advantage over quantile smoothing splines in the majority of scenarios.

2 General Methods

We formulate a general quantile regression problem over a function class $\mathcal{F}$. We assume that this function class satisfies

$$\|f\|_\infty \leq B, \quad \forall f \in \mathcal{F}, \tag{4}$$

where B represents a bounded constant. This property is easily satisfied for such continuous functions with bounded domains, which include applications of our theorems, trend filtering, KNN-fused lasso, and High-dimensional quantile regression, among others. We refer to Assumption 1 below for detailed technical conditions on the model.

We consider two cases for $\mathcal{F}$. First, suppose that the constraint set is convex. For $\lambda \in [0,1]$, the class $\mathcal{F}$ is closed under convex combinations, meaning that

$$\lambda f + (1-\lambda)g \in \mathcal{F}, \quad \forall f, g \in \mathcal{F}. \tag{5}$$

The estimator $\hat{f}$ can be viewed as a quantile analogue of constrained least squares. Convex constrained least squares in the Gaussian sequence model has a long and well-established history (see, e.g., Van de Geer (1990); Van Der Vaart et al. (1996); Hjort and Pollard (2011); Chatterjee et al. (2015)), and convex classes include many statistical models of interest.

We also consider the latter case in which $\mathcal{F}$ is nonconvex, as occurs in modern machine-learning models such as neural networks.

2.1 Assumptions

Assumption 1 *For our general method, we use the following assumptions.*

1. *The data* $\{(x_{ij}, y_{ij})\}_{i=1,j=1}^{n,m_i} \subseteq [0,1]^d \times \mathbb{R}$ *are generated based on model* (2).

2. *The design points* $\{x_{ij}\}_{i=1,j=1}^{n,m_i} \subseteq [0,1]^d$ *are sampled from a common density function* $v : [0,1]^d \to \mathbb{R}$, *such that* $0 < v_{\min} \leq v(x) \leq v_{\max}$ *for all* $x \in [0,1]^d$.

3. *For each fixed* $i \in \{1,\ldots,n\}$, *the* $\{x_{ij}\}_{j=1}^{m_i}$ *are independent.*

4. *Let* $\sigma_x(i) = \sigma(x_{ij}, j \in \{1,...,m_i\})$, *and suppose that* $\{\sigma_x(i)\}_{i=1}^n$ *is* β*-mixing with beta coefficients* $\beta(l)$. *See Appendix J for the* β*-mixing definition. The beta coefficients are assumed to have exponential decay,* $\beta(l) \leq e^{-C_\beta l}$ *for some constant* $C_\beta > 0$.

5. *The spatial noise* $\{\delta_i\}_{i=1}^n$ *is a collection of independently distributed random functions.*

6. *Moreover, let* $\sigma_\delta(i) := \sigma(\delta_i), i \in \{1,...,n\}$ *and suppose that* $\{\sigma_\delta(i)\}_{i=1}^n$ *are* β*-mixing with beta coefficients* $\beta_\delta(l)$. *The beta coefficients are assumed to have exponential decay,* $\beta_\delta(l) \leq e^{-C_\beta l}$.

7. *The measurement errors* $\{e_{ij}\}_{i=1,j=1}^{n,m_i}$ *are a collection of independently distributed random variables such that* $e_{ij} \perp\!\!\!\perp x_{i'j'}$ *for all* $(i',j') \neq (i,j)$.

8. *Moreover, let* $\sigma_e(i) = \sigma(e_i, j \in \{1,...,m_i\})$ *and suppose that* $\{\sigma_e(i)\}_{i=1}^n$ *are* β*-mixing with beta coefficients* $\beta(l)$. *The beta coefficients are assumed to have exponential decay,* $\beta_e(l) \leq e^{-C_\beta l}$.

9. *For any* $i \in \{1,...,n\}$, *we assume that* $cm \leq m_i \leq Cm$, *where* m *is the harmonic mean of* m_i *defined by*

$$m = \left(\frac{1}{n}\sum_{i=1}^n \frac{1}{m_i}\right)^{-1}. \tag{6}$$

Assumption 1.1 facilitates the representation of observed temporal-spatial data on random design points. Within the field of nonparametric statistics, it is customary to consider covariates $x_i \in \mathbb{R}^d$ uniformly distributed over $[0,1]^d$, as highlighted in Definition 3.4 of Györfi et al. (2002) and further discussed in Madrid Padilla et al. (2020) along with related references. Diverging from Assumption 1 in Padilla et al. (2023), our framework does not require the $\{x_{ij}\}_{i=1,j=1}^{n,m_i}$ to be identically distributed.

Our model acknowledges two primary sources of variation impacting the observed temporal-spatial data. The initial source, spatial noise, stems from the intrinsic variability within the temporal-spatial processes themselves. This implies that even in the absence of any measurement errors, observable variability would persist due to natural randomness, biological variation, or application-specific variation. This form of noise is delineated under Assumption 1.5 in our framework, a notion that finds parallel in Assumption 1.c of Padilla et al. (2023). A notable distinction in our approach is the absence of a constraint on δ_i to adhere to a subGaussian distribution, thereby accommodating the potential for heavy-tailed distributions.

The secondary source, measurement error, represented as $\{e_{ij}\}_{i=1,j=1}^{n,m}$, emerges from the limitations of measurement instruments or external disturbances. Such errors can impose systematic biases or random deviations in the observed dataset $\{y_{ij}\}_{i=1,j=1}^{n,m}$ that do not accurately represent the actual process. This aspect is encapsulated in Assumption 1.7 of our model. While a similar postulation regarding measurement error is present as 1.d in Padilla et al. (2023), our model diverges by permitting errors to follow heavy-tailed and non-identical distributions, thus encompassing a broader range of processes. Additionally, we stipulate independence among error sequences and data across different indices, aligning with conventional practices in nonparametric regression analysis.

Incorporating both spatial noise and measurement error, our model pragmatically accounts for discrepancies between observed data and the true underlying phenomena, acknowledging the inherent variability and errors essential for a comprehensive data analysis framework.

Assumption 1.9 is standard in the literature with spatial noise. For instance, Wang et al. (2022) focuses on the nonparametric estimation of covariance functions for functional data and employs a similar assumption, as shown on Page 3 of that paper.

Assumption 2 *There exists a constant $R > 0$ and $\underline{f} > 0$ and $\bar{f} > 0$ such that for any any function ϕ with $\|\phi\|_\infty \le R$ we have that*

$$\bar{f}\,|\phi(x_{ij})| \;\ge\; \Big|F_{y_{ij}|x_{ij}}(f^*(x_{ij}) + \phi(x_{ij})) - F_{y_{ij}|x_{ij}}(f^*(x_{ij})\Big| \;\ge\; \underline{f}\,|\phi(x_{ij})|\,, \tag{7}$$

where $F_{y_{ij}|x_{ij}}$ is the conditional CDF of y_{ij}.

The assumption that the conditional density of the response variable is bounded from below in a neighborhood is standard in quantile regression analysis. Similar conditions appear as D.1 in Belloni and Chernozhukov (2011) and as Condition 2 in He and Shi (1994).

Assumption 3 *For any $l = 1, \ldots, L$, for $i \in \mathcal{J}_{e,l}$, let $\delta_i : [0,1]^d \to \mathbb{R}$ be a random function, and let x be a vector following $x \in [0,1]^d$. Let us first introduce some convenient notation. Given vectors $x, x' \in \mathbb{R}^d$ and an index $k \in \{1, 2, \ldots, d\}$, we define a new vector $x^{\backslash k} \in \mathbb{R}^d$ via*

$$x_j^{\backslash k} := \begin{cases} x_j & \text{if } j \neq k, \\ x'_k & \text{if } j = k. \end{cases}$$

That is, $x^{\backslash k}$ is obtained by replacing the kth coordinate of x with the kth coordinate of x'. Let $u \in \mathbb{R}$. For $l = 1, \ldots, L$, define functions $\Pi_{e,l}$ on x and u, which satisfies $\Pi_{e,l}(x,u) = \Pi_{e,l}(x_S, u)$ for all x, where S is a subset of $\{1, \ldots, d\}$ such that $|S| = s$. Thus, we have

$$\begin{aligned}\Pi_{e,l}(x,u) \;=\; \Pi_{e,l}(x_S,u) \;&=\; \frac{1}{|\mathcal{J}_{e,l}|}\sum_{i\in\mathcal{J}_{e,l}}\left[\mathbb{P}(y_{i1} \;\le\; f^*(x_{i1}) + u \mid x_{i1} = x_S) - \mathbb{P}(y_{i1} \;\le\; f^*(x_{i1}) + u \mid \delta_i, x_{i1} = x_S)\right] \\ &=\; \frac{1}{|\mathcal{J}_{e,l}|}\sum_{i\in\mathcal{J}_{e,l}}\left[\mathbb{P}(\delta_i(x_{i1}) + e_{i1} \;\le\; u \mid x_{i1} = x_S) - \mathbb{P}(\delta_i(x_{i1}) + e_{i1} \;\le\; u \mid \delta_i, x_{i1} = x_S)\right].\end{aligned} \tag{8}$$

We assume that $\Pi_{e,l}$ satisfies the Lipschitz condition with a positive constant α almost surely for each coordinate with high probability in the multidimensional space. For any (x,u) and (x',u') in its domain, where $x, x' \in [0,1]^d$ and $u, u' \in \mathbb{R}$, we have

$$\begin{aligned}&|\Pi_{e,l}(x,u) - \Pi_{e,l}(x,u')| \;\le\; \alpha|u - u'| \\ &|\Pi_{e,l}(x,u) - \Pi_{e,l}(x^{\backslash k},u)| \;\le\; \alpha|x - x^{\backslash k}| \quad \forall k = 1, 2, \ldots, d.\end{aligned} \tag{9}$$

We also define $\Pi_{o,l}$ the same way in (8) and have condition (9) holds for $\Pi_{o,l}$.

Theorem 1 relies on Assumptions 3 and 4. We now provide examples satisfying these assumptions.

Example 1 (Examples for Assumption 3) *Consider the function $\Pi_{e,l}(x,u)$, which is Lipschitz continuous with respect to both x and u. This property holds as long as the mapping*

$$(x,u) \to \mathbb{P}(\delta_i(x_{i1}) + e_{i1} \;\le\; u \mid \delta_i, x_{i1} = x)$$

is also Lipschitz continuous in x and u.

Let us define $e_{i1} = v_{i1}g(x_{i1})$ where g is a specified function. Here, v_{i1} is a random variable with a distribution denoted by $F_{v_{i1}}$, and it is assumed to be independent of both δ_i and x_{i1}.

Consequently, it follows that

$$\mathbb{P}(\delta_i(x_{i1}) + e_{i1} \leq u \mid \delta_i, x_{i1} = x) = \mathbb{P}\left(v_{i1} \leq \frac{u - \delta_i(x)}{g(x)}\right),$$

which is equivalent to the cumulative distribution function (CDF) of v_{i1} evaluated at $\frac{u-\delta_i(x)}{g(x)}$.

It is well-known that the CDF is Lipschitz continuous if the corresponding probability density function (PDF) is bounded. Therefore, we require that v_{i1} possesses a bounded PDF, ensuring that $F_{v_{i1}}$ is a Lipschitz function with a Lipschitz constant denoted by K_1.

Furthermore, we impose the condition that the function

$$(x, u) \to \frac{u - \delta_i(x)}{g(x)}$$

is Lipschitz with respect to both u and x, with a Lipschitz constant K_2.

By imposing these conditions, we ensure that the family of classes under consideration adheres to the stipulated assumption, thus facilitating the application of the theoretical framework to this specific example.

Accordingly, we derive the following inequalities:

$$\begin{aligned}
\left|F_{v_{i1}}\left(\frac{u - \delta_i(x)}{g(x)}\right) - F_{v_{i1}}\left(\frac{u' - \delta_i(x)}{g(x)}\right)\right| &\leq K_1 \left|\frac{u - \delta_i(x)}{g(x)} - \frac{u' - \delta_i(x)}{g(x)}\right| \\
&\leq K_1 K_2 |u - u'| \quad \text{and} \\
\left|F_{v_{i1}}\left(\frac{u - \delta_i(x)}{g(x)}\right) - F_{v_{i1}}\left(\frac{u - \delta_i(x')}{g(x')}\right)\right| &\leq K_1 \left|\frac{u - \delta_i(x)}{g(x)} - \frac{u - \delta_i(x')}{g(x')}\right| \\
&\leq K_1 K_2 \left\|x - x'\right\|.
\end{aligned}$$

These inequalities are instrumental in confirming that Assumption 3 is satisfied. This is contingent upon v_{i1} possessing a bounded PDF, which encompasses a range of distributions, including but not limited to the Cauchy distribution and the t-distribution. Furthermore, the function $(x, u) \to \frac{u-\delta_i(x)}{g(x)}$ must be Lipschitz continuous with respect to both u and x, characterized by the Lipschitz constant K_2. This Lipschitz continuity is pivotal in ensuring the applicability of our theoretical results to the distributions mentioned, thereby substantiating the versatility and robustness of our framework.

Another assumption we make is about the tail probability our spatial noise function δ_i and our measurement error e_{ij}, stated as below.

Assumption 4 (Tail Probability) *For all e_{ij}, x_{ij}, and $\delta_i : [0,1]^d \to \mathbb{R}$, that satisfy all the existing Assumptions 1, there exists a constant T_n that depends on n, and a small constant $\tilde{c}_0$ such that the below tail probabilities are bounded with high probability.*

$$\max_{x\in[0,1]^d} \max_{i=1,\ldots,n} \left[\mathbb{P}(\delta_i(x_{i1}) + e_{i1} \geq T_n \mid \delta_i, x_{i1} = x) + \mathbb{P}(\delta_i(x_{i1}) + e_{i1} \geq T_n \mid x_{i1} = x)\right] \leq \tilde{c}_0 \text{ and} \tag{10}$$

$$\max_{x\in[0,1]^d} \max_{i=1,\ldots,n} \left[\mathbb{P}(\delta_i(x_{i1}) + e_{i1} \leq -T_n \mid \delta_i, x_{i1} = x) + \mathbb{P}(\delta_i(x_{i1}) + e_{i1} \leq -T_n \mid x_{i1} = x)\right] \leq \tilde{c}_0 \tag{11}$$

Theorem 1 relies on Assumptions 3 and 4. We now provide an example satisfying these assumptions.

Example 2 (Example for Assumption 4) *Below, we provide highlights of conditions under which Assumptions (10) and (11) from Assumption 4 are satisfied. The details of the conditions and derivations are in Appendix Example 3. Assume the event*

$$\Omega_1 = \left\{ \max_{i=1,\ldots,n} \|\delta_i\|_\infty \leq R_n \right\}$$

occurs with high probability. Furthermore, define $e_{i1} = g(x_{i1})V_{i1}$ where $\{V_{i1}\}$ is independent from $\{x_{i1}\}$, and consider the event

$$\Omega_2 = \left\{ \max_{i=1,\ldots,n} |g(x_{i1})| \leq s_n \right\}$$

also occurring with high probability. We then introduce T_n as

$$T_n = \max_{i=1,\ldots,n} \{2R_n, s_n\phi_n\},$$

with ϕ_n being a sequence tending to infinity. Suppose that

$$\mathbb{P}(\Omega_1^c) \to 0, \quad \mathbb{P}(\Omega_2^c) \to 0, \quad \max_{i=1,\ldots,n} \mathbb{P}(|V_{i1}| \geq \phi_n) \to 0, \quad \text{and } T_n \geq \max\{2R_n, \phi_n s_n\}. \tag{12}$$

hold. Then, on the multidimensional domain $[0,1]^d$,

$$\begin{aligned}
&\max_{x\in[0,1]^d} \max_{i=1,\ldots,n} [\mathbb{P}(\delta_i(x_{i1}) + e_{i1} \geq T_n \mid \delta_i, x_{i1} = x) \\
&\quad + \mathbb{P}(\delta_i(x_{i1}) + e_{i1} \geq T_n \mid x_{i1} = x)] \\
&\leq \tilde{c}_0,
\end{aligned}$$

and

$$\begin{aligned}
&\max_{x\in[0,1]^d} \max_{i=1,\ldots,n} [\mathbb{P}(\delta_i(x_{i1}) + e_{i1} \leq -T_n \mid \delta_i, x_{i1} = x) \\
&\quad + \mathbb{P}(\delta_i(x_{i1}) + e_{i1} \leq -T_n \mid x_{i1} = x)] \\
&\leq \tilde{c}_0.
\end{aligned}$$

for some positive constant $\tilde{c}_0$.

We then provide a concrete example of a class of functions that satisfy condition (12). For $i = 1, \ldots, n$, we let $\delta_i(x) = 0.5\delta_{i-1}(x) + \sum_{t=1}^{50} t^{-1} b_{i,t} h_t(x)$, where

$$h_t(x) = \prod_{j=1}^{d} \left(\frac{1}{\sqrt{2\pi}} \sin(t\pi x^{(j)}) \right),$$

are basis functions and $\{b_{i,t}\}_{t=1,i=1}^{50,n}$ are considered to be i.i.d. subGaussian. The δ_i is a linear combination of the sequence of functions in L^2 space, with coefficients are subGaussian or let δ_i to be a bounded function. Then by the general Hoeffiding inequalities for bounded functions, we can find many sequences of $\log(R_n) = o(n)$ to have the $\mathbb{P}(\Omega_1^c) \to 0$ as $n \to \infty$. Furthermore, suppose we consider the V_{i1} be t distributed or Cauchy distributed, at the same time, there are many sequences of $\log(\phi_n) = o(n)$ that satisfy the third condition in (12) for V_{i1}. And we can find $g(x_{i1})$ with a sequence of s_n, which does not go to infinity too fast, for example, let $\log(s_n) = o(n)$. Then we have the first three conditions in (12) satisfied. It is not difficult to find such $T_n \geq \max\{2R_n, \phi_n s_n\}$ such that $\log(T_n) = o(n)$ to satisfy the fourth condition. Then condition (12) is satisfied.

2.2 Results for General Methods

Consider $\{x_{ij}^*\}_{i=1,j=1}^{n,m_i}$ an i.i.d. copy ("ghost sample") of $\{x_{ij}\}_{i=1,j=1}^{n,m_i}$ constructed via the coupling scheme introduced in (71). Let $\{s_{ij}\}_{i=1,j=1}^{n,m_i}$ be i.i.d. Rademacher variables independent of $\{x_{ij}^*\}_{i,j}$.

Now, we present our main theoretical result on quantile temporal-spatial problem in (2) using the function class (5) and (4). First, let K be a function class with Rademacher width well defined. We introduce the following definitions.

$$R_{\mathcal{J}_{e,l}}(K) := \mathbb{E}_{x^*,s}\left[\sup_{f\in K}\frac{1}{\sum_{i\in\mathcal{J}_{e,l}} m_i}\sum_{i\in\mathcal{J}_{e,l}}\sum_{j=1}^{m_i} s_{ij}f(x_{ij}^*)\right], \quad R_{\mathcal{J}_{o,l}}(K) := \mathbb{E}_{x^*,s}\left[\sup_{f\in K}\frac{1}{\sum_{i\in\mathcal{J}_{o,l}} m_i}\sum_{i\in\mathcal{J}_{o,l}}\sum_{j=1}^{m_i} s_{ij}f(x_{ij}^*)\right]. \tag{13}$$

2.3 For $\mathcal{F}$ as Convex Set

Theorem 1 (Temporal-Spatial Data When $\mathcal{F}$ as Convex Set) *Consider observations $\{(x_{ij},y_{ij})\}_{i=1,j=1}^{n,m_i}$ satisfying Assumptions 1, 2, 3, and 4. And let $f^*(x_{ij})$ be the sequence of τ quantiles of y_{ij}. Let $\mathcal{F}$ the function class of our estimators to be convex. Additionally, we define*

$$L = \left\lceil \frac{1}{C_\beta} 2\log(n)\right\rceil, \quad \textit{for } L \textit{ in (3), for some positive constant } C_\beta. \tag{14}$$

Let the estimator be given by

$$\hat{f} \;=\; \arg\min_{f\in\mathcal{F}}\left\{\frac{1}{n}\sum_{i=1}^{n}\frac{1}{m_i}\sum_{j=1}^{m_i}\rho(y_{ij}-f(x_{ij}))\right\}. \tag{15}$$

It holds that

$$\begin{aligned}\left\|\hat{f}-f^*\right\|_2^2 \;=\; O_p\Bigg(b^2(B)\cdot\sum_{l=1}^{L}\Bigg(&R_{\mathcal{J}_{e,l}}\left(\mathcal{F}\cap\left\{f:\|f-f^*\|_2\,\le\,c_0^{-\frac{1}{2}}\cdot b(B)\cdot t\right\}\right)\\ &+R_{\mathcal{J}_{o,l}}\left(\mathcal{F}\cap\left\{f:\|f-f^*\|_2\,\le\,c_0^{-\frac{1}{2}}\cdot b(B)\cdot t\right\}\right)\;+\;\frac{b^2(B)\cdot s\cdot poly(\log n)\log(T_n)}{n}\Bigg)\Bigg),\end{aligned}$$

where $b(B)\;=\;\max\left\{B^{1/2},1\right\}$ is a constant that depends on B, and c_0 satisfies (84).

Theorem 1 establishes that risk bounds for the quantile spatial estimator within the convex function class can be derived from the localized Rademacher width term $\sum_{l=1}^{L}\left(R_{\mathcal{J}_{e,l}}\left(\mathcal{F}\cap\left\{f:\|f-f^*\|_2\,\le\,c_0^{-\frac{1}{2}}\cdot b(B)\cdot t\right\}\right)+R_{\mathcal{J}_{o,l}}\left(\mathcal{F}\cap\left\{f:\|f-f^*\|_2\,\le\,c_0^{-\frac{1}{2}}\cdot b(B)\cdot t\right\}\right)\right)$. For later applications to quantile trend filtering, quantile high-dimensional regression, etc., if we choose $|\mathcal{J}_{e,l}|$ and $|\mathcal{J}_{o,l}|$ to be the order $O(n/\log n)$, with L to be the order $O(\log n)$, we would achieve the convergence rate of $O(\text{poly}(\log n)(mn)^{-1})$, where $\text{poly}(\cdot)$ is the polynomial function, given the $\beta-$mixing assumption.

The rate also includes an additional term involving s, introduced in Assumption 3, and T_n, introduced in Assumption 4. It is standard to take $s = o(n)$, and we provide examples satisfying Assumption 4 with $\log(T_n) = o(n)$; under these conditions, this additional term is of order n^{-1}.

Historically, the Gaussian sequence model literature has extensively explored convex constrained least squares, a topic that is now well-documented (e.g., Van de Geer (1990); Wellner et al. (2013); Hjort and Pollard (2011); Chatterjee et al. (2015)). It posits that risk bounds, under the squared error loss, for convex constrained least squares estimators are inferred from the localized Gaussian width term $G(K \subset \mathbb{R}^n \cap \{\theta : \|\theta - \theta^*\| \leq t\})$, with the distinction between Gaussian G and Rademacher R being a constant factor.

In our analysis, the localized Rademacher complexity $R(\mathcal{F} \cap \{f : \|f - f^*\|_2 \leq \max\{B^{1/2}, 1\} \cdot t\})$ along with a term $O(sn^{-1})$ sets an upper bound for the loss function $\Delta^2(\hat{f} - f^*)$. Theorem 1 provides a nonparametric quantile bound that includes an additional term addressing the challenges posed by temporal-spatial data.

3 Application To Trend Filtering

Over the past decade, Trend Filtering has garnered considerable interest as a nonparametric method for nonparametric regression, aiming to fit a piecewise polynomial function under the assumption of independent errors Rudin et al. (1992); Mammen and Van De Geer (1997); Tibshirani et al. (2005); Kim et al. (2009). Specifically, in one-dimensional settings, Trend Filtering has been effective in estimating signals with a kth weak derivative that exhibits bounded total variation Tibshirani (2014).

Definition 2 *For the univariate trend-filtering application, set $d = 1$. Let F_k be the subspace of $\mathcal{F}$ consisting of functions that are $k-1$ times weakly differentiable and have finite J_k:*

$$\begin{aligned} F_k &= \{f \in \mathcal{F} : f \textit{ is } (k-1) \textit{ times weakly differentiable and } J_k(f) < \infty\}, \\ F_k(V) &= F_k \cap \{f : J_k(f) \leq V\}, \qquad J_k(f) = TV\left(f^{(k-1)}\right). \end{aligned} \tag{16}$$

For $g : [0,1] \to \mathbb{R}$, define

$$TV(g) = \sup_{0 \leq a_1 < \cdots < a_N \leq 1} \sum_{l=1}^{N-1} |g(a_l) - g(a_{l+1})|. \tag{17}$$

When g is differentiable, $TV(g) = \int_0^1 |g'(t)|\, dt$. Hence, when $f^{(k)}$ exists,

$$J_k(f) = TV\left(f^{(k-1)}\right) = \int_0^1 |f^{(k)}(t)|\, dt.$$

Under this indexing convention, the corresponding trend-filtering spline has degree $k-1$: $k = 1$ gives a piecewise-constant fit, $k = 2$ a piecewise-linear fit, and $k = 3$ a piecewise-quadratic fit Mammen and Van De Geer (1997).

For application to Trend filtering, we add more assumptions in below.

Assumption 5 *The univariate function $f^* : [0,1] \to \mathbb{R}$ satisfies the following:*

1. *It holds that $V^* := J_k(f^*) < \infty$, where V^* can depend on n, and with $k \geq 1$ a fixed integer, where $J_k(f) = TV(f^{(k-1)})$ for any function f.*
2. *Let $V \geq V^* = TV(f^{*(k-1)})$, $V > 0$ is a tuning parameter that controls the $k-1$th order total variation.*
3. *Let canonical scaling $V^* = \mathcal{O}(1)$.*
4. *Let $\int f^*(t)dt = 0$.*

Assumption 5.1 addresses the smoothness properties of the function f^* associated with the observed temporal-spatial data, positing that f^* is contained within a TV ball of radius V^*. This assumption allows for variations in the local smoothness of f^*, and encompasses more traditional function classes such as Hölder and Sobolev spaces as special instances.

Assumption 5.2, asserting that $f^* \in \mathcal{F}$, is a standard constraint, mirroring similar premises like that in Theorem 1 of Madrid Padilla and Chatterjee (2022) for univariate quantile trend filtering.

Assumption 5.3 specifies $V^* = \mathcal{O}(1)$. This condition is widely recognized in literature as the canonical scaling for functions within a total variation ball of radius V, as discussed in Madrid Padilla and Chatterjee (2022) and Sadhanala et al. (2016). Given this canonical scaling, setting $V = O(1)$ is logical, suggesting $TV(f^{(k-1)})$ to also be $O(1)$.

The condition that $\int f^*(t)dt = 0$ has been articulated in various forms across the literature for identifiability purposes. For instance, Sadhanala and Tibshirani (2017) employs a discrete version of this requirement, as detailed on Page 1 of their study. A similar condition is mentioned in Example 1 of Dalalyan et al. (2017), within the context of analyzing the univariate fixed design total variation estimator.

3.1 Result for Quantile Temporal-Spatial Trend Filtering

For $k \geq 1$, we introduce the Quantile Trend Filtering estimator for temporal-spatial data, defined through the optimization problem

$$\hat{f} = \arg\min_{f \in F_k(1)} \frac{1}{n}\sum_{i=1}^{n}\frac{1}{m_i}\sum_{j=1}^{m_i}\rho_\tau(y_{ij} - f(x_{ij})) \quad \text{subject to} \quad TV(f^{(k-1)}) \leq 1. \tag{18}$$

Equation (18) outlines a constrained version of Quantile Trend Filtering specifically for temporal-spatial data, analogous to the formulation presented in Equation (2). Our analysis begins with this constrained estimator.

Theorem 3 (Trend Filtering) *Consider $k \geq 1$ and observations $\{(x_{ij}, y_{ij})\}_{i=1,j=1}^{n,m_i}$ satisfying Assumptions 1–5. Let $f^*(x_{ij})$ be the conditional τ-quantiles of y_{ij}. Under the canonical scaling $V^* = \mathcal{O}(1)$, let $\hat{f} \in F_k(1)$ be the estimator defined in (18), with $J_k(f) = TV(f^{(k-1)})$. It holds that*

$$\left\|\hat{f} - f^*\right\|_2^2 = O_p\left(\frac{\max\left\{B^{\frac{6k+1}{2k+1}}, B^{\frac{2k-1}{2k+1}}\right\}\log(n)^{\frac{4k}{2k+1}}}{(nm)^{\frac{2k}{2k+1}}} + \frac{\max\{B,1\}\cdot s\cdot poly(\log n)\log(T_n)}{n}\right), \tag{19}$$

Theorem 1 elucidates that, within the function classes defined in (5) and (4), the estimator attains a convergence rate, disregarding logarithmic factors, characterized by a Huber type loss of order $(mn)^{-\frac{2k}{2k+1}} + sn^{-1}$, assuming $V \geq TV((f^*)^{(k-1)})$, and with the proper choice of $\log(T_n) = o(n)$ which is discussed in Example 2. Thus, (19) indicates a convergence rate for $\hat{f}$ of $(mn)^{-\frac{2k}{2k+1}} + sn^{-1}$, excluding logarithmic considerations and letting $L = o(mn)$.

Previous research Madrid Padilla and Chatterjee (2022) determined the convergence rate of the univariate quantile Trend Filtering estimator under fixed design and $m_i = O(1)$, and Zhang et al. (2023) developed the convergence rate for quantile regression in the context of additive Trend Filtering with $m_i = O(1)$. Our convergence rate (19) addresses a more intricate scenario that includes temporal-spatial dependence, thereby expanding upon prior rates to accommodate varying m_i's, dependent data, and temporal-spatial noise.

Moreover, Theorem 3 presents a rate of $(nm)^{-\frac{2k}{2k+1}} + sn^{-1}$, aligning with the lower bound in Theorem 1 of Padilla et al. (2023) for $s = o(n)$ and $L = o(mn)$, when estimating signals within $F_k(V)$

where $V = O(1)$. Unlike subsequent studies that assume subGaussian data, our findings suggest a minimax rate under broader conditions. Specifically, while the lower bound in their work applies to subGaussian errors, our upper bound accommodates arbitrary error distributions for measurement without necessitating moment conditions, thereby offering a more universally applicable rate.

4 Application To High-dimensional Quantile Linear Regression

Now we consider high-dimensional linear quantile regression. We study the constrained version of the ℓ_1-QR estimator defined in Fu and Knight (2000) and studied in Belloni and Chernozhukov (2011). ℓ_1-QR is commonly used as a robust tool for variable selection and prediction with high-dimensional covariates and is the quantile version of the constrained lasso estimator proposed in Tibshirani (1996).

Suppose that we are given $\{(x_{ij}, y_{ij})\}_{i=1,j=1}^{n,m_i} \subseteq [0,1]^d \times \mathbb{R}$ with the $\{x_{ij}\}_{i=1,j=1}^{n,m_i}$, and with $\{(y_{ij})\}_{i=1,j=1}^{n,m_i}$ random variables. And let $f^*(x_{ij})$ be the sequence of τ quantiles of y_{ij}.

Assumption 6 *The function $f^* : [0,1]^d \to \mathbb{R}$, Observations $\{(x_{ij}, y_{ij})\}_{i=1,j=1}^{n,m_i}$ and $\{\delta_i\}$, $\{e_{ij}\}_{i=1,j=1}^{n,m_i}$ satisfy the following Assumptions:*

1. *Suppose $f^* = x'\beta^*$ for some $\beta^* \in \mathbb{R}^d$. And $V^* \geq \|\beta^*\|_1$*
2. *Let $V \geq V^* \geq \|\beta^*\|_1$, $V > 0$ is a tuning parameter that controls the ℓ_1 norm of β.*

Assumption 6.1 specifies that the functions under estimation adhere to a linear regression framework. This assumption aligns with the premises of high-dimensional quantile linear regression with trend filtering as discussed in Madrid Padilla and Chatterjee (2022).

Based on these observations, we now consider the estimator to be:

$$\hat{f} = \arg\min_{f \in H(V)} \frac{1}{n} \sum_{i=1}^{n} \frac{1}{m_i} \sum_{j=1}^{m_i} \rho_\tau(y_{ij} - f(x_{ij})) \text{ where } H(V) := \{f : f(x_{ij}) = x_{ij}'\beta, ||\beta||_1 \leq V, \beta \in \mathbb{R}^d\}. \tag{20}$$

4.1 Result for High-dimensional Quantile Linear Regression

With the notation from above, we now present our convergence rate for High-dimensional Quantile Linear Regression.

Theorem 4 *Suppose $f^* = x'\beta^*$ for some $\beta^* \in \mathbb{R}^d$. Let $X \in \mathbb{R}^{\sum_{i=1}^{m_i} \times d}$, whose rows are $\left\{x_{ij}'\right\}_{i=1,j=1}^{n,m_i}$. Denote*

$$A := \mathbb{E}_x \left[\sum_{l=1}^{L} \left(\max_{j \in [d]} ||X_{e,l \cdot j}|| + \max_{j \in [d]} ||X_{o,l \cdot j}|| \right) \right].$$

There exist constants $C_6 > 0$ and $C_7 > 0$ such that

$$\left\| \hat{f} - f^* \right\|_2^2 = O_p \left(\frac{\max\{B, 1\} \log nV (\log d)^{1/2} A}{nm} + \frac{\max\{B, 1\}\, s \cdot poly(\log n) \log(T_n)}{n} \right),$$

where $X_{e,l \cdot j}$ is the jth column of $X_{e,l} \in \mathbb{R}^{\sum_{i \in \mathcal{J}_{e,l}} m_i \times d}$ and $\hat{f}$ is the estimator defined in (20).

Theorem 4 shows that with columns of X normalized—a standard assumption in high-dimensional regression captured by

$$\max_{j=1,\ldots,d} \|X_{\cdot j}\| \leq (mn)^{1/2},$$

we achieve a rate of $(\log d/(mn))^{1/2} + sn^{-1}$. This rate is comparable to the $(\log d/n)^{1/2}$ observed for univariate quantile trend filtering when $m_i = O(1)$, in the absence of data dependence and spatial noise, as reported in Madrid Padilla and Chatterjee (2022). Importantly, the $(\log d/n)^{1/2}$ rate is also applicable to the lasso, without any specific assumptions on the design matrix X and given $m_i = O(1)$, as demonstrated in Chatterjee (2013). Our findings extend these rates to scenarios involving dependent data and spatial noise, assuming $s = o(n)$. Although fast convergence rates have been obtained under restricted eigenvalue conditions on the design matrix in previous works (Belloni and Chernozhukov (2011); Fan et al. (2014); Sun et al. (2020)), our approach offers a broader applicability for dependent data without such assumptions.

5 Application To Quantile K-NN fused lasso

For dimensions $d > 1$, in addition to exploring High-dimensional Quantile Linear Regression, we examine the quantile temporal-spatial adaptation of the K-Nearest Neighbor fused lasso estimator (K-NN-FL), as investigated by Elmoataz et al. (2008), Ferradans et al. (2014), and Madrid Padilla et al. (2020). The K-NN-FL estimator employs a two-step approach. Initially, a K-nearest neighbor graph G_K is constructed with an edge set E_K, connecting two pairs (i_1, j_1) and (i_2, j_2) if $x_{i_1 j_1}$ is among the K-nearest neighbors of $x_{i_2 j_2}$, and vice versa. Subsequently, the quantile graph fused lasso estimator is applied to G_K. The work in Padilla et al. (2023) has shown the convergence rate of temporal-spatial K-NN-FL. Our implementation of quantile K-NN-FL, however, diverges from both works in terms of the loss function used for temporal-spatial data.

Let G_K represent the K-NN graph with edge set E_K, derived from $\{x_{ij}\}_{i=1,j=1}^{n,m_i}$. For a signal $\theta \in \mathbb{R}^{\sum_{i=1}^n m_i}$, its total variation along G_K is defined as

$$\|\nabla_{G_k}\theta\|_1 := \sum_{\{(i_1,j_1),(i_2,j_2)\}\in E_K} |\theta_{i_1,j_1} - \theta_{i_2,j_2}|. \tag{21}$$

This definition presupposes that if θ corresponds to function evaluations, $\|\nabla_{G_k}\theta\|_1$ would be significantly less than n, as G_K captures the domain geometry of the design points x_{ij}, rendering most differences $\|\theta_{i_1 j_1} - \theta_{i_2 j_2}\|$ minimal. Furthermore, drawing upon Von Luxburg et al. (2014), Madrid Padilla et al. (2020) demonstrated that under Assumption 7, it is probable that

$$\|\nabla_{G_k}\theta\|_1 \asymp (nm)^{-\frac{1}{d}}, \tag{22}$$

with $\theta^*_{ij} = f^*(x_{ij})$, assuming f^* either exhibits bounded variation or is piecewise Lipschitz, alongside a condition extending piecewise Lipschitz continuity.

Suppose that we are given $\{(x_{ij}, y_{ij})\}_{i=1,j=1}^{n,m_i} \subseteq [0,1]^d \times \mathbb{R}$ with the $\{x_{ij}\}_{i=1,j=1}^{n,m_i}$, and with $\{(y_{ij})\}_{i=1,j=1}^{n,m_i}$ random variables. And let $f^*(x_{ij})$ be the sequence of τ quantiles of y_{ij}.

Assumption 7 *For quantile K-NN fused lasso, we use the following assumptions.*

1. *Assume the data points are sampled from a common density function $v : [0,1]^d \to \mathbb{R}$, such that $0 < v_{\min} \leq v(x) \leq v_{\max}$ for all $x \in [0,1]^d$.*

2. *Suppose either f^* has bounded variation, along with an additional condition that generalizes piecewise Lipschitz continuity; or f^* is piecewise Lipschitz.*

3. *Let $V_n \geq \|\nabla_{G_k}\theta^*\|_1$, $V_n > 0$ is a tuning parameter that controls the ℓ_1 norm of $\nabla_{G_k}\theta$.*

4. *We also have for $s, l \in [1, \dots, L]$, $s \neq l$, $\{x_{ij}\}_{i\in\mathcal{J}_{e,l} j\in[m_i]}$, $\{x_{ij}\}_{i\in\mathcal{J}_{e,s} j\in[m_i]}$, $\{x_{ij}\}_{i\in\mathcal{J}_{o,l} j\in[m_i]}$ have the same distribution. Similarly, we have $\{e_{ij}\}_{i\in\mathcal{J}_{e,l} j\in[m_i]}$, $\{e_{ij}\}_{i\in\mathcal{J}_{e,s} j\in[m_i]}$, $\{e_{ij}\}_{i\in\mathcal{J}_{o,l} j\in[m_i]}$ have same distribution, and $\{\delta_i\}_{i\in\mathcal{J}_{e,l}}$, $\{\delta_i\}_{i\in\mathcal{J}_{e,s}}$, $\{\delta_i\}_{i\in\mathcal{J}_{o,l}}$ have the same distribution.*

Assumption 7.1 specifies that the distribution of covariates is bounded above and below by positive constants. This is analogous to the assumptions made by Györfi et al. (2002) and Meinshausen and Ridgeway (2006), where v represents the probability density function for the uniform distribution over $[0, 1]^d$.

Assumption 7.2 relates to the smoothness of the quantile function f^* associated with the observed temporal-spatial data. This assumption accommodates functions that may exhibit discontinuities, including those that are piecewise Lipschitz or conform to a broader definition than piecewise Lipschitz, subject to a bounded variation constraint. For further examples and discussions, see Madrid Padilla et al. (2020).

Assumption 7.3 mirrors the function of Assumption 5.2.

5.1 Result for Quantile K-NN fused lasso

This result extends the result shown in Padilla et al. (2023) to the quantile setting. Thus, a natural estimator for θ^* is $\hat{\theta}_n$, given as

$$\hat{\theta}_n = \arg \min_{\theta \in \mathbb{R}^{\sum_{i=1}^n m_i}} \left\{ \frac{1}{n}\sum_{i=1}^{n} \frac{1}{m_i}\sum_{j=1}^{m_i} \rho_\tau(y_{ij} - \theta_{ij}) \right\} \quad \text{s.t.} \quad \|\nabla_{G_k}\theta\|_1 \leq V_n, \tag{23}$$

for a tuning parameter $V_n > 0$. Once $\hat{\theta}_{V_n}$ is constructed, we let $\hat{f}_{V_n} : [0, 1]^d \to \mathbb{R}$ be given as

$$\hat{f}_{V_n}(x_{ij}) = (\hat{\theta}_{V_n})_{ij} \quad \text{for } i = 1, \dots, n,\ j = 1, \dots, m_i, \tag{24}$$

and

$$\hat{f}_{V_n}(x) = \frac{1}{\sum_{i=1}^n \sum_{j=1}^{m_i} I_{(x,x_{ij})}} \sum_{i=1}^{n}\sum_{j=1}^{m_i} I_{(x,x_{ij})}(\hat{\theta}_n)_{ij}, \tag{25}$$

for $x \in [0, 1]^d \backslash \{x_{ij}\}_{i=1,j=1}^{n,m_i}$, where $N_K(x)$ is the set of K-nearest neighbors of x among $\{x_{ij}\}_{i=1,j=1}^{n,m_i}$, and $I_{(x,x_{ij})} = 1$ if $x_{ij} \in N_K(x)$ and $I_{(x,x_{ij})} = 0$ otherwise. Notice in (24) we are using the same estimation rule described in Madrid Padilla et al. (2020) and Padilla et al. (2023).

With this notation, we are ready to state our main result regarding the constrained quantile K-NN fused lasso for temporal-spatial data (23)–(25).

Theorem 5 *Let $\{(x_{ij}, y_{ij})\}_{i=1,j=1}^{n,m_i}$ follow the Assumptions with $d > 1$. Suppose in addition that K has a choice of poly$(\log(nm))$. The estimator $\hat{\theta}_n$, defined in (23) satisfies that*

$$\left\|\hat{\theta}_{V_n} - \theta^*\right\|^2 = O_p\left(\frac{poly(B)poly(\log(nm))}{mn}V_n + \frac{poly(B)poly(\log(nm))}{n} + \frac{poly(B)\cdot d\cdot poly(\log n)\log(T_n)}{n}\right),$$

and provided that $V_n \geq \|V_{G^*}\|$. In addition, $\hat{f}_{V_n}$, defined in (24)–(25) satisfies that

$$\left\|\hat{f}_{V_n} - f^*\right\|_2^2 = O_p\left(\frac{poly(B)poly(\log(nm))}{mn}V_n + \frac{poly(B)d\cdot poly(\log n)\log(T_n)}{n} + \frac{poly(B)\left(poly(\log(nm))\right)^{1/d}}{(nm)^{1/d}}\right).$$

Based on (22), the optimal selection for V_n is $V_n = \|\nabla_{G_k}\theta^*\|_1 \asymp (nm)^{1-1/d}$. Theorem 5 then indicates that, for temporal-spatial data and in terms of the Huber-type loss defined in (63), the K-NN-FL estimator presented in (24) and (25) attains a convergence rate of $\text{poly}(nm)\text{poly}(B) \cdot (nm)^{-1/d} + \log(n)d \cdot n^{-1}$. With respect to the ℓ_2 loss, this rate is $\text{poly}(nm)\text{poly}(B) \cdot (nm)^{-1/d} + \log(n)d\text{poly}(B) \cdot n^{-1}$, where $\text{poly}(\cdot)$ denotes a polynomial function. This achieves an upper bound on the convergence rate, $(nm)^{-1/d}$, as detailed in Theorem 3 of Padilla et al. (2023) for the quantile K-NN fused lasso, but extends the applicability to include dependent measurements and spatial noise.

Our results also match the upper bound from Theorem 3 in Padilla et al. (2023), modulo some logarithmic factors, under the condition $d = o(n)$. However, in contrast to their analysis, our approach accommodates heavy-tailed errors without necessitating moment conditions. Drawing from Lemma 1 in Padilla et al. (2023), the rate we present is minimax optimal within the class of piecewise Lipschitz functions, aside from logarithmic factors, assuming $d = o(n)$.

6 For $\mathcal{F}$ as Non-convex Set and $f^* \notin \mathcal{F}$

We extend our function class $\mathcal{F}$ to be nonconvex, let $\hat{f}$, belong to $\mathcal{F}$. Furthermore, we let f^* not necessarily in $\mathcal{F}$. In this case, our analysis will project f^* into $\mathcal{F}$ and we will incorporate the projection error into our rate. Furthermore, to deal with nonconvex function class, we will introduce the star-shaped function class, which is a convex combination of f^* and any function in $\mathcal{F}$, with its definition as below,

Definition 6 *A function class $\mathcal{H}$ is star-shaped if for any $h \in \mathcal{H}$ and $\alpha \in [0,1]$, the rescaled function αh also belongs to $\mathcal{H}$.*

For $\mathcal{F}$, and f^* not necessarily in $\mathcal{F}$, we define a star-shaped set as:

$$\text{Star}(\mathcal{F}, f^*) \coloneqq \{(1-\mu)f^* + \mu f : f \in \mathcal{F}\} \quad \text{for any } \mu \in [0,1]. \tag{26}$$

With the all assumptions 1, 1.9, 2, 3, and 4 satisfied, no more assumptions needed as Theorem 1, we have the convergence rate for Temporal-Spatial Data presented in below Theorem.

Theorem 7 (Temporal-Spatial Data When $\mathcal{F}$ as Non-Convex Set) *Consider observations $\{(x_{ij}, y_{ij})\}_{i=1,j=1}^{n,m_i}$ satisfying Assumptions. And let $f^*(x_{ij})$ be the sequence of τ quantiles of y_{ij}. Let $\mathcal{F}$ the function class of our estimators to be convex. Let $\hat{f}$, belong to $\mathcal{F}$, and f^* not necessarily in $\mathcal{F}$, where $\mathcal{F}$ is not necessarily convex. Let the star-shaped set $Star(\mathcal{F}, f^*)$ be defined in (26). Suppose the distribution of $\{(x_{ij}, y_{i,j})\}_{i=1,j=1}^{n,m_i}$ obey Assumption 2, let $R(\mathcal{F})$ be the Rademacher width defined in 12, let $\tilde{c}_0$ be a small constant given in Assumption 4. Additionally, we define $L = \left\lceil \frac{1}{C_\beta} 2\log(n) \right\rceil$ for L in (3), for some positive constant C_β.*

Let the estimator be given by

$$\hat{f} = \arg\min_{f \in \mathcal{F}} \left\{ \frac{1}{n}\sum_{i=1}^{n} \frac{1}{m_i} \sum_{j=1}^{m_i} \rho(y_{ij} - f(x_{ij})) \right\} \tag{27}$$

It holds that

$$\left\|\hat{f} - f^*\right\|_2^2 = O_p\left(b^2(B) \cdot \sum_{l=1}^{L} \left(R_{\mathcal{J}_{e,l}}\left(Star(\mathcal{F}, f^*) \cap \left\{f : \|f - f^*\|_2 \le c_0^{-\frac{1}{2}} \cdot b(B) \cdot \tilde{t}\right\}\right)\right.\right.$$

$$+R_{\mathcal{J}_{o,l}}\left(Star(\mathcal{F}, f^*) \cap \left\{f : \|f - f^*\|_2 \leq c_0^{-\frac{1}{2}} \cdot b(B) \cdot \tilde{t}\right\}\right)$$
$$+ b^2(B)\mathcal{C}\phi_{nm}^2 + \frac{b^2(B) \cdot s \cdot poly(\log n)\log(T_n)}{n}\Bigg)\Bigg),$$

where $b(B) = \max\left\{B^{1/2}, 1\right\}$ is a constant that depends on B, $\mathcal{C}$ is a constant that depends on $\bar{f}$ and C_0, and c_0, C_0 satisfy (84). With r_{nm} is given as

$$\|f_{Proj} - f^*\|_\infty \leq \phi_{nm},$$

where the f_{Proj} denotes the projection of f^* in $\mathcal{F}$, and $\tilde{t}^2$ satisfying

$$\tilde{t}^2 := t^2 + \mathcal{C}\phi_{nm}^2.$$

Theorem 7 establishes that risk bounds for the quantile spatial estimator within the non convex function class can be derived from the localized Rademacher width term $\sum_{l=1}^{L} R_{\mathcal{J}_{e,l}}\left(\text{Star}(\mathcal{F}, f^*) \cap \left\{f : \|f - f^*\|_2 \leq c_0^{-\frac{1}{2}} \cdot b(B) \cdot t\right\}\right)$. Compared to the convex function class where $\mathcal{F}$ is used, the nonconvex use the $\text{Star}(\mathcal{F}, f^*)$. One of our key derivations involve conversion of Huber-type loss to the local-Rademacker Complexity, the $\text{Star}(\mathcal{F}, f^*)$ guarantees all the estimators we are focusing belong to $\text{Star}(\mathcal{F}, f^*)$, a larger set compared to $\mathcal{F}$, the details are in Corollary 52. The rate of convergence has an additional term $\mathcal{C}\phi_{nm}^2$ which is attributed to the approximation error $\phi_{nm} := \sup_{f_{\text{Proj}} \in \mathcal{F}} \|f_{\text{Proj}} - f^*\|_\infty$. In the next part, we will consider the non-convex Neural Network function class.

7 Application to Neural Networks

In recent times, multilayer (or deep) neural networks have demonstrated remarkable success in tackling complex tasks like image classification Krizhevsky et al. (2017). Deep learning has emerged as the leading approach in these domains, delivering state-of-the-art performance. However, despite its achievements, there remains a need for further mathematical comprehension in the field.

For several years now, neural networks have stood at the forefront of nonparametric statistical methods, particularly shining in areas such as pattern recognition and nonparametric regression, as discussed in various key texts (Anthony et al. (1999); Devroye et al. (2013); Györfi et al. (2002); Haykin (1998); Hertz (2018)). The advent of deep learning, characterized by the use of multi-layered feedforward neural networks with numerous hidden layers, has gained significant traction in practical applications, as documented in the recent literature ((Schmidhuber, 2015) and references therein). This wave of practical achievements has sparked a growing scholarly interest in the theoretical foundations and properties of neural networks, as evidenced by a burgeoning body of research (Schmidhuber (2015); Eldan and Shamir (2016); Elbrächter et al. (2021); Yarotsky (2018); Yarotsky and Zhevnerchuk (2020); Lu et al. (2021)).

The convergence rates for neural network-based regression estimates have been a subject of extensive study within the field of nonparametric statistics. Our paper aims to build a statistical theory that provides convergence rate of quantile regression model of neural networks for spatial-temporal data.

In order to construct quantile temporal spatial estimates with neural networks, the first step is to define a suitable space of functions $f : \mathbb{R}^d \to \mathbb{R}$ using neural networks. The starting point here is the choice of an activation function $\sigma : \mathbb{R} \to \mathbb{R}$. Traditionally, so-called squashing functions are chosen as activation function $\sigma : \mathbb{R} \to \mathbb{R}$, which are nondecreasing and satisfy $\lim_{x\to-\infty} \sigma(x) = 0$ and $\lim_{x\to\infty} \sigma(x) = 1$, e.g., the so-called sigmoidal or logistic squasher

$$\sigma(x) = \frac{1}{1+\exp(-x)}, \quad x \in \mathbb{R}. \tag{28}$$

Recently, also unbounded activation functions are used, e.g., the ReLU activation function

$$\sigma(x) = \max\{x, 0\}. \tag{29}$$

The network architecture $(\mathcal{L}, k)$ depends on a positive integer $\mathcal{L}$ called the number of hidden layers and a width vector $k = (k_1, \ldots, k_{\mathcal{L}}) \in \mathbb{N}^{\mathcal{L}}$ that describes the number of neurons in the first, second, ..., $\mathcal{L}$-th hidden layer. A multilayer feedforward neural network with network architecture $(\mathcal{L}, k)$ and ReLU activation function σ is a real-valued function defined on $\mathbb{R}^d$ of the form

$$f(x) = \sum_{i=1}^{k_{\mathcal{L}}} c_{i,1}^{(\mathcal{L})} \sigma\left(f_i^{(\mathcal{L}-1)}(x)\right) + c_{1,0}^{(\mathcal{L})}, \tag{30}$$

for some $c_{1,0}^{(\mathcal{L})}, \ldots, c_{1,k_{\mathcal{L}}}^{(\mathcal{L})} \in \mathbb{R}$ and for $f_i^{(\mathcal{L})}$'s recursively defined by

$$f_i^{(s)}(x) = \sigma\left(\sum_{j=1}^{k_{s-1}} c_{i,j}^{(s-1)} f_j^{(s-1)}(x) + c_{i,0}^{(s-1)}\right), \tag{31}$$
$$\text{for some } c_{i,0}^{(s-1)}, \ldots, c_{i,k_{s-1}}^{(s-1)} \in \mathbb{R}, s \in \{2, \ldots, L\},$$

and

$$f_i^{(1)}(x) = \sigma\left(\sum_{j=1}^{d} c_{i,j}^{(0)} x^{(j)} + c_{i,0}^{(0)}\right), \tag{32}$$

for some $c_{i,0}^{(0)}, \ldots, c_{i,d}^{(0)} \in \mathbb{R}$. The space of neural networks with $\mathcal{L}$ hidden layers and r neurons per layer is defined by

$$\mathcal{F}(\mathcal{L}, r) = \{f : f \text{ is of the form (30) with } k_1 = k_2 = \ldots = k_{\mathcal{L}} = r\}. \tag{33}$$

In order to judge the quality of such estimates theoretically, usually the rate of convergence of the L_2-error is considered. It is well-known, that smoothness assumptions on the regression function are necessary in order to derive non-trivial results on the rate of convergence (see, e.g., Theorem 7.2 and Problem 7.2 in Devroye et al. (2013) and Section 3 in Devroye and Wagner (1980)). For that purpose, we introduce the following definition of (p, C)-smoothness.

Definition 8 *Let $p = q + s$ for some $q \in \mathbb{N}_0$ and $0 < s \leq 1$. A function $m : \mathbb{R}^d \to \mathbb{R}$ is called (p, C)-smooth, if for every $a = (\alpha_1, \ldots, \alpha_d) \in \mathbb{N}_0^d$ with $\sum_{j=1}^d \alpha_j = q$ the partial derivative $\partial^q m/(\partial x_1^{\alpha_1} \ldots \partial x_d^{\alpha_d})$ exists and satisfies*

$$\left|\frac{\partial^q m}{\partial x_1^{\alpha_1} \ldots \partial x_d^{\alpha_d}}(x) - \frac{\partial^q m}{\partial x_1^{\alpha_1} \ldots \partial x_d^{\alpha_d}}(z)\right| \leq C\|x - z\|^s$$

for all $x, z \in \mathbb{R}^d$, where $\|\cdot\|$ denotes the Euclidean norm.

Stone (1982) showed that the optimal minimax rate of convergence in nonparametric regression for (p,C)-smooth functions is $n^{-2p/(2p+d)}$. This rate suffers from a characteristic feature in case of high-dimensional functions: If d is relatively large compared to p, then this rate of convergence can be extremely slow (so-called curse of dimensionality). As was shown in Stone (1985, 1994) it is possible to circumvent this curse of dimensionality by imposing structural assumptions like hierarchical composition on the regression function.

It was shown that for such models least squares estimators based on suitably defined multilayer neural networks (in which the number of hidden layers depends on the level of the generalized interaction model) achieve the rate of convergence $n^{-2p/(2p+d^*)}$ (up to some logarithmic factor) in case $p \leq 1$, and $d^* < d$. Bauer and Kohler (2019) showed that this result even holds for provided the squashing function is suitably chosen. Here slightly more general function spaces, which fulfill some composition assumption, were studied.

Thus, as mentioned above, the only possible way to avoid the so-called curse of dimensionality is to restrict the underlying function class. We therefore consider functions for f^*, which fulfill the following definition:

Definition 9 *Let $d \in \mathbb{N}$ and $m : \mathbb{R}^d \to \mathbb{R}$ and let $\mathcal{P}$ be a subset of $(0,\infty) \times \mathbb{N}$.*

a) We say that m satisfies a hierarchical composition model of level 0 with order and smoothness constraint $\mathcal{P}$, if there exists a $K \in \{1,\dots,d\}$ such that

$$m(\mathbf{x}) \;=\; x^{(K)} \textit{ for all } \mathbf{x} = (x^{(1)},\dots,x^{(d)})^\top \in \mathbb{R}^d.$$

b) We say that m satisfies a hierarchical composition model of level $l+1$ with order and smoothness constraint P, if there exist $(p,K) \in \mathcal{P}$, $C > 0$, $g : \mathbb{R}^K \to \mathbb{R}$, and $f_1,\dots,f_K : \mathbb{R}^d \to \mathbb{R}$, such that g is (p,C)-smooth, $f_1,\dots,f_K$ satisfy a hierarchical composition model of level l with order and smoothness constraint $\mathcal{P}$ and

$$m(\mathbf{x}) \;=\; g(f_1(\mathbf{x}),\dots,f_K(\mathbf{x})) \textit{ for all } \mathbf{x} \in \mathbb{R}^d.$$

For $l = 1$ and some order and smoothness constraint $\mathcal{P} \subseteq (0,\infty) \times \mathbb{N}$ our space of hierarchical composition models becomes

$$\begin{aligned}\mathcal{H}(1,\mathcal{P}) \;=\; \Big\{ & h : \mathbb{R}^d \to \mathbb{R} : h(\mathbf{x}) \;=\; g(\pi^{(1)}(\mathbf{x}),\dots,\pi^{(K)}(\mathbf{x})), \text{where} \\ & g : \mathbb{R}^K \to \mathbb{R} \text{ is } (p,C)\text{-smooth for some } (p,K) \in P \text{ and } \pi : \{1,\dots,K\} \to \{1,\dots,d\} \Big\}.\end{aligned}$$

For $l > 1$, we recursively define

$$\begin{aligned}\mathcal{H}(l,P) \;=\; \Big\{ & h : \mathbb{R}^d \to \mathbb{R} : h(\mathbf{x}) \;=\; g(f_1(\mathbf{x}),\dots,f_K(\mathbf{x})), \text{where} \\ & g : \mathbb{R}^K \to \mathbb{R} \text{ is } (p,C)\text{-smooth for some } (p,K) \in \mathcal{P} \text{ and } f_i \in \mathcal{H}(l-1,P) \Big\}.\end{aligned}$$

Thus, we add the following assumption.

Assumption 8 • *The f^* is a hierarchical composition model that satisfying Definition 9.*

Then, our estimator is defined as

$$\hat{f} \;=\; \arg\min_{f \in F(\mathcal{L},r)} \left\{ \frac{1}{n} \sum_{i=1}^{n} \frac{1}{m_i} \sum_{j=1}^{m_i} \rho(y_{ij} - f(x_{ij})) \right\}. \tag{34}$$

7.1 Results For Neural Networks

Theorem 10 (Neural Network) *Consider observations* $\{(x_{ij}, y_{ij})\}_{i=1,j=1}^{n,m_i}$ *satisfying Assumptions 1 to 4 and Assumption 8. And let* $f^*(x_{ij})$ *be the sequence of* τ *quantiles of* y_{ij}*. The* f^* *is not necessarily in* $\mathcal{F}$*. Based on these observations, consider the estimator* $\hat{f} \in F(\mathcal{L}, r)$ *be given by* (34)*. It holds that*

$$\begin{aligned}\left\|\hat{f} - f^*\right\|_2^2 = O_p\Bigg(&\max\left\{B^2, B\right\} \cdot \log^5(nm) \cdot \max_{(p,K)\in\mathcal{P}} (nm)^{\frac{-2p}{2p+K}} \\ &+ \max\{B, 1\} \cdot \mathcal{C} \cdot \max_{(p,K)\in\mathcal{P}} (nm)^{\frac{-2p}{2p+K}} + \max\{B, 1\} \frac{s \cdot poly(\log n)\log(T_n)}{n}\Bigg).\end{aligned} \tag{35}$$

Theorem 10 then indicates that, for temporal-spatial data and in terms of the $\mathcal{L}_2$ loss defined in (63), the fully connected neural network estimator presented in (34) which satisfies a hierarchical composition model with smoothness and order constraint $\mathcal{P}$ attains a convergence rate of $\left\{\max\left\{B^2, B\right\} \cdot \mathrm{poly}(\log(nm)) + \max\{B, 1\} \cdot \mathcal{C}\right\} \cdot \max_{(p,K)\in\mathcal{P}}(nm)^{\frac{-2p}{2p+K}} + \max\{B, 1\}\log(T_n)s \cdot n^{-1}$. This rate does not depend on d and which does therefore circumvent the so-called curse of dimensionality. The first term in our rate is the same as the rate $\max_{(p,K)\in\mathcal{P}}(nm)^{\frac{-2p}{2p+K}}$ in Kohler and Langer (2019) up to some logarithm factors and $\max\left\{B^2, B\right\}$, the difference is the second term approximation error and the last term, which comes from $f^* \notin \mathcal{F}$ and quantile temporal data. Our analysis first derives the rate for Quantile Temporal-Spatial Data with Neural Network as function class of estimators. We also have the result in terms of the Huber-type loss defined in (63) shown in (240).

8 Algorithms

This section provides algorithms to support our theoretical results. We start by building algorithms to evaluate the performance of the penalized estimators for Trend Filtering, High Dimensional Quantile Regression and K-NN-FL, respectively, for both settings $d = 1$ and $d > 1$. We develop an alternating direction method of multipliers (ADMM) specific to our objective function. See Boyd et al. (2011) for more details on ADMM algorithms.

8.1 Quantile Trend Filtering

As for quantile trend filtering, we consider the penalized estimator:

$$\hat{f} = \arg\min_{f\in F_k(1)} \frac{1}{n}\sum_{i=1}^{n}\frac{1}{m_i}\sum_{j=1}^{m_i}\rho_\tau(y_{ij} - f(x_{ij})) + \lambda TV^{(k)}(f). \tag{36}$$

We let $\theta_{ij} := f(x_{ij})$, and use $\left\|D^{(x,k+1)}(\theta)\right\|_1$ as a discrete operator representing $TV^{(k)}(\cdot)$ on unevenly spaced inputs. As in Tibshirani (2014), $D^{(x,k+1)}$ is defined recursively, with

$$D^{(x,1)} = D^{(1)} = \begin{bmatrix} -1 & 1 & 0 & \dots & 0 & 0 \\ 0 & -1 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & -1 & 1 \end{bmatrix} \in \mathbb{R}^{(M-1)\times M}, \tag{37}$$

$$D^{(x,k+1)} = D^{(1)} \cdot \mathrm{diag}\left(\frac{k}{\tilde{x}_{k+1}-\tilde{x}_1}, \frac{k}{\tilde{x}_{k+2}-\tilde{x}_2}, \ldots, \frac{k}{\tilde{x}_M - x_{M-k}}\right) \cdot D^{(x,k)}, \tag{38}$$

where $M = \sum_{i=1}^n m_i$, and $\tilde{x}_k (k = 1, ..., M)$ are the sorted x_{ij}.

So, we solve (39):

$$\hat{\theta} = \arg \min_{\theta \in \mathbb{R}^{\sum_{i=1}^n m_i}} \frac{1}{n}\sum_{i=1}^n \frac{1}{m_i}\sum_{j=1}^{m_i} \rho_\tau(y_{ij}-\theta_{ij}) + \lambda \left\| D^{(x,k+1)}(\theta) \right\|_1. \tag{39}$$

The alternating directions method of multipliers (ADMM) algorithm (Boyd et al., 2011) is a powerful tool for solving constrained optimization problems. We describe our approach in Algorithm 1. We first reformulate the optimization problem (39) as

$$\begin{aligned} \underset{\theta\in\mathbb{R}^{\sum_{i=1}^n m_i}, z\in\mathbb{R}^{\sum_{i=1}^n m_i}}{\text{minimize}} \quad & \frac{1}{n}\sum_{i=1}^n \frac{1}{m_i}\sum_{j=1}^{m_i} \rho_\tau(y_{ij}-\theta_{ij}) + \lambda \left\| D^{(x,k+1)}(z) \right\|_1 \\ \text{subject to} \quad & z = \theta, \end{aligned} \tag{40}$$

and the augmented Lagrangian can then be written as

$$L_P(\theta, z, u) = \frac{1}{n}\sum_{i=1}^n \frac{1}{m_i}\sum_{j=1}^{m_i} \rho_\tau(y_{ij}-\theta_{ij}) + \lambda \left\| D^{(x,k+1)}(z) \right\|_1 + \frac{P}{2}\|\theta - z + u\|^2, \tag{41}$$

where P is the penalty parameter that controls step size in the update. Thus we can solve (40) iteratively:

$$\textit{θ-minimization} \quad \theta = \arg \min_{\theta\in\mathbb{R}^{\sum_{i=1}^n m_i}} \left\{ \frac{1}{n}\sum_{i=1}^n \frac{1}{m_i}\sum_{j=1}^{m_i} \rho_\tau(y_{ij}-\theta_{ij}) + \frac{P}{2}\|\theta - z + u\|^2 \right\}, \tag{42}$$

$$\textit{z-minimization} \quad z = \arg \min_{z\in\mathbb{R}^{\sum_{i=1}^n m_i}} \left\{ \frac{1}{2}\|\theta + u - z\|^2 + \frac{\lambda}{P}\|D^{(x,k+1)}(z)\|_1 \right\}, \tag{43}$$

$$\textit{dual update} \quad u^{(t)} = u^{(t-1)} + \theta^{(t)} - z^{(t)}. \tag{44}$$

The θ-minimization (42) is separable and can be solved coordinate-wise. Define

$$q_{ij}^{(t-1)} = z_{ij}^{(t-1)} - u_{ij}^{(t-1)} \qquad \text{and} \qquad \gamma_i = \frac{1}{Pnm_i}.$$

The check-loss proximal update is

$$\theta_{ij}^{(t)} = \arg\min_{\theta_{ij}} \left\{ \frac{1}{n}\frac{1}{m_i}\rho_\tau(y_{ij}-\theta_{ij}) + \frac{P}{2}\left(\theta_{ij} - z_{ij}^{(t-1)} + u_{ij}^{(t-1)}\right)^2 \right\} \tag{45}$$

$$= \begin{cases} q_{ij}^{(t-1)} + \tau\gamma_i, & \text{if } y_{ij} > q_{ij}^{(t-1)} + \tau\gamma_i, \\ q_{ij}^{(t-1)} + (\tau-1)\gamma_i, & \text{if } y_{ij} < q_{ij}^{(t-1)} + (\tau-1)\gamma_i, \\ y_{ij}, & \text{otherwise.} \end{cases} \tag{46}$$

The z-minimization problem (43) is a univariate trend filtering problem, on unevenly spaced inputs. The solution to this problem has been well studied, and we solve it using methods described in Ramdas and Tibshirani (2016), specifically the `trendfilter` function in the `glmgen` package.

Lastly, the dual update is achieved via a simple gradient ascent:

$$u_{ij}^{(t)} = u_{ij}^{(t-1)} + \theta_{ij}^{(t)} - z_{ij}^{(t)}. \tag{47}$$

The entire procedure is presented in Algorithm 1. In practice, we can simply choose the penalty parameter P to be $\frac{1}{2}$. We require the procedure to stop if it reaches the maximum iterations or every coefficient $(\theta^{(t)} - \theta^{(t-1)})^2$ is within a tolerance κ, to which we set 10^{-3} in our computation.

Figure 1: Alternating Directions Method of Multipliers for quantile spatial temporal trend filtering in (36)

1: **Input:** responses $y_{ij} \in \mathbb{R}$ and input points $x_{ij} \in \mathbb{R}$, $i = 1 \dots n$, and $j = 1 \dots m_i$, penalty parameter $\lambda \in \mathbb{R}$, quantile $\tau \in (0,1)$, maximum iteration N_{iter}, tolerance κ.
2: **Output:** $\theta \in \mathbb{R}^{\sum_{i=1}^n m_i}$
3: **Set:** $t = 0$ and Initialize $\theta_{ij}^{(0)} = y_{ij}$, $z_{ij}^{(0)} = y_{ij}$, $u_{ij}^{(0)} = 0$ for all $i = 1, \dots, n$ and $j = 1, \dots, m_i$.
4: **for** $t = 1, 2, 3, \dots$ (until converge) $\|\theta^{(t)} - \theta^{(t-1)}\|_2 \leq \kappa$ or the procedure reaches N_{iter} **do**
5: (i) For $i = 1, 2, \cdots, n$, $j = 1, \cdots, m_i$,
update $\theta_{ij}^{(t)} \leftarrow \arg\min_{\theta_{ij}} \left\{ \frac{1}{n}\frac{1}{m_i}\rho_\tau(y_{ij} - \theta_{ij}) + \frac{P}{2}\left(\theta_{ij} - z_{ij}^{(t-1)} + u_{ij}^{(t-1)}\right)^2 \right\}$.
6: (ii) Update $z^{(t)} \leftarrow \arg\min_z \left\{ \frac{1}{2}\|(\theta^{(t)} + u^{(t-1)}) - z\|_2^2 + \frac{\lambda}{P}\left\|D^{(x,k+1)}(z)\right\|_1 \right\}$.
7: (iii) Update $u^{(t)} \leftarrow u^{(t-1)} + \theta^{(t)} - z^{(t)}$
8: **end for**
9: **Return** $\hat{\theta}^{(t)}$ at convergence.

8.2 Quantile K-NN Fused Lasso

For Quantile K-NN Fused Lasso, we consider the penalized version estimator:

$$\hat{\theta}_n = \arg\min_{\theta \in \mathbb{R}^{\sum_{i=1}^n m_i}} \frac{1}{n}\sum_{i=1}^n \frac{1}{m_i}\sum_{j=1}^{m_i} \rho_\tau(y_{ij} - \theta_{ij}) + \lambda\|\nabla_{G_k}\theta\|_1. \tag{48}$$

.

Here, ∇_{G_k} is the graph adjacency matrix, and $\|\nabla_{G_k}\theta\|_1$ is the total variation as in (21).

To solve (48), we use a very similar ADMM as in Algorithm 1. We provide a few details, and describe our approach in Algorithm 2. This approach is very similar to the ADMM algorithm in Ye and Padilla (2021), as well, though their method does not directly generalize to temporal-spatial data.

As before, we reformulate the optimization problem (48) into the augmented Lagrangian:

$$L_P(\theta, z, u) = \frac{1}{n}\sum_{i=1}^n \frac{1}{m_i}\sum_{j=1}^{m_i} \rho_\tau(y_{ij} - \theta_{ij}) + \lambda\|\nabla_{G_k} z\|_1 + \frac{P}{2}\|\theta - z + u\|^2, \tag{49}$$

where P is the penalty parameter that controls step size in the update.

We again solve (49) iteratively:

$$\theta\text{-minimization} \quad \theta = \arg \min_{\theta \in \mathbb{R}^{\sum_{i=1}^n m_i}} \left\{ \frac{1}{n} \sum_{i=1}^{n} \frac{1}{m_i} \sum_{j=1}^{m_i} \rho_\tau(y_{ij} - \theta_{ij}) + \frac{P}{2} \|\theta - z + u\|^2 \right\}, \tag{50}$$

$$z\text{-minimization} \quad z = \arg \min_{z \in \mathbb{R}^{\sum_{i=1}^n m_i}} \left\{ \frac{1}{2} \|\theta + u - z\|^2 + \frac{\lambda}{P} \|\nabla_{G_k} z\|_1 \right\}, \tag{51}$$

$$\text{dual update} \quad u^{(t)} = u^{(t-1)} + \theta^{(t)} - z^{(t)}. \tag{52}$$

The θ-minimization (50) uses the same coordinate-wise proximal update as above, and the dual variable uses the additive update in (52). The z-minimization problem (51) is a generalized lasso problem that can be solved with the parametric max-flow algorithm of Chambolle and Darbo (2009).

The entire procedure is presented in Algorithm 2. In practice, we can simply choose the penalty parameter P to be $\frac{1}{2}$. We require the procedure to stop if it reaches the maximum iterations or every coefficient $(\theta^{(t)} - \theta^{(t-1)})^2$ is within a tolerance κ, to which we set 10^{-3} in our computation.

Figure 2: Alternating direction method of multipliers for quantile K-NN fused lasso in (48)

1: **Input:** responses $y_{ij} \in \mathbb{R}$ and input points $x_{ij} \in \mathbb{R}^d$, $i = 1 \ldots n$, and $j = 1 \ldots m_i$, penalty parameter $\lambda \in \mathbb{R}$, quantile $\tau \in (0, 1)$, maximum iteration N_{iter}, tolerance κ.
2: **Output:** $\theta \in \mathbb{R}^{\sum_{i=1}^n m_i}$
3: **Set:** $t = 0$ and Initialize $\theta_{ij}^{(0)} = y_{ij}$, $z_{ij}^{(0)} = y_{ij}$, $u_{ij}^{(0)} = 0$ for all $i = 1, \ldots, n$ and $j = 1, \ldots, m_i$.
4: Compute the K-NN graph incidence matrix ∇_{G_k} from x_{ij}.
5: **for** $t = 1, 2, 3, \ldots$ (until converge) $\|\theta^{(t)} - \theta^{(t-1)}\|_2 \leq \kappa$ or the procedure reaches N_{iter} **do**
6: (i) For $i = 1, 2, \cdots, n$, $j = 1, \cdots, m_i$,
update $\theta_{ij}^{(t)} \leftarrow \arg\min_{\theta_{ij}} \left\{ \frac{1}{n} \frac{1}{m_i} \rho_\tau(y_{ij} - \theta_{ij}) + \frac{P}{2} \left(\theta_{ij} - z_{ij}^{(t-1)} + u_{ij}^{(t-1)} \right)^2 \right\}$.
7: (ii) Update $z^{(t)} \leftarrow \arg\min_z \left\{ \frac{1}{2} \|(\theta^{(t)} + u^{(t-1)}) - z\|_2^2 + \frac{\lambda}{P} \|\nabla_{G_k} z\|_1 \right\}$.
8: (iii) Update $u^{(t)} \leftarrow u^{(t-1)} + \theta^{(t)} - z^{(t)}$
9: **end for**
10: **Return** $\hat{\theta}^{(t)}$ at convergence.

9 Experiments

This section provides numerical evidence to support our theoretical results. We start by conducting empirical experiments to evaluate the performance of the penalized estimators for Trend Filtering, High Dimensional Quantile Regression and K-NN-FL, respectively, for both settings $d = 1$ and $d > 1$.

To establish a comparison, we include various competing methods as benchmarks.

9.1 Simulated data

To conduct our simulations, we consider various generative models, referred to as scenarios, for both settings $d = 1$ and $d > 1$. We describe these scenarios in the following subsections. Next, we outline the common aspects of our simulations for both the univariate and multivariate settings.

For the $\tau = 0.5$ setting, the performance of each method is evaluated by the MSE obtained when using the true quantile function. We then report the average MSE over 50 Monte Carlo simulations.

For other τ, we evaluate the performance using the test set coverage, averaged over 5 train-test splits. In both cases, we then report the average metric over 50 Monte Carlo simulations.

For each setting below, we vary n over $\{100, 200, 400\}$. For each fixed value of n, we then vary m_{mult} over $\{0.5, 1, 2\}$ to obtain different sets of values for $\{m_i\}_{i=1}^n$.

Specifically, we consider:

- $m_1 = \ldots = m_{n/4} = 16m_{mult}$,
- $m_{n/4+1} = \ldots = m_{n/2} = 24m_{mult}$,
- $m_{n/2+1} = \ldots = m_{3n/4} = 20m_{mult}$,
- $m_{3n/4+1} = \ldots = m_n = 10m_{mult}$.

The spatial noise is generated as

$$\delta_i(x) \ = \ 0.5\delta_{i-1}(x) + \sum_{t=1}^{25} t^{-1} b_{i,t} h_t(x), \tag{53}$$

where

$$h_t(x) \ = \ \left(\frac{1}{\sqrt{2}}\pi \sin(tx(j))\right)_{j=1}^{25}, \tag{54}$$

are basis functions and $b_{i,t} \sim G_i$ for some distributions G_i. The measurement error is generated as follows: For each $i \in [n]$, the vector $\tilde{e}_i \in \mathbb{R}^M$ is generated recursively as,

$$\tilde{e}_i = 0.3\tilde{e}_{i-1} + \xi_i, \tag{55}$$

where $\{\xi_i\}_{i=1}^n$ are independent errors with $\xi_i \sim F_i$ for some distributions F_i. To form the measurement errors, for each $i \in [n]$, we take the first m_i entries of the vector $\tilde{e}_i$ and denote the resulting vector as e_i, since m_i are varied across i. We note, also, that when $\tau \neq 0.5$, a normalization factor is included for both the spatial noise and measurement error to ensure the variance remains constant across i.

We observe the noisy temporal-spatial data $\{y_{ij}\}_{i=1,j=1}^{n,m_i}$ at design points $\{x_{ij}\}_{j=1}^{m_1}$ evenly sampled from $\text{Unif}([0,1]^d)$ as follows. First we generate $\{x_{1,j}\}_{j=1}^{m_1} \sim \text{Unif}([0,1]^d)$, then for any $1 < i \leq n$,

$$x_{ij} = \begin{cases} x_{i-1,j}, & \text{with probability } \phi = 0.1, \\ x_{ij} \sim \text{Unif}([0,1]^d), & \text{otherwise}, \end{cases} \tag{56}$$

for all $j = 1, \ldots, m_i$.

9.2 Univariate Study

For quantile trend filtering, we consider tuning parameters λ such that $\log_{10}(\lambda)$ is in a grid of 50 evenly spaced points between -7 and 1, where $a, b \in \mathbb{R}$. We then perform Algorithm 1 on λ/M.

For selecting the best λ, we utilize Bayesian Information Criteria (BIC; citeschwarz1978estimating). The BIC for quantile regression citeyu2001bayesian can be computed as

$$\text{BIC}(\tau) \ = \ \frac{2}{\sigma}\sum_{i=1}^{n} \rho_\tau(y_i - \hat{\theta}_i) + \nu \log n,$$

where ν denotes the degrees of freedom of the estimator, and $\sigma > 0$ can be empirically chosen as $\sigma = 1 - |1 - 2\tau|$. We estimate ν by the $d + k + 1$, where d is the number of elements of the k-th order difference operator larger than 0.05 in absolute value.

As benchmark methods, we consider the usual (mean regression) penalized trend filtering estimator of order $k = 1$ and $k = 2$ denoted as TF1 and TF2 respectively, and quantile cubic smoothing splines, denoted as QSS. Note that TF1 and TF2 only provide estimates for $\tau = 0.5$.

Mean regression trend filtering is implemented using the `trendfilter` function in the `glmgen` package, and the tuning parameter selected via BIC.

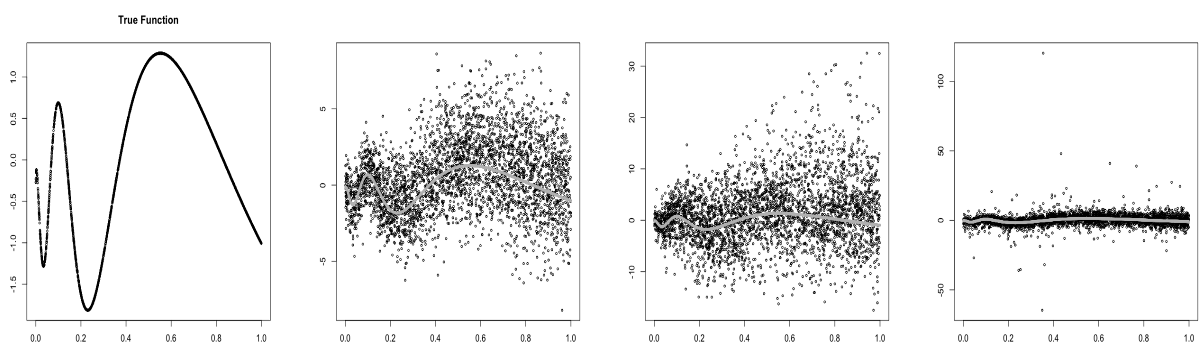
Quantile smoothing splines is implemented with a similar method as Algorithm 1, with step (ii) replaced by the `smooth.spline` function in the `fields` package. A full derivation of this Algorithm is presented in the appendix. The tuning parameter is selected via K-fold cross validation.

For all the following scenarios, we take $f^* = f - \int_0^1 f(x)\,dx$, where f is the following Doppler-type function with spatially inhomogeneous smoothness:

$$f(x) \;=\; 2x^{\frac{1}{4}} \sin\left(\frac{2\pi}{(x+0.1)^{\frac{1}{2}}}\right)$$

1. **Scenario 1**: We take $G_i \sim N(0,1)$ and $F_i \sim N(0, 0.5)$ This scenario may serve as a baseline.
2. **Scenario 2**: We take $G_i \sim \text{laplace}(0,2)$ and $F_i \sim t(3)$. Here $t(3)$ denotes the t-distribution with 3 degrees of freedom.
3. **Scenario 3**: We take $G_i \sim N(0,1)$, and $F_i \sim N(0,1)$. However, here we consider $\tau = 0.2$ and $\tau = 0.8$.

Other scenarios, with other underlying functions, are demonstrated in the Appendix.



(a) True function **(b)** Scenario 1 Noise **(c)** Scenario 2 Noise **(d)** Scenario 3 Noise

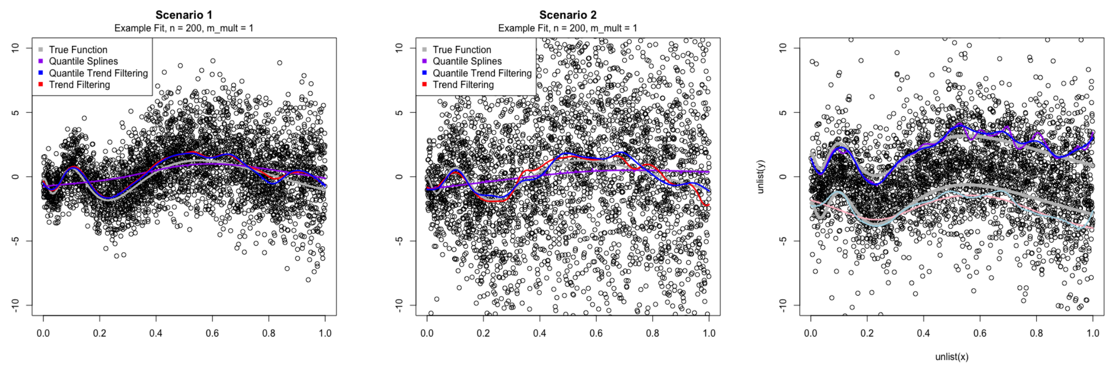
Figure 3: Panel a is the true function f^*. Panels b, c, and d are example plots of the noisy data with $n = 200$ and $m\_mult = 1$.

9.3 Multivariate Study

For all the following scenarios, we consider $f^* = f - \int_0^1 f(x)dx$ for the following piecewise constant function:

$$f_0(x) = \begin{cases} 1 & \text{if } \|x - \frac{1}{4}\mathbf{1}_d\|_2 < \|x - \frac{3}{4}\mathbf{1}_d\|_2, \\ -1 & \text{otherwise,} \end{cases}$$

1. **Scenario 1**: We take $G_i \sim N(0,1)$, $F_i \sim N(0, 0.5)$, and $d = 2$. This scenario may serve as a baseline.



(a) Scenario 1 Example Fit　**(b)** Scenario 2 Example Fit　**(c)** Scenario 3 Example Fit

Figure 4: Panels a, b, and c
are examples of the fitted curves of TF, QTF, and QSS with $n = 200$ and $m\_mult = 1$.

2. **Scenario 2**: We take $G_i \sim t(2)$, $F_i \sim \text{cauchy}(0, 1)$, and $d = 2$. $t(2)$ denotes the t-distribution with 2 degrees of freedom.

3. **Scenario 3**: We take $G_i \sim N(0, 1)$, $F_i \sim N(0, 1)$, and consider $\tau = 0.2$ and $\tau = 0.8$ $d = 2$.

4. **Scenario 4**: We take $G_i \sim N(0, 2)$, and $F_i \sim t(3)$. However, here we consider $d = 5$ and $d = 10$.

9.4 Non-Convex Study

We further study a non-convex setting by replacing the convex estimators in the multivariate study with neural networks. We consider two neural-network baselines with identical architectures, differing only in the training loss: (i) a squared-error network (SqErr Network) trained by minimizing the mean squared error loss, and (ii) a quantile network (Quantile Network) trained by minimizing the pinball loss for a set of target quantile levels $\mathcal{Q} = \{0.1, 0.3, 0.5, 0.7, 0.9\}$.

We consider three spatial-temporal scenarios (Scenarios 6–8) in the multivariate setting with $d = 2$. All three scenarios follow the common sampling scheme described in Section 9.1 (Simulated data), including the heterogeneous per-time sample sizes $\{m_i\}_{i=1}^n$, the temporally dependent design points $\{x_{ij}\}$, and the spatio-temporal noise components $\delta_i(\cdot)$ and e_i. Specifically, for each $i \in [n]$ and $j \in [m_i]$, the response is generated as

$$y_{ij} \ = \ \beta(x_{ij}) \ + \ \delta_i(x_{ij}) \ + \ e_{ij}, \tag{57}$$

where the signal $\beta(\cdot)$ is the piecewise-constant function

$$\beta(x) = \begin{cases} 1, & \text{if } \left\|x - \frac{1}{4}\mathbf{1}_d\right\|_2 < \left\|x - \frac{3}{4}\mathbf{1}_d\right\|_2, \\ -1, & \text{otherwise.} \end{cases} \tag{58}$$

Design points and flattening. For a fixed temporal length n, we first generate a padded array $\{x_{i\ell}\}_{\ell=1}^M$ with $M = \max_i m_i$ by sampling $x_{1\ell} \overset{\text{i.i.d.}}{\sim} \text{Unif}([0, 1]^d)$ and then, for $i \geq 2$, copying each previous location with probability $\phi = 0.1$ (as in (56) in Section 9.1); we only keep the first m_i locations at time i, i.e., $\{x_{ij}\}_{j=1}^{m_i}$. The training set used by neural networks is obtained by flattening the panel data $\{(x_{ij}, y_{ij})\}_{i=1,j=1}^{n,m_i}$ into flattened input–response pairs:

$$X_{\text{train}} = \left[x_{11}; \ldots; x_{1m_1}; x_{21}; \ldots; x_{nm_n}\right] \in \mathbb{R}^{(\sum_i m_i) \times d}, \qquad y_{\text{train}} = \left[y_{11}, \ldots, y_{nm_n}\right]^\top \in \mathbb{R}^{\sum_i m_i}.$$

The test set is formed in the same way using an independently generated panel of design points.

Spatial noise. The spatio-temporal component $\delta_i(\cdot)$ follows the recursive construction in (53):

$$\delta_i(x) \;=\; 0.5\,\delta_{i-1}(x) \;+\; \sum_{t=1}^{25} t^{-1} b_{i,t}\, h_t(x), \qquad b_{i,t} \sim G_i, \tag{59}$$

where $\{h_t\}_{t=1}^{25}$ are sine-basis functions as in (54). In implementation, for each location $x = (x(1), \dots, x(d)) \in [0,1]^d$, we use

$$h_t(x) \;=\; \prod_{r=1}^{d} \left(\frac{\pi}{\sqrt{2}} \sin\big(t\, x(r)\big) \right), \qquad t = 1, \dots, 25. \tag{60}$$

Measurement error. The measurement error is generated via the AR-type recursion in (55):

$$\tilde{e}_i \;=\; 0.3\,\tilde{e}_{i-1} \;+\; \xi_i, \qquad \xi_i \sim F_i, \tag{61}$$

where $\tilde{e}_i \in \mathbb{R}^M$, and we set e_{ij} to be the jth entry of $\tilde{e}_i$ for $j \le m_i$.

Scenario specifications. The scenarios differ only through the innovation distributions G_i and F_i and the quantile level τ:

- **Scenario 6 (light-tailed):** $b_{i,t} \overset{\text{i.i.d.}}{\sim} N(0,1)$ and $\xi_i(\ell) \overset{\text{i.i.d.}}{\sim} N(0, 0.5^2)$.
- **Scenario 7 (heavy-tailed):** $b_{i,t} \overset{\text{i.i.d.}}{\sim} t(5)$ and $\xi_i(\ell) \overset{\text{i.i.d.}}{\sim} t(5)$, leading to heavier tails in both $\delta_i(\cdot)$ and e_i.
- **Scenario 8 :** we consider $\tau = 0.5$ and define the target quantile signal as $\beta_\tau(x) = f_0(x) + \Phi^{-1}(\tau)$. The spatial-noise coefficients follow an AR-type recursion: for $t = 1, \dots, 25$, let $z_{i,t} \overset{\text{i.i.d.}}{\sim} N(0,1)$ and set

  $$b_{i,t} = \frac{0.5\, b_{i-1,t} + t^{-1} z_{i,t}}{s_\delta}, \qquad s_\delta = \sqrt{0.5^2 + 1^2},$$

  (with $s_\delta = 1$ at $i = 1$), and define $\delta_i(x) = \sum_{t=1}^{25} b_{i,t} h_t(x)$. The measurement error is generated by $u_{i,\ell} \overset{\text{i.i.d.}}{\sim} t(3)$ and

  $$\tilde{e}_i(\ell) = \frac{0.3\, \tilde{e}_{i-1}(\ell) + u_{i,\ell}}{s_e}, \qquad s_e = \sqrt{0.3^2 + 1^2},$$

  (with $s_e = 1$ at $i = 1$), and we take $e_{ij} = \tilde{e}_i(j)$ for $j \le m_i$. Finally, $y_{ij} = \beta_\tau(x_{ij}) + \delta_i(x_{ij}) + e_{ij}$.

Evaluation metric. For each quantile level $q \in \mathcal{Q}$, we evaluate the squared error between the learned quantile function and the oracle quantile function on a large test set, and report the average over Monte Carlo trials. We vary the temporal length $n \in \{100, 500, 1000\}$ (with the corresponding $\{m_i\}$ pattern described above).

Table 1: Non-convex study: test-set MSE between predicted and oracle quantile functions for five quantile levels $\mathcal{Q} = \{0.1, 0.3, 0.5, 0.7, 0.9\}$. We compare a squared-error network (SqErr) and a quantile network (Quantile). The best performance is **bolded**.

Scenario	Method (n)	$q = 0.1$	$q = 0.3$	$q = 0.5$	$q = 0.7$	$q = 0.9$
Scenario 6	SqErr (100)	21.717	6.522	3.060	5.422	19.034
	Quantile (100)	**3.702**	**2.486**	**2.457**	**3.097**	**7.141**
	SqErr (500)	23.058	7.071	3.059	4.870	17.690
	Quantile (500)	**2.625**	**2.243**	**2.205**	**2.333**	**2.799**
	SqErr (1000)	23.409	7.220	3.067	4.737	17.355
	Quantile (1000)	**2.516**	**2.251**	**2.150**	**2.136**	**2.308**
Scenario 7	SqErr (100)	37.480	9.810	4.067	7.505	30.190
	Quantile (100)	**4.913**	**3.982**	**4.513**	**5.058**	**7.696**
	SqErr (500)	37.040	9.618	4.031	7.630	30.553
	Quantile (500)	**3.864**	**3.730**	**3.810**	**4.186**	**5.119**
	SqErr (1000)	32.095	7.612	3.893	9.374	35.126
	Quantile (1000)	**4.742**	**3.965**	**3.687**	**3.792**	**4.796**
Scenario 8	SqErr (100)	19.419	5.226	2.750	7.078	26.661
	Quantile (100)	**2.409**	**2.079**	**1.854**	**1.883**	**2.237**
	SqErr (500)	22.925	6.557	2.557	5.264	22.440
	Quantile (500)	**1.803**	**1.631**	**1.644**	**1.667**	**1.805**
	SqErr (1000)	22.870	6.533	2.556	5.287	22.498
	Quantile (1000)	**2.018**	**1.671**	**1.643**	**1.637**	**1.696**

Summary of findings. Table 1 shows that the Quantile Network consistently outperforms the SqErr Network across all three scenarios and all quantile levels. The improvements are particularly pronounced for tail quantiles ($q = 0.1$ and $q = 0.9$), where minimizing squared error is misaligned with the quantile target and heavy-tailed errors exacerbate the discrepancy. In contrast, the pinball loss directly targets the conditional quantiles and yields substantially smaller errors, especially in Scenarios 6 and 8.

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Appendix A. Proof of Theorem 1

A.1 Definitions and Notation

To complete our proofs for the theorems, we provide important definitions.

Definition 11 *We introduce the Huber-type Huber (1992) loss. Let Δ^2_{nm} be a mapping of functions to real numbers, the Huber-type loss function is elucidated as*

$$\Delta^2_{nm}(f) \;=\; \frac{1}{n}\sum_{i=1}^{n}\frac{1}{m_i}\sum_{j=1}^{m_i}\min\{|f(x_{ij})|\,,(f(x_{ij}))^2\}, \tag{62}$$

which, modulo constants, corresponds to a Huber loss, as delineated in Huber (1992). The Huber loss serves as a natural lower bound for the quantile population loss, which provides the primary motivation for focusing on it. Let Q denote the design distribution on $[0,1]^d$. The population Huber loss is

$$\Delta^2(f) \;=\; \int_{[0,1]^d}\min\{|f(t)|\,,|f(t)|^2\}\,dQ(t). \tag{63}$$

We derive the rate of convergence by using the Huber-type loss and L_2 loss.

Definition 12 *Let K be a function class with Rademacher width well defined. We define its Empirical Rademacher width as*

$$R_{mn}(K) \;=\; \mathbb{E}_{\xi|\{x_{ij}\}}\left[\sup_{f\in K}\frac{1}{\sum_{i=1}^{n}m_i}\sum_{i=1}^{n}\sum_{j=1}^{m_i}\xi_{ij}f(x_{ij})\right].$$

Definition 13 *We define its Rademacher width as*

$$R(K) \;=\; \mathbb{E}_x\left[\mathbb{E}_{\xi|\{x_{ij}\}}\left[\sup_{f\in K}\frac{1}{\sum_{i=1}^{n}m_i}\sum_{i=1}^{n}\sum_{j=1}^{m_i}\xi_{ij}f(x_{ij})\right]\right] = \mathbb{E}_{x,\xi}\left[\sup_{f\in K}\frac{1}{\sum_{i=1}^{n}m_i}\sum_{i=1}^{n}\sum_{j=1}^{m_i}\xi_{ij}f(x_{ij})\right].$$

Definition 14 *We define $\hat{Z}_{ij}(f)$ and the empirical loss function $\hat{Z}(f)$ as*

$$\begin{aligned}\hat{Z}_{ij}(f) =&\rho_\tau(y_{ij}-f(x_{ij}))-\rho_\tau(y_{ij}-f^*(x_{ij}))\\ =&\rho_\tau(f^*(x_{ij})+\delta_i(x_{ij})+e_{ij}-f(x_{ij}))-\rho_\tau(\delta_i(x_{ij})+e_{ij}).\end{aligned} \tag{64}$$

$$\begin{aligned}\hat{Z}(f) \;&=\; \frac{1}{n}\sum_{i=1}^{n}\frac{1}{m_i}\sum_{j=1}^{m_i}\hat{Z}_{ij}(f)\\ &=\frac{1}{n}\sum_{i=1}^{n}\frac{1}{m_i}\sum_{j=1}^{m_i}\{\rho_\tau(y_{ij}-f(x_{ij}))-\rho_\tau(y_{ij}-f^*(x_{ij}))\}\\ &=\frac{1}{n}\sum_{i=1}^{n}\frac{1}{m_i}\sum_{j=1}^{m_i}\{\rho_\tau(f^*(x_{ij})+\delta_i(x_{ij})+e_{ij}-f(x_{ij}))-\rho_\tau(\delta_i(x_{ij})+e_{ij})\}.\end{aligned} \tag{65}$$

We also define $Z_{ij}(f)$ and $Z(f)$ as

$$\begin{aligned}Z_{ij}(f) \;:=&\mathbb{E}_{\delta,e|x_{ij}}\left[\rho_\tau(y_{ij}-f(x_{ij}))-\rho_\tau(y_{ij}-f^*(x_{ij}))\right]\\ =&\mathbb{E}_{\delta,e|x_{ij}}\left[\rho_\tau(f^*(x_{ij})+\delta_i(x_{ij})+e_{ij}-f(x_{ij}))-\rho_\tau(\delta_i(x_{ij})+e_{ij})\right].\end{aligned} \tag{66}$$

$$
\begin{aligned}
Z(f) &:= \mathbb{E}_{\delta,e|\{x_{ij}\}}\left[\hat{Z}(f)\right] \\
&= \frac{1}{n}\sum_{i=1}^{n}\frac{1}{m_i}\sum_{j=1}^{m_i}\mathbb{E}_{\delta,e|\{x_{ij}\}}\left[\rho_\tau(y_{ij}-f(x_{ij}))-\rho_\tau(y_{ij}-f^*(x_{ij}))\right] \\
&= \frac{1}{n}\sum_{i=1}^{n}\frac{1}{m_i}\sum_{j=1}^{m_i}\mathbb{E}_{\delta,e|\{x_{ij}\}}\left[\rho_\tau(f^*(x_{ij})+\delta_i(x_{ij})+e_{ij}-f(x_{ij}))-\rho_\tau(\delta_i(x_{ij})+e_{ij})\right],
\end{aligned}
\tag{67}
$$

where we use $\{x_{ij}\}$ as a shorthand for $\{x_{ij}\}_{i=1,j=1}^{n,m_{ij}}$ and $\{\delta_i\}$ as $\{\delta_i\}_{i=1}^{n}$.
We then define $\tilde{Z}_{ij}(f)$ and $\tilde{Z}(f)$ as

$$
\tilde{Z}_{ij}(f) := \mathbb{E}_{e|\delta_i,x_{ij}}\left[\hat{Z}_{ij}(f)\right].
$$

$$
\tilde{Z}(f) := \mathbb{E}_{e|\{\delta_i\},\{x_{ij}\}}\left[\hat{Z}(f)\right]. \tag{68}
$$

We define the population version $\bar{Z}_{ij}(f)$ and the population loss function as

$$
\bar{Z}_{ij}(f) := \mathbb{E}_{x,\delta,e}\left[\hat{Z}_{ij}(f)\right].
$$

$$
\bar{Z}(f) := \mathbb{E}_{x,\delta,e}\left[\hat{Z}(f)\right]. \tag{69}
$$

Note that the $\delta_i(x_{i1}),\ldots,\delta_i(x_{im_i})$ are not independent for $i\in\{1,\ldots,n\}$. Similarly, due to the definitions of $\hat{Z}_{ij}(f)$, $Z_{ij}(f)$, $\hat{Z}(f)$, and $Z(f)$, the terms $\hat{Z}_{i1}(f),\ldots,\hat{Z}_{im_i}(f)$ are also not independent. Because Z is an excess-risk functional relative to f^*, $Z(f^*)=0$ and $Z(f)\geq 0$.

Definition 15 *The constrained quantile trend filtering estimator for temporal-spatial data defined in (18) will be*

$$
\hat{f} = \underset{f\in F_k(V)}{\arg\min}\,\hat{Z}(f), \tag{70}
$$

where $F_k(V)$ is defined in (16).

A.2 Proof Sketches of Theorem 1

Step 1. To get the convergence rate of $\Delta^2(\hat{f}-f^*)$, we will control the quantity $\mathbb{P}\left(c_0\Delta^2(\hat{f}-f^*)>t^2\right)$, where c_0 satisfies (84).

We will first consider the temporal dependence. To control that, we would need to use the $\beta-$mixing concept that are constructed in Appendix H. We then get the setup of the indices for $|\mathcal{J}_{e,l}|,|\mathcal{J}_{o,l}|\asymp n/L$ in (3). We let $\mathcal{I}:=\bigcup_{l=1}^{L}(\mathcal{J}_{e,l}\cup\mathcal{J}_{o,l})$. For each fixed $l\in[L]$, we construct copies $x^*_{ij},e^*_{ij},\delta^*_i$ for $i\in\mathcal{J}_{e,l}$ (and similarly for $\mathcal{J}_{o,l}$) via the coupling in Appendix H. These copies have the same marginal distribution as x_{ij},e_{ij},δ_i, and are mutually independent within each block $\mathcal{J}_{e,l}$ (but not across different blocks). Next, we let

$$
X_i := (x_{i1},\ldots,x_{im_i}),\qquad E_i := (e_{i1},\ldots,e_{im_i}),\qquad \Delta_i := (\delta_{i1},\ldots,\delta_{im_i}),
$$

and their coupled copies

$$
X^*_i := (x^*_{i1},\ldots,x^*_{im_i}),\qquad E^*_i := (e^*_{i1},\ldots,e^*_{im_i}),\qquad \Delta^*_i := (\delta^*_{i1},\ldots,\delta^*_{im_i}),
$$

and then we define

$$
\begin{aligned}
\Omega_x &:= \{X_i = X_i^* \text{ for all } i \in \mathcal{I}\}, \\
\Omega_e &:= \{E_i = E_i^* \text{ for all } i \in \mathcal{I}\}, \\
\Omega_\delta &:= \{\Delta_i = \Delta_i^* \text{ for all } i \in \mathcal{I}\}, \\
\Omega_\beta &:= \Omega_x \cap \Omega_e \cap \Omega_\delta.
\end{aligned} \tag{71}
$$

Thus, we get

$$
\mathbb{P}\left(c_0\Delta^2(\hat{f} - f^*) > t^2\right) \le \mathbb{P}\left(\left\{c_0\Delta^2(\hat{f} - f^*) > t^2\right\} \cap \Omega_\beta\right) + \mathbb{P}\left(\left\{c_0\Delta^2(\hat{f} - f^*) > t^2\right\} \cap \Omega_\beta^c\right).
$$

Step 2. Then in Lemma 16, we show $\mathbb{P}\left(\left\{c_0\Delta^2(\hat{f} - f^*) > t^2\right\} \cap \Omega_\beta^c\right)$ is $O(\frac{1}{n})$.

Step 3. It remains to bound the joint probability $\mathbb{P}\Big(\{c_0\Delta^2(\hat{f} - f^*) > t^2\} \cap \Omega_\beta\Big)$. On the coupling event Ω_β (see Remark 17), the blockwise coupling reduces the original β-mixing sequence to an independent-block structure. Consequently, the arguments in Lemmas 20–60 (based on symmetrization and Rademacher complexities) yield an upper bound for $\mathbb{P}\Big(\{c_0\Delta^2(\hat{f} - f^*) > t^2\} \cap \Omega_\beta\Big)$ in terms of the corresponding blockwise Rademacher width terms.

Step 4. In details, with the above two steps, under Assumptions 1, 2, 3, and 4, the following results hold for any $t > 0$,

$$
\begin{aligned}
&\mathbb{P}\left(c_0\Delta^2(\hat{f} - f^*) > t^2\right) \\
&\le \frac{c_5}{t^2}\sum_{l=1}^{L}\Big(R_{\mathcal{J}_{e,l}}(\mathcal{F} \cap \{c_0\Delta^2(f - f^*) \le t^2\}) + R_{\mathcal{J}_{o,l}}(\mathcal{F} \cap \{c_0\Delta^2(f - f^*) \le t^2\})\Big) + \frac{c_4}{t^2}\frac{s\log n \log T_n}{n},
\end{aligned}
$$

where $s = |S|$ is the intrinsic dimension from Assumption 3 (the number of active coordinates of Π), T_n is from Assumption 4. Constant c_4 depends on $\tilde{c}_0$ in Assumption 4, C_β in Assumption 1.4, and c_1, c_2 in (75), and c_5 depends on constants C and c in Assumption 1.9.

Step 5. By using 35, we start by writing

$$
\begin{aligned}
R_{\mathcal{J}_{e,l}}\left(\mathcal{F} \cap \{f : c_0\Delta^2(f - f^*) \le t^2\}\right) &\le R_{\mathcal{J}_{e,l}}\left(\mathcal{F} \cap \left\{f : \|f - f^*\| \le c_0^{-1/2} \cdot b(B) \cdot t\right\}\right), \text{ and} \\
R_{\mathcal{J}_{o,l}}\left(\mathcal{F} \cap \{f : c_0\Delta^2(f - f^*) \le t^2\}\right) &\le R_{\mathcal{J}_{o,l}}\left(\mathcal{F} \cap \left\{f : \|f - f^*\| \le c_0^{-1/2} \cdot b(B) \cdot t\right\}\right),
\end{aligned}
$$

where $b(B) = \max\left\{B^{1/2}, 1\right\}$ is a constant that depends on B, with B introduced in (4).

Let

$$
\begin{aligned}
r_n^2 \asymp \sum_{l=1}^{L}\Bigg(&R_{\mathcal{J}_{e,l}}\left(\mathcal{F} \cap B_2\left(c_0^{-\frac{1}{2}} \cdot b(B) \cdot r_n\right)\right) \\
&+ R_{\mathcal{J}_{o,l}}\left(\mathcal{F} \cap B_2\left(c_0^{-\frac{1}{2}} \cdot b(B) \cdot r_n\right)\right)\Bigg) + \frac{s \cdot \mathrm{poly}(\log n)\log T_n}{n}.
\end{aligned}
$$

Then we get the convergence rate of

$$
\Delta^2(\hat{f} - f^*) = O_p\left(r_n^2\right). \tag{72}
$$

This follows from Lemma 60 and Corollary 38.

The details of the proofs of the above steps are provided as follows.

A.3 Technical Results for Proofs of Theorem 1

A.4 Proof of Step 1 and 2

To get the convergence rate of $\Delta^2(\hat{f} - f^*)$, we will control the quantity $\mathbb{P}\left(c_0\Delta^2(\hat{f} - f^*) > t^2\right)$, where c_0 satisfies (84), we will first consider the temporal dependence. To control that, we would need to use the $\beta-$mixing concept. In the following analysis, we will then deal with this dependence.

Specifically, we rely on the condition that $cm \le m_i \le Cm$ for some positive constants c, C. Additionally, we define

$$L = \left\lceil \frac{1}{C_\beta} 2\log(n) \right\rceil, \tag{73}$$

leading us to the conclusion that

$$\beta_x(L) \le \frac{1}{n^2}, \tag{74}$$

where β_x is defined in Assumption 1.4. This bound is equivalently satisfied by $\beta_\delta(L)$ and $\beta_e(L)$. Subsequently, throughout our discussion, we make frequent references to the copies of the random variables x_{ij}, e_{ij}, and δ_i as they are constructed in Appendix H. These copies are denoted as x^*_{ij}, e^*_{ij}, and δ^*_i, respectively. Furthermore, as delineated in Appendix H, during the formation of these copies the quantities $|\mathcal{J}_{e,l}|$ and $|\mathcal{J}_{o,l}|$ in (3), which represent the count of even and odd blocks respectively, satisfy $|\mathcal{J}_{e,l}|, |\mathcal{J}_{o,l}| \asymp n/L$. Thus, there exist positive constants c_1 and c_2 such that

$$c_1 n/L \le |\mathcal{J}_{e,l}|, |\mathcal{J}_{o,l}| \le c_2 n/L, \quad L = \left\lceil \frac{1}{C_\beta} 2\log(n) \right\rceil. \tag{75}$$

Lemma 16 *Let $\hat{f}$, f^* belong to $\mathcal{F}$, and let $\Delta^2(\hat{f} - f^*)$ be the Huber loss. Suppose Assumption 1 holds with the β-mixing condition satisfied in (74) and the block indices $\mathcal{J}_{e,l}, \mathcal{J}_{o,l}$ satisfying $|\mathcal{J}_{e,l}|, |\mathcal{J}_{o,l}| \asymp n/L$ as in (3). For each fixed $l \in [L]$, we construct identically distributed copies $\{x^*_{ij}, e^*_{ij}, \delta^*_i\}_{i\in\mathcal{J}_{e,l}}$ that are mutually independent within the block $\mathcal{J}_{e,l}$ (and similarly for $\mathcal{J}_{o,l}$), via the coupling in Appendix H. Let $\Omega_x, \Omega_e, \Omega_\delta, \Omega_\beta$ be defined in (71). Then we have*

$$\mathbb{P}\left(c_0\Delta^2(\hat{f} - f^*) > t^2\right) \le \mathbb{P}\left(\left\{c_0\Delta^2(\hat{f} - f^*) > t^2\right\} \cap \Omega_\beta\right) + \frac{\epsilon}{2}.$$

Proof Using a union bound argument and invoking Lemmas 69, 70, and 71, we note that the following sets, $\Omega_x, \Omega_e, \Omega_\delta$ in (71), satisfy the conditions

$$\mathbb{P}(\Omega_x^c) \le n\beta_x(L) \le \frac{1}{n}, \quad \mathbb{P}(\Omega_e^c) \le n\beta_e(L) \le \frac{1}{n}, \quad \text{and} \quad \mathbb{P}(\Omega_\delta^c) \le n\beta_\delta(L) \le \frac{1}{n}.$$

Therefore letting

$$\Omega_\beta = \Omega_x \cap \Omega_e \cap \Omega_\delta \tag{76}$$

we have that

$$\begin{aligned}
\mathbb{P}\left(c_0\Delta^2(\hat{f} - f^*) > t^2\right) &= \mathbb{P}\left(\left\{c_0\Delta^2(\hat{f} - f^*) > t^2\right\} \cap \Omega_\beta\right) + \mathbb{P}\left(\left\{c_0\Delta^2(\hat{f} - f^*) > t^2\right\} \cap \Omega_\beta^c\right) \\
&\le \mathbb{P}\left(\left\{c_0\Delta^2(\hat{f} - f^*) > t^2\right\} \cap \Omega_\beta\right) + \frac{3}{n} \\
&\le \mathbb{P}\left(\left\{c_0\Delta^2(\hat{f} - f^*) > t^2\right\} \cap \Omega_\beta\right) + \frac{\varepsilon}{2}
\end{aligned}$$

for any $\varepsilon > 0$ as n large enough. ∎

By previous lemma, in consequence, it is enough to bound the term

$$\mathbb{P}\left(\left\{c_0\Delta^2(\hat{f}-f^*) > t^2\right\}\cap\Omega_\beta\right). \tag{77}$$

To this end, we will work with the indicator of the coupling event. We write $\mathbf{1}_{\Omega_\beta}$ for the indicator of Ω_β.

We write $\mathbb{P}(\cdot\cap\Omega_\beta)$ for the probability restricted to Ω_β. We denote by $\mathbb{E}_{x,\delta,e}\left[\cdot\right]$ the expectation taken over (x_{ij},δ_i,e_{ij}). And we denote by $\mathbb{E}_{e|x_{ij}}\left[\cdot\right]$ the conditional expectation taken over $\{x_{ij}\}_{i=1,j=1}^{n,m_i}$ respectively.

In particular, for any nonnegative random variable X, we write

$$\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[X\,\mathbf{1}_{\Omega_\beta}\right]$$

to denote the expectation of X restricted to the coupling event Ω_β.

Remark 17 *Moreover, on Ω_β we work with the coupled sample $\{(x_{ij}^*,y_{ij}^*)\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$, which are i.i.d. within each fixed block $\mathcal{J}_{e,l}$. We now define the star versions of the loss quantities. The star empirical loss is*

$$\hat{Z}_{ij}^*(f) \coloneqq \rho_\tau(f^*(x_{ij}^*)+\delta_i^*(x_{ij}^*)+e_{ij}^*-f(x_{ij}^*))-\rho_\tau(\delta_i^*(x_{ij}^*)+e_{ij}^*), \tag{78}$$

and the star conditional expectations are

$$\begin{aligned}
Z_{ij}^*(f) &\coloneqq \mathbb{E}_{\delta^*,e^*|x_{ij}^*}\left[\hat{Z}_{ij}^*(f)\right],\\
\tilde{Z}_{ij}^*(f) &\coloneqq \mathbb{E}_{e^*|\delta_i^*,x_{ij}^*}\left[\hat{Z}_{ij}^*(f)\right],\\
\bar{Z}_{ij}^*(f) &\coloneqq \mathbb{E}_{x^*,\delta^*,e^*}\left[\hat{Z}_{ij}^*(f)\right].
\end{aligned}$$

By construction, for each fixed $l\in[L]$, the star quantities $\{\hat{Z}_{ij}^*(f)\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$ are built from the i.i.d. copies $(x_{ij}^*,e_{ij}^*,\delta_i^*)_{i\in\mathcal{J}_{e,l}}$, and hence the indices $i\in\mathcal{J}_{e,l}$ are independent (similarly for $\mathcal{J}_{o,l}$). However, the copies across different blocks l are not mutually independent. On the coupling event Ω_β we have $(x_{ij},e_{ij},\delta_i) = (x_{ij}^*,e_{ij}^*,\delta_i^*)$ for all $i\in\mathcal{I}$ and $j\in[m_i]$, so that

$$\hat{Z}_{ij}(f) = \hat{Z}_{ij}^*(f),\quad Z_{ij}(f) = Z_{ij}^*(f),\quad \tilde{Z}_{ij}(f) = \tilde{Z}_{ij}^*(f),\quad \bar{Z}_{ij}(f) = \bar{Z}_{ij}^*(f),\qquad \text{on } \Omega_\beta. \tag{79}$$

Results that do not invoke the coupling argument remain valid under the original probability measure and will be indicated explicitly when used.

A.5 Proof of Step 3

Lemma 18 – Corollary 22 are some fundamental lemmas support our convergence analysis.

Lemma 18 (Basic Inequality) *By Definitions in (27) and (65), we have*

$$\hat{Z}(\hat{f})-\hat{Z}(f^*) \leq 0$$

as the basic inequality.

Lemma 19 (Lipschitz condition of $\hat{Z}_{ij}(f)$, $Z_{ij}(f)$ and $r_{ij}(f)$) *The quantities $\hat{Z}_{ij}(f)$ (defined in (64)), $Z_{ij}(f)$ (defined in (66)), and $r_{ij}(f)$ (defined in (95)) are 1-Lipschitz in the following sense: for any $f, g \in \mathcal{F}$ and all (i,j),*

$$\begin{aligned}\left|\hat{Z}_{ij}(f) - \hat{Z}_{ij}(g)\right| &\leq \left|f(x_{ij}) - g(x_{ij})\right|, \\ \left|Z_{ij}(f) - Z_{ij}(g)\right| &\leq \left|f(x_{ij}) - g(x_{ij})\right|, \\ \left|r_{ij}(f) - r_{ij}(g)\right| &\leq \left|f(x_{ij}) - g(x_{ij})\right|.\end{aligned}$$

Proof
For $\hat{Z}_{ij}(f)$ and $Z_{ij}(f)$, their Lipschitz condition can be seen in the Lemma A.11 of Zhang et al. (2023). For

$$r_{ij}(f) = \int_0^{f(x_{ij})-f^*(x_{ij})} \left\{\mathbb{P}(y_{i,j} \leq f^*(x_{ij}) + t \mid x_{ij}) - \mathbb{P}(y_{i,j} \leq f^*(x_{ij}) + t | \delta_i, x_{ij})\right\} dt.$$

We have

$$\begin{aligned}&\left|r_{ij}(f) - r_{ij}(g)\right| \\ &= \int_0^{f(x_{ij})-g(x_{ij})} \left\{\mathbb{P}(y_{i,j} \leq f^*(x_{ij}) + t \mid x_{ij}) - \mathbb{P}(y_{i,j} \leq f^*(x_{ij}) + t | \delta_i, x_{ij})\right\} dt. \\ &\leq \left|f(x_{ij}) - g(x_{ij})\right|.\end{aligned}$$

■

Lemma 20 *For any $f, f^* \in \mathcal{F}$, let $h = f - f^*$, and let $h(x_{ij}) = (f - f^*)(x_{ij})$. Then we have*

$$Z_{ij}(f) - Z_{ij}(f^*) = \int_0^{h(x_{ij})} \left\{F_{y_{ij}|x_{ij}}(f^*(x_{ij}) + z) - F_{y_{ij}|x_{ij}}(f^*(x_{ij}))\right\} dz.$$

Proof Let I be the indicator function. By equation (B.3) in Belloni and Chernozhukov (2011),we have

$$\rho_\tau(w - v) - \rho_\tau(w) = -v(\tau - I(w \leq 0)) + \int_0^v \left\{I(w \leq z) - I(w \leq 0)\right\} dz. \tag{80}$$

Given any f and x_{ij}, let $w = y_{ij} - f^*(x_{ij})$, $v = f(x_{ij}) - f^*(x_{ij})$. Then taking conditional expectations on above equation with respect to y_{ij} conditioned on x_{ij} on both sides, we have

$$\begin{aligned}Z_{ij}(f) - Z_{ij}(f^*) &= \mathbb{E}_{\delta,e|x_{ij}}\left[\rho_\tau(y_{ij} - f(x_{ij})) - \rho_\tau(y_{ij} - f^*(x_{ij}))\right] \\ &= \mathbb{E}_{\delta,e|x_{ij}}\left[-h(x_{ij})(\tau - I(y_{ij} - f^*(x_{ij}) \leq 0))\right] \\ &+ \mathbb{E}_{\delta,e|x_{ij}}\left[\int_0^{h(x_{ij})} \left\{I(y_{ij} - f^*(x_{ij}) \leq z) - I(y_{ij} - f^*(x_{ij})) \leq 0\right\} dz\right] \\ &= 0 + \int_0^{h(x_{ij})} \left\{F_{y_{ij}|x_{ij}}(f^*(x_{ij}) + z) - F_{y_{ij}|x_{ij}}(f^*(x_{ij}))\right\} dz \\ &= \int_0^{h(x_{ij})} \left\{F_{y_{ij}|x_{ij}}(f^*(x_{ij}) + z) - F_{y_{ij}|x_{ij}}(f^*(x_{ij}))\right\} dz,\end{aligned} \tag{81}$$

for all $i = 1, \ldots, n$.

■

Lemma 21 *Under Assumption 2, with R and $\underline{f}$ from Assumption 2, let*

$$C_0 \coloneqq \min\left\{\frac{\underline{f}}{2},\ \frac{R\underline{f}}{4}\right\}. \tag{82}$$

Then for all $f, f^* \in \mathcal{F}$, we have

$$Z(f) \geq C_0 \Delta^2_{nm}(f - f^*).$$

Proof Let $h(x_{ij}) \coloneqq f(x_{ij}) - f^*(x_{ij})$. By Lemma 20, we have

$$Z_{ij}(f) - Z_{ij}(f^*) = \int_0^{h(x_{ij})} \Big\{F_{y_{ij}|x_{ij}}(f^*(x_{ij}) + z) - F_{y_{ij}|x_{ij}}(f^*(x_{ij}))\Big\}\, dz.$$

First, suppose $h(x_{ij}) \in [0, R]$. By Assumption 2, for any z with $|z| \leq R$, we have $F_{y_{ij}|x_{ij}}(f^*(x_{ij})+z) - F_{y_{ij}|x_{ij}}(f^*(x_{ij})) \geq \underline{f}\, z$. Therefore,

$$Z_{ij}(f) - Z_{ij}(f^*) \geq \int_0^{h(x_{ij})} \underline{f}\, z\, dz = \frac{\underline{f}\, h(x_{ij})^2}{2}.$$

Next, suppose $h(x_{ij}) \in [-R, 0]$. For $z \in [h(x_{ij}), 0]$ we have $|z| \leq R$, so by Assumption 2, $F_{y_{ij}|x_{ij}}(f^*(x_{ij}) + z) - F_{y_{ij}|x_{ij}}(f^*(x_{ij})) \leq \underline{f}\, z$ (noting $z \leq 0$). Therefore,

$$\begin{aligned} Z_{ij}(f) - Z_{ij}(f^*) &= \int_{h(x_{ij})}^0 \Big\{F_{y_{ij}|x_{ij}}(f^*(x_{ij})) - F_{y_{ij}|x_{ij}}(f^*(x_{ij}) + z)\Big\}\, dz \\ &\geq \int_{h(x_{ij})}^0 (-\underline{f}\, z)\, dz = \frac{\underline{f}\, h(x_{ij})^2}{2}. \end{aligned}$$

Now suppose $h(x_{ij}) > R$. Since $F_{y_{ij}|x_{ij}}$ is monotone non-decreasing, for $z \geq R/2$ we have $F_{y_{ij}|x_{ij}}(f^*(x_{ij}) + z) \geq F_{y_{ij}|x_{ij}}(f^*(x_{ij}) + R/2)$. By Assumption 2 (applied with $R/2 \leq R$), $F_{y_{ij}|x_{ij}}(f^*(x_{ij}) + R/2) - F_{y_{ij}|x_{ij}}(f^*(x_{ij})) \geq \underline{f}\, R/2$. Therefore,

$$\begin{aligned} Z_{ij}(f) - Z_{ij}(f^*) &\geq \int_{R/2}^{h(x_{ij})} \Big\{F_{y_{ij}|x_{ij}}(f^*(x_{ij}) + z) - F_{y_{ij}|x_{ij}}(f^*(x_{ij}))\Big\}\, dz \\ &\geq \int_{R/2}^{h(x_{ij})} \Big\{F_{y_{ij}|x_{ij}}(f^*(x_{ij}) + R/2) - F_{y_{ij}|x_{ij}}(f^*(x_{ij}))\Big\}\, dz \\ &= (h(x_{ij}) - R/2)\Big\{F_{y_{ij}|x_{ij}}(f^*(x_{ij}) + R/2) - F_{y_{ij}|x_{ij}}(f^*(x_{ij}))\Big\} \\ &\geq \frac{h(x_{ij})}{2} \cdot \frac{R\,\underline{f}}{2} = \frac{R\,\underline{f}}{4}\, |h(x_{ij})| \geq C_0\, |h(x_{ij})|, \end{aligned} \tag{83}$$

where the first inequality holds since the integrand is non-negative on $[0, R/2]$, the second by monotonicity of $F_{y_{ij}|x_{ij}}$, and the penultimate uses $h(x_{ij}) - R/2 \geq h(x_{ij})/2$ (since $h(x_{ij}) > R$).

The case $h(x_{ij}) < -R$ is analogous (by reversing the direction of integration) and yields the same bound $Z_{ij}(f) - Z_{ij}(f^*) \geq C_0\, |h(x_{ij})|$.

Combining the four cases, for all (i, j) we have

$$Z_{ij}(f) - Z_{ij}(f^*) \geq C_0 \min\left\{|h(x_{ij})|, h(x_{ij})^2\right\}.$$

Summing over all (i, j) and using $Z_{ij}(f^*) = 0$, we obtain

$$Z(f) \geq C_0 \Delta^2_{nm}(f - f^*),$$

where $\Delta^2_{nm}(\cdot)$ is the empirical Huber loss defined in (62). ∎

Corollary 22 *Assume Assumption 2 holds. Then for C_0 defined in (82) and for all $f, f^* \in \mathcal{F}$, we have*

$$\bar{Z}(f) \geq C_0 \Delta^2(f - f^*),$$

where $\bar{Z}(f) := \mathbb{E}_{x,\delta,e}\left[\hat{Z}(f)\right]$ is defined in (69) and $\Delta^2(\cdot)$ is the population Huber loss defined in (63).

Proof By Lemma 21, for each realization of $\{x_{ij}\}_{i=1,j=1}^{n,m_i}$, $Z(f) \geq C_0\Delta^2_{nm}(f - f^*)$. Taking expectation over the design points on both sides, and using $\bar{Z}(f) = \mathbb{E}_x[Z(f)]$ from (69) and $\Delta^2(f - f^*) = \mathbb{E}_x\left[\Delta^2_{nm}(f - f^*)\right]$ from (63), gives the result. ∎

Lemma 23–33 are used to bound the term (77).

Lemma 23 *Let $\hat{f}, f^* \in \mathcal{F}$ and let Ω_β be as defined in (76). Let C_0 be the constant from Corollary 22 (defined in (82)), satisfying $\bar{Z}(f) \geq C_0\Delta^2(f - f^*)$. Let c_0 be any constant satisfying*

$$0 < c_0 < C_0. \tag{84}$$

Then for any $t > 0$,

$$\mathbb{P}\Big(\{c_0\Delta^2(\hat{f} - f^*) > t^2\} \cap \Omega_\beta\Big) \leq \mathbb{P}\left(\left\{\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\leq t^2}} \left(\bar{Z}(f) - \hat{Z}(f) - \ell(f)\right) > t^2\right\} \cap \Omega_\beta\right),$$

where $\ell(f) := (C_0 - c_0)\Delta^2(f - f^*)$.

Proof Assume $c_0\Delta^2(\hat{f} - f^*) > t^2$, where c_0 satisfies (84). We know Δ^2 is continuous. Next, define $g : [0, 1] \to \mathbb{R}$ as $g(\mu) = c_0\Delta^2((1-\mu)f^* + \mu\hat{f} - f^*)$. Clearly, g is a continuous function with $g(0) = 0$ and $g(1) = c_0\Delta^2(\hat{f} - f^*)$. Therefore, there exists $\mu_{\hat{f}} \in [0, 1]$ such that $g(\mu_{\hat{f}}) = t^2$. We then let $\tilde{f} = (1 - \mu_{\hat{f}})f^* + \mu_{\hat{f}}\hat{f}$, so that by construction $c_0\Delta^2(\tilde{f} - f^*) = t^2$.

Then we have

$$\hat{Z}(\tilde{f}) = \hat{Z}\left\{(1 - \mu_{\hat{f}})f^* + \mu_{\hat{f}}\hat{f}\right\} \overset{(a)}{\leq} (1 - \mu_{\hat{f}})\hat{Z}(f^*) + \mu_{\hat{f}}\hat{Z}(\hat{f}) \overset{(b)}{\leq} 0. \tag{85}$$

Where (a) follows the convexity of $\hat{Z}$. The (b) follows by the $\hat{Z}(f^*) = 0$, and the basic inequality. First, let

$$\ell(\tilde{f}) := (C_0 - c_0)\Delta^2(\tilde{f} - f^*), \tag{86}$$

Since $\hat{f}, f^* \in \mathcal{F}$, and $\mathcal{F}$ is a convex set, thus, $\tilde{f} \in \mathcal{F}$, and $c_0\Delta^2(\tilde{f} - f^*) = t^2$, then we have

$$\begin{aligned}
t^2 &= c_0\Delta^2(\tilde{f} - f^*)\\
&= C_0\Delta^2(\tilde{f} - f^*) - \underbrace{(C_0 - c_0)\Delta^2(\tilde{f} - f^*)}_{:=\ell(\tilde{f})}\\
&\leq \bar{Z}(\tilde{f}) - \ell(\tilde{f})\\
&= \bar{Z}(\tilde{f}) - \hat{Z}(\tilde{f}) + \hat{Z}(\tilde{f}) - \ell(\tilde{f})
\end{aligned}$$

$$
\begin{aligned}
&\overset{(a)}{\leq} \bar{Z}(\tilde{f}) - \hat{Z}(\tilde{f}) - \ell(\tilde{f}) \\
&\leq \sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f - f^*) \leq t^2}} \left(\bar{Z}(f) - \hat{Z}(f) - \ell(f) \right).
\end{aligned}
$$

The first inequality holds by applying the Corollary 22. The (a) follows by expression in the line (85).

Thus, we have

$$
\mathbb{P}\left(\left\{ c_0 \Delta^2(\hat{f} - f^*) > t^2 \right\} \cap \Omega_\beta \right) \leq \mathbb{P}\left(\left\{ \sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f - f^*) \leq t^2}} \left(\bar{Z}(f) - \hat{Z}(f) - \ell(f) \right) > t^2 \right\} \cap \Omega_\beta \right).
$$

■

To control the quantity on the right-hand side, we apply Markov's inequality and then utilize symmetrization to control the empirical process with Rademacher complexity. We also need to account for the dependence among $\delta_i(x_{i1}), \ldots, \delta_i(x_{im_i})$ for $i \in \{1, \ldots, n\}$. The following theorem introduces the decomposition that handles this dependence.

Theorem 24 (Conversion of Huber loss) *Let $\hat{f}, f^* \in \mathcal{F}$, let Ω_β be defined in (76). Suppose the distribution of $\{(x_{ij}, y_{i,j})\}_{i=1,j=1}^{n,m_i}$ obey Assumption 2, where c_0 satisfies (84), and $\ell(f)$ is defined in*

(86), *the following inequality is true for any* $t > 0$,

$$
\begin{aligned}
&\mathbb{P}\left(\left\{c_0\Delta^2(\hat{f}-f^*) > t^2\right\} \cap \Omega_\beta\right) \\
&\leq \sum_{l=1}^{L} \frac{2L}{t^2} \underbrace{\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\Big(\sup_{\substack{f\in\mathcal{F} \\ c_0\Delta^2(f-f^*)\leq t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}} \frac{1}{m_i}\sum_{j=1}^{m_i} Z_{ij}(f) - \tilde{Z}_{ij}(f) - \frac{\ell(f)}{2L}\Big)\cdot \mathbf{1}_{\Omega_\beta}\right]}_{A_1} \\
&+ \sum_{l=1}^{L} \frac{2L}{t^2} \underbrace{\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\Big(\sup_{\substack{f\in\mathcal{F} \\ c_0\Delta^2(f-f^*)\leq t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}} \frac{1}{m_i}\sum_{j=1}^{m_i} \tilde{Z}_{ij}(f) - \hat{Z}_{ij}(f)\Big)\cdot \mathbf{1}_{\Omega_\beta}\right]}_{A_2} \\
&+ \sum_{l=1}^{L} \frac{2L}{t^2} \underbrace{\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\Big(\sup_{\substack{f\in\mathcal{F} \\ c_0\Delta^2(f-f^*)\leq t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}} \frac{1}{m_i}\sum_{j=1}^{m_i} \bar{Z}_{ij}(f) - Z_{ij}(f)\Big)\cdot \mathbf{1}_{\Omega_\beta}\right]}_{A_3} \qquad (87)\\
&+ \sum_{l=1}^{L} \frac{2L}{t^2} \underbrace{\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\Big(\sup_{\substack{f\in\mathcal{F} \\ c_0\Delta^2(f-f^*)\leq t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{o,l}} \frac{1}{m_i}\sum_{j=1}^{m_i} Z_{ij}(f) - \tilde{Z}_{ij}(f) - \frac{\ell(f)}{2L}\Big)\cdot \mathbf{1}_{\Omega_\beta}\right]}_{B_1} \\
&+ \sum_{l=1}^{L} \frac{2L}{t^2} \underbrace{\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\Big(\sup_{\substack{f\in\mathcal{F} \\ c_0\Delta^2(f-f^*)\leq t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{o,l}} \frac{1}{m_i}\sum_{j=1}^{m_i} \tilde{Z}_{ij}(f) - \hat{Z}_{ij}(f)\Big)\cdot \mathbf{1}_{\Omega_\beta}\right]}_{B_2} \\
&+ \sum_{l=1}^{L} \frac{2L}{t^2} \underbrace{\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\Big(\sup_{\substack{f\in\mathcal{F} \\ c_0\Delta^2(f-f^*)\leq t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{o,l}} \frac{1}{m_i}\sum_{j=1}^{m_i} \bar{Z}_{ij}(f) - Z_{ij}(f)\Big)\cdot \mathbf{1}_{\Omega_\beta}\right]}_{B_3},
\end{aligned}
$$

where L *is defined in* (73).

Proof By Lemma 23, it is true that

$$
\begin{aligned}
&\mathbb{P}\left(\left\{c_0\Delta^2(\hat{f}-f^*) > t^2\right\} \cap \Omega_\beta\right) \\
&\overset{(a)}{\leq} \mathbb{P}\left(\left\{\sup_{\substack{f\in\mathcal{F} \\ c_0\Delta^2(f-f^*)\leq t^2}} \left(\bar{Z}(f) - \hat{Z}(f) - \ell(f)\right) > t^2\right\} \cap \Omega_\beta\right) \\
&= \mathbb{P}\Bigg(\Bigg\{\sup_{\substack{f\in\mathcal{F} \\ c_0\Delta^2(f-f^*)\leq t^2}} \Bigg(\frac{1}{n}\sum_{l=1}^{L}\sum_{i\in\mathcal{J}_{e,l}} \frac{1}{m_i}\sum_{j=1}^{m_i} \bar{Z}_{ij}(f) - \hat{Z}_{ij}(f) - \frac{\ell(f)}{2}
\end{aligned}
$$

$$+\ \frac{1}{n}\sum_{l=1}^{L}\sum_{i\in\mathcal{J}_{o,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\bar{Z}_{ij}(f)-\hat{Z}_{ij}(f)-\frac{\ell(f)}{2}\Bigg)>t^2\Bigg\}\cap\Omega_\beta\Bigg)$$

$$\begin{aligned}
&\leq \mathbb{P}\left(\left\{\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\leq t^2}}\left(\frac{1}{n}\sum_{l=1}^{L}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\bar{Z}_{ij}(f)-\hat{Z}_{ij}(f)-\frac{\ell(f)}{2}\right)>\frac{t^2}{2}\right\}\cap\Omega_\beta\right)\\
&\quad+\mathbb{P}\left(\left\{\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\leq t^2}}\left(\frac{1}{n}\sum_{l=1}^{L}\sum_{i\in\mathcal{J}_{o,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\bar{Z}_{ij}(f)-\hat{Z}_{ij}(f)-\frac{\ell(f)}{2}\right)>\frac{t^2}{2}\right\}\cap\Omega_\beta\right)\\
&\leq \sum_{l=1}^{L}\mathbb{P}\left(\left\{\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\leq t^2}}\left(\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\bar{Z}_{ij}(f)-\hat{Z}_{ij}(f)-\frac{\ell(f)}{2L}\right)>\frac{t^2}{2L}\right\}\cap\Omega_\beta\right)\\
&\quad+\sum_{l=1}^{L}\mathbb{P}\left(\left\{\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\leq t^2}}\left(\frac{1}{n}\sum_{i\in\mathcal{J}_{o,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\bar{Z}_{ij}(f)-\hat{Z}_{ij}(f)-\frac{\ell(f)}{2L}\right)>\frac{t^2}{2L}\right\}\cap\Omega_\beta\right).
\end{aligned}$$

Now we analyze each of the terms above,

$$\begin{aligned}
&\mathbb{P}\left(\left\{\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\leq t^2}}\left(\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\bar{Z}_{ij}(f)-\hat{Z}_{ij}(f)-\frac{\ell(f)}{2L}\right)>\frac{t^2}{2L}\right\}\cap\Omega_\beta\right)\\
&\overset{(a)}{\leq}\frac{2L}{t^2}\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\Bigg(\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\leq t^2}}\left(\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\bar{Z}_{ij}(f)-\hat{Z}_{ij}(f)-\frac{\ell(f)}{2L}\right)\Bigg)\cdot\mathbf{1}_{\Omega_\beta}\right]\\
&\overset{(b)}{\leq}\frac{2L}{t^2}\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\Big[\Big(\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\leq t^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}Z_{ij}(f)-\tilde{Z}_{ij}(f)-\frac{\ell(f)}{2L}+\\
&\quad\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\leq t^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\tilde{Z}_{ij}(f)-\hat{Z}_{ij}(f)\\
&\quad+\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\leq t^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\bar{Z}_{ij}(f)-Z_{ij}(f)\Big)\cdot\mathbf{1}_{\Omega_\beta}\Big]\\
&=\frac{2L}{t^2}\underbrace{\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\Big(\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\leq t^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}Z_{ij}(f)-\tilde{Z}_{ij}(f)-\frac{\ell(f)}{2L}\Big)\cdot\mathbf{1}_{\Omega_\beta}\right]}_{A_1}\\
&\quad+\frac{2L}{t^2}\underbrace{\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\Big(\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\leq t^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\tilde{Z}_{ij}(f)-\hat{Z}_{ij}(f)\Big)\cdot\mathbf{1}_{\Omega_\beta}\right]}_{A_2}
\end{aligned}$$

$$
+ \frac{2L}{t^2} \underbrace{\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\Big(\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\le t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\bar{Z}_{ij}(f)-Z_{ij}(f)\Big)\cdot\mathbf{1}_{\Omega_\beta}\right]}_{A_3}
$$

$$
:= \frac{2LA_1}{t^2}+\frac{2LA_2}{t^2}+\frac{2LA_3}{t^2},
$$

The inequality (a) follows from Markov's inequality ,and the third inequality (b) follows from the triangle inequality.

Similarly, we have

$$
\mathbb{P}\left(\left\{\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\le t^2}}\left(\frac{1}{n}\sum_{i\in\mathcal{J}_{o,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\bar{z}_{ij}(f)-\hat{z}_{ij}(f)-\frac{\ell(f)}{2L}\right)>\frac{t^2}{2L}\right\}\cap\Omega_\beta\right)
$$

$$
\le \frac{2LB_1}{t^2}+\frac{2LB_2}{t^2}+\frac{2LB_3}{t^2}.
$$

■

Now we proceed to bound A_1, A_2 A_3 separately. For B_1, B_2 B_3, we will just give the conclusion, as the analysis is similar.

For this subsection, write

$$
\mathcal{F}_t := \{f\in\mathcal{F}: c_0\Delta^2(f-f^*)\le t^2\}.
$$

Lemma 25 *Let Ω_β be defined in (76). Let $\{s_{i,j}\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$ be independent Rademacher variables, independent of the data $\{(x_{ij},e_{ij},\delta_i)\}$ and their coupled copies $\{(x^*_{ij},e^*_{ij},\delta^*_i)\}$. Then we have*

$$
\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*,s}\left[\Big(\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\le t^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}s_{ij}f(x^*_{ij})\Big)\cdot\mathbf{1}_{\Omega_\beta}\right]
$$

$$
\le \frac{c_5}{2L}R_{\mathcal{J}_{e,l}}(\mathcal{F}_t),
$$

where c_5 is a constant depending only on the constants C and c in Assumption 1.9, and $R_{\mathcal{J}_{e,l}}$ is defined in (13).

Proof We have

$$
\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*,s}\left[\Big(\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\le t^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}s_{ij}f(x^*_{ij})\Big)\cdot\mathbf{1}_{\Omega_\beta}\right]
$$

$$
= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*,s}\Bigg[\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\le t^2}}\frac{1}{\sum_{i'\in\mathcal{J}_{e,l}}m_{i'}}\sum_{i\in\mathcal{J}_{e,l}} \tag{88}
$$

$$
\sum_{j=1}^{m_i}s_{ij}\frac{\sum_{i'\in\mathcal{J}_{e,l}}m_{i'}}{nm_i}f(x^*_{ij})\cdot\mathbf{1}_{\Omega_\beta}\Bigg].
$$

Next, by Assumption 1.9, for every $i \in \mathcal{J}_{e,l}$ we have $cm \le m_i \le Cm$. Combined with (75), which gives $|\mathcal{J}_{e,l}| \le c_2 n/L$ (where c_2 is from (75)), we obtain

$$\frac{\sum_{i' \in \mathcal{J}_{e,l}} m_{i'}}{n\, m_i} \le \frac{|\mathcal{J}_{e,l}| \cdot Cm}{n \cdot cm} \le \frac{c_2 C}{cL}. \tag{89}$$

Then by Ledoux Talagrand contraction inequality (see Proposition 5.28 in Wainwright (2019)), we have

$$\begin{aligned}
&\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*,s}\Bigg[\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\le t^2}} \frac{1}{\sum_{i'\in\mathcal{J}_{e,l}} m_{i'}} \sum_{i\in\mathcal{J}_{e,l}} \sum_{j=1}^{m_i} s_{ij} \frac{\sum_{i'\in\mathcal{J}_{e,l}} m_{i'}}{nm_i} f(x_{ij}^*)\\
&\quad \cdot \mathbf{1}_{\Omega_\beta}\Bigg]\\
&\le \mathbb{E}_{x^*,s}\Bigg[\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\le t^2}} \frac{1}{\sum_{i'\in\mathcal{J}_{e,l}} m_{i'}} \sum_{i\in\mathcal{J}_{e,l}} \sum_{j=1}^{m_i} s_{ij} \frac{\sum_{i'\in\mathcal{J}_{e,l}} m_{i'}}{nm_i} f(x_{ij}^*)\Bigg]\\
&\overset{\text{(a)}}{\le} \frac{c_2 C}{cL}\mathbb{E}_{x^*,s}\left[\left(\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\le t^2}} \frac{1}{\sum_{i'\in\mathcal{J}_{e,l}} m_{i'}} \sum_{i\in\mathcal{J}_{e,l}} \sum_{j=1}^{m_i} s_{ij} f(x_{ij}^*)\right)\right]\\
&= \frac{c_2 C}{cL} R_{\mathcal{J}_{e,l}}(\mathcal{F}_t) = \frac{c_5}{2L} R_{\mathcal{J}_{e,l}}(\mathcal{F}_t).
\end{aligned}$$

The (a) holds since (89). Equivalently, letting $c_5 := 2c_2C/c$, the prefactor becomes $\frac{c_5}{2L}$. ∎

A.5.1 Bound of A_2

Lemma 26 (Bound of A_2) *Let $\tilde{Z}(f)$, $\hat{Z}(f)$ be defined in (68) and (65), and let $\tilde{Z}^*(f)$, $\hat{Z}^*(f)$ be the star versions defined in (78). Let Ω_β be defined in (76). Let $\{s_{i,j}\}_{i\in\mathcal{J}_{e,l}, j\in[m_i]}$ be independent Rademacher variables, independent of the data $\{(x_{ij}, e_{ij}, \delta_i)\}$ and their coupled copies $\{(x_{ij}^*, e_{ij}^*, \delta_i^*)\}$. Let A_2 be the quantity introduced in (87). Then, for any $t > 0$, we have*

$$\begin{aligned}
A_2 &= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\left(\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\le t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}} \frac{1}{m_i}\sum_{j=1}^{m_i}\left\{\tilde{Z}_{ij}(f) - \hat{Z}_{ij}(f)\right\}\right)\cdot \mathbf{1}_{\Omega_\beta}\right]\\
&\le \frac{c_5}{L} R_{\mathcal{J}_{e,l}}(\mathcal{F}_t),
\end{aligned}$$

Here c_5 depends only on the constants C and c in Assumption 1.9, and $R_{\mathcal{J}_{e,l}}$ is defined in (13).

Proof We pass from A_2 to a star-world quantity A_2^* via the coupling, and then bound A_2^*.

By the construction of the coupling event Ω_β (see (71)), on Ω_β we have $(x_{ij}, e_{ij}, \delta_i) = (x_{ij}^*, e_{ij}^*, \delta_i^*)$ for all indices i in the coupled index set and all $j \in [m_i]$. Hence, by (79), for every $f \in \mathcal{F}$ and every such (i, j),

$$\hat{Z}_{ij}(f) = \hat{Z}_{ij}^*(f), \qquad \tilde{Z}_{ij}(f) = \tilde{Z}_{ij}^*(f) \qquad \text{on } \Omega_\beta. \tag{90}$$

Therefore,

$$\begin{aligned}&\Bigg(\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\le t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\Big\{\tilde{Z}_{ij}(f)-\hat{Z}_{ij}(f)\Big\}\Bigg)\cdot\mathbf{1}_{\Omega_\beta}\\ &= \Bigg(\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\le t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\Big\{\tilde{Z}^*_{ij}(f)-\hat{Z}^*_{ij}(f)\Big\}\Bigg)\cdot\mathbf{1}_{\Omega_\beta}.\end{aligned} \tag{91}$$

Since f^* belongs to the constraint set and $\tilde{Z}^*_{ij}(f^*)-\hat{Z}^*_{ij}(f^*)=0$, the supremum on the right-hand side is nonnegative. Hence $\mathbf{1}_{\Omega_\beta}\le 1$ implies

$$\begin{aligned}A_2 &= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\Bigg(\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\le t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\Big\{\tilde{Z}^*_{ij}(f)-\hat{Z}^*_{ij}(f)\Big\}\Bigg)\cdot\mathbf{1}_{\Omega_\beta}\right]\\ &\le \mathbb{E}_{x^*,\delta^*,e^*}\left[\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\le t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\Big\{\tilde{Z}^*_{ij}(f)-\hat{Z}^*_{ij}(f)\Big\}\right]\\ &=: A_2^*.\end{aligned} \tag{92}$$

It remains to bound A_2^*. By the tower property, conditioning on $\{\delta_i^*\}_{i\in\mathcal{J}_{e,l}}$ and $\{x^*_{ij}\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$, we have

$$\begin{aligned}A_2^* &= \mathbb{E}_{x^*,\delta^*,e^*}\left[\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\le t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\Big\{\tilde{Z}^*_{ij}(f)-\hat{Z}^*_{ij}(f)\Big\}\right]\\ &= \mathbb{E}_{\delta^*,x^*}\left[\mathbb{E}_{e^*|\delta^*,x^*}\left[\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\le t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\Big\{\tilde{Z}^*_{ij}(f)-\hat{Z}^*_{ij}(f)\Big\}\right]\right].\end{aligned}$$

Conditioned on δ^* and x^*, the quantity $\tilde{Z}^*_{ij}(f)=\mathbb{E}_{e^*|\delta_i^*,x_{ij}^*}\Big[\hat{Z}^*_{ij}(f)\Big]$ is deterministic, and $\{\hat{Z}^*_{ij}(f)-\tilde{Z}^*_{ij}(f)\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$ are independent mean-zero random variables (since within block $\mathcal{J}_{e,l}$, the e^*_{ij} are i.i.d. given δ^*,x^*). Applying the symmetrization inequality (Section 2.3.1 in Van Der Vaart et al. (1996)), we obtain

$$\begin{aligned}A_2^* &\le 2\,\mathbb{E}_{\delta^*,x^*}\left[\mathbb{E}_{e^*,s|\delta^*,x^*}\left[\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\le t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}s_{ij}\hat{Z}^*_{ij}(f)\right]\right]\\ &= 2\mathbb{E}_{x^*,\delta^*,e^*,s}\left[\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\le t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}s_{ij}\hat{Z}^*_{ij}(f)\right].\end{aligned}$$

We then apply the Lipschitz condition from Lemma 19 and the Ledoux-Talagrand contraction principle (Theorem 62) to obtain

$$A_2^* \le 2\mathbb{E}_{x^*,s}\left[\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\le t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}s_{ij}f(x^*_{ij})\right]$$

$$= \frac{c_5}{L} R_{\mathcal{J}_{e,l}}(\mathcal{F}_t),$$

The last equality follows from the definition of $R_{\mathcal{J}_{e,l}}$ in (13). ■

Lemma 27 (Simplify $\tilde{Z}_{ij}(f)$) *For any $f, f^* \in \mathcal{F}$, and $t \in \mathbb{R}$, we have*

$$\begin{aligned}\tilde{Z}_{ij}(f) \coloneqq \mathbb{E}_{e|x_{ij},\delta_i}\left[\hat{Z}_{i,j}(f)\right] = {} & (f^*(x_{ij}) - f(x_{ij}))\left(\tau - \mathbb{P}\left(y_{i,j} \le f^*(x_{ij}) \,\middle|\, x_{ij}, \delta_i\right)\right) \\ & + \int_0^{f(x_{ij})-f^*(x_{ij})} \Big(\mathbb{P}(y_{i,j} \le f^*(x_{ij}) + t \,|\, x_{ij}, \delta_i) \\ & \qquad - \mathbb{P}(y_{i,j} \le f^*(x_{ij}) \,|\, x_{ij}, \delta_i)\Big) dt.\end{aligned}$$

right hand side needs to be conditioning on x

Proof Let $I(\cdot)$ denote the indicator function. From equation (B.3) in Belloni and Chernozhukov (2011), for any scalars w, v:

$$\rho_\tau(w - v) - \rho_\tau(w) = -v(\tau - I(w \le 0)) + \int_0^v (I(w \le z) - I(w \le 0))\, dz.$$

Let $w = y_{ij} - f^*(x_{ij})$ and $v = f(x_{ij}) - f^*(x_{ij})$. Substituting these into the equation:

$$\begin{aligned}\hat{Z}_{ij}(f) &= \rho_\tau(y_{ij} - f(x_{ij})) - \rho_\tau(y_{ij} - f^*(x_{ij})) \\ &= (f^*(x_{ij}) - f(x_{ij}))\,(\tau - I(y_{ij} - f^*(x_{ij}) \le 0)) \\ &\quad + \int_0^{f(x_{ij})-f^*(x_{ij})} \{I(y_{ij} - f^*(x_{ij}) \le t) - I(y_{ij} - f^*(x_{ij}) \le 0)\}\, dt.\end{aligned}$$

Take conditional expectation $\mathbb{E}_{e|x_{ij},\delta_i}[\cdot]$ on both sides:
For linear term,

$$\begin{aligned}&\mathbb{E}_{e|x_{ij},\delta_i}\left[(f^*(x_{ij}) - f(x_{ij}))\,(\tau - I(y_{ij} \le f^*(x_{ij})))\right] \\ &= (f^*(x_{ij}) - f(x_{ij}))\left(\tau - \mathbb{P}(y_{ij} \le f^*(x_{ij}) \,|\, x_{ij}, \delta_i\right).\end{aligned}$$

For integral term, for $w = y_{ij} - f^*(x_{ij})$ and $v = f(x_{ij}) - f^*(x_{ij})$, define it as:

$$\begin{aligned}&\mathbb{E}_{e|x_{ij},\delta_i}\left[\int_0^v (I(w \le t) - I(w \le 0))\, dt\right] \\ &= \int_0^v \mathbb{E}_{e|x_{ij},\delta_i}\left[I(w \le t) - I(w \le 0)\right] dt \quad \text{(Fubini's theorem)} \\ &= \int_0^v \left(\mathbb{P}(w \le t \,|\, x_{ij}, \delta_i) - \mathbb{P}(w \le 0 \,|\, x_{ij}, \delta_i)\right) dt.\end{aligned}$$

Substituting $v = f(x_{ij}) - f^*(x_{ij})$ and $w = y_{ij} - f^*(x_{ij})$, we obtain:

$$\begin{aligned}\tilde{Z}_{ij}(f) = {} & (f^*(x_{ij}) - f(x_{ij}))\left(\tau - \mathbb{P}(y_{ij} \le f^*(x_{ij}) \,|\, x_{ij}, \delta_i\right) \\ & + \int_0^{f(x_{ij})-f^*(x_{ij})} \left(\mathbb{P}(y_{i,j} \le f^*(x_{ij}) + t \,|\, x_{ij}, \delta_i) - \mathbb{P}(y_{i,j} \le f^*(x_{ij}) \,|\, x_{ij}, \delta_i)\right) dt.\end{aligned}$$

■

Lemma 28 (Simplify $Z_{ij}(f)$) *For any $f, f^* \in \mathcal{F}$, and $t \in \mathbb{R}$, we have*

$$Z_{ij}(f) := \mathbb{E}\left[\hat{Z}_{i,j}(f)|x_{ij}\right] = \int_0^{f(x_{ij})-f^*(x_{ij})} \{\mathbb{P}(y_{i,j} \leq f^*(x_{ij})+t|x_{ij}) - \mathbb{P}(y_{i,j} \leq f^*(x_{ij})|x_{ij})\}\, dt.$$

Proof By Belloni and Chernozhukov (2011) (Eq. B.3),

$$\rho_\tau(w-v) - \rho_\tau(w) = -v(\tau - I(w \leq 0)) + \int_0^v (I(w \leq z) - I(w \leq 0))\, dz.$$

Let $w = y_{ij} - f^*(x_{ij})$, $v = f(x_{ij}) - f^*(x_{ij})$. Substituting gives

$$\begin{aligned}\hat{Z}_{ij}(f) &= \rho_\tau(y_{ij} - f(x_{ij})) - \rho_\tau(y_{ij} - f^*(x_{ij})) \\ &= (f^*(x_{ij}) - f(x_{ij}))\,(\tau - I(y_{ij} \leq f^*(x_{ij}))) \\ &\quad + \int_0^v (I(w \leq t) - I(w \leq 0))\, dt.\end{aligned}$$

By the definition of f^* as the conditional τ-quantile:

$$\begin{aligned}&\mathbb{E}\left[\tau - I(y_{ij} \leq f^*(x_{ij}))\,\middle|\, x_{ij}\right] = \tau - \mathbb{P}(y_{ij} \leq f^*(x_{ij})\,|\, x_{ij}) \\ &= 0.\end{aligned}$$

For the second term, apply Fubini's theorem and quantile property:

$$\begin{aligned}&\mathbb{E}\left[\int_0^v (I(w \leq t) - I(w \leq 0))\, dt \,\middle|\, x_{ij}\right] \\ &= \int_0^v \mathbb{E}\left[I(w \leq t) - I(w \leq 0)\,\middle|\, x_{ij}\right] dt \\ &= \int_0^v \left(\mathbb{P}(w \leq t\,|\, x_{ij}) - \mathbb{P}(w \leq 0\,|\, x_{ij})\right) dt \\ &= \int_0^{f(x_{ij})-f^*(x_{ij})} \{\mathbb{P}(y_{i,j} \leq f^*(x_{ij})+t|x_{ij}) - \mathbb{P}(y_{i,j} \leq f^*(x_{ij})|x_{ij})\}\, dt.\end{aligned}$$

The last equality holds for both $v \geq 0$ and $v < 0$ through sign-unification as in Lemma 27. Thus, in the end, we have

$$Z_{ij}(f) = \int_0^{f(x_{ij})-f^*(x_{ij})} \{\mathbb{P}(y_{i,j} \leq f^*(x_{ij})+t|x_{ij}) - \mathbb{P}(y_{i,j} \leq f^*(x_{ij})|x_{ij})\}\, dt.$$

■

A.5.2 Bound of A_1

Lemma 29 (Decomposition of A_1) *Let $\tilde{Z}(f)$ and $Z(f)$ be defined in (68) and (67), respectively. Let Ω_β be defined in (76). Let $\ell(f)$ be as in (86), and let $\{s_{i,j}\}_{i\in\mathcal{J}_{e,l},\, j\in[m_i]}$ be independent Rademacher random variables, independent of the data $\{(x_{ij}, e_{ij}, \delta_i)\}$ and their coupled copies $\{(x^*_{ij}, e^*_{ij}, \delta^*_i)\}$. Let A_1 be the quantity introduced in (87). Suppose $\tilde{c}_0$ is a small constant from Assumption 4, and let $c_4 > 0$ be any fixed positive constant. Let L be as defined in (73), and assume $s < d$ as in Assumption 3, and T_n as in Assumption 4. Define the remainder term as*

$$r_{ij}(f) = \int_0^{f(x_{ij})-f^*(x_{ij})} \{\mathbb{P}(y_{i,j} \leq f^*(x_{ij})+u \mid x_{ij}) - \mathbb{P}(y_{i,j} \leq f^*(x_{ij})+u|\delta_i, x_{ij})\}\, du,$$

for $u \in \mathbb{R}$*. Then for any* $t > 0$*, we have*

$$
\begin{aligned}
A_1 &:= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\left(\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\le t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i} Z_{ij}(f)-\tilde{Z}_{ij}(f)-\frac{\ell(f)}{2L}\right)\cdot \mathbf{1}_{\Omega_\beta}\right]\\
&\le 2\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*,s}\left[\left(\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\le t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i} s_{ij}r_{ij}\right)\cdot \mathbf{1}_{\Omega_\beta}\right]\\
&\quad + \frac{c_4}{L}\frac{s\log n\log T_n}{n}.
\end{aligned}
$$

Proof By previous Lemmas 27 and 28,

By Lemma 28 where $Z_{ij}(f) = \mathbb{E}_{\delta,e|x_{ij}}\left[\hat{Z}_{ij}(f)\right]$, then by Belloni and Chernozhukov (2011) (Eq. B.3), we have

$$
\begin{aligned}
Z_{ij}(f) &= \left(f^*(x_{ij})-f(x_{ij})\right)\left(\tau-\mathbb{P}(y_{i,j}\le f^*(x_{ij})\mid x_{ij})\right)\\
&\quad + \int_0^{f(x_{ij})-f^*(x_{ij})}\left\{\mathbb{P}(y_{i,j}\le f^*(x_{ij})+t\mid x_{ij})-\mathbb{P}(y_{i,j}\le f^*(x_{ij})\mid x_{ij})\right\}dt\\
&\overset{(a)}{=} \int_0^{f(x_{ij})-f^*(x_{ij})}\left\{\mathbb{P}(y_{i,j}\le f^*(x_{ij})+t\mid x_{ij})-\mathbb{P}(y_{i,j}\le f^*(x_{ij})\mid x_{ij})\right\}dt.
\end{aligned}
\tag{93}
$$

The equality (a) holds because $f^*(x_{ij})$ is the conditional τ-quantile given x_{ij}, so $\tau-\mathbb{P}(y_{i,j}\le f^*(x_{ij})\mid x_{ij}) = 0$.

By Lemma 27, since $\tilde{Z}_{ij}(f) = \mathbb{E}_{e|\delta_i,x_{ij}}\left[\hat{Z}_{ij}(f)\right]$ conditions on both x_{ij} and δ_i,

$$
\begin{aligned}
\tilde{Z}_{ij}(f) &= \left(f^*(x_{ij})-f(x_{ij})\right)\left(\tau-\mathbb{P}(y_{i,j}\le f^*(x_{ij})\mid x_{ij},\delta_i)\right)\\
&\quad + \int_0^{f(x_{ij})-f^*(x_{ij})}\left\{\mathbb{P}(y_{i,j}\le f^*(x_{ij})+t\mid x_{ij},\delta_i)-\mathbb{P}(y_{i,j}\le f^*(x_{ij})\mid x_{ij},\delta_i)\right\}dt.
\end{aligned}
\tag{94}
$$

Subtracting (94) and (93), we obtain

$$
\begin{aligned}
&Z_{ij}(f)-\tilde{Z}_{ij}(f)\\
&= \int_0^{f(x_{ij})-f^*(x_{ij})}\left\{\mathbb{P}(y_{i,j}\le f^*(x_{ij})+t\mid x_{ij})-\mathbb{P}(y_{i,j}\le f^*(x_{ij})+t\mid \delta_i,x_{ij})\right\}dt\\
&\quad + \int_0^{f(x_{ij})-f^*(x_{ij})}\left\{\mathbb{P}(y_{i,j}\le f^*(x_{ij})\mid \delta_i,x_{ij})-\mathbb{P}(y_{i,j}\le f^*(x_{ij})\mid x_{ij})\right\}dt\\
&\quad - \left(f^*(x_{ij})-f(x_{ij})\right)\left(\tau-\mathbb{P}(y_{i,j}\le f^*(x_{ij})\mid x_{ij},\delta_i)\right).
\end{aligned}
$$

Thus, we have

$$
\begin{aligned}
&Z_{ij}(f)-\tilde{Z}_{ij}(f)\\
&= \mathbb{E}_{\delta,e|x_{ij}}\left[\hat{Z}_{ij}(f)\right]-\mathbb{E}_{e|\delta_i,x_{ij}}\left[\hat{Z}_{ij}(f)\right]\\
&= \int_0^{f(x_{ij})-f^*(x_{ij})}\left\{\mathbb{P}(y_{i,j}\le f^*(x_{ij})+t\mid x_{ij})-\mathbb{P}(y_{i,j}\le f^*(x_{ij})+t\mid \delta_i,x_{ij})\right\}dt
\end{aligned}
$$

$$
\begin{aligned}
&+\int_0^{f(x_{ij})-f^*(x_{ij})} \left\{\mathbb{P}(y_{i,j} \le f^*(x_{ij}) \mid \delta_i, x_{ij}) - \mathbb{P}(y_{i,j} \le f^*(x_{ij}) \mid x_{ij})\right\} dt \\
&-\left(f^*(x_{ij}) - f(x_{ij})\right)\left(\tau - \mathbb{P}(y_{i,j} \le f^*(x_{ij}) \mid x_{ij}, \delta_i)\right). \\
=& -\left(f^*(x_{ij}) - f(x_{ij})\right)\left(\tau - \mathbb{P}\left(y_{i,j} - f^*(x_{ij}) \le 0 | \delta_i, x_{ij}\right)\right. \\
&+\int_0^{f(x_{ij})-f^*(x_{ij})} \left\{\mathbb{P}(y_{i,j} \le f^*(x_{ij}) + t | x_{ij}) - \mathbb{P}(y_{i,j} \le f^*(x_{ij}) + t | \delta_i, x_{ij})\right\} dt \\
&+\int_0^{f(x_{ij})-f^*(x_{ij})} \left\{-\mathbb{P}(y_{i,j} \le f^*(x_{ij}) | x_{ij}) + \mathbb{P}(y_{i,j} \le f^*(x_{ij}) | \delta_i, x_{ij})\right\} dt \\
=& -\left(f^*(x_{ij}) - f(x_{ij})\right)\left(\tau - \mathbb{P}\left(y_{i,j} - f^*(x_{ij}) \le 0 | \delta_i, x_{ij}\right)\right. \\
&+\int_0^{f(x_{ij})-f^*(x_{ij})} \left\{\mathbb{P}(y_{i,j} \le f^*(x_{ij}) + t | x_{ij}) - \mathbb{P}(y_{i,j} \le f^*(x_{ij}) + t | \delta_i, x_{ij})\right\} dt \\
&+\left(f(x_{ij}) - f^*(x_{ij})\right)\left(-\tau + \mathbb{P}(y_{i,j} \le f^*(x_{ij}) | \delta_i, x_{ij})\right) \\
=& \int_0^{f(x_{ij})-f^*(x_{ij})} \left\{\mathbb{P}(y_{i,j} \le f^*(x_{ij}) + t | x_{ij}) - \mathbb{P}(y_{i,j} \le f^*(x_{ij}) + t | \delta_i, x_{ij})\right\} dt.
\end{aligned}
$$

first equality seems wrong We define $r_{ij}(f)$ the as the integral expression given by the last line:

$$
r_{ij}(f) := \int_0^{f(x_{ij})-f^*(x_{ij})} \left\{\mathbb{P}(y_{i,j} \le f^*(x_{ij}) + t \mid x_{ij}) - \mathbb{P}(y_{i,j} \le f^*(x_{ij}) + t | \delta_i, x_{ij})\right\} dt. \tag{95}
$$

Then, we have

$$
\begin{aligned}
A_1 :=& \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\left(\sup_{\substack{f\in\mathcal{F} \\ c_0\Delta^2(f-f^*)\le t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}} \frac{1}{m_i}\sum_{j=1}^{m_i} Z_{ij}(f) - \tilde{Z}_{ij}(f) - \frac{\ell(f)}{2L}\right)\cdot \mathbf{1}_{\Omega_\beta}\right] \\
\le& \underbrace{\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\left(\sup_{\substack{f\in\mathcal{F} \\ c_0\Delta^2(f-f^*)\le t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}} \frac{1}{m_i}\sum_{j=1}^{m_i} r_{ij}(f) - \frac{\ell(f)}{2L}\right)\cdot \mathbf{1}_{\Omega_\beta}\right]}_{A_4}.
\end{aligned} \tag{96}
$$

Now we bound A_4,and we have

$$\begin{aligned}
A_4 \leq & \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\Big(\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\leq t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i} r_{ij}(f) - \mathbb{E}_{x,e|\delta_i}\left[r_{ij}(f)\right]\Big)\cdot \mathbf{1}_{\Omega_\beta}\right] \\
& + \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\Big(\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\leq t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i} \mathbb{E}_{x,e|\delta_i}\left[r_{ij}(f)\right] - \frac{\ell(f)}{2L}\Big)\cdot \mathbf{1}_{\Omega_\beta}\right] \\
= & \mathbb{E}_{\delta,\delta^*}\Bigg[\underbrace{\mathbb{E}_{x,e|\delta}\left[\left(\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\leq t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i} r_{ij}(f) - \mathbb{E}_{x,e|\delta_i}\left[r_{ij}(f)\right]\right)\cdot \mathbf{1}_{\Omega_\beta}\right]\Bigg]}_{A_5} \\
& + \underbrace{\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\Big(\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\leq t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i} \mathbb{E}_{x,e|\delta_i}\left[r_{ij}(f)\right] - \frac{\ell(f)}{2L}\Big)\cdot \mathbf{1}_{\Omega_\beta}\right]}_{A_6}.
\end{aligned} \tag{97}$$

where we denote

$$A_5 := \mathbb{E}_{x,e|\{\delta_i\}}\left[\Big(\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\leq t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i} r_{ij}(f) - \mathbb{E}_{x,e|\delta_i}\left[r_{ij}(f)\right]\Big)\cdot \mathbf{1}_{\Omega_\beta}\right], \tag{98}$$

and

$$A_6 := \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\Big(\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\leq t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i} \mathbb{E}_{x,e|\delta_i}\left[r_{ij}(f)\right] - \frac{\ell(f)}{2L}\Big)\cdot \mathbf{1}_{\Omega_\beta}\right]. \tag{99}$$

Then by Lemma 31 and 32, we get the bound for terms A_5 and A_6, thus, conclude the bound for A_1. ■

Corollary 30 (Bound of A_1) *Let $\tilde{Z}(f)$, $Z(f)$ been defined in (68) and (67). Let Ω_β be defined in (76). Let $\ell(f)$ be defined in (86), let $\{s_{i,j}\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$ be independent Rademacher variables, independent of the data $\{(x_{ij}, e_{ij}, \delta_i)\}$ and their coupled copies $\{(x^*_{ij}, e^*_{ij}, \delta^*_i)\}$, let A_1 be the quantity introduced in (87), then we have*

$$\begin{aligned}
A_1 &= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\Big(\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\leq t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i} Z_{ij}(f) - \tilde{Z}_{ij}(f) - \frac{\ell(f)}{2L}\Big)\cdot \mathbf{1}_{\Omega_\beta}\right] \\
&\leq 2\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*,s}\left[\Big(\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\leq t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i} s_{ij}r_{ij}\Big)\cdot \mathbf{1}_{\Omega_\beta}\right] \\
&\quad + \frac{c_4}{L}\frac{s\log n\log T_n}{n}
\end{aligned}$$

$$\leq \frac{c_5}{L} R_{\mathcal{J}_{e,l}}(\mathcal{F} \cap \{c_0 \Delta^2(f - f^*) \leq t^2\}) + \frac{c_4}{L} \frac{s \log n \log T_n}{n},$$

with a small constant c_4 that depends on $\tilde{c}_0$ in Assumption 4, C_β in Assumption 1.4, and c_1, c_2 in (75), and c_5 is a constant that depends on constants C and c in Assumption 1.9, and $R_{\mathcal{J}_{e,l}}$ is defined in (13).

Proof Let

$$\ell(f) = \frac{4\tilde{c}_0 C}{c} \Delta^2(f - f^*),$$

by Lemma 29 incomplete sentence we get the bound of A_1.

We have

$$\begin{aligned}
&\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*} \left[\Big(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f - f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} Z_{ij}(f) - \tilde{Z}_{ij}(f) - \frac{\ell(f)}{2L} \Big) \cdot \mathbf{1}_{\Omega_\beta} \right] \\
&\overset{\text{(a)}}{\leq} 2\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*,s} \left[\Big(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f - f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} s_{ij} r_{ij}(f) \Big) \cdot \mathbf{1}_{\Omega_\beta} \right] \\
&\quad + \frac{c_4}{L} \frac{s \log n \log T_n}{n} \\
&\overset{\text{(b)}}{\leq} 2\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*,s} \left[\Big(\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f - f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} s_{ij} f(x_{ij}^*) \Big) \cdot \mathbf{1}_{\Omega_\beta} \right] \\
&\quad + \frac{c_4}{L} \frac{s \log n \log T_n}{n} \\
&\overset{\text{(c)}}{\leq} \frac{c_5}{L} R_{\mathcal{J}_{e,l}}(\mathcal{F} \cap \{c_0 \Delta^2(f - f^*) \leq t^2\}) + \frac{c_4}{L} \frac{s \log n \log T_n}{n}.
\end{aligned}$$

Here (a) follows from the decomposition in Lemma 29; (b) follows from the Lipschitz condition of $r_{ij}(f)$ in Lemma 19 together with the Ledoux-Talagrand contraction principle in Theorem 62; and (c) follows from Lemma 25.

say which line is due to what. Oscar pauses here ■

Lemma 31 (Bound of A_5) *Let A_5 be the term defined in (98). Let $\{s_{ij}\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ be independent Rademacher variables, independent of the data $\{(x_{ij}, e_{ij}, \delta_i)\}$ and their coupled copies $\{(x_{ij}^*, e_{ij}^*, \delta_i^*)\}$. Let Ω_β be defined in (76). The remainder term $r_{ij}(f)$ is defined in Lemma 29. Then the following inequality holds:*

$$A_5 \leq 2\mathbb{E}_{x,e,s|\delta} \left[\sup_{\substack{f \in \mathcal{F} \\ c_0 \Delta^2(f - f^*) \leq t^2}} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} s_{ij} r_{ij}(f) \cdot \mathbf{1}_{\Omega_\beta} \right].$$

Proof By the coupling identity (79), on Ω_β we have $(x_{ij}, \delta_i) = (x_{ij}^*, \delta_i^*)$ for all $i \in \mathcal{J}_{e,l}$, $j \in [m_i]$. Since $\{x_{ij}^*\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ are i.i.d. by construction, and $\{\delta_i^*\}_{i \in \mathcal{J}_{e,l}}$ are i.i.d., the quantities $\{r_{ij}(f)\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$ are conditionally independent given $\{\delta_i^*\}_{i \in \mathcal{J}_{e,l}}$ on the event Ω_β.

Let $\{\tilde{x}_{ij}\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$ be an independent and identically distributed copy of $\{x^*_{ij}\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$, correspondingly, we let $\{\tilde{y}_{i,j}\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$ be the copy of y^*_{ij}, where each $\tilde{y}_{ij}$ and y^*_{ij} share the same $\{(\delta^*_{i1},\ldots,\delta^*_{im_i})\}_{i\in\mathcal{J}_{e,l}}$, $\{e^*_{i,j}\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$, but differ only in $\{\tilde{x}_{ij}\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$ and $\{x^*_{ij}\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$. Then we define the independent and identically distributed copy of $r_{ij}(f)$ as

$$\tilde{r}_{ij}(f,\delta^*_i,\tilde{x}_{ij}) := \int_0^{f(\tilde{x}_{ij})-f^*(\tilde{x}_{ij})} \{\mathbb{P}(\tilde{y}_{i,j} \le f^*(\tilde{x}_{ij})+t) - \mathbb{P}(\tilde{y}_{i,j} \le f^*(\tilde{x}_{ij})+t|\delta^*_i)\}\, dt.$$

Now we control the $\mathbb{E}_{\tilde{x},x,e|\{\delta^*_i\}}\left[r_{ij}(f)-\tilde{r}_{ij}(f)\right]$, the expectation takes on $\{\tilde{x}_{ij}\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$ and conditioned on $\{(\delta^*_{i1},\ldots,\delta^*_{im_i})\}_{i\in\mathcal{J}_{e,l}}$. So we have

$$\begin{aligned}
&\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\le t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\left\{\mathbb{E}_{x,e|\delta_i}\left[r_{ij}(f)\right]-\mathbb{E}_{\tilde{x}|\delta_i}\left[\tilde{r}_{ij}(f)\right]\right\}\\
&\cdot \mathbf{1}_{\Omega_\beta}\\
&\overset{(a)}{\le} \mathbb{E}_{\tilde{x},x,e|\delta_i}\Bigg[\Bigg(\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\le t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\left\{r_{ij}(f)-\tilde{r}_{ij}(f)\right\}\Bigg)\\
&\cdot \mathbf{1}_{\Omega_\beta}\Bigg].
\end{aligned}\tag{100}$$

The inequality (a) is held by Jensen's inequality. Then, the left-hand side of the above, can be further written as

$$\begin{aligned}
&\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\le t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\left\{\mathbb{E}_{x,e|\delta_i}\left[r_{ij}(f)\right]-\mathbb{E}_{\tilde{x}|\delta_i}\left[\tilde{r}_{ij}(f)\right]\right\}\\
&\cdot \mathbf{1}_{\Omega_\beta}\\
&\overset{(a)}{=} \sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\le t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\left\{r_{ij}(f)-\mathbb{E}_{x,e|\delta_i}\left[r_{ij}(f)\right]\right\}\cdot \mathbf{1}_{\Omega_\beta}.
\end{aligned}\tag{101}$$

The expectation $\mathbb{E}_{x,e|\delta_i}\left[r_{ij}(f)\right]$ takes on $\{x_{ij}\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$ (which equal $\{x^*_{ij}\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$ on Ω_β) and conditioned on $\{(\delta^*_{i1},\ldots,\delta^*_{im_i})\}_{i\in\mathcal{J}_{e,l}}$. The equality (a) holds since $\mathbb{E}_{x,e|\delta^*_i}\left[r_{ij}(f)\right]$ and $\mathbb{E}_{\tilde{x}|\delta^*_i}\left[\tilde{r}_{ij}(f)\right]$ are identically distributed.

Combining (100) and (101), taking expectation of $\{x^*_{ij}\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$ conditioning on $\{(\delta^*_{i1},\ldots,\delta^*_{im_i})\}_{i\in\mathcal{J}_{e,l}}$ we have

$$\begin{aligned}
A_5 &= \mathbb{E}_{x,e|\{\delta_i\}}\Bigg[\Bigg(\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\le t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\left\{r_{ij}(f)-\mathbb{E}_{x,e|\delta_i}\left[r_{ij}(f)\right]\right\}\Bigg)\\
&\quad\cdot \mathbf{1}_{\Omega_\beta}\Bigg]\\
&\le \mathbb{E}_{\tilde{x},x,e|\{\delta_i\}}\Bigg[\Bigg(\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\le t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\left\{r_{ij}(f)-\tilde{r}_{ij}(f)\right\}\Bigg)\\
&\quad\cdot \mathbf{1}_{\Omega_\beta}\Bigg]
\end{aligned}$$

$$\overset{\text{(a)}}{\leq} \mathbb{E}_{\tilde{x},x,e,s|\{\delta_i\}}\Bigg[\Big(\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\leq t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i} s_{ij}\left\{r_{ij}(f)-\tilde{r}_{ij}(f)\right\}\Big)$$

$$\cdot \mathbf{1}_{\Omega_\beta}\Bigg]$$

$$\overset{\text{(b)}}{\leq} 2\mathbb{E}_{x,e,s|\{\delta_i\}}\Bigg[\Big(\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\leq t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i} s_{ij}r_{ij}(f)\Big)$$

$$\cdot \mathbf{1}_{\Omega_\beta}\Bigg].$$

The (a) holds since on Ω_β we have $(x_{ij},\delta_i)=(x^*_{ij},\delta^*_i)$, and $\{x^*_{ij}\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$ are i.i.d. by construction. Hence, conditioned on $\{(\delta^*_{i1},\ldots,\delta^*_{im_i})\}_{i\in\mathcal{J}_{e,l}}$, the $r_{ij}(f)-\tilde{r}_{ij}(f)$ is independently distributed and symmetric with mean zero, and $\{s_{ij}(r_{ij}(f)-\tilde{r}_{ij}(f))\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$ and $\{r_{ij}(f)-\tilde{r}_{ij}(f)\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$ have the same distribution. The (b) follows because $\{-s_{ij}\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$ are also independent Rademacher variables.

Taking expectation of $\{(\delta^*_{i1},\ldots,\delta^*_{im_i})\}_{i\in\mathcal{J}_{e,l}}$ on both sides, we obtain the result. ■

Lemma 32 (Generalized Bound of A_6) *In the multi-dimensional setting where $x_{ij}\in[0,1]^d$, let Ω_β be defined in (76). Let A_6 be defined as*

$$A_6 := \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\Big(\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\leq t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\left(\mathbb{E}_{x,e|\delta_i}\left[r_{ij}(f)\right]-\frac{\ell(f)}{2L}\right)\Big)\cdot\mathbf{1}_{\Omega_\beta}\right],$$

$\ell(f)$ be defined in (86), and $r_{ij}(f)$ be defined in (95) as

$$r_{ij}(f) = \int_0^{f(x_{ij})-f^*(x_{ij})}\left\{\mathbb{P}(y_{ij}\leq f^*(x_{ij})+t\mid x_{ij})-\mathbb{P}(y_{ij}\leq f^*(x_{ij})+t\mid \delta_i,x_{ij})\right\}dt.$$

Under Assumptions 3 and 4, and if $\ell(f)=4\tilde{c}_0c_2\Delta^2(f-f^*)$, then the following bound holds:

$$A_6 \leq \frac{c_2}{L}\cdot\frac{c_3^2}{4\tilde{c}_0}\cdot\frac{s\log n\log T_n}{n} = \frac{c_4}{L}\cdot\frac{s\log n\log T_n}{n},$$

for some positive constant c_4 that depends on $\tilde{c}_0$, C_β, c_1, and c_2, where c_3 depends on C_β and c_1, and where s and T_n are as defined in Assumption 4.

Proof We first show the following bound holds in a multidimensional setting where $x_{ij}\in[0,1]^d$:

$$\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\mathbb{E}_{x,e|\delta_i}\left[r_{ij}(f)\right] \leq \frac{2\tilde{c}_0c_2}{L}\Delta^2(f-f^*)+\frac{c_2}{L}\frac{c_3^2}{4\tilde{c}_0}\frac{s\log n\log T_n}{n}.$$

We then obtain the result by taking expectations and setting

$$\ell(f) = 4\tilde{c}_0c_2\Delta^2(f-f^*).$$

Given that

$$\begin{aligned}\mathbb{E}\left(r_{ij}(f)|\delta_i\right) = &\int_{[0,1]^d}\left(\int_0^{f(x)-f^*(x)}\left(\mathbb{P}\left(y_{ij} \le f^*(x_{ij})+u \mid x_{ij}=x\right)\right.\right.\\ &\left.\left.-\mathbb{P}\left(y_{ij} \le f^*(x_{ij})+u \mid \delta_i, x_{ij}=x\right)\right)du\right)dx.\end{aligned} \tag{102}$$

Thus, we modify the function $\Pi(x,u)$ in (8) as follows:

$$\Pi(x,u) \coloneqq \frac{1}{|\mathcal{J}_{e,l}|}\sum_{i\in\mathcal{J}_{e,l}}\left[\mathbb{P}(y_{i1} \le f^*(x_{i1})+u \mid x_{i1}=x)-\mathbb{P}(y_{i1} \le f^*(x_{i1})+u \mid \delta_i, x_{i1}=x)\right]. \tag{103}$$

We then have that

$$\begin{aligned}&\left|\int_{[0,1]^d}\int_0^{f(x)-f^*(x)}\Pi(x,u)\,du\,dx\right|\\ &=\left|\int_{[0,1]^d}\left[\int_0^{f(x)-f^*(x)}\Pi(x,u)\,du\right]\cdot\mathbb{I}_{[\{|f(x)-f^*(x)|\le 1\}]}dx\right.\\ &\quad\left.+\int_{[0,1]^d}\left[\int_0^{f(x)-f^*(x)}\Pi(x,u)\,du\right]\cdot\mathbb{I}_{[|f(x)-f^*(x)|>1]}\,dx\right|\\ &\le\int_{[0,1]^d}|f(x)-f^*(x)|\cdot\max_{x\in[0,1]^d}\max_{u\in[-1,1]}|\Pi(x,u)|\cdot\mathbb{I}_{[|f(x)-f^*(x)|\le 1]}\,dx\\ &\quad+\int_{[0,1]^d}|f(x)-f^*(x)|\cdot\max_{x\in[0,1]^d}\max_{u\in[-|f(x)-f^*(x)|,|f(x)-f^*(x)|]}|\Pi(x,u)|\\ &\quad\cdot\mathbb{I}_{[|f(x)-f^*(x)|>1]}\,dx\end{aligned} \tag{104}$$

Step 1: Now, for the $\max_{x\in[0,1]^d}\max_{u\in[-|f(x)-f^*(x)|,|f(x)-f^*(x)|]}|\Pi(x,u)|$ term above, we have

$$\begin{aligned}\max_{x\in[0,1]^d}\max_u\left|\frac{1}{|\mathcal{J}_{e,l}|}\sum_{i\in\mathcal{J}_{e,l}}\right.&\left[\mathbb{P}(y_{i1} \le f^*(x_{i1})+u \mid x_{i1}=x)\right.\\ &\left.\left.-\mathbb{P}(y_{i1} \le f^*(x_{i1})+u \mid \delta_i, x_{i1}=x)\right]\right|.\end{aligned}$$

This term represents the maximum discrepancy between the probabilities of y_{i1} not exceeding the threshold defined by $f^*(x_{i1})+u$, with and without the consideration of δ_i.

Let $T_n>0$ be a constant that depends on n, then we have

$$\begin{aligned}&\max_{u>T_n}\max_{x\in[0,1]^d}\left|\frac{1}{|\mathcal{J}_{e,l}|}\sum_{i\in\mathcal{J}_{e,l}}\left[\mathbb{P}(\delta_i(x_{i1})+e_{i1} \le u \mid \delta_i, x_{i1}=x)-\mathbb{P}(\delta_i(x_{i1})+e_{i1} \le u \mid x_{i1}=x)\right]\right|\\ &\le\max_{u>T_n}\max_{x\in[0,1]^d}\max_{i\in\mathcal{J}_{e,l}}\left[\mathbb{P}(\delta_i(x_{i1})+e_{i1} \ge u \mid \delta_i, x_{i1}=x)+\mathbb{P}(\delta_i(x_{i1})+e_{i1} \ge u \mid x_{i1}=x)\right]\\ &\le\max_{x\in[0,1]^d}\max_{i\in\mathcal{J}_{e,l}}\left[\mathbb{P}(\delta_i(x_{i1})+e_{i1} \ge T_n \mid \delta_i, x_{i1}=x)+\mathbb{P}(\delta_i(x_{i1})+e_{i1} \ge T_n \mid x_{i1}=x)\right]\\ &\overset{(a)}{\le}\tilde{c}_0 \text{ with high probability.}\end{aligned}$$

The (a) holds by Assumption 4. Since

$$\begin{aligned}
&\max_{x\in[0,1]^d}\max_{i\in\mathcal{J}_{e,l}}\left[\mathbb{P}(\delta_i(x_{i1})+e_{i1}\ \ge\ T_n\mid \delta_i,x_{i1}=x)+\mathbb{P}(\delta_i(x_{i1})+e_{i1}\ \ge\ T_n\mid x_{i1}=x)\right]\\
&\le \max_{x\in[0,1]^d}\max_{i=1,\ldots,n}\left[\mathbb{P}(\delta_i(x_{i1})+e_{i1}\ \ge\ T_n\mid \delta_i,x_{i1}=x)+\mathbb{P}(\delta_i(x_{i1})+e_{i1}\ \ge\ T_n\mid x_{i1}=x)\right]\ \le\ \tilde{c}_0
\end{aligned}$$

Similarly, we have

$$\begin{aligned}
&\max_{u<-T_n}\max_{x\in[0,1]^d}\left|\frac{1}{|\mathcal{J}_{e,l}|}\sum_{i\in\mathcal{J}_{e,l}}\left[\mathbb{P}(\delta_i(x_{i1})+e_{i1}\ \le\ u\mid \delta_i,x_{i1}=x)-\mathbb{P}(\delta_i(x_{i1})+e_{i1}\ \le\ u\mid x_{i1}=x)\right]\right|\\
&\le \max_{u<-T_n}\max_{x\in[0,1]^d}\max_{i\in\mathcal{J}_{e,l}}\left[\mathbb{P}(\delta_i(x_{i1})+e_{i1}\ \le\ u\mid \delta_i,x_{i1}=x)+\mathbb{P}(\delta_i(x_{i1})+e_{i1}\ \le\ u\mid x_{i1}=x)\right]\\
&\le \max_{x\in[0,1]^d}\max_{i\in\mathcal{J}_{e,l}}\left[\mathbb{P}(\delta_i(x_{i1})+e_{i1}\ \le\ -T_n\mid \delta_i,x_{i1}=x)+\mathbb{P}(\delta_i(x_{i1})+e_{i1}\ \le\ -T_n\mid x_{i1}=x)\right]\\
&\overset{(a)}{\le} \tilde{c}_0 \text{ with high probability.}
\end{aligned}$$

The (a) holds by Assumption 4. Since

$$\begin{aligned}
&\max_{x\in[0,1]^d}\max_{i\in\mathcal{J}_{e,l}}\left[\mathbb{P}(\delta_i(x_{i1})+e_{i1}\ \ge\ -T_n\mid \delta_i,x_{i1}=x)+\mathbb{P}(\delta_i(x_{i1})+e_{i1}\ \ge\ -T_n\mid x_{i1}=x)\right]\\
&\le \max_{x\in[0,1]^d}\max_{i=1,\ldots,n}\left[\mathbb{P}(\delta_i(x_{i1})+e_{i1}\ \ge\ -T_n\mid \delta_i,x_{i1}=x)+\mathbb{P}(\delta_i(x_{i1})+e_{i1}\ \ge\ -T_n\mid x_{i1}=x)\right]\ \le\ \tilde{c}_0.
\end{aligned}$$

Step 2: Next, for $u\in[-T_n,T_n]$, we define:

$$\max_{x_S\in[0,1]^s}\max_{u\in[-T_n,T_n]}\left\{\underbrace{\left[\mathbb{P}(y_{i1}\le f^*(x_{i1})+u\mid x_{i1}=x_S)-\mathbb{P}(y_{i1}\le f^*(x_{i1})+u|\delta_i,x_{i1}=x_S)\right]}_{:=\Pi(x_S,u)}\right\}.$$

Consider the space $[0,1]^s$. For the i-th dimension of $[0,1]^s$, let $0=a_{1,i}<\cdots<a_{M,i}\ \le\ a_{M+1,i}=1$ be an evenly spaced partition. Let $c_{k,i}$ be the midpoint of $[a_{k,i},a_{k+1,i}]$ for $k=1,\ldots,M$. Define the multi-dimensional grid points in $[0,1]^s$ as $c_k=(c_{k,1},c_{k,2},\ldots,c_{k,s})$. Similarly, let $-T_n=b_1<\cdots<b_M\ \le\ b_{M+1}=T_n$, partition the interval $[-T_n,T_n]$ into M equal parts with midpoints d_l. For each $k=1,\ldots,M$, given vectors $x_S,c_k\in\mathbb{R}^s$ and an index $i\in\{1,2,\ldots,s\}$, we define a new vector $x_S^{\backslash ki}\in\mathbb{R}^s$ via

$$x_{Sj}^{\backslash ki}:=\begin{cases}x_{Sj} & \text{if } j\neq i,\\ c_{k,i} & \text{if } j=i.\end{cases}$$

The $x_S^{\backslash ki}$, $i=1,\ldots,s$ indicates with i-th entry of x_S replaced with c_{ki}.

Given a Lipschitz function $\Pi:[0,1]^s\times[-T_n,T_n]\to\mathbb{R}$, we approximate $\Pi(x_S,u)$ as follows:

$$\begin{aligned}
&\max_{x_S\in[0,1]^s}\max_{u\in[-T_n,T_n]}|\Pi(x_S,u)|\\
&\le \max_{k1=1,\ldots,M}\max_{k2=1,\ldots,M}\cdots\max_{ks=1,\ldots,M}\max_{l=1,\ldots,M}|\Pi((c_{k1},c_{k2},\ldots,c_{ks}),d_l)|\\
&\quad+\max_{x_S\in[0,1]^s}\max_{u\in[-T_n,T_n]}\min_{l=1,\ldots,M}\min_{k1=1,\ldots,M}|\Pi(x_S^{\backslash k1},d_l)-\Pi(x_S,u)|\\
&\quad+\cdots\\
&\quad+\max_{x_S\in[0,1]^s}\max_{u\in[-T_n,T_n]}\min_{l=1,\ldots,M}\min_{ks=1,\ldots,M}|\Pi(x_S^{\backslash ks},d_l)-\Pi(x_S,u)|.
\end{aligned}$$

For $x_S \in [a_{k,1}, a_{k+1,1}] \times \cdots \times [a_{k,s}, a_{k+1,s}]$ and $u \in [b_l, b_{l+1}]$, we estimate:

$$\begin{aligned}
&|\Pi(x_S^{\setminus ki}, d_l) - \Pi(x_S, u)| \\
&\leq\ |\Pi(x_S^{\setminus ki}, u) - \Pi(x_S^{\setminus ki}, d_l)| + |\Pi(x_S^{\setminus ki}, u) - \Pi(x_S, u)| \\
&\leq\ \frac{\alpha \cdot 2T_n}{M} + \frac{\alpha}{M} \lesssim \frac{T_n}{M}.
\end{aligned}$$

The last inequality follows by (9). By Assumption 3, Π is Lipschitz, leading to the following probability bound:

$$\begin{aligned}
&\mathbb{P}\left(\max_{x_S \in [0,1]^s} \max_{u \in [-T_n, T_n]} |\Pi(x_S, u)| - \frac{sT_n}{M}\ \geq\ t \right) \\
&\leq\ \mathbb{P}\left(\max_{k1=1,\ldots,M} \max_{k2=1,\ldots,M} \cdots \max_{ks=1,\ldots,M} \max_{l=1,\ldots,M} |\Pi((c_{k1}, c_{k2}, \ldots, c_{ks}), d_l)|\ \geq\ t \right) \\
&\leq\ M^s e^{-2|\mathcal{J}_{e,l}|t^2}\ \leq\ M^s e^{-\frac{2c_1 C_\beta n t^2}{\log n}},
\end{aligned}$$

where the second last inequality holds since $|\Pi(\cdot)| \leq 1$ uniformly, and then using the Hoeffding's inequality and union bound for the $\Pi(\cdot)$ defined in (233). Take $M = \left(sT_n \cdot \sqrt{\frac{n}{\log n}}\right)$, therefore:

$$\max_{x_S \in [0,1]^s} \max_{u \in [-T_n, T_n]} |\Pi(x, u)|\ \leq\ c_3 \sqrt{\frac{s \log n \log T_n}{n}}\ =\ O_p\left(\sqrt{\frac{s \log n \log T_n}{n}}\right). \tag{105}$$

where the c_3 is a constant that depends on c_1 and C_β.

Step 3 : Returning to (232), using Assumption $|\mathcal{J}_{e,l}| \leq c_2 n/L$ in (75), we have:

$$\begin{aligned}
&\frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} |\mathbb{E}[r_{ij}(f)|\delta_i]| \overset{\text{(a)}}{=} \frac{1}{n} \sum_{i \in \mathcal{J}_{e,l}} |\mathbb{E}[r_{ij}(f)|\delta_i]| \\
&\overset{\text{(b)}}{\leq}\ \frac{c_2}{L} \Bigg| \frac{1}{|\mathcal{J}_{e,l}|} \sum_{i \in \mathcal{J}_{e,l}} \int_{[0,1]^d} \int_0^{f(x)-f^*(x)} \Big[\mathbb{P}(y_{i1}\ \leq\ f^*(x_{i1}) + u \mid x_{i1} = x) \\
&\qquad\qquad - \mathbb{P}(y_{i1}\ \leq\ f^*(x_{i1}) + u \mid \delta_i, x_{i1} = x)\Big] du\, dx \Bigg| \\
&\leq\ \frac{c_2}{L} \int_{[0,1]^d} |f(x) - f^*(x)| \cdot \max_{x \in [0,1]^d} \max_{u \in [-1,1]} |\Pi(x, u)| \cdot \nVdash_{\{f(x)-f^*(x) \leq 1\}}\, dx \\
&\quad + \frac{c_2}{L} \int_{[0,1]^d} |f(x) - f^*(x)| \cdot \max_{x \in [0,1]^d} \max_{u \in [-|f(x)-f^*(x)|, |f(x)-f^*(x)|]} |\Pi(x, u)| \cdot \nVdash_{\{f(x)-f^*(x) > 1\}}\, dx \\
&\leq\ \tilde{c}_0 \frac{c_2}{L} \int_{[0,1]^d} |f(x) - f^*(x)| \cdot \nVdash_{\{f(x)-f^*(x) > 1\}}\, dx \\
&\quad + \frac{c_2}{L} \int_{[0,1]^d} |f(x) - f^*(x)| \cdot \nVdash_{\{f(x)-f^*(x) \leq 1\}}\, dx \cdot c_3 \sqrt{\frac{s \log n \log T_n}{n}} \\
&\leq\ \tilde{c}_0 \frac{c_2}{L} \Delta^2(f - f^*) + \frac{c_2}{L} \Delta(f - f^*) \cdot c_3 \sqrt{\frac{s \log n \log T_n}{n}} \\
&\overset{\text{(c)}}{\leq}\ \tilde{c}_0 \frac{c_2}{L} \Delta^2(f - f^*) + \tilde{c}_0 \frac{c_2}{L} \Delta^2(f - f^*) + \frac{c_3^2 c_2}{L} \frac{1}{4\tilde{c}_0} \frac{s \log n \log T_n}{n}
\end{aligned}$$

$$\leq \frac{2\tilde{c}_0 c_2}{L}\Delta^2(f-f^*) + \frac{c_2}{L}\frac{c_3^2}{4\tilde{c}_0}\frac{s\log n\log T_n}{n},$$

where the equality (a) uses the fact for fixed i, $\mathbb{E}[r_{ij}(f)|\delta_i] = \mathbb{E}[r_{i1}(f)|\delta_i]$ for all $j = 1, \ldots, m_i$, where the equality (b) follows by $|\mathcal{J}_{e,l}| \leq c_2 n/L$, where the inequality (c) uses the bound $ab \leq \frac{pa^2}{2} + \frac{b^2}{2p}$ for any $p > 0$, setting $p = 2\tilde{c}_0$.

Thus, we have

$$\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\mathbb{E}_{x,e|\delta_i}\left[r_{ij}(f)\right] \leq \frac{2\tilde{c}_0 c_2}{L}\Delta^2(f-f^*) + \frac{c_2}{L}\frac{c_3^2}{4\tilde{c}_0}\frac{s\log n\log T_n}{n}.$$

Adding expectation and supremum on both sides, we get

$$\begin{aligned}
&\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\left(\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\leq t^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\mathbb{E}_{x,e|\delta_i}\left[r_{ij}(f)\right]\right)\cdot\mathbf{1}_{\Omega_\beta}\right]\\
&\quad\leq \frac{2\tilde{c}_0 c_2}{L}\Delta^2(f-f^*) + \frac{c_2}{L}\frac{c_3^2}{4\tilde{c}_0}\frac{s\log n\log T_n}{n}.
\end{aligned}$$

■

Below, we provide examples under which Assumptions (10) and (11) from Assumption 4 are satisfied.

Example 3 *Assume the event*

$$\Omega_1 = \left\{\max_{i=1,\ldots,n}\|\delta_i\|_\infty \leq R_n\right\}$$

occurs with high probability. Furthermore, define $e_{i1} = g(x_{i1})V_{i1}$ where $\{V_{i1}\}$ is independent from $\{x_{i1}\}$, and consider the event

$$\Omega_2 = \left\{\max_{i=1,\ldots,n}|g(x_{i1})| \leq s_n\right\}$$

also occurring with high probability. We then introduce T_n as

$$T_n = \max_{i=1,\ldots,n}\{2R_n, s_n\phi_n\},$$

with ϕ_n being a sequence tending to infinity. Then we have

$$\begin{aligned}
&\max_{x\in[0,1]^d}\max_{i=1,\ldots,n}\left[\mathbb{P}(\delta_i(x_{i1}) + e_{i1} \geq T_n \mid \delta_i, x_{i1} = x)\right.\\
&\quad\left.+\mathbb{P}(\delta_i(x_{i1}) + e_{i1} \geq T_n \mid x_{i1} = x)\right]\\
&\overset{(a)}{\leq} \max_{x\in[0,1]^d}\max_{i=1,\ldots,n}\left[2\mathbb{P}\left(\|\delta_i\|_\infty \geq R_n\right)\right.\\
&\quad+2\mathbb{P}(\Omega_2^c)\\
&\quad\left.+2\mathbb{P}\left(|V_{i1}| \geq \max\left\{\frac{R_n}{s_n}, \frac{\phi_n}{2}\right\}\right)\right]
\end{aligned}$$

$$\leq \tilde{c}_0.$$

The (a) *holds since we have*

$$\begin{aligned}
&\max_{x\in[0,1]^d}\max_{i=1,\dots,n}\left[\mathbb{P}(\delta_i(x_{i1})+e_{i1}\ \geq\ T_n \mid \delta_i, x_{i1}=x)\right.\\
&\quad\left.+\mathbb{P}(\delta_i(x_{i1})+e_{i1}\ \geq\ T_n \mid x_{i1}=x)\right]\\
&\leq \max_{x\in[0,1]^d}\max_{i=1,\dots,n}\left[\mathbb{P}\left(g(x_{i1})V_{i1}\ \geq\ \max\left\{R_n, \frac{s_n\phi_n}{2}\right\}|x_{i1}=x\right)\right.\\
&\quad+2\mathbb{P}\left(\|\delta_i\|_\infty\ \geq\ \max\left\{R_n, \frac{s_n\phi_n}{2}\right\}\right)\\
&\quad\left.+\mathbb{P}\left(\left\{g(x_{i1})V_{i1}\geq \max\left\{R_n, \frac{s_n\phi_n}{2}\right\}\right\}|x_{i1}=x\right)\right]\\
&\leq \max_{x\in[0,1]^d}\max_{i=1,\dots,n}\left[2\mathbb{P}\left(\|\delta_i\|_\infty\ \geq\ \max\left\{R_n, \frac{s_n\phi_n}{2}\right\}\right)\right.\\
&\quad+2\mathbb{P}(\Omega_2^c \mid x_{i1}=x)\\
&\quad\left.+2\mathbb{P}\left(g(x_{i1})V_{i1}\ \geq\ \max\left\{R_n, \frac{s_n\phi_n}{2}\right\}|\Omega_2, x_{i1}=x\right)\right]\\
&\leq \tilde{c}_0.
\end{aligned}$$

Provided

1. $\mathbb{P}(\Omega_1^c)\to 0$.
2. $\mathbb{P}(\Omega_2^c)\to 0$.
3. $\max_{i=1,\dots,n}\mathbb{P}(|V_{i1}|\geq\phi_n)\to 0$.
4. $T_n\geq\max\{2R_n, \phi_n s_n\}$.

Similarly, we can get for the multidimensional domain $[0,1]^d$

$$\begin{aligned}
&\max_{x\in[0,1]^d}\max_{i=1,\dots,n}\left[\mathbb{P}(\delta_i(x_{i1})+e_{i1}\ \leq\ -T_n \mid \delta_i, x_{i1}=x)\right.\\
&\quad\left.+\mathbb{P}(\delta_i(x_{i1})+e_{i1}\ \leq\ -T_n \mid x_{i1}=x)\right]\\
&\leq \tilde{c}_0.
\end{aligned}$$

So in summary, (10) *and* (11) *of Assumption 4 hold if the above conditions hold for the multidimensional domain* $[0,1]^d$.

Lemma 33 (Bound of A_3) *Let $\bar{Z}(f)$ and $Z(f)$ be defined in* (69) *and* (67)*, respectively. Let Ω_β be defined in* (76)*. Let $\{s_{ij}\}_{i\in\mathcal{J}_{e,l},\, j\in[m_i]}$ be independent Rademacher random variables, independent of the data $\{(x_{ij}, e_{ij}, \delta_i)\}$ and their coupled copies $\{(x^*_{ij}, e^*_{ij}, \delta^*_i)\}$. Let A_3 be the quantity introduced in* (87)*. Then, for any $t>0$, we have*

$$A_3\ =\ \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\left(\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\leq t^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\left(\bar{Z}_{ij}(f)-Z_{ij}(f)\right)\right)\cdot\mathbf{1}_{\Omega_\beta}\right]$$

$$\le 2\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*,s}\left[\Big(\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\le t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i} s_{ij}Z_{ij}(f)\Big)\cdot \mathbf{1}_{\Omega_\beta}\right]$$
$$= \frac{c_5}{L}R_{\mathcal{J}_{e,l}}\left(\mathcal{F}\cap\left\{c_0\Delta^2(f-f^*)\ \le\ t^2\right\}\right),$$

where c_5 depends only on the constants C and c in Assumption 1.9, and $R_{\mathcal{J}_{e,l}}$ is defined in (13).

Proof We pass from A_3 to a star-world quantity A_3^* via the coupling, and then bound A_3^*.

By the coupling identity (79), on Ω_β we have $(x_{ij}, e_{ij}, \delta_i) = (x_{ij}^*, e_{ij}^*, \delta_i^*)$, so that

$$\bar{Z}_{ij}(f) = \bar{Z}_{ij}^*(f), \qquad Z_{ij}(f) = Z_{ij}^*(f) \qquad \text{on } \Omega_\beta.$$

Therefore,

$$\begin{aligned}&\Big(\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\le t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\left(\bar{Z}_{ij}(f)-Z_{ij}(f)\right)\Big)\cdot\mathbf{1}_{\Omega_\beta}\\ &=\Big(\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\le t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\left(\bar{Z}_{ij}^*(f)-Z_{ij}^*(f)\right)\Big)\cdot\mathbf{1}_{\Omega_\beta}.\end{aligned} \tag{106}$$

Since f^* belongs to the constraint set and $\bar{Z}_{ij}^*(f^*) - Z_{ij}^*(f^*)\ =\ 0$, the supremum is nonnegative. Hence $\mathbf{1}_{\Omega_\beta}\ \le\ 1$ implies

$$A_3 \le \mathbb{E}_{x^*,\delta^*,e^*}\left[\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\le t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\left(\bar{Z}_{ij}^*(f)-Z_{ij}^*(f)\right)\right] =: A_3^*. \tag{107}$$

It remains to bound A_3^*. Let $\{\check{x}_{ij}\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$ be an independent and identically distributed copy of $\{x_{ij}^*\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$, then we define the independent and identically distributed copy of $Z_{ij}^*(f)$ as

$$\check{Z}_{ij}(f) \coloneqq \mathbb{E}_{\delta^*,e^*|\check{x}_{ij}}\left[\rho_\tau(f^*(\check{x}_{ij})+\delta_i^*(\check{x}_{ij})+e_{ij}^*-f(\check{x}_{ij}))-\rho_\tau(\delta_i^*(\check{x}_{ij})+e_{ij}^*)\right].$$

Now we control $\mathbb{E}_{\check{x}|x^*,\delta^*,e^*}\left[\check{Z}_{ij}(f)-Z_{ij}^*(f)\right]$, the expectation takes on $\{\check{x}_{ij}\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$. So we have

$$\begin{aligned}&\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\le t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\left\{\mathbb{E}_{\check{x}}\left[\check{Z}_{ij}(f)\right]-\mathbb{E}_{\check{x}}\left[Z_{ij}^*(f)\right]\right\}\\ &\overset{\text{(a)}}{\le}\mathbb{E}_{\check{x}}\left[\Big(\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\le t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\left\{\check{Z}_{ij}(f)-Z_{ij}^*(f)\right\}\Big)\right].\end{aligned} \tag{108}$$

The inequality (a) is held by Jensen's inequality. Then, the left-hand side of the above, can be further written as

$$\begin{aligned}&\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\le t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\left\{\mathbb{E}_{\check{x}}\left[\check{Z}_{ij}(f)\right]-Z_{ij}^*(f)\right\}\\ &\overset{\text{(a)}}{=}\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\le t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\left\{\bar{Z}_{ij}^*(f)-Z_{ij}^*(f)\right\}.\end{aligned} \tag{109}$$

The equality (a) holds since given δ_i^*, $\check{Z}_{ij}(f)$ is independent of x^*, e^* and has the same conditional distribution as $Z_{ij}^*(f)$, so $\mathbb{E}_{\check{x}}\left[\check{Z}_{ij}(f)\right]$ and $\bar{Z}_{ij}^*(f) = \mathbb{E}_{x^*,\delta^*,e^*}\left[\hat{Z}_{ij}^*(f)\right]$ are the same quantity.

Combining (229) and (235), taking expectation of $\{x_{ij}^*\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$ we have

$$\begin{aligned}
A_3^* &= \mathbb{E}_{x^*,\delta^*,e^*}\left[\left(\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\le t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\left\{\bar{Z}_{ij}^*(f) - Z_{ij}^*(f)\right\}\right)\right] \\
&\le \mathbb{E}_{\check{x},x^*,\delta^*,e^*}\left[\left(\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\le t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\left\{\check{Z}_{ij}(f) - Z_{ij}^*(f)\right\}\right)\right] \\
&\overset{(a)}{=} \mathbb{E}_{\check{x},x^*,\delta^*,e^*,s}\left[\left(\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\le t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}s_{ij}\left\{\check{Z}_{ij}(f) - Z_{ij}^*(f)\right\}\right)\right] \\
&\overset{(b)}{\le} 2\mathbb{E}_{x^*,\delta^*,e^*,s}\left[\left(\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\le t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}s_{ij}Z_{ij}^*(f)\right)\right].
\end{aligned}$$

The (a) holds since $\{x_{ij}^*\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$ are i.i.d. by construction, and $\{\check{x}_{ij}\}$ is an independent copy with the same distribution. Hence the $\left\{\check{Z}_{ij}(f) - Z_{ij}^*(f)\right\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$ is independently distributed and symmetric with mean zero, and $\left\{s_{ij}(\check{Z}_{ij}(f) - Z_{ij}^*(f))\right\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$ and $\left\{\check{Z}_{ij}(f) - Z_{ij}^*(f)\right\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$ have the same distribution. The (b) follows because $\{-s_{ij}\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$ are also independent Rademacher variables.

We apply the Lipschitz condition of $\hat{Z}_{ij}(f)$, $Z_{ij}(f)$ and $r_{ij}(f)$ in Lemma 19, and then apply the Ledoux-Talagrand contraction principle in Theorem 62.

Then we have

$$\begin{aligned}
A_3 \le A_3^* &\le 2\mathbb{E}_{x^*,\delta^*,e^*,s}\left[\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\le t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}s_{ij}Z_{ij}^*(f)\right] \\
&\le 2\mathbb{E}_{x^*,s}\left[\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\le t^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}s_{ij}f(x_{ij}^*)\right] \\
&= \frac{c_5}{L}R_{\mathcal{J}_{e,l}}(\mathcal{F}\cap\left\{c_0\Delta^2(f-f^*)\le t^2\right\}).
\end{aligned} \tag{110}$$

Where the second inequality follows from the Lipschitz condition and the Ledoux-Talagrand contraction principle, and the last equality follows from Lemma 25. ■

A.6 Proof of Step 4

Lemma 34 (Bound of A_1, A_2, and A_3 by Rademacher width) *Let $\hat{f}, f^*\in\mathcal{F}$. Let Ω_β be defined in (76). Suppose the distribution of $\{(x_{ij},y_{ij})\}_{i=1,j=1}^{n,m_i}$ satisfies Assumption 2.*

Let c_0 satisfy (84). Let $R(\mathcal{F})$ denote the Rademacher width defined in 12, and let $\tilde{c}_0$ be the small constant given in Assumption 4.

Then, for any $t > 0$, the following inequality holds:

$$\mathbb{P}\left(c_0\Delta^2(\hat{f}-f^*) > t^2\right) \le \frac{c_5}{t^2}\sum_{l=1}^{L}\Big(R_{\mathcal{J}_{e,l}}\left(\mathcal{F}\cap\left\{c_0\Delta^2(f-f^*)\le t^2\right\}\right) + R_{\mathcal{J}_{o,l}}\left(\mathcal{F}\cap\left\{c_0\Delta^2(f-f^*)\le t^2\right\}\right)\Big) + \frac{c_4}{t^2}\cdot\frac{s\log n\log T_n}{n}, \tag{111}$$

where c_5 depends only on the constants C and c in Assumption 1.9, c_4 depends on $\tilde{c}_0$ in Assumption 4, C_β in Assumption 1.4, and c_1, c_2 in (75), and $R_{\mathcal{J}_{e,l}}$, $R_{\mathcal{J}_{o,l}}$ are defined in (13).

Proof

We apply the Lipschitz condition of $\hat{Z}_{ij}(f)$, $Z_{ij}(f)$ and $r_{ij}(f)$ in Lemma 19, and then apply the Ledoux-Talagrand contraction principle in Theorem 62. We apply the Lemmas 30, 26 and 33. let A_1 be the quantity introduced in (87), By Lemma 30, we have

$$\begin{aligned} A_1 &= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\Big(\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\le t^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}Z_{ij}(f)-\tilde{Z}_{ij}(f)-\frac{\ell(f)}{2L}\Big)\cdot\mathbf{1}_{\Omega_\beta}\right] \\ &\le \frac{c_5}{L}R_{\mathcal{J}_{e,l}}(\mathcal{F}\cap\left\{c_0\Delta^2(f-f^*)\le t^2\right\}) + \frac{c_4}{L}\frac{s\log n\log T_n}{n}. \end{aligned}$$

Similarly, we have for B_1 in (87)

$$B_1 \le \frac{c_5}{L}R_{\mathcal{J}_{o,l}}(\mathcal{F}\cap\left\{c_0\Delta^2(f-f^*)\le t^2\right\}) + \frac{c_4}{L}\frac{s\log n\log T_n}{n}.$$

Let A_2 be the quantity introduced in (87), by Lemma 26, we have

$$\begin{aligned} A_2 &= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\Big(\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\le t^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\left\{\tilde{Z}_{ij}(f)-\hat{Z}_{ij}(f)\right\}\Big)\cdot\mathbf{1}_{\Omega_\beta}\right] \\ &\le \frac{c_5}{L}R_{\mathcal{J}_{e,l}}(\mathcal{F}\cap\left\{c_0\Delta^2(f-f^*)\le t^2\right\}). \end{aligned}$$

Similarly, we have

$$B_2 \le \frac{c_5}{L}R_{\mathcal{J}_{o,l}}(\mathcal{F}\cap\left\{c_0\Delta^2(f-f^*)\le t^2\right\}).$$

By Lemma 33 we have

$$\begin{aligned} A_3 &= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\Big(\sup_{\substack{f\in\mathcal{F}\\ c_0\Delta^2(f-f^*)\le t^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\left\{\bar{Z}_{ij}(f)-Z_{ij}(f)\right\}\Big)\cdot\mathbf{1}_{\Omega_\beta}\right] \\ &\le \frac{c_5}{L}R_{\mathcal{J}_{e,l}}(\mathcal{F}\cap\left\{c_0\Delta^2(f-f^*)\le t^2\right\}). \end{aligned} \tag{112}$$

Similarly, we have

$$B_3 \le \frac{c_5}{L} R_{\mathcal{J}_{o,l}}(\mathcal{F} \cap \{c_0\Delta^2(f - f^*) \le t^2\}).$$

Thus, by Theorem 1, with Ω_β defined in (76), we have

$$\begin{aligned}
&\mathbb{P}\left(c_0\Delta^2(\hat{f} - f^*) > t^2\right)\\
&\overset{(a)}{\le} \sum_{l=1}^{L}\Bigg(6c_5\frac{R_{\mathcal{J}_{e,l}}\left(\mathcal{F} \cap \{c_0\Delta^2(f - f^*) \le t^2\}\right) + R_{\mathcal{J}_{o,l}}\left(\mathcal{F} \cap \{c_0\Delta^2(f - f^*) \le t^2\}\right)}{t^2}\\
&\quad + \frac{2c_4}{Lt^2}\frac{s\log n \log T_n}{n}\Bigg)\\
&= 6c_5\frac{\sum_{l=1}^{L} R_{\mathcal{J}_{e,1}}\left(\mathcal{F} \cap \{c_0\Delta^2(f - f^*) \le t^2\}\right) + R_{\mathcal{J}_{o,1}}\left(\mathcal{F} \cap \{c_0\Delta^2(f - f^*) \le t^2\}\right)}{t^2}\\
&\quad + \frac{2c_4}{t^2}\frac{s\log n \log T_n}{n}\\
&= \frac{c_5\left(\sum_{l=1}^{L} R_{\mathcal{J}_{e,1}}\left(\mathcal{F} \cap \{c_0\Delta^2(f - f^*) \le t^2\}\right) + R_{\mathcal{J}_{o,1}}\left(\mathcal{F} \cap \{c_0\Delta^2(f - f^*) \le t^2\}\right)\right)}{t^2}\\
&\quad + \frac{c_4}{t^2}\frac{s\log n \log T_n}{n}
\end{aligned}$$

■

Based on the results in Lemma 60 and Lemma 16, we have on event $\Omega_\beta \cup \Omega_\beta^c$,

$$\begin{aligned}
&\mathbb{P}\left(c_0\Delta^2(\hat{f} - f^*) > t^2\right)\\
&= \frac{c_5}{t^2}\Bigg(\sum_{l=1}^{L} R_{\mathcal{J}_{e,1}}\left(\mathcal{F} \cap \{c_0\Delta^2(f - f^*) \le t^2\}\right)\\
&\qquad + R_{\mathcal{J}_{o,1}}\left(\mathcal{F} \cap \{c_0\Delta^2(f - f^*) \le t^2\}\right)\Bigg) + \frac{c_4}{t^2}\frac{s\log n \log T_n}{n} + \frac{3}{nt^2} \qquad (113)\\
&= \frac{c_5}{t^2}\Bigg(\sum_{l=1}^{L} R_{\mathcal{J}_{e,1}}\left(\mathcal{F} \cap \{c_0\Delta^2(f - f^*) \le t^2\}\right)\\
&\qquad + R_{\mathcal{J}_{o,1}}\left(\mathcal{F} \cap \{c_0\Delta^2(f - f^*) \le t^2\}\right)\Bigg) + \frac{c_4}{t^2}\frac{s\log n \log T_n}{n}.
\end{aligned}$$

In the last line, we combine the c_4 with some other numeric constant into one constant c_4.

A.7 Proof of Step 5

Lemma 35 *Let $f, f^* \in \mathcal{F}$. Let $B > 0$ be the constant defined in (4), and let the Huber-type loss $\Delta^2(f)$ be defined in (63). Then the following inequality holds:*

$$\|f - f^*\|_2^2 \le \max\{B, 1\} \cdot \Delta^2(f - f^*).$$

Proof We notice that

$$\begin{aligned}
\|f-f^*\|_2^2 &= \int (f-f^*)^2(x)dx \\
&= \int_{\{t:|(f-f^*)(t)\le 1|\}} |(f-f^*)(x)|^2 dx + \int_{\{t:|(f-f^*)(t)>1|\}} |(f-f^*)(x)|^2 dx \\
&\le \int_{\{t:|(f-f^*)(t)\le 1|\}} |(f-f^*)(x)|^2 dx + B\cdot \int_{\{t:|(f-f^*)(t)>1|\}} |(f-f^*)(x)| dx \\
&\le \max\{B,1\}\cdot \Delta^2(f-f^*).
\end{aligned}$$

■

Definition 36 *We provide addition definitions used in the below proofs. Let*

$$B_2(t) := \{f : \|f-f^*\|_2 \le t\},$$

$$B_{mn}(t) := \{f : \|f-f^*\|_{nm} \le t\}.$$

Lemma 37 *For $f, f^* \in \mathcal{F}$, there exists a constant $b(B) := \max\left\{B^{1/2}, 1\right\}$ that depends on B, such that*

$$R_{\mathcal{J}_{e,l}}\left(\mathcal{F}\cap\left\{c_0\Delta^2(f-f^*)\le t^2\right\}\right) \le R_{\mathcal{J}_{e,l}}\left(\mathcal{F}\cap B_2\left(c_0^{-\frac{1}{2}}\cdot b(B)\cdot t\right)\right) \tag{114}$$

Proof By Lemma 35,

$$\Delta^2(f-f^*) \le t^2 \implies \|f-f^*\|_2^2 \le \max\{B,1\}\cdot\Delta^2(f-f^*) \le \max\{B,1\}\cdot t^2.$$

Then we have

$$R_{\mathcal{J}_{e,l}}\left(\mathcal{F}\cap\left\{f : c_0\Delta^2(f-f^*)\le t^2\right\}\right) \le R_{\mathcal{J}_{e,l}}\left(\mathcal{F}\cap\left\{f : \|f-f^*\|_2^2 \le c^{-1}\cdot\max\{B,1\}\cdot t^2\right\}\right).$$

■

Corollary 38 *Let $\kappa > 0$ be a constant and if $\{r_n\}$ is a sequence such that*

$$\begin{aligned}
&\lim_{\kappa\to\infty}\sup_{n\ge 1}\left\{\frac{c_5\sum_{l=1}^{L} R_{\mathcal{J}_{e,1}}\left(\mathcal{F}\cap B_2\left(c_0^{-\frac{1}{2}}\cdot b(B)\cdot\kappa r_n\right)\right)}{\kappa^2 r_n^2}\right. \\
&\left.+\frac{c_5\sum_{l=1}^{L} R_{\mathcal{J}_{o,1}}\left(\mathcal{F}\cap B_2\left(c_0^{-\frac{1}{2}}\cdot b(B)\cdot\kappa r_n\right)\right)}{\kappa^2 r_n^2} + \frac{\frac{c_4 s\log n\log T_n}{n}+\frac{3}{n}}{\kappa^2 r_n^2}\right\} = 0,
\end{aligned} \tag{115}$$

then

$$\Delta^2(\hat{f}-f^*) = O_p\left(r_n^2\right). \tag{116}$$

Proof Let $\varepsilon > 0$ be given. Notice that for any $\kappa > 0$ we have that, from Lemma 60 and Lemma 16,

$$\begin{aligned}
&\mathbb{P}\left(c_0\Delta^2(\hat{f}-f^*) > \kappa^2 r_n^2\right)\\
&\leq \frac{c_5\sum_{l=1}^{L} R_{\mathcal{J}_{e,1}}\left(\mathcal{F}\cap B_2\left(c_0^{-\frac{1}{2}}\cdot b(B)\cdot \kappa r_n\right)\right)}{\kappa^2 r_n^2}\\
&\quad + \frac{c_5\sum_{l=1}^{L} R_{\mathcal{J}_{o,1}}\left(\mathcal{F}\cap B_2\left(c_0^{-\frac{1}{2}}\cdot b(B)\cdot \kappa r_n\right)\right)}{\kappa^2 r_n^2} + \frac{\frac{c_4 s\log n\log T_n}{n}+\frac{3}{n}}{\kappa^2 r_n^2}\\
&\overset{(a)}{<} \varepsilon,
\end{aligned} \tag{117}$$

where the inequality (a) holds and the last by choosing κ large enough.

Then we have

$$\Delta^2(\hat{f}-f^*) \;=\; O_p\left(r_n^2\right). \tag{118}$$

And by (117) setting that

$$r_n^2 \asymp \sum_{l=1}^{L}\left(R_{\mathcal{J}_{e,l}}\left(\mathcal{F}\cap B_2\left(c_0^{-\frac{1}{2}}\cdot b(B)\cdot r_n\right)\right) + R_{\mathcal{J}_{o,l}}\left(\mathcal{F}\cap B_2\left(c_0^{-\frac{1}{2}}\cdot b(B)\cdot r_n\right)\right)\right) + \frac{s\cdot \mathrm{poly}(\log n)\log T_n}{n}.$$

■

A.8 Proof of Theorem 1

Finally, we present the proof of Theorem 1. To proceed in the proof of Theorems 1, by using our Lemma 35, we start by writing

$$R_{\mathcal{J}_{e,l}}\left(\mathcal{F}\cap\left\{c_0\Delta^2(f-f^*)\leq t^2\right\}\right) \;\leq\; R_{\mathcal{J}_{e,1}}\left(\mathcal{F}\cap\left\{f : \|f-f^*\|\leq c_0^{-1/2}\cdot b(B)\cdot t\right\}\right),$$

where $b(B) \;=\; \max\left\{B^{1/2},1\right\}$ is a constant that depends on B.

From Lemma 60, we get

$$\begin{aligned}
&\mathbb{P}\left(c_0\Delta^2(\hat{f}-f^*) > t^2\right)\\
&\leq \frac{c_5}{t^2}\sum_{l=1}^{L}\left(R_{\mathcal{J}_{e,l}}(\mathcal{F}\cap\left\{c_0\Delta^2(f-f^*)\leq t^2\right\}) + R_{\mathcal{J}_{o,l}}(\mathcal{F}\cap\left\{c_0\Delta^2(f-f^*)\leq t^2\right\})\right) + \frac{c_4}{t^2}\frac{s\log n\log T_n}{n}.
\end{aligned}$$

By using our Lemma 35, we have

$$\begin{aligned}
&\mathbb{P}\left(c_0\Delta^2(\hat{f}-f^*) > t^2\right)\\
&\leq \frac{c_5}{t^2}\sum_{l=1}^{L}\left(R_{\mathcal{J}_{e,l}}\left(\mathcal{F}\cap B_2\left(c_0^{-\frac{1}{2}}\cdot b(B)\cdot t^2\right)\right)\right.\\
&\qquad\left. + R_{\mathcal{J}_{o,l}}\left(\mathcal{F}\cap B_2\left(c_0^{-\frac{1}{2}}\cdot b(B)\cdot t^2\right)\right)\right) + \frac{c_4}{t^2}\frac{s\log n\log T_n}{n}.
\end{aligned}$$

Then by Corollary 38, we have

$$\Delta^2(\hat{f} - f^*) \, = \, O_p\left(t^2\right), \tag{119}$$

where

$$t^2 \asymp \sum_{l=1}^{L} \Bigg(R_{\mathcal{J}_{e,l}} \left(\mathcal{F} \cap B_2 \left(c_0^{-\frac{1}{2}} \cdot b(B) \cdot t \right) \right) \\ + R_{\mathcal{J}_{o,l}} \left(\mathcal{F} \cap B_2 \left(c_0^{-\frac{1}{2}} \cdot b(B) \cdot t \right) \right) \Bigg) + \frac{s \cdot \mathrm{poly}(\log n) \log T_n}{n},$$

for some $\mu > 0$.

Appendix B. Trend Filtering

B.1 Proof sketch of Theorem 3

Recall Definition 2 that $F_k(1)$ denote function space

$$F_k(1) \, := \, \big\{ f \in \mathcal{F} : [0,1] \to \mathbb{R} : f \text{ is } (k-1) \text{ times weakly differentiable} \\ \text{and } J_k(f) \, = \, TV(f^{(k-1)}) < 1 \big\}. \blacksquare$$

By Corollary 38 or Theorem 1, we would need to get $R_{\mathcal{J}_{e,l}}\left(F_k(1) \cap B_2\left(c^{-\frac{1}{2}} \cdot b(B) \cdot t\right)\right)$ and $R_{\mathcal{J}_{o,l}}\left(F_k(1) \cap B_2\left(c^{-\frac{1}{2}} \cdot b(B) \cdot t\right)\right)$, to get that, we will use an existing Theorem that bounds local Rademacher complexity using the L_2 set with the empirical set.

$$R_{\mathcal{J}_{e,l}}\left(F_k(1) \cap \{f : \|f - f^*\| \le b\}\right) \, \le \, R_{\mathcal{J}_{e,l}}\left(F_k(1) \cap \left\{ f : \|f - f^*\|_{nm} \le \sqrt{2}b \right\}\right), \text{ and} \\ R_{\mathcal{J}_{o,l}}\left(F_k(1) \cap \{f : \|f - f^*\| \le b\}\right) \, \le \, R_{\mathcal{J}_{o,l}}\left(F_k(1) \cap \left\{ f : \|f - f^*\|_{nm} \le \sqrt{2}b \right\}\right).$$

where $b > 0$ is any constant, where we denote $\|g\|_{mn} = \left(\frac{1}{\sum_{i \in \mathcal{J}_{e,l}} m_i} \sum_{i \in \mathcal{J}_{e,l}} \sum_{j=1}^{m_i} g(x_{ij})^2 \right)^{1/2}$

Lastly, we will apply Dudley's entropy integral (see Dudley (1967)) to get the rate. The Lemma 39 provides the details of the last two steps.

B.2 Technical Results for Proof of Theorem 3

Lemma 39 (Trend Filtering) *Let $J_k(\cdot) \, = \, TV(\cdot^{k-1})$, the k-th order trend filtering operator, for $F_k(1)$, we have*

$$\Delta^2(\hat{f} - f^*) \, = \, O_p\left(\frac{\min\left\{ B^{\frac{4k}{2k+1}}, B^{\frac{6k-1}{2k+1}} \right\}}{(nm)^{\frac{2k}{2k+1}}} + \frac{\log n}{n} \right).$$

Proof By Corollary 38, we look for the radius to satisfy (115), By using (113), we first analyze the below quantity,

$$\frac{R_{\mathcal{J}_{e,l}}\left(\mathcal{F} \cap \left\{ c\Delta^2(f - f^*) \le t^2 \right\}\right) + R_{\mathcal{J}_{o,l}}\left(\mathcal{F} \cap \left\{ c\Delta^2(f - f^*) \le t^2 \right\}\right)}{t^2},$$

Also, let $b = c^{-1/2} \max\left\{B^{1/2}, 1\right\} \cdot t$.

For any function class $\mathcal{F} \cap B_2(b)$, we have

$$R_{\mathcal{J}_{e,1}}\left(\mathcal{F} \cap B_2(b)\right) \;=\; \mathbb{E}_{x^*}\left[\mathbb{E}_s\Big[\sup_{g \in \mathcal{F} \cap B_2(b)} \frac{1}{\sum_{i \in \mathcal{J}_{e,l}} m_i} \sum_{i \in \mathcal{J}_{e,l}} \sum_{j=1}^{m_i} s_{ij} g(x_{ij}^*) \,\Big|\, \{x_{ij}^*\}\Big]\right]$$

Denote

$$R_{\mathcal{J}_{elmn}}\left(\mathcal{F} \cap B_2(b)\right) \;=\; \mathbb{E}_s\left[\sup_{g \in \mathcal{F} \cap B_2(b)} \frac{1}{\sum_{i \in \mathcal{J}_{e,l}} m_i} \sum_{i \in \mathcal{J}_{e,l}} \sum_{j=1}^{m_i} s_{ij} g(x_{ij}^*) \,\Big|\, \{x_{ij}^*\}\right] \tag{120}$$

Also denote by $\|g\|_{mn} = \left(\frac{1}{\sum_{i \in \mathcal{J}_{e,l}} m_i} \sum_{i \in \mathcal{J}_{e,l}} \sum_{j=1}^{m_i} g(x_{ij}^*)^2\right)^{1/2}$

Corollary 2.2 of Bartlett et al. (2005) gives

$$\mathcal{F} \cap B_2(b) \subseteq \mathcal{F} \cap B_{mn}(\sqrt{2}b),$$

with probability at least $1 - \frac{1}{\sum_{i \in \mathcal{J}_{e,l}} m_i} \geq 1 - \frac{1}{C_\beta} \frac{2 \log n}{c c_1 n m}$, where c appears in Assumption 1.9. Define this good event as $\mathcal{E}$. On $\mathcal{E}$ using Dudley's entropy integral (see Dudley (1967)), Equation (75), and the assumption $m_i \asymp m$ specified in the statement of the Lemma, with c defined in Assumption 1.9, we bound the empirical Rademacher complexity defined in Equation (120) as follows

$$\begin{aligned}
R_{\mathcal{J}_{elmn}}(\mathcal{F} \cap B_2(b)) \;&\leq\; R_{\mathcal{J}_{elmn}}\left(\left\{g \in F : ||g||_{mn}^2 \;\leq\; 2\tilde{b}^2\right\}\right) \\
&\leq \frac{C_{\text{Dud}}}{\sqrt{\sum_{i \in \mathcal{J}_{e,1}} m_i}} \int_0^{\sqrt{2}b} \sqrt{\log N(\epsilon, ||\cdot||_{mn}, \mathcal{F})} d\epsilon \\
&\leq \frac{C_{\text{Dud}}}{\sqrt{C_\beta c c_1 n m}} \int_0^{\sqrt{2}b} \sqrt{\log N(\epsilon, ||\cdot||_{mn}, \mathcal{F})} d\epsilon.
\end{aligned}$$

Under the event $\mathcal{E}^c$, given that $F \subseteq B_\infty(B)$, we have that $R_{\mathcal{J}_{elmn}}(\mathcal{F} \cap B_2(b)) \leq B$. Therefore, splitting the expectation over $\mathcal{E}$ and $\mathcal{E}^c$,

Set that $\mathcal{F} \equiv F_k(1)$, we obtain that for some positive constant c_7, and $A > 0$ from Lemma 40.

The Rademacher complexity defined in Equation (120) is bounded by

$$\begin{aligned}
R_{\mathcal{J}_{e,1}}(F_k(1) \cap B_2(b)) \;&=\; \mathbb{E}_{x^*}[R_{\mathcal{J}_{elmn}}(F_k(1) \cap B_2(b))] \\
&\leq \frac{C_{\text{Dud}}}{\sqrt{C_\beta c c_1 n m}} \int_0^{\sqrt{2}b} \sqrt{\log N(\epsilon, ||\cdot||_{mn}, F_k(1))} d\epsilon + \frac{2B \log(n)}{C_\beta c c_1 n m} \\
&\leq \frac{A C_{\text{Dud}} B^{1/2k}}{\sqrt{C_\beta c c_1 n m}} \int_0^{\sqrt{2}b} \epsilon^{-1/(2k)} d\epsilon + \frac{2B \log(n)}{C_\beta c c_1 n m} \\
&= c_6 \frac{B^{1/2k} \sqrt{2 \log(n)}}{\sqrt{C_\beta c c_1 n m}} b^{1 - 1/(2k)} + \frac{2B \log(n)}{C_\beta c c_1 n m}.
\end{aligned}$$

Throughout all the proofs, we will use c_6 to denote some absolute constant that only depends on A, C_{Dud}, C_β, c, c_1, which may change according to the context.

Without loss of generality assume that $b \geq (nm)^{-1/2}$. Since $k/(2k-1) \geq 1/2$, we have that $b \geq (nm)^{-k/(2k-1)}$ and in consequence $b^{(2k-1)/k} \geq (nm)^{-1}$. Therefore $b^{(2k-1)/k} \geq (nm)^{-1/2}$, which can be written as $b^{1-1/(2k)} \geq (nm)^{-1/2}$, which implies $b^{1-1/(2k)}/\sqrt{nm} \geq (nm)^{-1}$. Thus we have that

$$c_6 \frac{B^{1/2k}\sqrt{2\log(n)}}{\sqrt{C_\beta c c_1 nm}} b^{1-1/(2k)} + \frac{2B\log(n)}{C_\beta c c_1 \sqrt{nm}} b^{1-1/(2k)} < c_6 B \frac{2\log(n)}{\sqrt{C_\beta c c_1 nm}} b^{1-1/(2k)}. \tag{121}$$

Here we use Now, using (121), we see that

$$\begin{aligned} &\frac{R_{\mathcal{J}_{e},1}\left(\mathcal{F} \cap \left\{c\Delta^2(f-f^*) \leq t^2\right\}\right) + R_{\mathcal{J}_{o},1}\left(\mathcal{F} \cap \left\{c\Delta^2(f-f^*) \leq t^2\right\}\right)}{t^2} \\ &\leq \frac{c_6 B\log(n)}{t^2\sqrt{C_\beta c c_1 nm}} b^{1-1/(2k)} = \frac{c_6 c^{-1}\max\left\{B, B^2\right\}\log(n) b^{1-1/(2k)}}{c^{-1}\max\left\{B,1\right\} t^2 \sqrt{C_\beta c c_1 nm}} = \frac{c_6 \max\left\{B, B^2\right\}\log(n)}{b^{(2k+1)/(2k)}\sqrt{C_\beta c c_1 nm}}. \end{aligned} \tag{122}$$

If $B \geq 1$, from Inequality (122), we get

$$\frac{c_6 \max\left\{B, B^2\right\}\log(n)}{b^{(2k+1)/(2k)}\sqrt{C_\beta c c_1 nm}} = \frac{c_6 B^2 \log(n)}{b^{(2k+1)/(2k)}\sqrt{C_\beta c c_1 nm}}. \tag{123}$$

Then we let

$$b^2 = \frac{B^{\frac{8k}{2k+1}}\log(n)^{\frac{4k}{2k+1}}}{(nm)^{\frac{2k}{2k+1}}}$$

to make the (124) be $O\left(\frac{1}{nm}\right)$.

Furthermore, we know $b^2 = c^{-1}\max\left\{B,1\right\} t^2 = c^{-1}Bt^2$ when $B > 1$, so we get

$$t^2 = \frac{cB^{\frac{6k-1}{2k+1}}\log(n)^{\frac{4k}{2k+1}}}{(nm)^{\frac{2k}{2k+1}}}$$

If $B < 1$, from Inequality (122),

$$\frac{c_6 \max\left\{B, B^2\right\}\log(n)}{b^{(2k+1)/(2k)}\sqrt{C_\beta c c_1 nm}} = \frac{c_6 B \log(n)}{b^{(2k+1)/(2k)}\sqrt{C_\beta c c_1 nm}} \tag{124}$$

then we let

$$b^2 = \frac{B^{\frac{4k}{2k+1}}\log(n)^{\frac{4k}{2k+1}}}{(nm)^{\frac{2k}{2k+1}}}$$

The (124) is $O\left(\frac{1}{nm}\right)$.

Furthermore, we know $b^2 = c^{-1}\max\left\{B,1\right\} t^2 = c^{-1}t^2$ when $B \leq 1$, so we get

$$t^2 = \frac{cB^{\frac{4k}{2k+1}}\log(n)^{\frac{4k}{2k+1}}}{(nm)^{\frac{2k}{2k+1}}}$$

Overall, we have

$$t^2 = \frac{c \max\left\{B^{\frac{4k}{2k+1}}, B^{\frac{6k-1}{2k+1}}\right\} \log(n)^{\frac{4k}{2k+1}}}{(nm)^{\frac{2k}{2k+1}}}.$$

and

$$R_{\mathcal{J}_{e,l}}\left(F_k(1) \cap B_2\left(c^{-\frac{1}{2}} \cdot b(L) \cdot t\right)\right) \le t^2 = \frac{c^2 \max\left\{L^{\frac{4k}{2k+1}}, L^{\frac{2k-1}{2k+1}}\right\} \log(n)^{\frac{4k}{2k+1}}}{(nm)^{\frac{2k}{2k+1}}}. \tag{125}$$

Similarly, we have

$$R_{\mathcal{J}_{o,l}}\left(F_k(1) \cap B_2\left(c^{-\frac{1}{2}} \cdot b(L) \cdot t\right)\right) \le \frac{c^2 \max\left\{L^{\frac{4k}{2k+1}}, L^{\frac{2k-1}{2k+1}}\right\} \log(n)^{\frac{4k}{2k+1}}}{(nm)^{\frac{2k}{2k+1}}}.$$

Then by Theorem 1 and (125), and the choice of $L = \left\lceil \frac{1}{C_\beta} 2\log(n) \right\rceil$ in (73), we have

$$\Delta^2(\hat{f} - f^*) = O_p\left(r_n^2\right),$$

by setting that

$$r_n^2 \asymp \frac{\max\left\{B^{\frac{4k}{2k+1}}, B^{\frac{6k-1}{2k+1}}\right\} \log(n)^{\frac{6k+1}{2k+1}}}{(nm)^{\frac{2k}{2k+1}}} + \frac{s \log^{1+\mu}(n) \log(T_n)}{n},$$

for some $\mu > 0$. ■

Lemma 40 *Let $\{x_{ij}\}_{i=1,j=1}^{n,m_j}$ be a collection of random variables in $[0,1]$. Denote*

$$\|f\|_{mn}^2 = \frac{1}{n}\sum_{i=1}^{n} \frac{1}{m_i} \sum_{j=1}^{m_i} f^2(x_{ij}).$$

Then, there exists an absolute constant A such that for $k \ge 1$,

$$\log N\left(\varepsilon, \|\cdot\|_{mn}, F_k(1)\right) \le AB^{1/k} \varepsilon^{-1/k}.$$

Proof This is a direct consequence of Sadhanala et al. (2016), see page 3050 of the mentioned reference. ■

B.3 Proof of Theorem 3

Proof By Lemma 39, we have

$$\Delta^2(\hat{f} - f^*) = O_p\left(\frac{\max\left\{B^{\frac{6k+1}{2k+1}}, B^{\frac{2k-1}{2k+1}}\right\} \log(n)^{\frac{6k+1}{2k+1}}}{(nm)^{\frac{2k}{2k+1}}} + \frac{s \cdot \mathrm{poly}(\log n) \log(T_n)}{n}\right),$$

So we get

$$\left\|\hat{f} - f^*\right\|_2^2 = O_p\left(\frac{\max\left\{B^{\frac{6k+1}{2k+1}}, B^{\frac{2k-1}{2k+1}}\right\} \log(n)^{\frac{4k}{2k+1}}}{(nm)^{\frac{2k}{2k+1}}} + \frac{\max\{B, 1\} \cdot s \cdot \mathrm{poly}(\log n) \log(T_n)}{n}\right).$$

Appendix C. High-dimensional Quantile Linear Regression

C.1 Proof sketch of Theorem 4

In this case, the constraint set is

$$H(V) = \{f : f(x_{ij}) = x_{ij}'\beta, ||\beta||_1 \leq V, \beta \in \mathbb{R}^d\}$$

Since this is a compact set, we directly use Theorem 1 to get an expectation bound by letting $\mathcal{F} \equiv H(V)$. This means that we need to simply bound $R(H(V))$ which can be done using standard existing results.

C.2 Proof of Theorem 4

Proof By Corollary 38,

It suffices to get $R_{\mathcal{J}_{e,l}}\left(\mathcal{F} \cap B_2\left(c^{-\frac{1}{2}} \cdot b(B) \cdot t\right)\right)$ and $R_{\mathcal{J}_{o,l}}\left(\mathcal{F} \cap B_2\left(c^{-\frac{1}{2}} \cdot b(B) \cdot t\right)\right)$ for $l = 1, \ldots, n$. Then we have

$$\begin{aligned}
R_{\mathcal{J}_{e,l}}\left(\mathcal{F} \cap B_2\left(c^{-\frac{1}{2}} \cdot b(B) \cdot t\right)\right) &= \mathbb{E}_{x^*,s}\left[\sup_{\substack{f \in H(V) \\ f \in B_2\left(c^{-\frac{1}{2}} \cdot b(B) \cdot t\right)}} \frac{1}{\sum_{i \in \mathcal{J}_{e,l}} m_i} \sum_{i \in \mathcal{J}_{e,l}} \sum_{j=1}^{m_i} s_{ij} f(x_{ij}^*)\right] \\
&\leq \mathbb{E}_{x^*,s}\left[\sup_{f \in H(V)} \frac{1}{\sum_{i \in \mathcal{J}_{e,l}} m_i} \sum_{i \in \mathcal{J}_{e,l}} \sum_{j=1}^{m_i} s_{ij} f(x_{ij}^*)\right] \\
&= R_{\mathcal{J}_{e,l}}\left(H(V)\right)
\end{aligned}$$

Let

$$K_V := \{\beta \in \mathbb{R}^d : ||\beta||_1 \leq V\}.$$

Also, recall $X_{e,l} \in \mathbb{R}^{\sum_{i \in \mathcal{J}_{e,l}} m_i \times d}$ is the matrix with its rows are $\{x_{ij}\}_{i \in \mathcal{J}_{e,l}, j \in [m_i]}$. Then we have

$$\begin{aligned}
\mathbb{E}_{x^*,s}\left[\sup_{f \in H(V)} \frac{1}{\sum_{i \in \mathcal{J}_{e,l}} m_i} \sum_{i \in \mathcal{J}_{e,l}} \sum_{j=1}^{m_i} s_{ij} f(x_{ij}^*)\right] &\leq \mathbb{E}\left[\frac{V}{\sum_{i \in \mathcal{J}_{e,l}} m_i} \sup_{\beta \in K_1} \sum_{i \in \mathcal{J}_{e,l}} \sum_{j=1}^{m_i} s_{ij} {x_{ij}^*}' \beta\right] \\
&= \frac{V}{\sum_{i \in \mathcal{J}_{e,l}} m_i} \mathbb{E}\left[\sup_{v \in XK_1} s'v\right].
\end{aligned} \tag{126}$$

where

$$K_1 = \{v : ||v||_1 \leq 1\}.$$

By the proof of Theorem 2.4 in Rigollet and Hütter (2023), there exists a constant $h_2 > 0$ such that for a given fixed design matrix $X_{e,l} \in \mathbb{R}^{\sum_{i \in \mathcal{J}_{e,l}} m_i \times d}$.

$$\mathbb{E}\left[\sup_{v \in XK_1} s'v\right] \leq h_2 (\log d)^{1/2} \max_{j \in [d]} ||X_{e,l \cdot j}||.$$

Taking expectations on both sides, thus we have

$$\begin{aligned} R_{\mathcal{J}_{e,l}}\left(H(V)\right) &\leq \frac{V}{\sum_{i\in\mathcal{J}_{e,l}} m_i} \cdot h_2 (\log d)^{1/2}\, \mathbb{E}\left[\max_{j\in[d]} ||X_{e,l\cdot j}||\right] \\ &\leq \frac{L h_2 V (\log d)^{1/2}\, \mathbb{E}\left[\max_{j\in[d]} ||X_{e,l\cdot j}||\right]}{c c_1 n m} \\ &= \frac{L h_3 V (\log d)^{1/2}\, \mathbb{E}\left[\max_{j\in[d]} ||X_{e,l\cdot j}||\right]}{nm}. \end{aligned}$$

Similarly, we have

$$\begin{aligned} R_{\mathcal{J}_{o,l}}\left(H(V)\right) &\leq \frac{V}{\sum_{i\in\mathcal{J}_{e,l}} m_i} \cdot h_2 (\log d)^{1/2}\, \mathbb{E}\left[\max_{j\in[d]} ||X_{o,l\cdot j}||\right] \\ &\leq \frac{L h_3 V (\log d)^{1/2}\, \mathbb{E}\left[\max_{j\in[d]} ||X_{o,l\cdot j}||\right]}{nm}. \end{aligned}$$

Thus, we get

$$\begin{aligned} & c_5 \sum_{l=1}^{L} \left(R_{\mathcal{J}_{e,l}}\left(H(V)\right) + R_{\mathcal{J}_{o,l}}\left(H(V)\right)\right) + \frac{c_4 s \log n \log T_n}{n} \\ \leq\ & \frac{L c_5 h_3 V (\log d)^{1/2}\, \mathbb{E}\left[\sum_{l=1}^{L}\left(\max_{j\in[d]} ||X_{e,l\cdot j}|| + \max_{j\in[d]} ||X_{o,l\cdot j}||\right]\right)}{nm} + \frac{c_4 s \log n \log T_n}{n} \\ =\ & \frac{h_4 \log n V (\log d)^{1/2}\, \mathbb{E}\left[\sum_{l=1}^{L}\left(\max_{j\in[d]} ||X_{e,l\cdot j}|| + \max_{j\in[d]} ||X_{o,l\cdot j}||\right]\right)}{nm} + \frac{c_4 s \log n \log T_n}{n} \end{aligned}$$

The claim of the theorem then follows and we get

$$\Delta^2(\hat{f} - f^*) = O_p\left(\frac{\log n V (\log d)^{1/2} A}{nm} + \frac{s \cdot \mathrm{poly}(\log n) \log(T_n)}{n}\right), \quad \text{and}$$

$$\left\|\hat{f} - f^*\right\|_2^2 = O_p\left(\frac{\max\{B,1\} \log n V (\log d)^{1/2} A}{nm} + \frac{\max\{B,1\}\, s \cdot \mathrm{poly}(\log n) \log(T_n)}{n}\right).$$

■

Appendix D. Quantile K-NN fused lasso

D.1 Proof of Theorem 5

Proof We will apply the result from Theorem 1, and also incorporate our problem in 23, to get

$$\begin{aligned} & R_{\mathcal{J}_{e,l}}\left(\|\nabla G_K \theta\|_1 \leq V_n \cap \left\{\theta : \|\theta - \theta^*\| \leq \max\left\{1, B^{1/2}\right\} t\right\}\right), \\ & R_{\mathcal{J}_{o,l}}\left(\|\nabla G_K \theta\|_1 \leq V_n \cap \left\{\theta : \|\theta - \theta^*\| \leq \max\left\{1, B^{1/2}\right\} t\right\}\right) \end{aligned}$$

for all $l = 1, \dots, L$,based on the results from Lemma 42.

Then we can show that Suppose in addition that $K \asymp \log^{2+\mu}(nm)$ for some $\mu > 0$. The estimator $\hat{\theta}_{V_n}$, defined in (23) satisfies that

$$\Delta^2(\hat{\theta}_{V_n} - \theta^*) \;=\; O_p\left(\frac{\max\{B,1\}\log^{5+\mu}(nm)V_n}{nm} + \frac{\log^2(nm)\max\{B,1\}}{n} + \frac{d\log^{1+\mu}(n)\log(T_n)}{n}\right) \tag{17}$$

$$\left\|\hat{\theta}_{V_n} - \theta^*\right\|^2 \;=\; O_p\Bigg(\frac{\max\left\{B^2,B\right\}\log^{5+\mu}(nm)V_n}{nm} + \frac{\log^2(nm)\max\left\{B^2,B\right\}}{n} + \frac{\max\{B,1\}\,d\log^{1+\mu}(n)\log(T_n)}{n}\Bigg),$$

and provided that $V_n \;\geq\; \|V_{G^*}\|$.

In addition, based on the result from Lemma 44, $\hat{f}_{V_n}$, defined in (24)–(25) satisfies that

$$\Delta^2(\hat{f}_{V_n} - f^*) \;=\; O_p\Bigg(\frac{\max\{B,1\}\log^{5+\mu}(nm)V_n}{nm} + \frac{\log^2(nm)\max\{B,1\}}{n} + \frac{d\log^{1+\mu}(n)\log(T_n)}{n} + \frac{\log^{2+\mu}(nm)}{(nm)^{1/d}}\Bigg)$$

$$\left\|\hat{f}_{V_n} - f^*\right\|_2^2 \;=\; O_p\Bigg(\max\{B,B^2\}\left(\frac{\log^{5+2\mu}(nm)}{nm} + \frac{\log^{3+\mu}(nm)V_n}{n}\right) + \max\{B,1\}\cdot\left(\frac{d\log n}{n} + \frac{\log^{2+\mu/d}(nm)}{(nm)^{1/d}}\right)\Bigg).$$

■

D.2 Additional Definitions for proof of Theorem 5

We give some definitions that will be encountered in the proof of the lemma.

D.2.1 The lattice graph

We introduce the Lattice graph, for more details, we refer readers to Madrid Padilla et al. (2020). Given $N \in \mathbb{N}$, denote the equally spaced lattice graph in $[0,1]^d$ as $G_{\text{lat}} = (V_{\text{lat}}, E_{\text{lat}})$, where the distance between any two neighbor lattice points is $1/N$. Therefore, the number of nodes $|V_{\text{lat}}| = N^d$. Without loss of generality, we assume that the nodes of the grid correspond to the points

$$I = P_{\text{lat}}(N) \;=\; \left\{\left(\frac{i_1}{N}, \frac{i_2}{N}, \dots, \frac{i_d}{N}\right) : i_1, \dots, i_d \in \{1, \dots, N\}\right\} \quad (271) \tag{127}$$

Let $\{c_r\}_{r=1}^{N^d}$ be the corresponding centers and define a collection of cells, $\{C(c_r)\}_{r=1}^{N^d}$ as

$$\mathcal{I}_r = C(c_r) \;=\; \left\{z \in [0,1]^d : c_r = \arg\min_{r' \in P_{\text{lat}}(N)} \|z - r'\|_\infty\right\}. \tag{128}$$

Now, let $\theta \in \mathbb{R}^{\sum_{i=1}^{n} m_i}$ be a collection of signals observed on the sample points $\{x_{ij}\}_{i=1,j=1}^{n,m}$.

Let $\theta_I \in \mathbb{R}^{\sum_{i=1}^{n} m_i}$ be the modified version of θ according to the grid induced by the lattice graph of size N. Let $\theta^I \in \mathbb{R}^{N^d}$ be the modified version of θ that takes values on the centers of the lattice graph. See section E of Madrid Padilla et al. (2020) for definitions of θ_I and θ^I. For completion, the definitions are included as follows.

Definition 2. Given $\theta \in \mathbb{R}^{\sum_{i=1}^{n} m_i}$, let $\{\mathcal{I}_r\}_{r=1}^{N^d}$ be the collection of the cells in the lattice graph G_{lat} and let $\{c_r\}_{r=1}^{N^d}$ be the corresponding centers. Let $\theta_I \in \mathbb{R}^{\sum_{i=1}^{n} m_i}$ be such that for any $1 \le \alpha \le n, 1 \le \beta \le m_\alpha$,

$$(\theta_I)_{\alpha,\beta} = \theta_{ij},$$

if $x_{\alpha,\beta}$ and x_{ij} are in the same cell $\mathcal{I}_r$, and that $(i,j) = \arg\min_{1 \le i' \le n, 1 \le j' \le m} \|x_{i'j'} - c_r\|$. Denote by

$$I_r = \{(i,j) \in [n] \times [m] : P_I(x_{ij}) = c_r\} \tag{129}$$

for $r = 1, \ldots, N^d$, where $P_I(x)$ denotes the point in $P_{\text{lat}}(N)$ such that $x \in C(P_I(x))$.

Definition 3. Let $\theta \in \mathbb{R}^{\sum_{i=1}^{n} m_i}$ be a collection of signals observed on the sample points $\{x_{ij}\}_{i=1,j=1}^{n,m_i}$. Let θ^I in $\mathbb{R}^{N^d}$ be such that, for any $r \in [N^d]$

$$\theta^I_r = \begin{cases} \theta_{i_r j_r}, & \text{if } I_r \neq \emptyset \text{ and } (i_r j_r) = \arg\min_{1 \le i' \le n, 1 \le j' \le m_{i'}} \|x_{i'j'} - c_r\|; \\ 0, & \text{if } I_r = \emptyset. \end{cases}$$

This means that θ^I_r take the value $\theta_{i_r j_r}$, if $x_{i_r j_r}$ is the closest point to the center of c_r among all $\{x_{ij}\}_{i=1,j=1}^{n,m_i}$.

D.2.2 Piecewise Lipschitz functions

Definition 41 *For a set $A \subset [0,1]^d$, we write $B_\varepsilon(A) = \{a : \exists a' \in A, \text{ with } \|a - a'\|_2 \le \varepsilon\}$. We let ∂A denote the boundary of the set A.*

Let $\Omega_\varepsilon := [0,1]^d \backslash B_\varepsilon(\partial[0,1]^d)$. We say that a bounded function $f : [0,1]^d \to \mathbb{R}$ is piecewise Lipschitz if there exists a set $S \subseteq (0,1)^d$ that has the following properties:

- *The set S has Lebesgue measure zero.*
- *For some constants $C_S, \varepsilon_0 > 0$, we have that $Vol(B_\varepsilon(S) \cup ([0,1]^d \backslash \Omega_\varepsilon)) \le C_S \varepsilon$ for all $0 < \varepsilon < \varepsilon_0$.*
- *There exists a positive constant L_f such that if x and x' belong to the same connected component of $\Omega_\varepsilon \backslash B_\varepsilon(S)$, then $|g(x) - g(x')| \le L_f \|x - x'\|_2$.*

D.3 Technical Results for proof of Theorem 5

The remaining proofs leading to the derivations of

$$\begin{aligned}\Delta^2(\hat{\theta}_{V_n} - \theta^*) = O_p\Bigg(\sum_{l=1}^{L}\Bigg(&R_{\mathcal{J}_{e,l}}\left(\|\nabla G_K \theta_{e,l}\|_1 \le V_n \cap \left\{\theta : \left\|\theta_{e,l} - \theta^*_{e,l}\right\| \le c^{-\frac{1}{2}} \cdot b(B) \cdot t\right\}\right) \\ &+ R_{\mathcal{J}_{o,l}}\left(\|\nabla G_K \theta_{o,l}\|_1 \le V_n \cap \left\{\theta : \left\|\theta_{o,l} - \theta^*_{o,l}\right\| \le c^{-\frac{1}{2}} \cdot b(B) \cdot t\right\}\right) \\ &+ \frac{s \log n \log T_n}{n}\Bigg)\Bigg), \text{ and}\end{aligned}$$

follows the similar proofs in Corollary 26 to Lemma 35.

Lemma 42 *The estimator $\hat{\theta}_{V_n}$, defined in (23), Suppose in addition that $K \asymp \log^{2+\mu}(nm)$ for some $\mu > 0$ satisfies that*

$$\Delta^2(\hat{\theta}_{V_n} - \theta^*) \,=\, O_p\left(r_n^2\right),$$

by setting that

$$r_n^2 \asymp \frac{\max\{B,1\}\log^{5+\mu}(nm)V_n}{nm} + \frac{\log^2(nm)\max\{B,1\}}{n} + \frac{d\log^{1+\mu}(n)\log(T_n)}{n}.$$

Proof By definition, we have

$$R_{\mathcal{J}_{e,l}}\left(\|\nabla G_K\theta\|_1 \le V_n \cap \{\theta : \|\theta-\theta^*\| \le b\}\right) = \mathbb{E}_{x^*,s}\left[\sup_{\substack{\|\theta-\theta^*\|\le b \\ \|\nabla G_K\theta\|_1 \le V_n}} \frac{1}{\sum_{i\in\mathcal{J}_{e,l}} m_i}\sum_{i\in\mathcal{J}_{e,l}}\sum_{j=1}^{m_i} s_{ij}\left(\theta_{ij}-\theta^*_{ij}\right)\right]. \quad \blacksquare$$

Define:

$$(\tilde{s})_{i,j} = \begin{cases}(s)_{i,j}, & \text{for } i\in\mathcal{J}_{e,l}, \text{and } j\in m_i,\\ 0, & \text{for } i\notin\mathcal{J}_{e,l}, \text{and } j\notin m_i.\end{cases} \tag{130}$$

One can show that $\{\tilde{s}_{ij}\}_{i=1,j=1}^{n,m_i}$ are independent SubGaussian random variables with SubGaussian parameter 1.

Now by triangle inequality, and let $\Lambda = \theta - \theta^*$, we have

$$\begin{aligned}
&R_{\mathcal{J}_{e,1}}\left(\|\nabla G_K\theta\|_1 \le V_n \cap \{\theta : \|\theta-\theta^*\| \le b\}\right)\\
&= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*,s}\left[\left(\sup_{\substack{\|\theta-\theta^*\|\le b \\ \|\nabla G_K\theta\|_1 \le V_n}} \frac{1}{\sum_{i\in\mathcal{J}_{e,l}} m_i}\sum_{i\in\mathcal{J}_{e,l}}\sum_{j=1}^{m_i} s_{ij}(\theta_{ij}-\theta^*_{ij}+\theta^*_{ij})\right)\cdot \mathbf{1}_{\Omega_\beta}\right]\\
&= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*,s}\left[\left(\sup_{\substack{\|\theta-\theta^*\|\le b \\ \|\nabla G_K\theta\|_1 \le V_n}} \frac{1}{\sum_{i\in\mathcal{J}_{e,l}} m_i}\sum_{i\in\mathcal{J}_{e,l}}\sum_{j=1}^{m_i} s_{ij}(\theta_{ij}-\theta^*_{ij})\right)\cdot \mathbf{1}_{\Omega_\beta}\right]\\
&\quad + \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*,s}\left[\frac{1}{\sum_{i\in\mathcal{J}_{e,l}} m_i}\sum_{i\in\mathcal{J}_{e,l}}\sum_{j=1}^{m_i} s_{ij}\theta^*_{ij}\right]\\
&= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*,s}\left[\left(\sup_{\substack{\|\theta-\theta^*\|\le b, \|\nabla G_K(\theta-\theta^*)\|_1 \le \|\nabla G_K\theta\|_1 + \|\nabla G_K\theta^*\|_1 \le 2V_n \\ \|\nabla G_K\theta\|_1 \le V_n, \|\nabla G_K\theta^*\|_1 \le V_n}} \frac{1}{\sum_{i\in\mathcal{J}_{e,l}} m_i}\sum_{i\in\mathcal{J}_{e,l}}\sum_{j=1}^{m_i} s_{ij}(\theta_{ij}-\theta^*_{ij})\right)\cdot \mathbf{1}_{\Omega_\beta}\right]\\
&\le \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*,s}\left[\left(\sup_{\substack{\|\Lambda\|\le b \\ \|\nabla G_K\Lambda\|_1 \le \|\nabla G_K\theta\|_1 + \|\nabla G_K\theta^*\|_1 \le 2V_n}} \frac{1}{\sum_{i\in\mathcal{J}_{e,l}} m_i}\sum_{i\in\mathcal{J}_{e,l}}\sum_{j=1}^{m_i} \Lambda_{ij}s_{ij}\right)\cdot \mathbf{1}_{\Omega_\beta}\right]\\
&= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*,s}\left[\left(\sup_{\substack{\|\Lambda\|\le b \\ \|\nabla G_K\Lambda\|_1 \le \|\nabla G_K\theta\|_1 + \|\nabla G_K\theta^*\|_1 \le 2V_n}} \frac{1}{\sum_{i\in\mathcal{J}_{e,l}} m_i}\sum_{i=1}^{n}\sum_{j=1}^{m_i} \Lambda_{ij}s_{ij}\right)\cdot \mathbf{1}_{\Omega_\beta}\right]
\end{aligned}$$

$$= \frac{\sum_{i=1}^n m_i}{\sum_{i\in\mathcal{J}_{e,l}} m_i}\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*,s}\left[\left(\sup_{\substack{\|\Lambda\|\le b\\ \|\nabla G_K\Lambda\|_1\le\|\nabla G_K\theta\|_1+\|\nabla G_K\theta^*\|_1\le 2V_n}}\frac{1}{\sum_{i=1}^n m_i}\sum_{i=1}^n\sum_{j=1}^{m_i}\Lambda_{ij}\tilde{s}_{ij}\right)\cdot\mathbf{1}_{\Omega_\beta}\right]$$

$$\asymp \frac{nm}{|\mathcal{J}_{e,l}|\, m}\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*,s}\left[\left(\sup_{\substack{\|\Lambda\|\le b\\ \|\nabla G_K\Lambda\|_1\le\|\nabla G_K\theta\|_1+\|\nabla G_K\theta^*\|_1\le 2V_n}}\frac{1}{\sum_{i=1}^n m_i}\sum_{i=1}^n\sum_{j=1}^{m_i}\Lambda_{ij}\tilde{s}_{ij}\right)\cdot\mathbf{1}_{\Omega_\beta}\right]$$

$$\le \frac{nL}{c_1 n}\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*,s}\left[\left(\sup_{\substack{\|\Lambda\|\le b\\ \|\nabla G_K\Lambda\|_1\le\|\nabla G_K\theta\|_1+\|\nabla G_K\theta^*\|_1\le 2V_n}}\frac{1}{\sum_{i=1}^n m_i}\sum_{i=1}^n\sum_{j=1}^{m_i}\Lambda_{ij}\tilde{s}_{ij}\right)\cdot\mathbf{1}_{\Omega_\beta}\right]$$

$$\le \frac{2\log(n)}{c_1 C_\beta}\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*,s}\left[\left(\sup_{\substack{\|\Lambda\|\le b\\ \|\nabla G_K\Lambda\|_1\le\|\nabla G_K\theta\|_1+\|\nabla G_K\theta^*\|_1\le 2V_n}}\frac{1}{\sum_{i=1}^n m_i}\sum_{i=1}^n\sum_{j=1}^{m_i}\Lambda_{ij}\tilde{s}_{ij}\right)\cdot\mathbf{1}_{\Omega_\beta}\right],$$

Where the first equality holds by the fact that $\|\nabla G_K\theta\|_1 \le V_n$ and $\|\nabla G_K\theta^*\|_1 \le V_n$ implies $\|\nabla G_K\Lambda\|_1 \le 2V_n$. We use the assumption $m_i \asymp m$ specified in the statement of the Lemma.

In this step the term on the right hand side, we get

$$\begin{aligned}&R_{\mathcal{J}_{e,l}}\left(\|\nabla G_K\theta\|_1 \le V_n \cap \{\theta : \|\theta-\theta^*\|\le b\}\right)\\ &\quad\le \frac{2\log(n)}{c_1 C_\beta}\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*,s}\left[\left(\sup_{\substack{\|\Lambda\|\le b\\ \|\nabla G_K\Lambda\|_1\le 2V_n}}\frac{1}{\sum_{i=1}^n m_i}\sum_{i=1}^n\sum_{j=1}^{m_i}\Lambda_{ij}\tilde{s}_{ij}\right)\cdot\mathbf{1}_{\Omega_\beta}\right],\end{aligned}$$

For the rest of the proofs we give the sketch, the exact proof is provided in Lemma 43.

From Lemma 43 and the observation after Equation (134), with a probability of at least

$$1-\left(\frac{2}{nm}+\frac{4}{C_\beta}C\log(n)nm\exp\left(-\frac{\tilde{C}K}{24\log(n)}\right)+\frac{3}{n}\quad+\frac{4}{C_\beta}\log(n)\frac{nm}{K}\exp\left(-\frac{C_\beta\tilde{b}_2K}{48\log(n)}\right)+\frac{1}{nm}\right),$$

it is satisfied that

$$\sup_{\|\Lambda\|_2\le b,\|\nabla G_K\Lambda\|_1\le 2V_n}\frac{1}{\sum_{i=1}^n m_i}\sum_{i=1}^n\sum_{j=1}^{m_i}\Lambda_{ij}s_{ij}\le C_3K\frac{\log^{\frac{1}{2}}(nm)}{\sum_{i=1}^n m_i}b+C_4K\frac{\log(nm)}{\sum_{i=1}^n m_i}V_n,$$

where $\tilde{b}_2 \ge 0$ is a constant depending on d, C_3 and v_{max}. Moreover, $\tilde{C}$ is $\frac{c_1 c}{c_2 C}$ with c_1 and c_2 positive constants. Here v_{max}, v_{min}, C and c are from Assumptions 1.9 and 7. The positive constants C_3 and C_4 only depend on d. To find the quantity $R_{\mathcal{J}_{e,1}}\left(\|\nabla G_K\theta\|_1 \le V_n\cap\{\theta:\|\theta-\theta^*\|\le b\}\right)$, we will analyze

$$\frac{R_{\mathcal{J}_{e,1}}\left(\|\nabla G_K\theta\|_1 \le V_n\cap\{\theta:\|\theta-\theta^*\|\le b\}\right)}{t^2}+\frac{R_{\mathcal{J}_{e,1}}\left(\|\nabla G_K\theta\|_1 \le V_n\cap\{\theta:\|\theta-\theta^*\|\le b\}\right)}{t^2}.$$

See the equation (13.17) in Wainwright (2019). Hence,

$$\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*,s}\left[\left(\sup_{\|\Lambda\|\le b,\|\nabla G_K\Lambda\|_1\le V_n}\frac{1}{\sum_{i=1}^n m_i}\sum_{i=1}^n\sum_{j=1}^{m_i}\Lambda_{ij}\tilde{s}_{ij}\right)\cdot\mathbf{1}_{\Omega_\beta}\right]$$

$$
\begin{aligned}
&\leq C_3 K \frac{\log^{\frac{1}{2}}(nm)}{\sum_{i=1}^n m_i} b + C_4 K \frac{\log(nm)}{\sum_{i=1}^n m_i} V_n \\
&\quad + B \cdot \left(\frac{2}{nm} + \frac{4}{C_\beta} C \log(n) nm \exp\left(-\frac{\tilde{C} K}{24 \log(n)} \right) \right. \\
&\quad \left. + \frac{3}{n} + \frac{4}{C_\beta} \log(n) \frac{nm}{K} \exp\left(-\frac{C_\beta \tilde{b}_2 K}{48 \log(n)} \right) + \frac{1}{nm} \right).
\end{aligned}
$$

Also with Lemma 37, we have $b^2 = c^{-1} \max\{B, 1\} \cdot t^2$,

Let denote $\frac{1}{\sum_{i=1}^n m_i}$ as C_6. Now, we analyze

$$
\frac{R_{\mathcal{J}_{e,1}}\left(\mathcal{F} \cap \left\{c\Delta^2(f - f^*) \leq t^2\right\}\right) + R_{\mathcal{J}_{o,1}}\left(\mathcal{F} \cap \left\{c\Delta^2(f - f^*) \leq t^2\right\}\right)}{t^2} \tag{131}
$$

We assume $b^2 = O\left(\frac{1}{n} + \frac{1}{nm}\right)$, then we have

$$
\begin{aligned}
&\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*,s} \left[\left(\sup_{\substack{\|\varLambda\| \leq b \\ \|\nabla G_K \varLambda\|_1 \leq \|\nabla G_K \theta\|_1 + \|\nabla G_K \theta^*\|_1 \leq 2V_n}} \frac{1}{\sum_{i=1}^n m_i} \sum_{i=1}^n \sum_{j=1}^{m_i} \varLambda_{ij} s_{ij} \right) \cdot \mathbf{1}_{\Omega_\beta} \right] \\
&\leq C_3 K \frac{\log^{\frac{1}{2}}(nm)}{\sum_{i=1}^n m_i} b + C_4 K \frac{\log(nm)}{\sum_{i=1}^n m_i} V_n \\
&\quad + B \cdot \left(\frac{2}{nm} + \frac{4}{C_\beta} C \log(n) nm \exp\left(-\frac{\tilde{C} K}{24 \log(n)} \right) \right. \\
&\quad \left. + \frac{3}{n} + \frac{4}{C_\beta} \log(n) \frac{nm}{K} \exp\left(-\frac{C_\beta \tilde{b}_2 K}{48 \log(n)} \right) + \frac{1}{nm} \right) \\
&\overset{(a)}{\leq} \frac{1}{\sum_{i=1}^n m_i} C_4 K \log(nm) V_n \\
&\quad + B \cdot \left(\frac{4}{C_\beta} C \log(n) nm \exp\left(-\frac{\tilde{C} K}{24 \log(n)} \right) \right. \\
&\quad \left. + \frac{3}{n} + \frac{4}{C_\beta} \log(n) \frac{nm}{K} \exp\left(-\frac{C_\beta \tilde{b}_2 K}{48 \log(n)} \right) \right) \\
&\overset{(b)}{\leq} \frac{1}{\sum_{i=1}^n m_i} C_4 K \log(nm) V_n \\
&\quad + B C_7 \log(n) nm \exp\left(-\frac{\tilde{C} K}{24 \log(n)} \right) + \frac{B C_8}{n}.
\end{aligned}
$$

The (a) holds since we assume $b^2 = O\left(\frac{1}{n} + \frac{1}{nm}\right)$. The (b) holds since we assume $K = \text{poly}(log(nm))$.

Case 1, if the $B \geq 1$ so $b = c^{-1/2} B^{1/2} \cdot t$, Then we let

$$
K = 2\log(n)\log(nm) \leq \log(nm)^{2+\mu}
$$

for some $\mu > 0$, we have

$$
\frac{1}{\sum_{i=1}^n m_i} C_4 K \log(nm) V_n + B C_7 \log(n) nm \exp\left(-\frac{\tilde{C} K}{24 \log(n)} \right) + \frac{B C_8}{n}
$$

$$\leq \frac{1}{\sum_{i=1}^{n} m_i} C_9 BK \log(nm) V_n + \frac{BC_8}{n}.$$

So we have

$$\begin{aligned}
&R_{\mathcal{J}_{e,1}}\left(\mathcal{F} \cap \left\{c\Delta^2(f - f^*) \leq t^2\right\}\right) + R_{\mathcal{J}_{o,1}}\left(\mathcal{F} \cap \left\{c\Delta^2(f - f^*) \leq t^2\right\}\right) \\
&\leq \frac{1}{\sum_{i=1}^{n} m_i} C_9 BK \log(nm) V_n + \frac{BC_8}{n} \\
&\leq \frac{1}{\sum_{i=1}^{n} m_i} C_9 B \log(nm)^{3+l} V_n + \frac{BC_8}{n}.
\end{aligned}$$

Thus, we seek a

$$t^2 = \frac{C_9 B \log(nm)^{4+l} V_n}{nm} + \frac{\log(nm) BC_8}{n},$$

to get

$$\frac{R_{\mathcal{J}_{e,1}}\left(\mathcal{F} \cap \left\{c\Delta^2(f - f^*) \leq t^2\right\}\right) + R_{\mathcal{J}_{o,1}}\left(\mathcal{F} \cap \left\{c\Delta^2(f - f^*) \leq t^2\right\}\right)}{t^2} = O\left(\frac{1}{\log(nm)}\right).$$

Case 2, if the $B < 1$ so $b = c^{-1/2} \cdot t$,
So we have

$$\begin{aligned}
&R_{\mathcal{J}_{e,1}}\left(\mathcal{F} \cap \left\{c\Delta^2(f - f^*) \leq t^2\right\}\right) + R_{\mathcal{J}_{o,1}}\left(\mathcal{F} \cap \left\{c\Delta^2(f - f^*) \leq t^2\right\}\right) \\
&\leq \frac{1}{\sum_{i=1}^{n} m_i} C_9 BK \log(nm) V_n + \frac{BC_8}{n} \\
&\leq \frac{1}{\sum_{i=1}^{n} m_i} C_{10} \log(nm)^{3+l} V_n + \frac{C_8}{n}.
\end{aligned}$$

Thus, we seek a

$$t^2 = \frac{C_{10} \log(nm)^{4+l} V_n}{nm} + \frac{\log(nm) C_8}{n},$$

to get

$$\frac{R_{\mathcal{J}_{e,1}}\left(\mathcal{F} \cap \left\{c\Delta^2(f - f^*) \leq t^2\right\}\right) + R_{\mathcal{J}_{o,1}}\left(\mathcal{F} \cap \left\{c\Delta^2(f - f^*) \leq t^2\right\}\right)}{t^2} = O\left(\frac{1}{\log(nm)}\right).$$

Overall, we get

$$t^2 = \frac{C_{11} \max\{B, 1\} \log(nm)^{4+l} V_n}{nm} + \frac{\log(nm) \max\{B, 1\} C_8}{n}.$$

Furthermore, by Assumption 1.9, $\frac{1}{\sum_{i=1}^{n} m_i} \leq \frac{1}{cmn}$, and Equation (75), the result The above results hold uniformly for $R_{\mathcal{J}_{e,l}}$ and $R_{\mathcal{J}_{o,l}}$ for all $l = 1, \ldots, L$, then we can show that

$$\frac{1}{t^2} \sum_{l=1}^{L} \left(R_{\mathcal{J}_{e,l}}\left(\|\nabla G_K \theta\|_1 \leq V_n \cap \left\{\theta : \|\theta - \theta^*\| \leq \max\left\{1, B^{1/2}\right\} t\right\}\right)\right.$$

$$+R_{\mathcal{J}_{o,l}}\left(\|\nabla G_K \theta\|_1 \,\leq\, V_n \cap \left\{ \theta : \|\theta - \theta^*\| \leq \max\left\{1, B^{1/2}\right\} t \right\} \right) \;=\; O\left(\frac{1}{\log(nm)} \right),$$

by letting

$$t^2 = \frac{C_{12} \max\{B,1\} \log^{5+\mu}(nm) V_n}{nm} + \frac{\log^2(nm) \max\{B,1\} C_{13}}{n}.$$

Suppose in addition that $K \asymp \log^{2+\mu}(nm)$ for some $\mu > 0$.
The estimator $\hat{\theta}_{V_n}$, defined in (23) satisfies that
Then by Theorem 1, The estimator $\hat{\theta}_{V_n}$, defined in (23) satisfies that

$$\Delta^2(\hat{\theta}_{V_n} - \theta^*) \;=\; O_p\left(r_n^2\right),$$

by setting that

$$r_n^2 \asymp \frac{\max\{B,1\} \log^{5+\mu}(nm) V_n}{nm} + \frac{\log^2(nm) \max\{B,1\}}{n} + \frac{d \log^{1+\mu}(n) \log(T_n)}{n}.$$

■

Lemma 43 *Let Ω_β be defined in (76). Let $\{s_{ij}\}_{i=1,j=1}^{n,m_i} \in \mathbb{R}^{\sum_{i=1}^n m_i}$ be independent Rademacher variables, independent of the data $\{(x_{ij}, e_{ij}, \delta_i)\}$ and their coupled copies $\{(x^*_{ij}, e^*_{ij}, \delta^*_i)\}$, and use the notation $\Lambda = \theta_1 - \theta_2$, for $\theta_1, \theta_2 \in \mathbb{R}^{\sum_{i=1}^n m_i}$. For some positive constants C_3 and C_4 we have that*

$$\begin{aligned}
&\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*,s}\left[\left(\sup_{\|\Lambda\| \leq b, \|\nabla G_K \Lambda\|_1 \leq 2V_n} \frac{1}{\sum_{i=1}^n m_i} \sum_{i=1}^n \sum_{j=1}^{m_i} \Lambda_{ij} s_{ij} \right) \cdot \mathbf{1}_{\Omega_\beta} \right] \\
&\leq C_3 K \frac{\log^{\frac{1}{2}}(nm)}{\sum_{i=1}^n m_i} b + C_4 K \frac{\log(nm)}{\sum_{i=1}^n m_i} V_n \\
&\quad + B \cdot \left(\frac{2}{nm} + \frac{4}{C_\beta} C \log(n) nm \exp\left(-\frac{\tilde{C} K}{24 \log(n)} \right) \right. \\
&\quad \left. + \frac{3}{n} + \frac{4}{C_\beta} \log(n) \frac{nm}{K} \exp\left(-\frac{C_\beta \tilde{b}_2 K}{48 \log(n)} \right) + \frac{1}{nm} \right).
\end{aligned}$$

Here $C_3 \geq 0$ is constant depending on d, v_{min} and v_{max}, and C_5 is constant depending on d, v_{min} and v_{max} and L. Moreover, $\tilde{b}_2 > 0$ is a constant depending on d, v_{min}, and v_{max}. Furthermore, $\tilde{C} = \frac{C_\beta c c_1}{c_2 C}$ with c_1 and c_2 positive constants, and and $\tilde{C}$ is a constant in $(0,1)$. Here v_{max}, v_{min}, C and c' are from Assumptions 1.9 and 7.

Proof Let I_r be given in (129), and let N in the construction of $P_{\text{lat}}(N)$ of (127) is chosen as

$$N = \frac{\left\lceil 3\sqrt{d}(nm)^{1/d} \right\rceil}{aK^{1/d}}, \qquad \text{where} \qquad a = \frac{1}{(4Cc_{2,d}c_2 v_{\max})^{1/d}},$$

for a positive constants c_2, and, $v_{\max}$, and C are from Assumptions 1.9 and 7. Moreover, $c_{2,d}$ is a positive constants that depends on d. Then there exist positive constants $\bar{b}_1$ and $\bar{b}_2$ depending on d,

$v_{\min}$, and $v_{\max}$ with $v_{\min}$ from Assumptions 1.9 and 7. Let set $C(\cdot)$ be defined in (128). We start by introducing the notation $x'_{ij} = P_I(x_{ij})$.Then define the event Ω as

$$\Omega: \quad \text{“If } x_{ij} \in C(x'_{ij}) \text{ and } x_{st} \in C(x'_{rs}) \text{for } x'_{ij}, x'_{st} \in P_{\mathrm{lat}}(N) \text{ with } \|x'_{ij}, x'_{st}\|_2 \leq N^{-1}, \text{ then } x_{ij} \text{ and } x_{st} \text{ are connected in the } K\text{-NN graph.”} \tag{132}$$

Also, define the following events,

$$E_1 = \left\{ \min_{r=1,\ldots,N^d} |I_r| \geq \frac{1}{2} b_2 K \right\}, \tag{310}$$

and

$$E_2 = \left\{ \max_{r=1,\ldots,N^d} |I_r| \leq \frac{3}{2} b_2 K \right\}. \tag{311}$$

hold. Observe that by Lemma 47 event Ω hold with probability at least,

$$1 - \frac{4}{C_\beta} C \log(n) nm \exp\left(-\frac{\tilde{C} K}{24 \log(n)} \right) - \frac{1}{n},$$

and events $E_1 \cap E_2$ hold with probability at least

$$1 - \frac{4}{C_\beta} C \log(n) nm^d \exp\left(-\frac{C_\beta \tilde{b}_2 K}{48 \log(n)} \right) - \frac{1}{n},$$

respectively, where $\tilde{C} = \frac{C_\beta c_1 c}{c_2 C}$ and $\tilde{C} \in (0,1)$, with c_1 and c_2 are some constant. Next, by (2.67) in Wainwright (2019) we have that the event

$$E_3 = \left\{ \max_{1 \leq i \leq n, 1 \leq j \leq m_i} |s_{ij}| \leq \sqrt{2 \log\left(\sum_{i=1}^n m_i \right)} \right\}$$

happens with probability at least$1 - \frac{1}{\sum_{i=1}^n m_i}$.

Then let $\tilde{\Pi}$ be the orthogonal projection onto the span of $1_{N^d} \in \mathbb{R}^{N^d}$, and $D^\dagger$ is the pseudoinverse of the incidence matrix D from Lemma 48.

Further, using the proof of Theorem 2 in Hütter and Rigollet (2016) (see page 21 in that paper), if $\tilde{s} \in \mathbb{R}^{N^d}$ is a mean zero vector whose coordinates are Rademacher variables or zero values as introduced in (130). it yields that the following two inequalities hold simultaneously on an event E_4 of probability at least $1 - \frac{1}{nm}$,

$$E_4 = \left\{ \left\| (D^\dagger)^\top \tilde{s} \right\|_\infty \leq \rho \sqrt{2 \log(eN^d nm)}, \quad \left\| \tilde{\Pi} \tilde{s} \right\|_2 \leq 2\sqrt{2 \log(enm)} \right\},$$

where $\rho = \max_{1 \leq v \leq |E_{\mathrm{lat}}|} \left\| D^\dagger_{,v} \right\|_2$. Additionally, using propositions 4 and 6 in Hütter and Rigollet (2016), we have that $\rho \leq \widetilde{C(d)} \sqrt{\log(N)}$, where $\widetilde{C(d)}$ is a positive constant depending on d. Hence,

$$\left\| (D^\dagger)^\top \tilde{s} \right\|_\infty \leq C(d) \sqrt{\log\left(\frac{nm}{K} \right) \log\left(\frac{nm}{K} nm \right)} \quad \text{and} \quad \|\tilde{\Pi} \tilde{s}\|_2 \leq 2\sqrt{2 \log(enm)},$$

with probability at least $1 - 2 \cdot \frac{1}{nm}$. Here, $C(d) > 0$ is a constant depending on d, C, and v_{max}.

Then denote $\mathcal{E} \equiv \Omega \cap E_1 \cap E_2 \cap E_3 \cap E_4$, on the event $\mathcal{E}$, the following is satisfied.

Let $\theta_1, \theta_2 \in \mathbb{R}^{\sum_{i=1}^n m_i}$ such that $\|\varLambda\|_2 \leq b$ and $\|\nabla G_K \varLambda\|_1 \leq 2V_n$. And $\{s_{ij}\}_{i=1,j=1}^{n,m_i} = s \in \mathbb{R}^{\sum_{i=1}^n m_i}$ are independent Rademacher variables, independent of the data $\{(x_{ij}, e_{ij}, \delta_i)\}$ and their coupled copies $\{(x_{ij}^*, e_{ij}^*, \delta_i^*)\}$. First, by Lemma 49 and E_3,

$$s^\top(\theta_1 - \theta_2) \leq 4\sqrt{2\log(nm)}V_n + s^\top(\theta_{1I} - \theta_{2I}).$$

Then, by Lemma 50, E_2, and E_4, we have that

$$s^\top(\theta_{1I} - \theta_{2I}) \leq \max_{r=1,\dots,N^d} |I_r| \left(\|\tilde{\Pi}\tilde{s}\|_2 \|\theta_1^* - \theta_2^*\|_2 + \|(D^\dagger)^\top\|_\infty \left(\|\nabla_{G_K}\theta_{1I}\|_1 + \|\nabla_{G_K}\theta_{2I}\|_1\right)\right), \quad (133)$$

Thus, we have

$$\begin{aligned}
\frac{1}{\sum_{i=1}^n m_i} s^\top \varLambda \leq\ & \frac{1}{\sum_{i=1}^n m_i} 4\sqrt{2\log(nm)}V_n \\
& + \max_{r=1,\dots,N^d} |I_r| \frac{1}{\sum_{i=1}^n m_i} \left(\left\|\tilde{\Pi}\tilde{s}\right\|_2 b + \left\|(D^\dagger)^\top \tilde{s}\right\|_\infty \cdot 4V_n \right) \\
\leq\ & 4\frac{\sqrt{2\log(\sum_{i=1}^n m_i)}}{\sum_{i=1}^n m_i} V_n + \frac{3}{2}\tilde{b}_1 K \left(2b\frac{\sqrt{2\log(enm)}}{\sum_{i=1}^n m_i} + C(d)\frac{\log(\frac{(nm)^2}{K})}{\sum_{i=1}^n m_i}(4V_n)\right) \\
\leq\ & C_3 K \frac{\log^{\frac{1}{2}}(nm)}{\sum_{i=1}^n m_i} b + C_4 K \frac{\log(nm)}{\sum_{i=1}^n m_i} V_n,
\end{aligned}$$

for some positive constants C_3 and C_4. Here C_4 depends on d.

Finally, on the event $\mathcal{E}$ the following is satisfied. Thus,

$$\sup_{\|\varLambda\|_2 \leq b, \|\nabla G_K \varLambda\|_1 \leq 2V_n} \frac{1}{\sum_{i=1}^n m_i} \sum_{i=1}^n \sum_{j=1}^{m_i} \varLambda_{ij} s_{ij} \leq C_3 K \frac{\log^{\frac{1}{2}}(nm)}{\sum_{i=1}^n m_i} b + C_4 K \frac{\log(nm)}{\sum_{i=1}^n m_i} V_n. \quad (134)$$

Under the event $\mathcal{E}^c$, we have that $R_{\mathcal{J}_{elmn}}(\|\nabla G_K\theta\|_1 \leq V_n \cap \{\theta : \|\theta - \theta^*\| \leq b\}) \leq B$. Therefore, splitting the expectation over $\mathcal{E}$ and $\mathcal{E}^c$,

$$\begin{aligned}
& \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*,s}\left[\left(\sup_{\substack{\|\varLambda\| \leq b \\ \|\nabla G_K \varLambda\|_1 \leq \|\nabla G_K \theta\|_1 + \|\nabla G_K \theta^*\|_1 \leq 2V_n}} \frac{1}{\sum_{i=1}^n m_i} \sum_{i=1}^n \sum_{j=1}^{m_i} \varLambda_{ij} s_{ij} \right) \cdot \mathbf{1}_{\Omega_\beta} \right] \\
&= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*,s}\left[R_{\mathcal{J}_{elmn}}(\|\nabla G_K \theta\|_1 \leq V_n \cap \{\theta : \|\theta - \theta^*\| \leq b\}) \right] \\
&\leq C_3 K \frac{\log^{\frac{1}{2}}(nm)}{\sum_{i=1}^n m_i} b + C_4 K \frac{\log(nm)}{\sum_{i=1}^n m_i} V_n \\
&\quad + B \cdot \left(\frac{2}{nm} + \frac{4}{C_\beta} C \log(n) nm \exp\left(-\frac{\tilde{C}K}{24\log(n)}\right) \right. \\
&\quad \left. + \frac{3}{n} + \frac{4}{C_\beta}\log(n)\frac{nm}{K}\exp\left(-\frac{C_\beta \tilde{b}_2 K}{48\log(n)}\right) + \frac{1}{nm}\right)
\end{aligned}$$

∎

The below lemma gives the error rate for $\hat{f}_{V_n}$ defined in (25).

Lemma 44 *In addition, $\hat{f}_{V_n}$, defined in (25) satisfies that*

$$\Delta^2(\hat{f}_{V_n} - f^*) \,=\, O_p(\tilde{r}_{nm})$$

for

$$\tilde{r}_{nm} = \frac{\max\{B,1\}\log^{5+\mu}(nm)V_n}{nm} + \frac{\log^2(nm)\max\{B,1\}}{n} + \frac{d\log^{1+\mu}(n)\log(T_n)}{n} + \frac{\log^{2+\mu}(nm)}{(nm)^{1/d}}.$$

Proof Through this step, notation from Appendix section D.2.2 is used. We start noticing that,

$$\|f^* - \hat{f}_{V_n}\|_2^2 = \int \Big(f^*(x) - \hat{f}_{V_n}(x)\Big)^2 v(x)dx.$$

Then, adding and subtracting $\frac{1}{|N_K(x)|}\sum_{x_{ij}\in N_K(x)} f^*(x_{ij})$, it can be written as

$$\begin{aligned}
&\int \Bigg(f^*(x) - \frac{1}{|N_K(x)|}\sum_{x_{ij}\in N_K(x)} f^*(x_{ij}) + \frac{1}{|N_K(x)|}\sum_{x_{ij}\in N_K(x)} f^*(x_{ij}) \\
&\quad - \frac{1}{|N_K(x)|}\sum_{x_{ij}\in N_K(x)} \hat{f}_{V_n}(x_{ij})\Bigg)^2 v(x)dx \\
&\le 2\int \Bigg(f^*(x) - \frac{1}{|N_K(x)|}\sum_{x_{ij}\in N_K(x)} f^*(x_{ij})\Bigg)^2 v(x)dx \\
&\quad + 2\int \Bigg(\frac{1}{|N_K(x)|}\sum_{x_{ij}\in N_K(x)} f^*(x_{ij}) - \frac{1}{|N_K(x)|}\sum_{x_{ij}\in N_K(x)} \hat{f}_{V_n}(x_{ij})\Bigg)^2 v(x)dx.
\end{aligned}$$

Now we proceed to bound each of the terms

$$\mathrm{Int}_1 = \int \Bigg(f^*(x) - \frac{1}{|N_K(x)|}\sum_{x_{ij}\in N_K(x)} f^*(x_{ij})\Bigg)^2 v(x)dx, \tag{135}$$

and,

$$\mathrm{Int}_2 = \int \Bigg(\frac{1}{|N_K(x)|}\sum_{x_{ij}\in N_K(x)} f^*(x_{ij}) - \frac{1}{|N_K(x)|}\sum_{x_{ij}\in N_K(x)} \hat{f}_{V_n}(x_{ij})\Bigg)^2 v(x)dx. \tag{136}$$

For the second term Int_2 in Equation (136), we observe that

$$\begin{aligned}
&\int \Bigg(\frac{1}{|N_K(x)|}\sum_{x_{ij}\in N_K(x)} f^*(x_{ij}) - \frac{1}{|N_K(x)|}\sum_{x_{ij}\in N_K(x)} \hat{f}_{V_n}(x_{ij})\Bigg)^2 v(x)dx \\
&= \sum_{r=1}^{N^d}\int_{\mathcal{I}_r} \Bigg(\frac{1}{|N_K(x)|}\sum_{x_{ij}\in N_K(x)} f^*(x_{ij}) - \frac{1}{|N_K(x)|}\sum_{x_{ij}\in N_K(x)} \hat{f}_{V_n}(x_{ij})\Bigg)^2 v(x)dx.
\end{aligned}$$

Moreover, for $x' \in [0,1]^d$, there exists $c_r(x') \in P_{\text{lat}}(N)$ that satisfies $x' \in C(c_r(x'))$ and

$$\|x' - P_I(x')\|_2 \,\leq\, \frac{1}{2N}.$$

Even more, by Lemma 47, with high probability, there exists $(i(x'), j(x')) \in [n] \times [m_i]$ such that $x_{i(x')j(x')} \in C(c_r(x'))$. Hence,

$$\|x' - x_{i(x')j(x')}\|_2 \,\leq\, \frac{1}{N}.$$

Denote by $x_{i_1j_1}, \ldots, x_{i_Kj_K}$ $[i_1, \ldots, i_K] \in [n]$ and $(j_1, \ldots, j_K) \in [m_{i_1}] \times \ldots \times [m_{i_K}]$ K-nearest neighbors corresponding to $x_{i'(x')j(x')}$. Observe that Observe that

$$\|x' - x_{i_rj_r}\|_2 \,\leq\, \frac{1}{N} + R_{K,\max}, \quad r = 1, \ldots, K$$

where

$$R_{K,\max} = \max_{1 \leq i \leq n, 1 \leq j \leq m_i} R_K(x_{ij}),$$

with $R_K(x)$ denoting the distance from $x \in [0,1]^d$ to its Kth nearest neighbor in the set $\{x_{11}, \ldots, x_{nm_n}\}$. Thus, with high probability, for any $x' \in [0,1]^d$,

$$N_K(x') \subset \{x_{ij} : \|x' - x_{ij}\|_2 \,\leq\, \frac{1}{N} + R_{K,\max}\} \subset \{x_{ij} : \|c_r(x') - x_{ij}\|_2 \,\leq\, \left(\frac{1}{2} + \frac{1}{N}\right) + R_{K,\max}\}.$$

Hence, we set

$$\tilde{N}(c_r(x')) \,=\, \{x_{ij} : \|c_r(x') - x_{ij}\|_2 \,\leq\, \left(\frac{1}{2} + \frac{1}{N}\right) + R_{K,\max}\}.$$

As a result, for $r \in [N]^d$,

$$\int_{C(c_r)} \left(\frac{1}{|N_K(x)|} \sum_{x_{ij} \in N_K(x)} f^*(x_{ij}) - \frac{1}{|N_K(x)|} \sum_{x_{ij} \in N_K(x)} \hat{f}_{V_n}(x_{ij}) \right)^2 v(x)dx,$$

can be bounded as,

$$\begin{aligned}
&\int_{C(c_r)} \left(\frac{1}{|N_K(x)|} \sum_{x_{ij} \in N_K(x)} f^*(x_{ij}) - \frac{1}{|N_K(x)|} \sum_{x_{ij} \in N_K(x)} \hat{f}_{V_n}(x_{ij}) \right)^2 v(x)dx \\
&\leq \frac{1}{K} \int_{C(c_r)} \left(\sum_{x_{ij} \in \tilde{N}(c_r)} \left\{ f^*(x_{ij}) - \hat{f}_{V_n}(x_{ij}) \right\}^2 \right) v(x)dx \\
&= \frac{1}{K} \sum_{x_{ij} \in \tilde{N}(c_r)} \left\{ f^*(x_{ij}) - \hat{f}_{V_n}(x_{ij}) \right\}^2 \int_{C(c_r)} v(x)dx.
\end{aligned}$$

The above leads to an upper bound for the term

$$\int \left(\frac{1}{|N_K(x)|} \sum_{x_{ij} \in N_K(x)} f^*(x_{ij}) - \frac{1}{|N_K(x)|} \sum_{x_{ij} \in N_K(x)} \hat{f}_{V_n}(x_{ij}) \right)^2 v(x)dx.$$

This upper bound has the form,

$$\sum_{r=1}^{N^d}\left(\frac{1}{K}\sum_{x_{ij}\in\tilde{N}(c_r)}\left\{f^*(x_{ij})-\hat{f}_{V_n}(x_{ij})\right\}^2\right)\int_{C(c_r)} v(x)dx,$$

which can be bounded by

$$\begin{aligned}
&\left[\sum_{r=1}^{N^d}\frac{1}{K}\sum_{x_{ij}\in N(r)}\left\{f^*(x_{ij})-\hat{f}_{V_n}(x_{ij})\right\}^2\right]\max_{1\le r\le N^d}\int_{C(c_r)} v(x)\,dx\\
&\le \sum_{i=1}^{n}\sum_{j=1}^{m_i}\frac{1}{K}\sum_{\substack{r\in[N^d]:\\ \|x_{ij}-c_r\|_2\le(\frac{1}{2}+1)N+R_{k,\max}}}\left\{f^*(x_{ij})-\hat{f}_{V_n}(x_{ij})\right\}^2\max_{1\le r\le N^d}\int_{C(c_r)} v(x)\,dx\\
&\le \left[\sum_{i=1}^{n}\sum_{j=1}^{m_i}\left\{f^*(x_{ij})-\hat{f}_{V_n}(x_{ij})\right\}^2\right]\left[\frac{1}{K}\max_{1\le r\le N^d}\int_{C(c_r)} v(x)\,dx\right]\\
&\quad\times\max_{(i,j)\in[n]\times[m_i]}\left\{k:\|x_{ij}-c_r\|_2\ \le\ \left(\frac{1}{2}+1\right)\frac{1}{N}+R_{k,\max}\right\}.
\end{aligned}$$

Next, for a set $A\subset\mathbb{R}^d$ and positive constant $\bar{r}$, we define the packing and external covering numbers as

$$N_{\bar{r}}^{\text{pack}}(A):=\max\left\{r\in\mathbb{N}:\exists q_1,\ldots,q_r\in A,\ \text{such that}\ \|q_i-q_j\|_2>\bar{r}\,\forall i\neq j\right\},$$

$$N_{\bar{r}}^{\text{ext}}(A):=\min\left\{r\in\mathbb{N}:\exists q_1,\ldots,q_r\in\mathbb{R}^d,\ \text{such that}\ \forall x\in A \text{ there exists } r_x \text{ with } \|x-q_r\|_2<\bar{r_x}\right\}.$$

Furthermore, by Lemma 46, there exists a constant $\tilde{c}$ such that $R_{K,\max}\le\frac{\tilde{c}}{N}$ with high probability. This implies that with high probability, for any $x'\in[0,1]^d$,

$$\begin{aligned}
&\max_{(i,j)\in[n]\times[m_i]}\left|\left\{r\in[N^d]:\|x_{ij}-c_r\|_2\ \le\ \left(\frac{1}{2}+\frac{1}{N}\right)+R_{K,\max}\right\}\right|\\
&\le \max_{(i,j)\in[n]\times[m_i]}\left|\left\{r\in[N^d]:\|x_{ij}-c_r\|_2\ \le\ \frac{\tilde{C}}{N}\right\}\right|\le \max_{(i,j)\in[n]\times[m_i]}N_{\frac{1}{N}}^{\text{pack}}\left(B_{\frac{C}{N}}(x_{ij})\right)\\
&\le N_{\frac{1}{N}}^{\text{ext}}\left(B_{\frac{\tilde{C}}{N}}(0)\right)=N_{1}^{\text{ext}}\left(B_{\tilde{C}}^{N}(0)\right)<C',
\end{aligned}$$

for some positive constant C', where the first inequality follows from $R_{K,\max}\le\frac{\tilde{c}}{N}$, the second from the definition of packing number, and the remaining inequalities from basic properties of packing and external covering numbers. Therefore, there exists a constant $C_1>0$ such that

$$\begin{aligned}
\text{Int}_2&=\int\left(\frac{1}{|N_K(x)|}\sum_{x_{ij}\in N_K(x)}f^*(x_{ij})-\frac{1}{|N_K(x)|}\sum_{x_{ij}\in N_K(x)}\hat{f}_{V_n}(x_{ij})\right)^2 v(x)\mu(dx)\\
&\le\left[\sum_{i=1}^{n}\sum_{j=1}^{m_i}\frac{1}{K}\left(f^*(x_{ij})-\hat{f}_{V_n}(x_{ij})\right)^2\right]\frac{C'}{K}\max_{1\le r\le N^d}\int_{C(c_r)} v(x)dx
\end{aligned}$$

$$\leq \left[\sum_{i=1}^{n}\sum_{j=1}^{m_i}\left(f^*(x_{ij})-\hat{f}_{V_n}(x_{ij})\right)^2\right]\frac{C' c_2 d v_{\max}}{K}\mathrm{Vol}\left(B_{\frac{\tilde{b}_1^{1/d}}{(2c_{2,d}Cv_{\max}c_2)^{1/d}}\left(\frac{K}{nm}\right)^{1/d}}(x)\right)$$

$$\leq \left[\frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m_i}\left(\theta_{i,j}-(\hat{\theta}_{V_n})_{i,j}\right)^2\right]\tilde{C}_1 v_{\max},$$

where, the second inequality is followed by Equation (169). Then, making use of (**??**), we conclude that

$$\mathrm{Int}_2 = \int\left(\frac{1}{|N_K(x)|}\sum_{x_{ij}\in N_K(x)}f^*(x_{ij})-\frac{1}{|N_K(x)|}\sum_{x_{ij}\in N_K(x)}\hat{f}_{V_n}(x_{ij})\right)^2 v(x)dx \;=\; O_p(r_{nm}^2), \quad (137)$$

where

$$r_{nm}^2 = \frac{\max\left\{B^2,B\right\}\log^{5+\mu}(nm)V_n}{nm}+\frac{\log^2(nm)\max\left\{B^2,B\right\}}{n}+\frac{\max\left\{B,1\right\}d\log^{1+\mu}(n)\log(T_n)}{n}.$$

It remains to analyze the expression in Equation (135), this is,

$$\mathrm{Int}_1 = \int\left(f^*(x)-\frac{1}{|N_K(x)|}\sum_{x_{ij}\in N_K(x)}f^*(x_{ij})\right)^2 v(x)dx.$$

To this end, we use the notation from Appendix section D.2.1. Observe that, if S corresponds to the piecewise definition of f^* in Definition (41), the term

$$\int\left(f^*(x)-\frac{1}{|N_K(x)|}\sum_{x_{ij}\in N_K(x)}f^*(x_{ij})\right)^2 v(x)dx,$$

can be written as,

$$\sum_{r\in[N]^d}\int_{C(c_r)}\left(f^*(x)-\frac{1}{|N_K(x)|}\sum_{x_{ij}\in N_K(x)}f^*(x_{ij})\right)^2 v(x)dx,$$

which by convexity can be bounded by,

$$\sum_{r\in[N]^d}\int_{C(c_r)}\sum_{x_{ij}\in N_K(x)}\frac{1}{K}\left(f^*(x)-f^*(x_{ij})\right)^2 v(x)dx$$

$$= \sum_{r\in[N]^d:C(c_r)\cap S=\emptyset}\int_{C(c_r)}\sum_{x_{ij}\in N_K(x)}\frac{1}{K}\left(f^*(x)-f^*(x_{ij})\right)^2 v(x)dx$$

$$+ \sum_{r\in[N]^d:C(c_r)\cap S\neq\emptyset}\int_{C(c_r)}\sum_{x_{ij}\in N_K(x)}\frac{1}{K}\left(f^*(x)-f^*(x_{ij})\right)^2 v(x)dx.$$

Therefore, the term

$$\int\left(f^*(x)-\frac{1}{|N_K(x)|}\sum_{x_{ij}\in N_K(x)}f^*(x_{ij})\right)^2 v(x)dx,$$

is bounded by

$$4\left|\left\{r \in [N]^d : C(c_r)\cap S \neq \emptyset\right\}\right| \|f_0\|_\infty^2 \max_{1\leq r\leq N^d} \int_{C(c_r)} v(x)dx$$

$$+ \sum_{r\in[N]^d: C(c_r)\cap S=\emptyset} \int_{C(c_r)} \sum_{x_{ij}\in N_K(x)} \frac{1}{K}\left(f_0(x)-f_0(x_{ij})\right)^2 v(x)dx,$$

which by Equation (169) is smaller than

$$\mathrm{Vol}\left(B_{\frac{\tilde{b}_1^{1/d}}{(2c_{2,d}Cv_{\max}c_2)^{1/d}}\left(\frac{K}{nm}\right)^{1/d}}(x)\right)\left(4v_{\max}\|f_0\|_\infty^2\right)\left|\left\{r\in[N]^d : C(c_r)\cap S\neq\emptyset\right\}\right|$$

$$+ \sum_{r\in[N]^d: C(c_r)\cap S=\emptyset} \int_{C(c_r)} \sum_{x_{ij}\in N_K(x)} \frac{1}{K}\left(f^*(x)-f^*(x_{ij})\right)^2 v(x)dx.$$

Then, using the bound for volume and the piece-wise Lipschitz condition, it is bounded by

$$\left(\frac{4\tilde{b}_1\|f^*\|_\infty^2}{nm}\right) K\left|\left\{r\in[N]^d : C(c_r)\cap S\neq\emptyset\right\}\right|$$

$$\begin{aligned}
&+ \sum_{r\in[N]^d: C(c_r)\cap S=\emptyset} \int_{C(c_r)} \sum_{x_{ij}\in N_K(x)} \frac{L_0}{K}\left(\|x-x_{ij}\|_2^2\right) v(x)dx \\
&\leq \left(\frac{4\tilde{b}_1^d\|f^*\|_\infty^2}{nm}\right) K\left|\left\{r\in[N]^d : C(c_r)\cap S\neq\emptyset\right\}\right| \\
&+L_0\left(\sum_{r\in[N]^d: C(c_r)\cap S=\emptyset}\int_{C(c_r)} v(x)dx\right)\left(\frac{1}{N}+R_{K,\max}\right)^2 \\
\leq&\left(\frac{4\tilde{b}_1^d\|f^*\|_\infty^2}{nm}\right) K\left|\left\{r\in[N]^d : C(c_r)\cap S\neq\emptyset\right\}\right| + L_0\left(\frac{1}{N}+R_{K,\max}\right)^2.
\end{aligned}$$

Finally, by the choice of N, Lemma 46 and following the conclusion on Lemma 15 of Madrid Padilla et al. (2020), we get that

$$\mathrm{Int}_2 = \int\left(f^*(x) - \frac{1}{|N_K(x)|}\sum_{x_{ij}\in N_K(x)} f^*(x_{ij})\right)^2 v(x)dx \;=\; O_p\left(\left(\frac{K}{nm}\right)^{\frac{1}{d}}\right). \tag{138}$$

From Equation (137) and (138), and the fact that

$$\Delta^2(\hat{f}_{V_n}-f^*) \;\leq\; \|f^*-\hat{f}_{V_n}\|_2^2,$$

we then conclude the claim.

■

Lemma 45 *Let m be a Binomial (M,q) random variable. Then, for all $\delta \in (0,1)$,*

$$\mathbb{P}(m \,\leq\, (1-\delta)Mq) \,\leq\, \exp\left(-\frac{1}{3}\delta^2 Mq\right),$$

$$\mathbb{P}(m \,\geq\, (1+\delta)Mq) \,\leq\, \exp\left(-\frac{1}{3}\delta^2 Mq\right)$$

Lemma 46 *(See Lemma 6 in Madrid Padilla et al. (2020)). Under Assumption 1. Denote by $R_k(x)$ the distance from $\in [0,1]^d$ to its Kth nearest neighbor in the set $\{x_{ij}\}_{i=1,j=1}^{n,m}$. Setting $R_{k,\max} = \max\limits_{1 \leq i \leq n, 1\leq j \leq m_i} R_k(x_{ij})$, and $R_{k,\min} = \min\limits_{1 \leq i \leq n, 1\leq j \leq m_i} R_k(x_{ij})$, we have that,*

$$\begin{aligned}
&\mathbb{P}\left(a\left(\frac{K}{nm}\right)^{1/d} \,\leq\, R_{k,\min} \,\leq\, R_{k,\max} \,\leq\, \tilde{a}\left(\frac{K}{nm}\right)^{1/d}\right)\\
&\geq 1-\frac{4}{C_\beta}\left(\log(n) n \exp\left(-\frac{\tilde{C}K}{24\log(n)}\right)\right)-\frac{4}{C_\beta}\left(\log(n) n \exp\left(-\frac{C_\beta K}{24\log(n)}\right)\right)-\frac{2}{n},
\end{aligned}$$

where $\tilde{C} = \frac{C_\beta c_1 c}{c_2 C}$, $\alpha = 1/(4(Cc_{2,d}c_2 v_{\max})^{1/d})$, and $\tilde{a} = (1/(c_1 c_{1,d} c_1 v_{\min}))^{1/d}$ for some positive constants c_2, and c_1, $v_{\max}$, $v_{\min}$, C, C_β, and c from Assumption 1. Moreover, $c_{1,d}$ and $c_{2,d}$ are positive constants that depends on d.

Proof First note that for any $x \in [0,1]^d$, by Assumptions 1 we have

$$\mathbb{P}\left(x_{ij} \in B_r(x)\right) \,=\, \int_{B_r(x)} v(t)dt \,\leq\, c_2 r^d v_{\max} =: \mu_{\max} < 1, \tag{139}$$

(for some small enough r and a positive constant c_2 that depends on d. Let $L = \frac{2}{C_\beta}\log(n)$ and $\{x_{ij}^*\}_{i=1,j=1}^{n,m_i}$ defined as in Appendix J and consider the event $\Omega^c = \bigcap\limits_{1\leq i \leq n} \{x_{ij} = x_{ij}^* \,\forall 1 \,\leq\, j \,\leq\, m_i\}$. It is satisfied that $|\mathcal{J}_{e,l}|, |\mathcal{J}_{o,l}| \,\geq\, n/L$, from where, there exist positive constants c_1 and c_2 such that

$$c_1 n/L \,\leq\, |\mathcal{J}_{e,l}|, |\mathcal{J}_{o,l}| \,\leq\, c_2 n/L. \tag{140}$$

Moreover, by Lemma 69 and Assumption 1, we have that $\mathbb{P}(\Omega^c) \,\leq\, \frac{1}{n}$. Then,

$$\mathbb{P}\left(\left\{R_{k,\min} \,\leq\, a\left(\frac{K}{(nm)}\right)^{1/d}\right\}\right) \,\leq\, \mathbb{P}\left(\left\{R_{k,\min} \,\leq\, a\left(\frac{K}{(nm)}\right)^{1/d}\right\}\cap\Omega\right)+\frac{1}{n}. \tag{141}$$

To analyze the term $\mathbb{P}\left(\left\{R_{k,\min} \,\leq\, a\left(\frac{K}{(nm)}\right)^{1/d}\right\}\cap\Omega\right)$, we observe that

$$\mathbb{P}\left(\left\{R_{k,\min} \,\leq\, a\left(\frac{K}{(nm)}\right)^{1/d}\right\}\cap\Omega\right) \,\leq\, \mathbb{P}\left(\left\{(\exists i,j) \in [n]\times[m_i] : R_k(x_{ij}) \,\leq\, a(K/mn)^{1/d}\right\}\cap\tilde{\Omega}\right) \tag{142}$$

We also have that $R_k(x) \,\leq\, r$ if and only if there are at least K observations among $\{x_{ij}\}$ in $B_r(x)$. Therefore, letting

$$r = a(K/(nm))^{1/d}, \tag{143}$$

we have that

$$
\begin{aligned}
&\mathbb{P}\left(\{(i,j) \in [n] \times [m_i] : R_k(x_{ij}) \,\leq\, a(K/(nm))^{1/d}\} \cap \tilde{\Omega}\right) \\
&\leq \sum_{i=1}^{n}\sum_{j=1}^{m_i} \mathbb{P}\left(\left\{\sum_{t=1}^{n}\sum_{s=1}^{m_t} 1_{x_{t,s}\in B_r(x_{ij})} \,\geq\, K\right\} \cap \tilde{\Omega}\right) \\
&\leq Cnm\mathbb{P}\left(\sum_{t=1}^{n}\sum_{s=1}^{m_t} 1_{x^*_{t,s}\in B_r(x_{ij})} \,\geq\, K\right), \qquad (144)
\end{aligned}
$$

where the first inequality is followed by a union bound argument together with the last observation. The second inequality is followed from Assumption 1.9 coupled with the definition of the event Ω at the beginning of the proof. Then, using a union bound argument

$$
\begin{aligned}
&\mathbb{P}\left(\sum_{i=1}^{n}\sum_{s=1}^{m_t} 1_{x^*_{t,s}\in B_r(x_{ij})} \,\geq\, K\right) \qquad (145) \\
&\leq \sum_{l=1}^{L}\mathbb{P}\left(\sum_{t\in\mathcal{J}_{e,l}}\sum_{s=1}^{m_t} 1_{x^*_{t,s}\in B_r(x_{ij})} > K/(2L)\right) + \sum_{l=1}^{L}\mathbb{P}\left(\sum_{t\in\mathcal{J}_{o,l}}\sum_{s=1}^{m_t} 1_{x^*_{t,s}\in B_r(x_{ij})} > K/(2L)\right) \\
&= I_1 + I_2. \qquad (146)
\end{aligned}
$$

We now bound each of the terms I_1 and I_2. The analysis employed for the second term follows the same line of arguments as for I_1 and is omitted. Using Lemma 69 for any $1 \,\leq\, l \,\leq\, L$ we have that

$$
\sum_{t\in\mathcal{J}_{e,l}}\sum_{s=1}^{m_t} 1_{x^*_{t,s}\in B_r(x_{ij})} \sim \text{Binomial}\left(\sum_{t\in\mathcal{J}_{e,l}} m_t, \mathbb{P}(x^*_{1,1} \in B_r(x_{ij}))\right). \qquad (147)
$$

Let Bin $\sim$ Binomial $\left(\sum_{t\in\mathcal{J}_{e,l}} m_t, \mu_{\max}\right)$. By Inequality (139),

$$
\mathbb{P}\left(\sum_{t\in\mathcal{J}_{e,l}}\sum_{s=1}^{m_t} 1_{x_{t,s}\in B_r(x_{ij})} \,\geq\, K/(2L)\right) \,\leq\, \mathbb{P}(\text{Bin} \,\geq\, K/(2L)). \qquad (148)
$$

By Inequality (140), Equation (143), Assumption 1.9 and the definition of $\mu_{\max}$ in Inequality (139), observe that

$$
2\mu_{\max}\sum_{t\in\mathcal{J}_{e,l}} m_t \,\leq\, \frac{2c_{2,d}v_{\max}KCc_2nm}{4Cc_{2,d}c_2v_{\max}nmL} = \frac{K}{2L}. \qquad (149)
$$

From Inequality (149) for any $1 \,\leq\, L \,\leq\, L$,

$$
\mathbb{P}(\text{Bin} \,\geq\, K/(2L)) \,\leq\, \mathbb{P}(\text{Bin} \,\geq\, 2\mu_{\max}\sum_{t\in\mathcal{J}_{e,l}} m_t). \qquad (150)
$$

Therefore, by 45, Inequality (148) and Inequality (150)

$$
\begin{aligned}
\mathbb{P}\Bigg(\sum_{t\in\mathcal{J}_{e,l}}\sum_{s=1}^{m_t} 1_{x^*_{t,s}\in B_r(x_{ij})} \,\geq\, K/(2L)\Bigg) &\leq \exp\left(-\frac{1}{3}\mu_{\max}\sum_{t\in\mathcal{J}_{e,l}} m_t\right) \\
&\leq \exp\left(-\frac{c_1c}{12c_2C}\frac{K}{L}\right) \,=\, \exp\left(-\frac{c_1cC_\beta}{24c_2C}\frac{K}{\log(n)}\right). \qquad (151)
\end{aligned}
$$

where the second inequality is followed by Inequality (140), the choice of r in Equation (143), Assumption 1.9, and the choice of $\mu_{\max}$ in Inequality (139). The Equality is attained by the choice of L at the beginning of the proof. Therefore, using Inequality (148), (150), (151) and the choice of L at the beginning of the proof, the term I_1 is upper bounded as follows

$$\begin{aligned} I_1 \,\le\, & \frac{2}{C_\beta}\log(n)\exp\left(-\frac{c_1 c}{24 c_2 C}\frac{K}{L}\right) \\ =\, & \frac{2}{C_\beta}\log(n)\exp\left(-\frac{c_1 c C_\beta}{24 c_2 C_2}\frac{K}{\log(n)}\right). \end{aligned} \tag{152}$$

Similarly we obtain that

$$I_2 \,\le\, \frac{2}{C_\beta}\log(n)\exp\left(-\frac{C_\beta c_1 c}{24 C_2}\frac{K}{\log(n)}\right). \tag{153}$$

In consequence, by Inequality (144), (146), (152) and (153) we obtain that

$$\mathbb{P}\left(\{(i,j)\in[n]\times[m_i] : R_k(x_{ij}) \,\le\, a(K/(nm))^{1/d}\}\cap\Omega\right) \,\le\, \frac{4}{C_\beta}Cnm\log(n)\exp\left(-\frac{c_1 c C_\beta}{24 c_2 C}\frac{K}{\log(n)}\right). \tag{154}$$

From Equation (141), (142) and (154) it follows that

$$\mathbb{P}\left(\{R_{k,\min} \,\le\, a(K/(nm))^{1/d}\}\right) \,\le\, \frac{4}{C_\beta}\log(n)\exp\left(-\frac{c_1 c C_\beta}{24 c_2 C}\frac{K}{\log(n)}\right) + \frac{1}{n}. \tag{155}$$

Now we analyze $P(R_{k,\max} \,\ge\, \tilde{a}(K/nm)^{1/d})$. To this end, we first observe that,

$$\mathbb{P}(x_{ij}\in B_r(x)) \,=\, \int_{B_r(x)} v(t)dt \,\ge\, c_{1,d}r^d v_{\min} \,:=\, \mu_{\min} > 0, \tag{156}$$

for a positive constant $c_{1,d}$ that depends on d. Following a similar argument as above, we arrive at

$$\mathbb{P}(R_{k,\max} \,\ge\, \tilde{a}(K/nm)^{1/d}) \,\le\, \frac{4}{C_\beta}\log(n)nm\exp\left(-\frac{C_\beta}{24}\frac{K}{\log(n)}\right) + \frac{1}{n}, \tag{157}$$

concluding the proof.

We provide the following lemma to characterize the minimum and maximum number of observations $\{x_{ij}\}$ that fall within each cell $\mathcal{I}_r$, for $r = 1,\ldots,N^d$. Remembering that $I_r = \{(i,j)\in[n]\times[m_i] : P_{I_r}(x_{ij}) = c_r\}$ for $r = 1,\ldots,N^d$, where $P_{I_r}(x)$ denotes the point in $P_{\text{lat}}(N)$ such that $x\in C(P_{I_r}(x))$. ■

Lemma 47 *(see Lemma 7 in Madrid Padilla et al. (2020)). Assume that N in the construction of $P_{lat}(N)$ of (127) is chosen as*

$$N = \left\lceil \frac{3\sqrt{d(nm)^{1/d}}}{aK^{1/d}} \right\rceil, \qquad \textit{where} \qquad a = \frac{1}{(4Cc_{2,d}c_2 v_{\max})^{1/d}}, \tag{158}$$

for a positive constants c_2, and $v_{\max}$, and C are from Assumptions 1.9 and 1.2. Moreover, $c_{2,d}$ is a positive constant that depends on d. Then there exist positive constants $\tilde{b}_1$ and $\tilde{b}_2$ depending on d, $v_{\min}$, and $v_{\max}$ with $v_{\min}$ from Assumption 1, such that for C_β from Assumption 1,

$$\mathbb{P}\left(\max_{r=1,\ldots,N^d} |I_r| \geq (1+\delta)\tilde{b}_1 K \right) \leq \frac{4}{C_\beta}\left(\log(n) n \exp\left(-\frac{C_\beta}{12}\delta^2 \tilde{b}_2 \frac{\tilde{b}_2 K}{\log(n)} \right) + \frac{1}{n} \right), \tag{159}$$

$$\mathbb{P}\left(\min_{r=1,\ldots,N^d} |I_r| \leq (1-\delta)\tilde{b}_1 K \right) \leq \frac{4}{C_\beta}\left(\log(n) n \exp\left(-\frac{C_\beta}{12}\delta^2 \tilde{b}_2 \frac{\tilde{b}_2 K}{\log(n)} \right) + \frac{1}{n} \right), \tag{160}$$

for all $\delta \in (0,1)$. Moreover, the symmetric K-NN graph has maximum degree $d_{\max}$ satisfying

$$\begin{aligned} &\mathbb{P}(d_{\max} \geq 3Cc_2 c_{2,d} v_{max} \tilde{\alpha}^d K) \\ &\leq C_{nm}\left(\frac{4}{C_\beta}\log(n)\exp\left(-\frac{C_\beta}{24}\frac{K}{\log(n)} \right) + \frac{4}{C_\beta} C \log(n) nm \exp\left(-\frac{C_\beta}{24}\frac{K}{\log(n)} \right) + \frac{1}{n} \right), \end{aligned} \tag{161}$$

where $\tilde{a} = \frac{1}{(cc_1 c_{1,d} v_{\min})^{1/d}}$, with c_1 is a positive constant. Moreover, $c_{1,d}$ is a positive constants that depends on d. Let $C(\cdot)$ be defined in (128). Define the event Ω as: If $x_{ij} \in C(x'_{i,j})$ and $x_{s,t} \in C(x'_{s,t})$ for $x'_{i,j}, x'_{s,t} \in P_{lat}(N)$ with $\|x'_{i,j} - x'_{s,t}\|_2 \leq N^{-1}$, then x_{ij} and $x_{s,t}$ are connected in the K-NN graph. Then

$$\mathbb{P}(\Omega) \geq 1 - \frac{4}{C_\beta} C \log(n) n \exp\left(-\frac{\tilde{C}K}{24\log(n)} \right) - \frac{1}{n}. \tag{162}$$

where $\tilde{C} = \frac{C_\beta c_1 c}{c_2 C}$.

Proof By observation, the following holds,

$$\begin{aligned} \|x_{ij} - x_{s,t}\|_2 &\leq \sqrt{d}\|x_{ij} - x_{s,t}\|_\infty \\ &\leq \sqrt{d}\left\{ \|x_{ij} - x'_{i,j}\|_\infty + \|x'_{i,j} - x'_{s,t}\|_\infty + \|x_{s,t} - x'_{s,t}\|_\infty \right\} \\ &< 3\sqrt{d} N^{-1} \leq a\left(\frac{K}{nm} \right)^{1/d}, \end{aligned} \tag{163}$$

with $a = 1/(2Cc_2 v_{\max})^{1/d}$ and $v_{\max}$ and C are from Assumptions 1.9 and 1.2. The first inequality is followed by a basic inequality between the l_2 and l_∞ norms. The second is due to the triangle inequality, while the third inequality comes after using that $\|x'_{i,j} - x'_{s,t}\|_\infty \leq N^{-1}$ and $x'_{i,j}, x'_{s,t} \in P_{\text{lat}}(N)$.

Proof. Through this proof, we make use of the event $\tilde{\Omega}$ defined as $\tilde{\Omega} = \bigcap_{1 \leq i \leq n} \{x_{ij} = x^*_{ij}\}$, where x_{ij} are defined in Appendix H. We use

$$L = \frac{2}{C_\beta}\log(n), \tag{164}$$

and for $l \in [L]$

$$\frac{c_1 n}{L} \leq |\mathcal{J}_{e,l}|, |\mathcal{J}_{o,l}| \leq \frac{c_2 n}{L}, \tag{165}$$

with c_1 and c_2 positive constants. First, we analyze the event Ω. To this end, let $x_{ij} \in C(x'_{i,j})$ and $x_{s,t} \in C(x'_{s,t})$ for $x^*_{ij}, x'_{s,t} \in P_{\text{lat}}(N)$ with

$$\|x^*_{ij} - x'_{s,t}\|_2 \leq N^{-1}. \tag{166}$$

Then the following holds,

$$\begin{aligned}
\|x_{ij} - x_{s,t}\|_2 &\leq \sqrt{d}\|x_{ij} - x_{s,t}\|_\infty \\
&\leq \sqrt{d}(\|x_{ij} - x^*_{ij}\|_\infty + \|x^*_{ij} - x'_{s,t}\|_\infty + \|x'_{s,t} - x_{s,t}\|_\infty) \\
&< 3\sqrt{d}N^{-1} \\
&\leq a\left(\frac{K}{nm}\right)^{1/d},
\end{aligned}$$

where $a = 1/(2Cc_2 v_{\max})^{1/d}$ and $v_{\max}$ and C are from Assumption 1. The first inequality is followed by a basic inequality between the ℓ_2 and ℓ_∞ norms. The second is due to the triangle inequality, while the third inequality comes after using that $\|x^*_{ij} - x'_{s,t}\|_2 \leq N^{-1}$ and $x^*_{ij}, x'_{s,t} \in P_{\text{lat}}(N)$. Using the definition of N in the statement of the Lemma provides the last inequality. From Inequality (166) we observe that $\left\{\tilde{a}\left(\frac{K}{nm}\right)^{1/d} \leq R_{k,\min}\right\} \subseteq \Omega$. Therefore, using the result from the proof of Lemma 46,

$$\mathbb{P}(\Omega) \geq \mathbb{P}\left(a\,(K/nm)^{1/d} \leq R_{k,\min}\right) \geq 1 - \frac{4}{C_\beta} C\log(n) n \exp\left(-\frac{\tilde{C}}{24}\frac{K}{\log(n)}\right) - \frac{1}{n}, \tag{167}$$

Next, we proceed to derive an upper bound on the counts $\{I_l\}_{l=1}^{N^d}$. Assume that $x \in P_{\text{lat}}(N)$, and $x' \in C(x)$. Since $x' \in C(x)$, by the definition of $P_{\text{lat}}(N)$ we have that $\|x' - x\|_\infty \leq \frac{1}{2N}$. Therefore, using the basic inequality between the l_2 and l_∞ norms,

Assume that $x \in P_{\text{lat}}(N)$, and $x' \in C(x)$. Since $x' \in C(x)$, by the definition of $P_{\text{lat}}(N)$ we have that $\|x - x'\|_\infty \leq \frac{1}{2N}$. Therefore, using the basic inequality between the l_2 and l_∞ norms,

$$\|x - x'\|_2 \leq \sqrt{d}\|x - x'\|_\infty \leq \frac{\sqrt{d}}{2N} \leq \frac{a}{6}\left(\frac{K}{nm}\right)^{1/d} = \frac{\tilde{b}_1^{1/d}}{(2c_{2,d}v_{\max}2Cc_2)^{1/d}}\left(\frac{K}{nm}\right)^{1/d}, \tag{168}$$

where the third inequality is obtained from the choice of N in the statement of the Lemma, and b_1 is an appropriate constant. Here $c_{2,d} > 0$ is a constant that depends on d. Therefore, from Inequality (168).

$$C(x) \subseteq B_{\frac{(\tilde{b}_1)^{1/d}}{(2c_{2,d}Cv_{\max}c_2)^{1/d}}\left(\frac{K}{nm}\right)^{1/d}}(x). \tag{169}$$

On the other hand, let $\tilde{b}_2 > 0$ a constant such that $\left(\frac{(\tilde{b}_2)^{1/d}}{(2c_{1,d}cc_1v_{\min})^{1/d}}\right) \leq \frac{a}{2}\frac{1}{3\sqrt{d}}$, with $c_{1,d} > 0$ a constant that depends on d. Observe that if the following holds

$$\|x - x'\|_2 \leq \left(\frac{\tilde{b}_2}{(2cc_1v_{\min})}\right)^{1/d}\left(\frac{K}{nm}\right)^{1/d}, \tag{170}$$

we have that,

$$\|x - x'\|_\infty \leq \|x - x'\|_2 \leq \left(\frac{\tilde{b}_2}{(2cc_{1,d}c_1 v_{\min})}\right)^{1/d}\left(\frac{K}{nm}\right)^{1/d}$$
$$\leq \frac{2}{3}\sqrt{d}\left(\frac{K}{nm}\right)^{1/d} \leq \frac{1}{2N}. \tag{171}$$

The third inequality is followed by the choice of $\tilde{b}_2$ and the Equality due to the choice of N in the statement of the Lemma. Therefore,

$$B_{\frac{(\tilde{b}_1)^{1/d}}{(2c_{2,d}Cv_{\max}c_2)^{1/d}}\left(\frac{K}{nm}\right)^{1/d}}(x) \subseteq C(x). \tag{172}$$

We are now ready to obtain the bound for $\mathbb{P}\left\{\max_{r=1,\ldots,N^d}|I_r| \geq (1+\delta)\tilde{b}_1 K\right\}$. Notice that

$$\mathbb{P}\left\{\max_{r=1,\ldots,N^d}|I_r| \geq (1+\delta)\tilde{b}_1 K\right\} \leq \mathbb{P}\left(\left\{\max_{r=1,\ldots,N^d}|I_r| \geq (1+\delta)\tilde{b}_1 K\right\}\cap\tilde{\Omega}\right) + \frac{1}{n}, \tag{173}$$

with $\tilde{\Omega}$ defined at the beginning of the proof. Then, by a union bound argument

$$\mathbb{P}\left\{\max_{r=1,\ldots,N^d}|I_r| \geq (1+\delta)\tilde{b}_1 K\right\} \tag{174}$$
$$\leq \sum_{r=1}^{N^d}\mathbb{P}\left\{|I_1| \geq (1+\delta)\tilde{b}_1 K\right\} \tag{175}$$
$$\leq \sum_{x\in P_{\mathrm{lat}}(N)}\mathbb{P}\left[\left\{|C(x)| \geq (1+\delta)\tilde{b}_1 K\right\}\cap\tilde{\Omega}\right], \tag{176}$$

where $C(x) = \{(i,j)\in[n]\times[m_i] : P_I(x_{ij}) = x\}$. Moreover, for any $x \in P_{\mathrm{lat}}(N)$ using Equation (169) we have that

$$\mathbb{P}(x_{1,1}\in C(x)) \leq \mathbb{P}\left(x_{1,1}\in B_{\frac{(\tilde{b}_1)^{1/d}}{(2c_{2,d}Cv_{\max}c_2)^{1/d}}\left(\frac{K}{nm}\right)^{1/d}}(x)\right)$$
$$\leq \frac{\left(\tilde{b}_1\right)}{(2c_{2,d}v_{\max}Cc_2)}\left(\frac{K}{nm}\right)c_{2,d}v_{\max}, \tag{177}$$

where the last inequality follows the same line of arguments as in Inequality (139) in the proof of Lemma 46. Therefore,

$$2Cc_2nm\mathbb{P}(x_{1,1}\in C(x)) \leq 2Cc_2nm\frac{\left(\tilde{b}_1\right)}{(2c_{2,d}v_{\max}Cc_2)}\left(\frac{K}{nm}\right)c_{2,d}v_{\max} = \tilde{b_1}K$$

The above Inequality and Inequality (176) lead to

$$\sum_{x\in P_{\mathrm{lat}}(N)}\mathbb{P}\left[\left\{|C(x)| \geq (1+\delta)\tilde{b}_1 K\right\}\cap\tilde{\Omega}\right] \leq \sum_{x\in P_{\mathrm{lat}}(N)}\mathbb{P}\left[\left\{|C(x)| \geq (1+\delta)2Cc_2nm\mathbb{P}(x^*_{1,1}\in C(x))\right\}\right]. \tag{178}$$

■

Then, using the definition of the event Ω at the beginning of the proof and a union bound argument,

$$\begin{aligned}
&\mathbb{P}\left[\left\{|C(x)| \,\geq\, (1+\delta)2Cc_2nm\mathbb{P}(x^*_{1,1}\in C(x))\right\}\cap\tilde{\Omega}\right]\\
&\leq \sum_{l=1}^{L}\mathbb{P}\left(\sum_{i\in\mathcal{J}_{e,l}}\sum_{j=1}^{m_i}1_{x_{t,s}\in C(x)} \,\geq\, (1+\delta)2Cc_2nm\mathbb{P}(x^*_{1,1}\in C(x))/(2L)\right)\\
&\quad+\sum_{l=1}^{L}\mathbb{P}\left(\sum_{i\in\mathcal{J}_{e,l}}\sum_{j=1}^{m_i}1_{x_{t,s}\in C(x)} \,\geq\, (1+\delta)2Cc_2nm\mathbb{P}(x^*_{1,1}\in C(x))/(2L)\right).
\end{aligned}\tag{179}$$

Now we analyze each of the terms

$$\mathbb{P}\Big(\sum_{i\in\mathcal{J}_{e,l}}\sum_{j=1}^{m_i}1_{x^*_{t,s}\in C(x)} \,\geq\, (1+\delta)2Cc_2nm\mathbb{P}(x^*_{1,1}\in C(x))/L\Big).$$

By Inequality (165) and Assumption 1.9

$$\mathbb{P}\left(\sum_{i\in\mathcal{J}_{e,l}}\sum_{j=1}^{m_i}1_{x^*_{t,s}\in C(x)} \,\geq\, (1+\delta)2Cc_2nm\mathbb{P}(x^*_{1,1}\in C(x))/L\right)\tag{180}$$

$$\leq \mathbb{P}\left(\sum_{i\in\mathcal{J}_{e,l}}\sum_{j=1}^{m_i}1_{x^*_{t,s}\in C(x)} \,\geq\, (1+\delta)\left(\sum_{i\in\mathcal{J}_{e,l}}m_i\right)\mathbb{P}(x^*_{1,1}\in C(x))\right).\tag{181}$$

By Lemma 45 it follows that

$$\begin{aligned}
&\mathbb{P}\Big(\sum_{i\in\mathcal{J}_{e,l}}\sum_{j=1}^{m_i}1_{x^*_{t,s}\in C(x)} \,\geq\, (1+\delta)\Big(\sum_{i\in\mathcal{J}_{e,l}}m_i\Big)\mathbb{P}(x^*_{1,1}\in C(x))\Big)\\
&\leq \exp\Big(-\frac{1}{3}\delta^2\Big(\sum_{i\in\mathcal{J}_{e,l}}m_i\Big)\mathbb{P}(x^*_{1,1}\in C(x))\Big).
\end{aligned}\tag{182}$$

Now Equation (172) implies that

$$\mathbb{P}(x^*_{1,1}\in C(x)) \,\geq\, \mathbb{P}\left(x^*_{1,1}\in B_{\frac{(\tilde{b}_1)^{1/d}}{(2c_{2,d}Cv_{\max}c_2)^{1/d}}\left(\frac{K}{nm}\right)^{1/d}}(x)\right) \,\geq\, c_{1,d}v_{\min}\left(\frac{\tilde{b}_2}{(2cc_{1,d}c_1v_{\min})}\right)\left(\frac{K}{nm}\right),\tag{183}$$

from where, using Inequality (165) and Assumption 1.9 we get that

$$\left(\sum_{i\in\mathcal{J}_{e,l}}m_i\right)\left(\mathbb{P}(x^*_{1,1}\in C(x))\right) \,\geq\, cc_1\frac{n}{L}mc_{1,d}v_{\min}\left(\frac{\tilde{b}_2}{(2cc_{1,d}c_1v_{\min})}\right)\left(\frac{K}{nm}\right).\tag{184}$$

Thus, by our choice of L in Equation (164) and Inequality (181)

$$\mathbb{P}\left(\sum_{i\in\mathcal{J}_{e,l}}\sum_{j=1}^{m_i}1_{x^*_{t,s}\in C(x)} \,\geq\, (1+\delta)2Cc_2nm\mathbb{P}(x^*_{1,1}\in C(x))/L\right)\tag{185}$$

$$\leq \exp\left(-\frac{1}{3}\delta^2cc_1\frac{n}{L}mc_{1,d}v_{\min}\frac{\tilde{b}_2}{(2cc_{1,d}c_1v_{\min})}\frac{K}{nm}\right) \,=\, \exp\left(-\frac{C_\beta}{12}\delta^2\tilde{b}_2\frac{K}{\log(n)}\right).\tag{186}$$

A similarly analysis is done for

$$\mathbb{P}\left(\sum_{i\in\mathcal{J}_{e,l}}\sum_{j=1}^{m_i} 1_{x^*_{t,s}\in C(x)} \,\geq\, (1+\delta)2Cc_2nm\mathbb{P}(x^*_{1,1}\in C(x))/(2L)\right). \tag{187}$$

Thus, by Equation (173), (176), (178), (179), and (183), we obtain that

$$\mathbb{P}\left(\max_{r=1,\ldots,N^d}|I_r| \,\geq\, (1+\delta)\tilde{b}_1K\right) \,\leq\, \frac{4}{C_\beta}\log(n)N^d\exp\left(-\frac{C_\beta}{12}\delta^2\tilde{b}_2\frac{K}{\log(n)}\right)+\frac{1}{n}. \tag{188}$$

On the other hand, with a similar argument, we have that

$$\begin{aligned}
\mathbb{P}\left(\min_{r=1,\ldots,N^d}|I_r| \,\leq\, (1-\delta)\tilde{b}_2K\right) \,&\leq\, \sum_{x\in P_{\mathrm{lat}}(N)}\mathbb{P}\left(\left\{|C(x)| \,\leq\, (1-\delta)\tilde{b}_2K\right\}\cap\tilde{\Omega}\right)\\
&\leq\, \sum_{x\in P_{\mathrm{lat}}(N)}\mathbb{P}\left(\left\{|C(x)| \,\leq\, (1-\delta)2cc_1nm\mathbb{P}(x^*_{1,1}\in C(x))\right\}\cap\tilde{\Omega}\right)+\frac{1}{N}\\
&\leq\, \frac{4}{C_\beta}\log(n)N^d\exp\left(-\frac{C_\beta}{12}\delta^2\tilde{b}_2\frac{K}{\log(n)}\right)+\frac{1}{n}.
\end{aligned} \tag{189}$$

$\blacksquare$

Next we proceed to find an upper bound on the maximum degree of the K-NN graph. To this end, we proceed as in the proof of Lemma 46. First, define the sets

$$\begin{aligned}
B_{i,j}(x) \,&=\, \left\{(i',j')\in[n]\times[m_i] \mid (i',j')\neq(i,j); x_{i',j'}\in B_{\tilde{a}(K/nm)^{1/d}}(x)\right\}\\
B_{i,j} \,&=\, \left\{(i',j')\in[n]\times[m_i] \mid (i',j')\neq(i,j); x_{i',j'}\in B_{\tilde{a}(K/nm)^{1/d}}(x_{ij})\right\},
\end{aligned} \tag{190}$$

for $(i,j)\in[n]\times[m_i]$, and $x\in[0,1]^d$. Observe that

$$\begin{aligned}
&\mathbb{P}(|B_{i,j}(a)|>3c_2Cc_{2,d}v_{\max}\tilde{a}^dK)\\
&\leq \mathbb{P}\Bigg(\sum_{(i',j')\in[n]\times[m_i]\backslash\{(i,j)\}} 1_{x_{i',j'}\in B_{\tilde{a}(K/nm)^{1/d}}(x)} \,\geq\, \frac{3}{2}c_2Cc_{2,d}v_{\max}\tilde{a}^dK\Bigg).
\end{aligned} \tag{191}$$

To bound the term

$$\mathbb{P}\left(\sum_{(i',j')\in[n]\times[m_i]\backslash\{(i,j)\}} 1_{x_{i',j'}\in B_{\tilde{a}(K/nm)^{1/d}}(x)} > 3c_2Cc_{2,d}v_{\max}\tilde{a}^dK\right), \tag{192}$$

we use a union bound arguments to get

$$\mathbb{P}\left(\sum_{(i',j')\in[n]\times[m_i]\backslash\{(i,j)\}} 1_{x_{i',j'}\in B_{\tilde{a}(K/nm)^{1/d}}(x)} > 3c_2Cc_{2,d}v_{\max}\tilde{a}^dK\right) \tag{193}$$

$$\leq \sum_{l=1}^{L}\mathbb{P}\left(\sum_{(i',j')\in\mathcal{J}_{e,l}\times[m_i]\backslash\{(i,j)\}} 1_{x_{i',j'}\in B_{\tilde{a}(K/nm)^{1/d}}(x)} > 3c_2Cc_{2,d}v_{\max}\tilde{a}^dK/(2L)\right) \tag{194}$$

$$+\sum_{l=1}^{L}\mathbb{P}\left(\sum_{(i',j')\in\mathcal{J}_{o,l}\times[m_i]\backslash\{(i,j)\}} 1_{x_{i',j'}\in B_{\tilde{a}(K/nm)^{1/d}}(x)} > 3c_2Cc_{2,d}v_{\max}\tilde{a}^dK/(2L)\right). \tag{195}$$

Further, using Lemma 69 for any $l \in [L]$ we have that

$$\sum_{(i',j') \in [n]\times[m_{i'}]\setminus\{(i,j)\}} 1_{x_{i',j'} \in B_{\tilde{a}(K/nm)^{1/d}}(x)} \sim \text{Binomial}\left(\sum_{i' \in J_l} m_{i'} - 1, \mathbb{P}(x_{i',1} \in B_{\tilde{a}(K/nm)^{1/d}}(x))\right). \tag{196}$$

By Equation (164), Inequality (165), Assumption 1.9 and the Inequality (139) in the proof of Lemma 46, we have that

$$\frac{3}{2}\mathbb{P}\left(x^*_{1,1} \in B_{\tilde{a}(K/nm)^{1/d}}(x)\right)\left(\sum_{i' \in J_l} m_{i'} - 1\right) \leq \frac{3}{2}c_2 \mu_{\max} a^d \frac{K}{nm} C \frac{2nm}{L}. \tag{197}$$

Thus, for any $1 \leq l \leq L$,

$$\begin{aligned}
&\mathbb{P}\left(\sum_{(i',j') \in \mathcal{J}_{e,l}\times[m_{i'}]\setminus\{(i,j)\}} 1_{x_{i',j'} \in B_{\tilde{a}(K/nm)^{1/d}}(x)} \geq 3c_2 C c_{2,d} v_{\max} \tilde{a}^d K/(2L)\right) \\
&\leq \mathbb{P}\Bigg(\sum_{(i',j') \in \mathcal{J}_{e,l}\times[m_{i'}]\setminus\{(i,j)\}} 1_{x_{i',j'} \in B_{\tilde{a}(K/nm)^{1/d}}(x)} \\
&\qquad \geq 3\left(\frac{\tilde{b}_2}{(2cc_{1,d}c_1 v_{\min})}\right)^{1/d}\left(\frac{K}{nm}\right)^{1/d}\Bigg) \times \left(\sum_{i' \in \mathcal{J}_{e,l}} m_{i'} - 1\right)
\end{aligned} \tag{198}$$

Now, from Equation (164), Inequality (165), and Assumption 1.9 we have that

$$\frac{cc_1}{2}c_{1,d} v_{\min} \tilde{a}^d \frac{K}{L} \leq c_{1,d} v_{\min} \tilde{a}^d \frac{K}{nm} - \frac{c_1}{L} \leq \mathbb{P}\left(\{x^*_{1,1} \in B_{\tilde{a}(K/nm)^{1/d}}(x)\}\right)\left(\sum_{i' \in \mathcal{J}_{e,l}} m_{i'} - 1\right) \tag{199}$$

Thus, using 45 and the definition of $\tilde{a}$ in the statement of the Lemma,

$$\begin{aligned}
&\mathbb{P}\left(\sum_{(i',j') \in \mathcal{J}_{e,l}\times[m_{i'}]\setminus\{(i,j)\}} 1_{x_{i',j'} \in B_{\tilde{a}(K/nm)^{1/d}}(x)} \geq 3\left(\frac{\tilde{b}_2}{(2cc_{1,d}c_1 v_{\min})}\right)^{1/d}\left(\frac{K}{nm}\right)^{1/d}\right) \\
&\leq \exp\left(-\frac{1}{12}\left(\frac{\tilde{b}_2}{(2cc_{1,d}c_1 v_{\min})}\right)^{1/d}\left(\frac{K}{nm}\right)^{1/d}\left(\sum_{i' \in \mathcal{J}_{e,l}} m_{i'} - 1\right)\right) \\
&\leq \exp\left(-\frac{C_\beta}{24}\frac{K}{\log(n)}\right)
\end{aligned} \tag{200}$$

Analogously, the exact same bound can be achieved when the case of $\mathcal{J}_{o,l}$. Hence, using the above inequality together with Equation (164), Inequality (191) and Inequality (198)

$$\mathbb{P}\left(|B_{i,j}(a)| > 3c_2 C c_{2,d} v_{\max} \tilde{a}^d K\right) \leq \frac{4}{C_\beta}\log(n)\exp\left(-\frac{C_\beta}{24}\frac{K}{\log(n)}\right) \tag{201}$$

Therefore,

$$\begin{aligned}
\mathbb{P}\left(|B_{i,j}| \geq 3c_2 C c_{2,d} v_{\max} \tilde{a}^d K\right) &= \int_{[0,1]^d} \mathbb{P}\left(|B_{i,j}(x)| \geq 3c_2 C c_{2,d} v_{\max} \tilde{a}^d K\right) p(x)\, \mu(dx) \\
&\leq \frac{4}{C_\beta}\log(n) N^d \exp\left(-\frac{C_\beta K}{24\log(n)}\right).
\end{aligned} \tag{202}$$

Finally let $d_{i,j}$ be the degree associated with x_{ij}. Using a union bound argument

$$\begin{aligned}\mathbb{P}\left(d_{\max} \geq 3c_2Cc_{2,d}v_{\max}\tilde{a}^dK\right) &\leq \mathbb{P}\left(\{d_{\max} \geq 3c_2Cc_{2,d}v_{\max}\tilde{a}^dK\} \cap \tilde{\Omega}\right) + \frac{1}{n} \\ &\leq Cn\mathbb{P}\left(d_{i,j} \geq 3c_2Cc_{2,d}v_{\max}\tilde{a}^dK\right) \cap \tilde{\Omega} + \frac{1}{n}. \end{aligned} \tag{203}$$

Moreover,

$$\begin{aligned}&\mathbb{P}\big(\left\{d_{i,j} \leq 3c_2Cc_{2,d}v_{\max}\tilde{a}^dK\right\} \cap \tilde{\Omega}\big) \\ &\geq \mathbb{P}\left(\left\{(i',j') \in [n]\times[m_{i'}], (i',j') \neq (i,j); \|x_{i,j} - x_{i',j'}\|_2 \leq R_{k,\max}\right\} \leq 3v_{\max}Cc_2c_{2,d}\tilde{a}^dK\right) \cap \tilde{\Omega} \\ &\geq \mathbb{P}\left(\left\{(i',j') \in [n]\times[m_{i'}], (i',j') \neq (i,j); \|x_{i,j} - x_{i',j'}\|_2 \leq R_{k,\max}\right\} \leq \tilde{a}\left(\frac{K}{nm}\right)^{1/d}\right) \cap \tilde{\Omega} \\ &\geq 1 - \mathbb{P}\left(R_{k,\max} > \tilde{a}\left(\frac{K}{nm}\right)^{1/d}\right) \cap \tilde{\Omega} - \mathbb{P}\Big(|B_{i,j}| > 3v_{\max}Cc_2c_{2,d}\tilde{a}^dK\Big), \end{aligned} \tag{204}$$

and the claim follows from Inequality (202) and the proof of Lemma 46.

The 47 demonstrated that the count of observations x_{ij} that fall within each cell I_i is proportional to K, with high probability. We leverage this understanding to derive an upper limit for the MSE in the following result. ■

Lemma 48 (see Lemma 8 in Madrid Padilla et al. (2020)) *Consider the notation in D.2.1. Assume that the event Ω in (132) intersected with*

$$\min_{r=1,\ldots,N^d} |I_r| \geq \frac{1}{2}\tilde{b}_2K, \tag{205}$$

holds. Then for all $s \in \mathbb{R}^{\sum_{i=1}^n m_i}$, it holds that

$$|s^\top(\theta - \theta_1)| \leq 2\,\|s\|_\infty\,\|\nabla_{G_k}\theta\|_1, \quad \forall\theta \in \mathbb{R}^{\sum_{i=1}^n m_i}. \tag{206}$$

Moreover,

$$\|D\theta^I\| \leq \|\nabla_{G_k}\theta\|_1, \quad \forall\theta \in \mathbb{R}^{\sum_{i=1}^n m_i}, \tag{207}$$

where D is the incidence matrix of a d-dimensional grid graph $G_{grid} = (V_{grid}, E_{grid})$ with $V_{grid} = [N^d]$, where $(r, r') \in E_{grid}$ if and only if

$$\|P_I(x_{i_rj_r}) - P_I(x_{i_{r'}j_{r'}})\|_2 = \|c_r - c_{r'}\|_2 = \frac{1}{N},$$

where $x_{i_rj_r} \in C(c_r)$ and $x_{i_{r'}j_{r'}} \in C(c_{r'})$.

Proof We start by introducing the notation $x'_{ij} = P_I(x_{ij})$. To prove (206) we proceed in cases. *Case 1.* If $(\theta_I)_{11} = \theta_{11}$. Then clearly $|s_{(11)}|\,|\theta_{11} - (\theta_I)_{11}| = 0$. *Case 2.* If $(\theta_I)_{11} = \theta_{ij}$ for $(i,j) \neq (1,1)$. Then

$$\|x'_{11} - x_{ij}\|_\infty \leq \|x'_{11} - x_{11}\|_\infty \leq \frac{1}{2N}.$$

Thus, $x'_{11} = x'_{ij}$, and so $((1,1),(i,j)) \in E_K$ by the assumption that the event Ω holds. Therefore for every $(i,j) \in [n] \times [m_i]$, there exists $(s_{(ij)}, t_{(ij)}) \in [n] \times [m]$ such that $(\theta_I)_{ij} = \theta_{s_{(ij)},t_{(ij)}}$ and either $(i,j) = (s_{(ij)}, t_{(ij)})$ or $(i,j), (s_{(ij)}, t_{(ij)}) \in E_K$. Hence,

$$|s^\top(\theta - \theta_1)| \leq \sum_{i=1}^{n}\sum_{j=1}^{m_i} |s_{ij}||\theta_{i,j} - \theta_{s_{(ij)}t_{(ij)}}| \leq 2\,\|s\|_\infty\,\|\nabla_{G_k}\theta\|_1.$$

To verify (207), we observe that

$$\|D\theta^I\|_1 = \sum_{(r,r')\in E_{\text{grid}}} |\theta_{i_r,j_r} - \theta_{i_{r'}j_{r'}}|.$$

Now, if $(r,r') \in E_{\text{grid}}$, then x_{i_r,j_r} and $x_{i_{r'}j_{r'}}$ are neighboring cells in I. This implies that $((i_r,j_r),(i_{r'}j_{r'}))$ is an edge in the K-NN graph. Thus, every edge in the grid graph G_{grid} corresponds to an edge in the K-NN graph and the mapping is injective. Note that here we have used the fact that (205) ensures that every cell has at least one point, provided that K is large enough. The claim is then followed. ■

Lemma 49 (see Lemma 69 in Madrid Padilla et al. (2020).) *Consider the notation in D.2.1 and assume that that event Ω in (132) happens. We have that for s, θ_1, $\theta_2 \in \mathbb{R}^{\sum_{i=1}^n m_i}$, it holds that*

$$s^\top(\theta_1 - \theta_2) \leq 2\|s\|_\infty(\|\nabla_{G_k}\theta_1\|_2 + \|\nabla_{G_k}\theta_2\|_2) = s^\top(\theta_{1I} - \theta_{2I}).$$

Proof We observe that

$$s^\top(\theta_1 - \theta_2) = s^\top(\theta_1 - \theta_{1I}) + s^\top(\theta_{1I} - \theta_{2I}) + s^\top(\theta_{2I} - \theta_2),$$

and the claim is followed by Lemma 48. ■

Lemma 50 (see Lemma 10 in Madrid Padilla et al. (2020).) *Assume that the event Ω from Lemma 47 intersected with*

$$\left\{\min_{r=1,\ldots,N^d} |I_r| > \frac{1}{2}\tilde{b}_2 K\right\}, \tag{208}$$

holds. Let $\theta_1, \theta_2 \in \mathbb{R}^{\sum_{i=1}^n m_i}$ and $\{s_{i,j}\}_{i=1,j=1}^{n,m_i} = s \in \mathbb{R}^{\sum_{i=1}^n m_i}$ sub-Gaussian with parameter 1. Then, we have that

$$\|s^\top(\theta_1 - \theta_2)\| \leq \max_{r=1,\ldots,N^d} |I_r| \left(\|\tilde{\Pi}, \tilde{s}\|_2\|\theta_1 - \theta_2\|_2 + \|(D^\dagger)^\top\tilde{s}\|_\infty \times (\|(\nabla_{G_K}\theta_{1I})\|_1 + \|(\nabla_{G_K}\theta_{2I})\|_1)\right), \tag{209}$$

where $\tilde{s} \in \mathbb{R}^{N^d}$ is a mean zero vector whose coordinates are sub-Gaussian with the same constants 1. Here, $\tilde{\Pi}$ is the orthogonal projection onto the span of $1_{N^d} \in \mathbb{R}^{N^d}$, and $D^\dagger$ is the pseudoinverse of the incidence matrix D from Lemma 48.

Proof Using the notation from the proof of Lemma 48, we have that

$$\begin{aligned} s^\top(\theta_{1I} - \theta_{2I}) \;&=\; \sum_{r=1}^{N^d} \sum_{(i',j')\in I_r} s_{[i',j']}(\theta_{1[i_r,j_r]} - \theta_{2[i_r,j_r]}) \\ &=\; \max_{r=1,\ldots,N^d} |I_r|\, \tilde{s}^\top(\theta_1^I - \theta_2^I), \end{aligned} \tag{210}$$

where

$$\tilde{s}_j \;=\; \left(\max_{r=1,\ldots,N^d} |I_r|\right)^{-1} \sum_{(i',j')\in I_r} s_{[i',j']}. \tag{211}$$

Clearly, the $\tilde{s}_1,\ldots,\tilde{s}_{N^d}$ are also sub-Gaussian with the same constants as the original errors $\{s_{i,j}\}_{i=1,j=1}^{n,m_i}$.

Moreover, let $\tilde{\Pi}$ be the orthogonal projection onto the span of $1_{N_d} \in \mathbb{R}^{N_d}$. By Hölder's inequality and by the triangle inequality,

$$\begin{aligned} s^\top(\theta_1 - \theta_2) \;&\leq\; \left\{\max_{r=1,\ldots,N_d} |I_r|\right\} \left\{\|\tilde{\Pi}\tilde{s}\|_2 \|\theta_1^I - \theta_2^I\|_2\right\} + \|(D^+)^\top \tilde{s}\|_\infty \|D(\theta_1 - \theta_2)\|_1 \\ &\leq\; \left\{\max_{r=1,\ldots,N_d} |I_r|\right\} \left\{\|\tilde{\Pi}\tilde{s}\|_2 \|\hat{\theta}^I - \theta^{*,I}\|_2 + \|(D^+)^\top \tilde{s}\|_\infty (\|D\theta_1^I\|_1 + \|D\theta_2^I\|_1)\right\}. \end{aligned} \tag{212}$$

Next we observe that

$$\|\theta_1^I - \theta_2^I\|_2 \;=\; \left(\sum_{r=1}^{N_d} (\theta_{1 i_r, j_r} - \theta_{2 i_r, j_r})^2\right)^{1/2} \;\leq\; \left(\sum_{i=1}^{n} \sum_{j=1}^{m_i} (\theta_{1,i,j} - \theta_{2,i,j})^2\right)^{1/2}. \tag{213}$$

Therefore, combining (213), (212) and Lemma 48 we arrive at

$$s^\top(\theta_{1I} - \theta_{2I}) \;\leq\; \left\{\max_{r=1,\ldots,N_d} |I_r|\right\} \left\{\|\tilde{\Pi}\tilde{s}\|_2 \|\theta_1 - \theta^*\|_2\right\} + \|(D^+)^\top \tilde{s}\|_\infty \left(\|\nabla_{G_k}\theta_1\|_1 + \|\nabla_{G_k}\theta_2\|_1\right).$$

■

Appendix E. For $f^* \notin \mathcal{F}$ And $\mathcal{F}$ is not convex

E.1 Technical Results for proof of Theorem 7

Lemma 51 *Assume Assumption 2 holds, then there exists c_0 such that for all $h = f - f^*$, we have*

$$Z_{ij}(f) \;\leq\; \frac{\bar{f} h^2(x_{ij})}{2}.$$

Proof Supposing that $|h(x_{ij})| = |f(x_{ij}) - f^*(x_{ij})|$, by Assumption 2 and Lemma 20, we have

$$\begin{aligned} Z_{ij}(f) - Z_{ij}(f^*) &= \int_0^{h(x_{ij})} \left\{ F_{y_{ij}|x_{ij}}(f^*(x_{ij}) + z) - F_{y_{ij}|x_{ij}}(f^*(x_{ij})) \right\} dz \\ &\leq \int_0^{h(x_{ij})} \bar{f} z dz \\ &\leq \frac{\bar{f} h^2(x_{ij})}{2}. \end{aligned}$$

Thus, by the fact $Z_{ij}(f^*) = 0$, we have $Z_{ij}(f) \leq \frac{\bar{f}h^2(x_{ij})}{2}$. ■

Corollary 52 *On event Ω, let $\hat{f}$, belong to $\mathcal{F}$, and f^* not necessarily in $\mathcal{F}$, where $\mathcal{F}$ is not necessarily convex, and let $\Delta^2(\hat{f} - f^*)$ be the huber loss. Let f_{Proj} in $\mathcal{F}$ be*

$$f_{Proj} \coloneqq \inf_{f \in \mathcal{F}(L,r)} \|f - f^*\|_\infty, \tag{214}$$

such that

$$\|f_{Proj} - f^*\|_\infty \leq \phi_{nm}.$$

Let

$$\tilde{t}^2 \coloneqq t^2 + 2\left(C_0 + \frac{\bar{f}}{2}\right)\phi_{nm}^2, \tag{215}$$

Then we have

$$\begin{aligned} &\mathbb{P}\left(c_0\Delta^2(\hat{f} - f^*) > \tilde{t}^2\right) \\ &\leq \mathbb{P}\left(\sup_{\substack{f \in Star(\mathcal{F}, f^*) \\ c_0\Delta^2(f - f^*) \leq \tilde{t}^2}} \left(\bar{Z}(f) - \hat{Z}(f) - \ell(f)\right) + \sup_{\substack{f \in Star(\mathcal{F}, f^*) \\ c_0\Delta^2(f - f^*) \leq \tilde{t}^2}} \left(\hat{Z}(f) - \bar{Z}(f) - \ell(f)\right) > \tilde{t}^2 \right). \end{aligned}$$

The c_0 satisfies (84), and $\ell(f) \coloneqq (C_0 - c_0)\Delta^2(f - f^*)$.

Proof Assume

$$c_0\Delta^2(\hat{f} - f^*) > \tilde{t}^2 = t^2 + 2\left(C_0 + \frac{\bar{f}}{2}\right)\phi_{nm}^2 \tag{216}$$

where c_0 satisfies (84), we know Δ^2 is continuous. Moreover let f_{Proj} in $\mathcal{F}$ such that

$$\|f_{\text{Proj}} - f^*\|_\infty \leq \phi_{nm}$$

Consider $g : [0,1] \to \mathbb{R}$ such that

$$g(\mu) = c_0\Delta^2((1-\mu)f^* + \mu\hat{f} - f^*)$$

Clearly, g is continuous with

$$g(0) = 0, \quad g(1) = c_0\Delta^2(\hat{f} - f^*)$$

Therefore, there is $\mu_{\hat{f}} \in [0,1]$ such that

$$g(\mu_{\hat{f}}) \,=\, \tilde{t}^2.$$

Hence, for any $\tilde{f}$ between $\hat{f}$ and f^*, by letting $\tilde{f} = (1-\mu_{\hat{f}})f^* + \mu_{\hat{f}}\hat{f}$,
By construction we have

$$c_0\Delta^2(\tilde{f} - f^*) \,=\, \tilde{t}^2.$$

So far we have just use the continuity of Δ^2.
Furthermore, we have

$$\tilde{f} \in \mathrm{Star}(\mathcal{F}, f^*),$$

note that $\tilde{f}$ may not stay in $\mathcal{F}$ anymore.
Then by the convexity of $\hat{Z}$, we have that

$$\begin{aligned}\hat{Z}(\tilde{f}) \,&\leq\, (1-\mu_{\hat{f}})\hat{Z}(f^*) + \mu_{\hat{f}}\hat{Z}(\hat{f}) \\ &\overset{(a)}{=}\, \mu_{\hat{f}}\hat{Z}(\hat{f}) \\ &\overset{(b)}{\leq}\, \mu_{\hat{f}}\hat{Z}(f_{\mathrm{Proj}}),\end{aligned} \tag{217}$$

Where (a) holds since $\hat{Z}(f^*) = 0$, and (b) holds since $\hat{f}$ is the minimizer of the $\hat{Z}$ on $\mathcal{F}$ and we have $f, f_{\mathrm{Proj}} \in \mathcal{F}$.
Let

$$\ell(\tilde{f}) \,:=\, (C_0 - c_0)\Delta^2(\tilde{f} - f^*), \tag{218}$$

observe that

$$\begin{aligned}\tilde{t}^2 \,&=\, c_0\Delta^2(\tilde{f} - f^*) \\ &=\, C_0\Delta^2(\tilde{f} - f^*) - (C_0 - c_0)\Delta^2(\tilde{f} - f^*) \\ &\leq\, \bar{Z}(\tilde{f}) - \ell(\tilde{f})\end{aligned}$$

by Corollary 22 and definition of $\ell(f)$ in (218).
Then, adding and subtracting $\hat{Z}(\tilde{f})$ we get that

$$\tilde{t}^2 \,\leq\, \bar{Z}(\tilde{f}) - \hat{Z}(\tilde{f}) + \hat{Z}(\tilde{f}) - \ell(\tilde{f})$$

and by equation (217),

$$\tilde{t}^2 \,\leq\, \bar{Z}(\tilde{f}) - \hat{Z}(\tilde{f}) - \ell(\tilde{f}) + \hat{Z}(f),$$

implies

$$\tilde{t}^2 \,\leq\, \bar{Z}(\tilde{f}) - \hat{Z}(\tilde{f}) - \ell(\tilde{f}) + \mu_{\hat{f}}\hat{Z}(f_{\mathrm{Proj}}).$$

Then, adding and subtracting $\mu_{\hat{f}}\bar{Z}(f_{\mathrm{Proj}})$, we get

$$\begin{aligned}\tilde{t}^2 \,\leq\, &\bar{Z}(\tilde{f}) - \hat{Z}(\tilde{f}) - \ell(\tilde{f}) + \mu_{\hat{f}}\hat{Z}(f_{\mathrm{Proj}}) - \mu_{\hat{f}}\bar{Z}(f_{\mathrm{Proj}}) + \mu_{\hat{f}}\bar{Z}(f_{\mathrm{Proj}}) \\ &- \mu_{\hat{f}}l(f_{\mathrm{Proj}}) + \mu_{\hat{f}}l(f_{\mathrm{Proj}}).\end{aligned}$$

which implies

$$\begin{aligned}\tilde{t}^2 \le\; & \bar{Z}(\tilde{f}) - \hat{Z}(\tilde{f}) - \ell(\tilde{f}) + \mu_{\hat{f}}\left[\hat{Z}(f_{\text{Proj}}) - \bar{Z}(f_{\text{Proj}}) - l(f_{\text{Proj}})\right] \\ & + \mu_{\hat{f}} l(f_{\text{Proj}}) + \mu_{\hat{f}}\bar{Z}(f_{\text{Proj}}). \\ \overset{\text{(a)}}{\le} & \sup_{\substack{f\in \text{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le \tilde{t}^2}} \left[\bar{Z}(f) - \hat{Z}(f) - \ell(f)\right] \\ & + \mu_{\hat{f}} \sup_{\substack{f\in \text{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le \tilde{t}^2}} \left[\hat{Z}(f) - \bar{Z}(f) - \ell(f)\right] \\ & + \mu_{\hat{f}} l(f_{\text{Proj}}) + \mu_{\hat{f}}\bar{Z}(f_{\text{Proj}})\end{aligned}$$

The (a) holds since $f_{\text{Proj}} \in \text{Star}(\mathcal{F}, f^*)$ and

$$c_0\Delta^2(f_{\text{Proj}} - f^*) \le c_0 \|f_{\text{Proj}} - f^*\|_2^2 \le c_0 \|f_{\text{Proj}} - f^*\|_\infty^2 \le c_0\phi_{nm}^2 \le C_0\phi_{nm}^2 \le \tilde{t}^2,$$

where the last inequality holds by (216).

Then, observe that $f^* \in \text{Star}(\mathcal{F}, f^*)$, and $c_0\Delta^2(f^* - f^*) = 0$.

Thus

$$\sup_{\substack{f\in \text{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le \tilde{t}^2}} \left[\hat{Z}(f) - \bar{Z}(f) - \ell(f)\right] \ge \hat{Z}(f^*) - \bar{Z}(f^*) - l(f^*) = 0,$$

and

$$\sup_{\substack{f\in \text{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le \tilde{t}^2}} \left[\bar{Z}(f) - \hat{Z}(f) - \ell(f)\right] \ge \bar{Z}(f^*) - \hat{Z}(f^*) - l(f^*) = 0. \tag{219}$$

Furthermore, $0 \le \mu_{\hat{f}} \le 1$, and

$$\begin{aligned} 0 \overset{\text{(a)}}{\le} l(f_{\text{Proj}}) &= (C_0 - c_0)\Delta^2(f_{\text{Proj}} - f^*) \\ &\le C_0 \|f_{\text{Proj}} - f^*\|_2^2 \le C_0\phi_{nm}^2 \end{aligned}$$

where (a) holds by (218).

Therefore,

$$\begin{aligned}\tilde{t}^2 \le & \sup_{\substack{f\in \text{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le \tilde{t}^2}} \left[\hat{Z}(f) - \hat{Z}(f) - \ell(f)\right] \\ & + \mu_{\hat{f}} \sup_{\substack{f\in \text{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le \tilde{t}^2}} \left[\hat{Z}(f) - \bar{Z}(f) - \ell(f)\right] \\ & + C_0\phi_{nm}^2 + \mu_{\hat{f}}\bar{Z}(f_{\text{Proj}}).\end{aligned}$$

Then, we use Lemma 51, we get that

$$\bar{Z}(f_{\text{Proj}}) \le \frac{\bar{f}}{2}\|f_{\text{Proj}} - f^*\|_2^2$$

$$
\begin{aligned}
&\leq \frac{\bar{f}}{2}\|f_{\text{Proj}} - f^*\|_\infty^2 \\
&\leq \frac{\bar{f}}{2}\phi_{nm}^2.
\end{aligned}
$$

Thus, since $0 \leq \mu_{\hat{f}} \leq 1$, we have

$$
\begin{aligned}
\tilde{t}^2 \leq & \sup_{\substack{f \in \mathrm{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f - f^*) \leq \tilde{t}^2}} \left[\bar{Z}(f) - \hat{Z}(f) - \ell(f)\right] \\
& + \sup_{\substack{f \in \mathrm{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f - f^*) \leq \tilde{t}^2}} \left[\hat{Z}(f) - \bar{Z}(f) - \ell(f)\right] \\
& + \left(C_0 + \frac{\bar{f}}{2}\right)\phi_{nm}^2,
\end{aligned}
$$

where the first inequality is by (219) and hence,

$$
\begin{aligned}
\tilde{\tilde{t}}^2 := t^2 + \left(C_0 + \frac{\bar{f}}{2}\right)\phi_{nm}^2 \leq & \sup_{\substack{f \in \mathrm{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f - f^*) \leq \tilde{t}^2}} \left[\bar{Z}(f) - \hat{Z}(f) - \ell(f)\right] \\
& + \sup_{\substack{f \in \mathrm{Star}(\mathcal{F}, f^*) \\ c_0 \Delta^2(f - f^*) \leq \tilde{t}^2}} \left[\hat{Z}(f) - \bar{Z}(f) - \ell(f)\right].
\end{aligned}
$$

■

Theorem 53 (Conversion of Huber loss) *Let $\hat{f}, f^* \in \mathcal{F}$, let Ω_β be defined in (76). Suppose the distribution of $\{(x_{ij}, y_{i,j})\}_{i=1,j=1}^{n,m_i}$ obey Assumption 2, where c_0 satisfies (84), and $\ell(f)$ is defined in (86), the following inequality is true for any $t > 0$, let*

$$
\begin{aligned}
\tilde{A}_1 &:= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\left(\sup_{\substack{f \in Star(\mathcal{F}, f^*) \\ c_0 \Delta^2(f - f^*) \leq \tilde{t}^2}} \frac{1}{n}\sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i}\sum_{j=1}^{m_i} Z_{ij}(f) - \tilde{Z}_{ij}(f) - \frac{\ell(f)}{2L}\right) \cdot \mathbf{1}_{\Omega_\beta}\right] \\
\tilde{A}_2 &:= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\left(\sup_{\substack{f \in Star(\mathcal{F}, f^*) \\ c_0 \Delta^2(f - f^*) \leq \tilde{t}^2}} \frac{1}{n}\sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i}\sum_{j=1}^{m_i} \tilde{Z}_{ij}(f) - \hat{Z}_{ij}(f)\right) \cdot \mathbf{1}_{\Omega_\beta}\right] \qquad (220) \\
\tilde{A}_3 &:= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\left(\sup_{\substack{f \in Star(\mathcal{F}, f^*) \\ c_0 \Delta^2(f - f^*) \leq \tilde{t}^2}} \frac{1}{n}\sum_{i \in \mathcal{J}_{e,l}} \frac{1}{m_i}\sum_{j=1}^{m_i} \bar{Z}_{ij}(f) - Z_{ij}(f)\right) \cdot \mathbf{1}_{\Omega_\beta}\right],
\end{aligned}
$$

$$
\begin{aligned}
\tilde{B}_1 &:= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\Big(\sup_{\substack{f\in Star(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{o,l}}\frac{1}{m_i}\sum_{j=1}^{m_i} Z_{ij}(f)-\tilde{Z}_{ij}(f)-\frac{\ell(f)}{2L}\Big)\cdot\mathbf{1}_{\Omega_\beta}\right]\\
\tilde{B}_2 &:= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\Big(\sup_{\substack{f\in Star(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{o,l}}\frac{1}{m_i}\sum_{j=1}^{m_i} \tilde{Z}_{ij}(f)-\hat{Z}_{ij}(f)\Big)\cdot\mathbf{1}_{\Omega_\beta}\right]\\
\tilde{B}_3 &:= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\Big(\sup_{\substack{f\in Star(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{o,l}}\frac{1}{m_i}\sum_{j=1}^{m_i} \bar{Z}_{ij}(f)-Z_{ij}(f)\Big)\cdot\mathbf{1}_{\Omega_\beta}\right]
\end{aligned}
\tag{221}
$$

$$
\begin{aligned}
\tilde{E}_1 &:= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\Big(\sup_{\substack{f\in Star(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i} \tilde{Z}_{ij}(f)-Z_{ij}(f)-\frac{\ell(f)}{2L}\Big)\cdot\mathbf{1}_{\Omega_\beta}\right]\\
\tilde{E}_2 &:= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\Big(\sup_{\substack{f\in Star(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i} \hat{Z}_{ij}(f)-\tilde{Z}_{ij}(f)\Big)\cdot\mathbf{1}_{\Omega_\beta}\right]\\
\tilde{E}_3 &:= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\Big(\sup_{\substack{f\in Star(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i} Z_{ij}(f)-\bar{Z}_{ij}(f)\Big)\cdot\mathbf{1}_{\Omega_\beta}\right],
\end{aligned}
\tag{222}
$$

$$
\begin{aligned}
\tilde{F}_1 &:= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\Big(\sup_{\substack{f\in Star(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{o,l}}\frac{1}{m_i}\sum_{j=1}^{m_i} \tilde{Z}_{ij}(f)-Z_{ij}(f)-\frac{\ell(f)}{2L}\Big)\cdot\mathbf{1}_{\Omega_\beta}\right]\\
\tilde{F}_2 &:= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\Big(\sup_{\substack{f\in Star(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{o,l}}\frac{1}{m_i}\sum_{j=1}^{m_i} \hat{Z}_{ij}(f)-\tilde{Z}_{ij}(f)\Big)\cdot\mathbf{1}_{\Omega_\beta}\right]\\
\tilde{F}_3 &:= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\Big(\sup_{\substack{f\in Star(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{o,l}}\frac{1}{m_i}\sum_{j=1}^{m_i} Z_{ij}(f)-\bar{Z}_{ij}(f)\Big)\cdot\mathbf{1}_{\Omega_\beta}\right],
\end{aligned}
\tag{223}
$$

$$
\begin{aligned}
&\mathbb{P}\left(c_0\Delta^2(\hat{f}-f^*)>\tilde{t}^2\right)\\
&\le \sum_{l=1}^{L}\frac{4L}{\overset{\approx}{t}^2}\left(\tilde{A}_1+\tilde{A}_2+\tilde{A}_3+\tilde{B}_1+\tilde{B}_2+\tilde{B}_3+\tilde{E}_1+\tilde{E}_2+\tilde{E}_3+\tilde{F}_1+\tilde{F}_2+\tilde{F}_3\right).
\end{aligned}
\tag{224}
$$

Proof By Corollary 52, it is true that

$$
\begin{aligned}
&\mathbb{P}\left(c_0\Delta^2(\hat{f}-f^*) > \tilde{t}^2\right) \\
&\overset{(a)}{\leq} \mathbb{P}\left(\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\leq\tilde{t}^2}} \left(\bar{Z}(f)-\hat{Z}(f)-\ell(f)\right) + \sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\leq\tilde{t}^2}} \left(\hat{Z}(f)-\bar{Z}(f)-\ell(f)\right) > \tilde{t}^2\right) \\
&\leq \underbrace{\mathbb{P}\left(\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\leq\tilde{t}^2}} \left(\bar{Z}(f)-\hat{Z}(f)-\ell(f)\right) > \frac{\tilde{\tilde{t}}^2}{2}\right)}_{I_1} \\
&\quad + \underbrace{\mathbb{P}\left(\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\leq\tilde{t}^2}} \left(\hat{Z}(f)-\bar{Z}(f)-\ell(f)\right) > \frac{\tilde{\tilde{t}}^2}{2}\right)}_{I_2}.
\end{aligned}
$$

Then for the first term I_1 above, we have

$$
\begin{aligned}
&\mathbb{P}\left(\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\leq\tilde{t}^2}} \left(\bar{Z}(f)-\hat{Z}(f)-\ell(f)\right) > \frac{\tilde{\tilde{t}}^2}{2}\right) \\
&= \mathbb{P}\left(\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\leq\tilde{t}^2}} \left(\frac{1}{n}\sum_{l=1}^{L}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\bar{Z}_{ij}(f)-\hat{Z}_{ij}(f)-\frac{\ell(f)}{2}\right.\right. \\
&\qquad \left.\left. + \frac{1}{n}\sum_{l=1}^{L}\sum_{i\in\mathcal{J}_{o,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\bar{Z}_{ij}(f)-\hat{Z}_{ij}(f)-\frac{\ell(f)}{2}\right) > \frac{\tilde{\tilde{t}}^2}{2}\right) \\
&\leq \mathbb{P}\left(\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\leq\tilde{t}^2}} \left(\frac{1}{n}\sum_{l=1}^{L}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\bar{Z}_{ij}(f)-\hat{Z}_{ij}(f)-\frac{\ell(f)}{2}\right) > \frac{\tilde{\tilde{t}}^2}{4}\right) \\
&\quad + \mathbb{P}\left(\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\leq\tilde{t}^2}} \left(\frac{1}{n}\sum_{l=1}^{L}\sum_{i\in\mathcal{J}_{o,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\bar{Z}_{ij}(f)-\hat{Z}_{ij}(f)-\frac{\ell(f)}{2}\right) > \frac{\tilde{\tilde{t}}^2}{4}\right) \\
&\leq \sum_{l=1}^{L}\mathbb{P}\left(\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\leq\tilde{t}^2}} \left(\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\bar{Z}_{ij}(f)-\hat{Z}_{ij}(f)-\frac{\ell(f)}{2L}\right) > \frac{\tilde{\tilde{t}}^2}{4L}\right)
\end{aligned}
$$

$$+\sum_{l=1}^{L}\mathbb{P}\left(\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}}\left(\frac{1}{n}\sum_{i\in\mathcal{J}_{o,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\bar{Z}_{ij}(f)-\hat{Z}_{ij}(f)-\frac{\ell(f)}{2L}\right)>\frac{\overset{\approx}{t}^2}{4L}\right).$$

Now we analyze each of the terms above,

$$\begin{aligned}
&\mathbb{P}\left(\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}}\left(\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\bar{Z}_{ij}(f)-\hat{Z}_{ij}(f)-\frac{\ell(f)}{2L}\right)>\frac{\overset{\approx}{t}^2}{4L}\right)\\
&\overset{\text{(a)}}{\le}\frac{4L}{\overset{\approx}{t}^2}\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\Big(\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}}\Big(\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\bar{Z}_{ij}(f)-\hat{Z}_{ij}(f)-\frac{\ell(f)}{2L}\Big)\Big)\cdot\mathbf{1}_{\Omega_\beta}\right]\\
&\overset{\text{(b)}}{\le}\frac{4L}{\overset{\approx}{t}^2}\mathbb{E}\left[\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\tilde{Z}_{ij}(f)-Z_{ij}(f)-\frac{\ell(f)}{2L}+\right.\\
&\quad\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\tilde{Z}_{ij}(f)-\hat{Z}_{ij}(f)\\
&\quad\left.+\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\bar{Z}_{ij}(f)-Z_{ij}(f)\right]\\
&=\frac{4L}{\overset{\approx}{t}^2}\underbrace{\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\Big(\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}Z_{ij}(f)-\tilde{Z}_{ij}(f)-\frac{\ell(f)}{2L}\Big)\cdot\mathbf{1}_{\Omega_\beta}\right]}_{\tilde{A}_1}\\
&\quad+\frac{4L}{\overset{\approx}{t}^2}\underbrace{\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\Big(\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\tilde{Z}_{ij}(f)-\hat{Z}_{ij}(f)\Big)\cdot\mathbf{1}_{\Omega_\beta}\right]}_{\tilde{A}_2}\\
&\quad+\frac{4L}{\overset{\approx}{t}^2}\\
&\quad\underbrace{\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\Big(\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\bar{Z}_{ij}(f)-Z_{ij}(f)\Big)\cdot\mathbf{1}_{\Omega_\beta}\right]}_{\tilde{A}_3}\\
&:=\frac{4L}{\overset{\approx}{t}^2}\left(\tilde{A}_1+\tilde{A}_2+\tilde{A}_3\right).
\end{aligned}$$

Similarly, we have

$$\mathbb{P}\left(\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\leq\tilde{t}^2}}\left(\frac{1}{n}\sum_{i\in\mathcal{J}_{o,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\bar{z}_{ij}(f)-\hat{z}_{ij}(f)-\frac{\ell(f)}{2L}\right)>\frac{\overset{\approx}{t}^{2}}{4L}\right)$$

$$=\frac{4L}{\overset{\approx}{t}^{2}}\underbrace{\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\left(\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\leq\tilde{t}^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{o,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}Z_{ij}(f)-\tilde{Z}_{ij}(f)-\frac{\ell(f)}{2L}\right)\cdot\mathbf{1}_{\Omega_\beta}\right]}_{\tilde{B}_1}$$

$$+\frac{4L}{\overset{\approx}{t}^{2}}\underbrace{\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\left(\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\leq\tilde{t}^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{o,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\tilde{Z}_{ij}(f)-\hat{Z}_{ij}(f)\right)\cdot\mathbf{1}_{\Omega_\beta}\right]}_{\tilde{B}_2}$$

$$+\frac{4L}{\overset{\approx}{t}^{2}}$$

$$\underbrace{\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\left(\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\leq\tilde{t}^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{o,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\bar{Z}_{ij}(f)-Z_{ij}(f)\right)\cdot\mathbf{1}_{\Omega_\beta}\right]}_{\tilde{B}_3}$$

$$:=\frac{4L}{\overset{\approx}{t}^{2}}\left(\tilde{B}_1+\tilde{B}_2+\tilde{B}_3\right).$$

where the first inequality (a) follows from Lemma 23, The inequality (a) follows from Markov's inequality ,and the third inequality (b) follows from the triangle inequality.

Thus, we have

$$I_1\ \leq\ \sum_{l=1}^{L}\frac{4L}{\overset{\approx}{t}^{2}}\left(\tilde{A}_1+\tilde{A}_2+\tilde{A}_3+\tilde{B}_1+\tilde{B}_2+\tilde{B}_3\right).$$

Then similarly, we have

$$I_2=\mathbb{P}\left(\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\leq\tilde{t}^2}}\left(\hat{Z}(f)-\bar{Z}(f)-\ell(f)\right)>\frac{\overset{\approx}{t}^{2}}{2}\right)$$

$$\leq\sum_{l=1}^{L}\mathbb{P}\Bigg(\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\leq\tilde{t}^2}}\Bigg(\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\hat{Z}_{ij}(f)-\bar{Z}_{ij}(f)$$

$$-\frac{\ell(f)}{2L}\Bigg)>\frac{\overset{\approx}{t}^{2}}{4L}\Bigg)$$

$$+\sum_{l=1}^{L}\mathbb{P}\Bigg(\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\leq\tilde{t}^2}}\Bigg(\frac{1}{n}\sum_{i\in\mathcal{J}_{o,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\hat{Z}_{ij}(f)-\bar{Z}_{ij}(f)$$

$$
\begin{aligned}
&- \frac{\ell(f)}{2L}\Big) > \frac{\overset{\approx^2}{t}}{4L}\Big) \\
&\leq \sum_{l=1}^{L} \frac{4L}{\overset{\approx^2}{t}} \underbrace{\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*} \left[\Big(\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\leq\tilde{t}^2}} \frac{1}{n} \sum_{i\in\mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \tilde{Z}_{ij}(f) - Z_{ij}(f) - \frac{\ell(f)}{2L} \Big) \cdot \mathbf{1}_{\Omega_\beta} \right]}_{\tilde{E}_1} \\
&+ \sum_{l=1}^{L} \frac{4L}{\overset{\approx^2}{t}} \underbrace{\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*} \left[\Big(\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\leq\tilde{t}^2}} \frac{1}{n} \sum_{i\in\mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \hat{Z}_{ij}(f) - \tilde{Z}_{ij}(f) \Big) \cdot \mathbf{1}_{\Omega_\beta} \right]}_{\tilde{E}_2} \\
&+ \sum_{l=1}^{L} \frac{4L}{\overset{\approx^2}{t}} \underbrace{\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*} \left[\Big(\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\leq\tilde{t}^2}} \frac{1}{n} \sum_{i\in\mathcal{J}_{e,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} Z_{ij}(f) - \bar{Z}_{ij}(f) \Big) \cdot \mathbf{1}_{\Omega_\beta} \right]}_{\tilde{E}_3} \\
&+ \sum_{l=1}^{L} \frac{4L}{\overset{\approx^2}{t}} \underbrace{\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*} \left[\Big(\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\leq\tilde{t}^2}} \frac{1}{n} \sum_{i\in\mathcal{J}_{o,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \tilde{Z}_{ij}(f) - Z_{ij}(f) - \frac{\ell(f)}{2L} \Big) \cdot \mathbf{1}_{\Omega_\beta} \right]}_{\tilde{F}_1} \\
&+ \sum_{l=1}^{L} \frac{4L}{\overset{\approx^2}{t}} \underbrace{\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*} \left[\Big(\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\leq\tilde{t}^2}} \frac{1}{n} \sum_{i\in\mathcal{J}_{o,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} \hat{Z}_{ij}(f) - \tilde{Z}_{ij}(f) \Big) \cdot \mathbf{1}_{\Omega_\beta} \right]}_{\tilde{F}_2} \\
&+ \sum_{l=1}^{L} \frac{4L}{\overset{\approx^2}{t}} \underbrace{\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*} \left[\Big(\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\leq\tilde{t}^2}} \frac{1}{n} \sum_{i\in\mathcal{J}_{o,l}} \frac{1}{m_i} \sum_{j=1}^{m_i} Z_{ij}(f) - \bar{Z}_{ij}(f) \Big) \cdot \mathbf{1}_{\Omega_\beta} \right]}_{\tilde{F}_3}.
\end{aligned}
$$

We thus get

$$
I_2 \leq \sum_{l=1}^{L} \frac{4L}{\overset{\approx^2}{t}} \left(\tilde{E}_1 + \tilde{E}_2 + \tilde{E}_3 + \tilde{F}_1 + \tilde{F}_2 + \tilde{F}_3 \right).
$$

To proceed, we would need to bound $\tilde{A}_1$, $\tilde{A}_2$ $\tilde{A}_3$, $\tilde{B}_1$, $\tilde{B}_2$ $\tilde{B}_3$, and $\tilde{E}_1$, $\tilde{E}_2$ $\tilde{E}_3$, $\tilde{F}_1$, $\tilde{F}_2$ $\tilde{F}_3$ separately. We only need to analyze the $\tilde{A}_1$, $\tilde{A}_2$, and $\tilde{A}_3$, as for $\tilde{B}_1$, $\tilde{B}_2$ $\tilde{B}_3$, the analysis is similar to $\tilde{A}_1$, $\tilde{A}_2$ $\tilde{A}_3$, we will just give the conclusion.

Similarly, we only need to analyze the $\tilde{E}_1$, $\tilde{E}_2$, and $\tilde{E}_3$, as for $\tilde{F}_1$, $\tilde{F}_2$ $\tilde{F}_3$, the analysis is similar to $\tilde{E}_1$, $\tilde{E}_2$, and $\tilde{E}_3$, we will just give the conclusion. ■

Corollary 54 (Bound of $\tilde{A}_2$, $\tilde{B}_2$) *Let $\tilde{Z}(f)$, $\hat{Z}(f)$ been defined in (68) and (65). Let Ω_β be defined in (76). Let $\{s_{i,j}\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$ be independent Rademacher variables, independent of the data $\{(x_{ij}, e_{ij}, \delta_i)\}$ and their coupled copies $\{(x^*_{ij}, e^*_{ij}, \delta^*_i)\}$, let $\tilde{A}_2$ be the quantity introduced in (220), on event Ω, then we have*

$$
\begin{aligned}
\tilde{A}_2 &= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\left(\sup_{\substack{f\in Star(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\left\{\tilde{Z}_{ij}(f)-\hat{Z}_{ij}(f)\right\}\right)\cdot\mathbf{1}_{\Omega_\beta}\right]\\
&\le 2\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\left(\sup_{\substack{f\in Star(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}s_{ij}\hat{Z}_{ij}(f)\right)\cdot\mathbf{1}_{\Omega_\beta}\right]\\
&= \frac{c_5}{L}R_{\mathcal{J}_{e,l}}(Star(\mathcal{F},f^*)\cap\{c_0\Delta^2(f-f^*)\le\tilde{t}^2\}).
\end{aligned}
$$

Similarly, for $\tilde{B}_2$ in (221) we have

$$
\tilde{B}_2 \le \frac{c_5}{L}R_{\mathcal{J}_{o,l}}(Star(\mathcal{F},f^*)\cap\{c_0\Delta^2(f-f^*)\le\tilde{t}^2\}).
$$

Proof The proof is a direct application of Corollary 26 by replacing $\mathcal{F}$ with Star$(\mathcal{F}, f^*)$. ■

Corollary 55 (Bound of $\tilde{A}_1$, $\tilde{B}_1$) *Let Ω_β be defined in (76). Let $\tilde{Z}(f)$, $Z(f)$ been defined in (68) and (67), let $\ell(f)$ be defined in (218), let $\{s_{i,j}\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$ be independent Rademacher variables, independent of the data $\{(x_{ij}, e_{ij}, \delta_i)\}$ and their coupled copies $\{(x^*_{ij}, e^*_{ij}, \delta^*_i)\}$, let $\tilde{A}_1$ be the quantity introduced in (220), then we have*

$$
\begin{aligned}
\tilde{A}_1 &= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\left(\sup_{\substack{f\in Star(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\tilde{Z}_{ij}(f)-Z_{ij}(f)-\frac{\ell(f)}{2L}\right)\cdot\mathbf{1}_{\Omega_\beta}\right]\\
&\le 2\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\left(\sup_{\substack{f\in Star(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}s_{ij}u_{ij}\right)\cdot\mathbf{1}_{\Omega_\beta}\right]+\frac{c_4}{L}\frac{s\log n\log T_n}{n}\\
&= \frac{c_5}{L}R_{\mathcal{J}_{e,l}}(Star(\mathcal{F},f^*)\cap\{c_0\Delta^2(f-f^*)\le\tilde{t}^2\})+\frac{c_4}{L}\frac{s\log n\log T_n}{n},
\end{aligned}
$$

with a small constant c_4 that depends on $\tilde{c}_0$ is given in Assumption 4, and c_1, c_2, C_β. Similarly, for $\tilde{B}_1$ in (221), we have

$$
\tilde{B}_1 \le \frac{c_5}{L}R_{\mathcal{J}_{o,l}}(Star(\mathcal{F},f^*)\cap\{c_0\Delta^2(f-f^*)\le\tilde{t}^2\})+\frac{c_4}{L}\frac{s\log n\log T_n}{n},
$$

Proof The proof is a direct application of Corollary 30 by replacing $\mathcal{F}$ with Star$(\mathcal{F}, f^*)$. ■

Corollary 56 (Bound of $\tilde{A}_3$ and $\tilde{B}_3$) *Let Ω_β be defined in (76). Let $\bar{Z}(f)$, $Z(f)$ been defined in (69) and (67), let $\{s_{i,j}\}_{i\in\mathcal{J}_{e,l}, j\in[m_i]}$ be independent Rademacher variables, independent of the data $\{(x_{ij}, e_{ij}, \delta_i)\}$ and their coupled copies $\{(x^*_{ij}, e^*_{ij}, \delta^*_i)\}$, let $\tilde{A}_3$ be the quantity introduced in expression (220), then we have*

$$\begin{aligned}
\tilde{A}_3 &= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\left(\sup_{\substack{f\in Star(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\left\{\bar{Z}_{ij}(f)-Z_{ij}(f)\right\}\right)\cdot \mathbf{1}_{\Omega_\beta}\right] \\
&\le 2\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\left(\sup_{\substack{f\in Star(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i} s_{ij}Z_{ij}(f)\right)\cdot \mathbf{1}_{\Omega_\beta}\right] \\
&= \frac{c_5}{L}R_{\mathcal{J}_{e,l}}(Star(\mathcal{F},f^*)\cap\left\{c_0\Delta^2(f-f^*)\le\tilde{t}^2\right\}).
\end{aligned}$$

Similarly, for $\tilde{B}_3$ in (221) we have

$$\begin{aligned}
\tilde{B}_3 &= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\left(\sup_{\substack{f\in Star(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{o,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\left\{\bar{Z}_{ij}(f)-Z_{ij}(f)\right\}\right)\cdot \mathbf{1}_{\Omega_\beta}\right] \\
&\le \frac{c_5}{L}R_{\mathcal{J}_{o,l}}(Star(\mathcal{F},f^*)\cap\left\{c_0\Delta^2(f-f^*)\le\tilde{t}^2\right\}).
\end{aligned}$$

Proof The proof is a direct application of Corollary 33 by replacing $\mathcal{F}$ with $\mathrm{Star}(\mathcal{F}, f^*)$. ■

Corollary 57 (Bound of $\tilde{E}_2$, $\tilde{F}_2$) *Let $\tilde{Z}(f)$, $\hat{Z}(f)$ been defined in (68) and (65). Let Ω_β be defined in (76). Let $\{s_{i,j}\}_{i\in\mathcal{J}_{e,l}, j\in[m_i]}$ be independent Rademacher variables, independent of the data $\{(x_{ij}, e_{ij}, \delta_i)\}$ and their coupled copies $\{(x^*_{ij}, e^*_{ij}, \delta^*_i)\}$, let $\tilde{E}_2$ be the quantity introduced in (222), on event Ω, then we have*

$$\begin{aligned}
\tilde{E}_2 &= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\left(\sup_{\substack{f\in Star(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\left\{\hat{Z}_{ij}(f)-\tilde{Z}_{ij}(f)\right\}\right)\cdot \mathbf{1}_{\Omega_\beta}\right] \\
&\le 2\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\left(\sup_{\substack{f\in Star(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i} s_{ij}\hat{Z}_{ij}(f)\right)\cdot \mathbf{1}_{\Omega_\beta}\right] \\
&= \frac{c_5}{L}R_{\mathcal{J}_{e,l}}(Star(\mathcal{F},f^*)\cap\left\{c_0\Delta^2(f-f^*)\le\tilde{t}^2\right\}).
\end{aligned}$$

Similarly, for $\tilde{F}_2$ in (223) we have

$$\tilde{F}_2 \le \frac{c_5}{L}R_{\mathcal{J}_{o,l}}(Star(\mathcal{F},f^*)\cap\left\{c_0\Delta^2(f-f^*)\le\tilde{t}^2\right\}).$$

Proof In the proof, we can see that the $\tilde{E}_2$ and $\tilde{A}_2$ have same type of upper bound. The proof is similar to the proof of Corollary 26.

First, by Tower property, we have

$$
\begin{aligned}
A_2 &:= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\left(\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\left\{\hat{Z}_{ij}(f)-\tilde{Z}_{ij}(f)\right\}\right)\cdot\mathbf{1}_{\Omega_\beta}\right] \\
&= \mathbb{E}\left[\left(\mathbb{E}\left[\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\left\{\hat{Z}_{ij}(f)-\mathbb{E}\left[\hat{Z}_{ij}(f)\mid\{\delta_i\}\right]\right\}\mid\{\delta_i\}\right]\right)\cdot\mathbf{1}_{\Omega_\beta}\right].
\end{aligned}
$$

To facilitate our analysis, we define more terms. Let $\{\check{e}_{i,j}\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$, $\{\check{x}_{ij}\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$ be an independent and identically distributed copy of $\{e_{i,j}\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$ $\{x_{ij}\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$, then we define $\check{Z}_{ij}(f)$ as

$$
\check{Z}_{ij}(f) := \rho_\tau(f^*(\check{x}_{ij})+\delta_i(\check{x}_{ij})+\check{e}_{ij}-f(\check{x}_{ij}))-\rho_\tau(\delta_i(\check{x}_{ij})+\check{e}_{ij}).
$$

Now we control the $\mathbb{E}\left[\hat{Z}_{ij}(f)-\check{Z}_{ij}(f)\mid\{\delta_i\},\{e_{i,j}\}_{i\in\mathcal{J}_{e,l},j\in[m_i]},\{x_{i,j}\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}\right]$, the expectation takes on $\{\check{e}_{i,j}\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$, $\{\check{x}_{ij}\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$, and conditioned on $\{(\delta_{i1},\dots,\delta_{im_i})\}_{i\in\mathcal{J}_{e,l}}$, $\{e_{i,j}\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$ and $\{x_{ij}\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$.

So we have

$$
\begin{aligned}
&\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\Big\{\mathbb{E}\left[\hat{Z}_{ij}(f)\mid\{\delta_i\},\{e_{i,j}\},\{x_{ij}\}\right] \\
&\qquad -\mathbb{E}\left[\check{Z}_{ij}(f)\mid\{\delta_i\},\{e_{i,j}\},\{x_{ij}\}\right]\Big\} \\
&\overset{(a)}{\le}\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*,s}\left[\left(\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\left\{\hat{Z}_{ij}(f)-\check{Z}_{ij}(f)\right\}\right)\cdot\mathbf{1}_{\Omega_\beta}\mid\{\delta_i\},\{e_{i,j}\},\{x_{ij}\}\right].
\end{aligned} \tag{225}
$$

The equality (a) is held by Jensen's inequality. Then, the left-hand side of the above, can be further written as

$$
\begin{aligned}
&\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\Big\{\mathbb{E}\left[\hat{Z}_{ij}(f)\mid\{\delta_i\},\{e_{i,j}\},\{x_{ij}\}\right] \\
&\qquad -\mathbb{E}\left[\check{Z}_{ij}(f)\mid\{\delta_i\},\{e_{i,j}\},\{x_{ij}\}\right]\Big\} \\
&\overset{(a)}{=}\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\left\{\hat{Z}_{ij}(f)-\mathbb{E}\left[\hat{Z}_{ij}(f)\mid\{\delta_i\}\right]\right\}.
\end{aligned} \tag{226}
$$

The second expectation takes on $\{e_{i,j}\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$, $\{x_{ij}\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$, and conditioned on $\{(\delta_{i1},\dots,\delta_{im_i})\}_{i\in\mathcal{J}_{e,l}}$, then the equality (a) holds since $\mathbb{E}\left[\check{Z}_{ij}(f)\mid\{\delta_i\},\{e_{i,j}\},\{x_{ij}\}\right]$ and $\mathbb{E}\left[\hat{Z}_{ij}(f)\mid\{\delta_i\}\right]$ are the same quantity.

Combining (225) and (226), taking expectation of $\{e_{i,j}\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$, $\{x_{ij}\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$ conditioning on $\{(\delta_{i1},\ldots,\delta_{im_i})\}_{i\in\mathcal{J}_{e,l}}$ we have

$$\begin{aligned}
&\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\left(\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\left\{\hat{Z}_{ij}(f)-\mathbb{E}\left[\hat{Z}_{ij}(f)\mid\{\delta_i\}\right]\right\}\mid\{\delta_i\}\right)\cdot\mathbf{1}_{\Omega_\beta}\right]\\
\le&\mathbb{E}\left[\left(\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\left\{\hat{Z}_{ij}(f)\right\}-\check{Z}_{ij}(f)\right)\cdot\mathbf{1}_{\Omega_\beta}\mid\{\delta_i\}\right]\\
\overset{(a)}{=}&\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\left(\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}s_{ij}\left\{\hat{Z}_{ij}(f)-\check{Z}_{ij}(f)\right\}\right)\cdot\mathbf{1}_{\Omega_\beta}\mid\{\delta_i\}\right]\\
\overset{(b)}{\le}&2\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\left(\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}s_{ij}\hat{Z}_{ij}(f)\right)\cdot\mathbf{1}_{\Omega_\beta}\mid\{\delta_i\}\right].
\end{aligned}$$

The (a) holds since conditioned on $\{(\delta_{i1},\ldots,\delta_{im_i})\}_{i\in\mathcal{J}_{e,l}}$, the $\left\{\hat{Z}_{ij}(f)-\check{Z}_{ij}(f)\right\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$ is independently distributed and symmetric with mean zero and $\left\{s_{ij}(\hat{Z}_{ij}(f)-\check{Z}_{ij}(f))\right\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$ and $\left\{\hat{Z}_{ij}(f)-\check{Z}_{ij}(f)\right\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$ have the same distribution. The (b) follows because $\{-s_{ij}\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$ are also independent Rademacher variables. Taking expectation of $\{(\delta_{i1},\ldots,\delta_{im_i})\}_{i\in\mathcal{J}_{e,l}}$ on both sides, we obtain the result.

We apply the Lipschitz condition of $\hat{Z}_{ij}(f)$, $Z_{ij}(f)$ and $u_{ij}(f)$ in Lemma 19, and then apply the Ledoux-Talagrand contraction principle in Theorem 62.

$$\begin{aligned}
&\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\left(\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\left\{\hat{Z}_{ij}(f)-\tilde{Z}_{ij}(f)\right\}\right)\cdot\mathbf{1}_{\Omega_\beta}\right]\\
\le&\ 2\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\left(\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}s_{ij}\hat{Z}_{ij}(f)\right)\cdot\mathbf{1}_{\Omega_\beta}\right]\\
\overset{(a)}{\le}&\ 2\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\left(\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}s_{ij}f(x^*_{ij})\right)\cdot\mathbf{1}_{\Omega_\beta}\right]\\
=&\ \frac{c_5}{L}R_{\mathcal{J}_{e,l}}(\mathrm{Star}(\mathcal{F},f^*)\cap\left\{c_0\Delta^2(f-f^*)\le\tilde{t}^2\right\}).
\end{aligned}$$

■

Lemma 58 (Bound of $\tilde{E}_1$ and $\tilde{F}_1$) *Let Ω_β be defined in (76). Let $\tilde{Z}(f)$, $Z(f)$ been defined in (68) and (67), and $\ell(f)$ be defined in (218), let $\{s_{i,j}\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$ be independent Rademacher variables, independent of the data $\{(x_{ij},e_{ij},\delta_i)\}$ and their coupled copies $\{(x^*_{ij},e^*_{ij},\delta^*_i)\}$, let $\tilde{E}_1$ be the quantity introduced in (222), with a small constant $\tilde{c}_0$ is given in Assumption 4, then we have*

$$\begin{aligned}\tilde{E}_1 &:= \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\Big(\sup_{\substack{f\in Star(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\tilde{Z}_{ij}(f)-Z_{ij}(f)-\frac{\ell(f)}{2L}\Big)\cdot\mathbf{1}_{\Omega_\beta}\right]\\ &\le\frac{c_5}{L}R_{\mathcal{J}_{e,l}}(Star(\mathcal{F},f^*)\cap\{c_0\Delta^2(f-f^*)\le\tilde{t}^2\})+\frac{c_4}{L}\frac{s\log n\log T_n}{n}.\end{aligned}$$

Similarly, for $\tilde{F}_1$ in (223) we have

$$\tilde{F}_1\le\frac{c_5}{L}R_{\mathcal{J}_{o,l}}(Star(\mathcal{F},f^*)\cap\{c_0\Delta^2(f-f^*)\le\tilde{t}^2\})+\frac{c_4}{L}\frac{s\log n\log T_n}{n}.$$

Proof

Step 1 : By previous Lemmas 27 and 28,

$$\tilde{Z}_{ij}(f)-Z_{ij}(f)=\int_0^{f(x_{ij})-f^*(x_{ij})}\{\mathbb{P}(y_{i,j}\le f^*(x_{ij})+t|\delta_i,x_{ij})-\mathbb{P}(y_{i,j}\le f^*(x_{ij})+t|x_{ij})\}\,dt$$

We define $u_{ij}(f)$ as the integral expression given by the last line:

$$u_{ij}(f):=\int_0^{f(x_{ij})-f^*(x_{ij})}\{\mathbb{P}(y_{i,j}\le f^*(x_{ij})+t|\delta_i,x_{ij})-\mathbb{P}(y_{i,j}\le f^*(x_{ij})+t|x_{ij})\}\,dt.\tag{227}$$

Then, we have

$$\begin{aligned}\tilde{E}_1 :=&\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\Big(\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\tilde{Z}_{ij}(f)-Z_{ij}(f)-\frac{\ell(f)}{2L}\Big)\cdot\mathbf{1}_{\Omega_\beta}\right]\\ =&\underbrace{\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\Big(\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}u_{ij}(f)-\frac{\ell(f)}{2L}\Big)\cdot\mathbf{1}_{\Omega_\beta}\right]}_{\tilde{E}_4}.\end{aligned}\tag{228}$$

Now we bound $\tilde{E}_4$,and we have

$$\begin{aligned}\tilde{E}_4\le&\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\Big(\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}u_{ij}(f)-\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*,s}[u_{ij}(f)|\delta_i]\Big)\cdot\mathbf{1}_{\Omega_\beta}\right]\\ &+\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\Big(\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*,s}[u_{ij}(f)|\delta_i]-\frac{\ell(f)}{2L}\Big)\cdot\mathbf{1}_{\Omega_\beta}\right]\end{aligned}$$

$$
\begin{aligned}
=&\mathbb{E}_{\delta,\delta^*}\Bigg[\underbrace{\mathbb{E}_{x,e|\delta}\Bigg[\Big(\sup_{\substack{f\in\text{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}u_{ij}(f)-\mathbb{E}_{x,e|\delta_i}\left[u_{ij}(f)\right]\Big)\cdot\mathbf{1}_{\Omega_\beta}\Bigg]\Bigg]}_{E_5(\{\delta_i\}_{i\in\mathcal{J}_{e,l}})}\\
&+\underbrace{\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\Bigg[\Big(\sup_{\substack{f\in\text{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\mathbb{E}_{x,e|\delta_i}\left[u_{ij}(f)\right]-\frac{\ell(f)}{2L}\Big)\cdot\mathbf{1}_{\Omega_\beta}\Bigg]}_{E_6}.
\end{aligned}
$$

We will bound $E_5(\{\delta_i\}_{i\in\mathcal{J}_{e,l}})$ and E_6 separately.

Step 2 : First, we will show that

$$
E_5(\{\delta_i\}_{i\in\mathcal{J}_{e,l}})\le 2\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\Bigg[\Big(\sup_{\substack{f\in\text{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}s_{ij}u_{ij}(f)\Big)\cdot\mathbf{1}_{\Omega_\beta}\mid\{\delta_i\}\Bigg].
$$

Let $\{\tilde{x}_{ij}\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$ be an independent and identically distributed copy of $\{x_{ij}\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$, correspondingly, we let $\{\tilde{y}_{i,j}\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$ be the copy of y_{ij}, where each $\tilde{y}_{ij}$ and y_{ij} share the same $\{(\delta_{i1},\dots,\delta_{im_i})\}_{i\in\mathcal{J}_{e,l}}$, $\{e_{i,j}\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$, but differ only in $\{\tilde{x}_{ij}\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$ and $\{x_{ij}\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$. Then we define the independent and identically distributed copy of $u_{ij}(f)$ as

$$
\tilde{u}_{ij}(f,\delta_i,\tilde{x}_{ij}):=\int_0^{f(\tilde{x}_{ij})-f^*(\tilde{x}_{ij})}\left\{\mathbb{P}(\tilde{y}_{i,j}\le f^*(\tilde{x}_{ij})+t|\delta_i)-\mathbb{P}(\tilde{y}_{i,j}\le f^*(\tilde{x}_{ij})+t)\right\}dt.
$$

Now we control the $\mathbb{E}\left[u_{ij}(f)-\tilde{u}_{ij}(f)\mid\{\delta_i\}\right]$, the expectation takes on $\{\tilde{x}_{ij}\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$ and conditioned on $\{(\delta_{i1},\dots,\delta_{im_i})\}_{i\in\mathcal{J}_{e,l}}$. So we have

$$
\begin{aligned}
&\sup_{\substack{f\in\text{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\left\{\mathbb{E}\left[u_{ij}(f)\mid\{\delta_i\}\right]-\mathbb{E}\left[\tilde{u}_{ij}(f)\mid\{\delta_i\}\right]\right\}\\
\overset{\text{(a)}}{\le}&\ \mathbb{E}\Bigg[\Big(\sup_{\substack{f\in\text{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\left\{u_{ij}(f)-\tilde{u}_{ij}(f)\right\}\Big)\cdot\mathbf{1}_{\Omega_\beta}\mid\{\delta_i\}\Bigg].
\end{aligned}\tag{229}
$$

The equality (a) is held by Jensen's inequality. Then, the left-hand side of the above, can be further written as

$$
\begin{aligned}
&\sup_{\substack{f\in\text{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\left\{\mathbb{E}\left[u_{ij}(f)\mid\{\delta_i\}\right]-\mathbb{E}\left[\tilde{u}_{ij}(f)\mid\{\delta_i\}\right]\right\}\\
\overset{\text{(a)}}{=}&\sup_{\substack{f\in\text{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\left\{u_{ij}(f)-\mathbb{E}\left[u_{ij}(f)\mid\{\delta_i\}\right]\right\}.
\end{aligned}\tag{230}
$$

The equality (a) holds since $\mathbb{E}\left[u_{ij}(f)\mid\{\delta_i\}\right]$ and $\mathbb{E}\left[\tilde{u}_{ij}(f)\mid\{\delta_i\}\right]$ are identically distributed.

Combining (229) and (235), taking expectation of $\{x_{ij}\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$ conditioning on $\{(\delta_{i1},\ldots,\delta_{im_i})\}_{i\in\mathcal{J}_{e,l}}$ we have

$$\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\left(\sup_{\substack{f\in\text{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\{u_{ij}(f)-\mathbb{E}\left[u_{ij}(f)\mid\{\delta_i\}\right]\}\mid\{\delta_i\}\right)\cdot\mathbf{1}_{\Omega_\beta}\right]$$

$$\le\mathbb{E}\left[\left(\sup_{\substack{f\in\text{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\{u_{ij}(f)-\tilde{u}_{ij}(f)\}\right)\cdot\mathbf{1}_{\Omega_\beta}\mid\{\delta_i\}\right]$$

$$\overset{\text{(a)}}{\le}\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\left(\sup_{\substack{f\in\text{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}s_{ij}\{u_{ij}(f)-\tilde{u}_{ij}(f)\}\right)\cdot\mathbf{1}_{\Omega_\beta}\mid\{\delta_i\}\right]$$

$$\overset{\text{(b)}}{\le}2\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\left(\sup_{\substack{f\in\text{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}s_{ij}u_{ij}(f)\right)\cdot\mathbf{1}_{\Omega_\beta}\mid\{\delta_i\}\right].$$

The (a) holds since conditioned on $\{(\delta_{i1},\ldots,\delta_{im_i})\}_{i\in\mathcal{J}_{e,l}}$, the $u_{ij}(f)-\tilde{u}_{ij}(f)$ is symmetric with mean zero and $\{s_{ij}(u_{ij}(f)-\tilde{u}_{ij}(f))\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$ and $\{\tilde{u}_{ij}(f)-\hat{u}_{ij}(f)\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$ have the same distribution. The (b) follows because $\{-s_{ij}\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$ are also independent Rademacher variables.

Taking expectation of $\{(\delta_{i1},\ldots,\delta_{im_i})\}_{i\in\mathcal{J}_{e,l}}$ on both sides, we obtain the result.

Step 3 : Next, we will show that under the Assumption 3, and if $\ell(f)=4\tilde{c}_0\Delta^2(f-f^*)$, we have

$$E_6:=\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\left(\sup_{\substack{f\in\text{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\mathbb{E}_{x,e|\delta_i}\left[u_{ij}(f)\right]-\frac{\ell(f)}{2L}\right)\cdot\mathbf{1}_{\Omega_\beta}\right].\tag{231}$$

We will show that

$$E_6\;\le\;\frac{c_2}{L}\frac{c_3^2}{4\tilde{c}_0}\frac{s\log n\log T_n}{n}=\frac{c_4}{L}\frac{s\log n\log T_n}{n},$$

for a positive constant c_4, where c_3 is a constant depends on C_β and c_1.

We first show the following bound holds in a multidimensional setting where $x_{ij}\in[0,1]^d$:

$$\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\mathbb{E}\left[u_{ij}(f)|\delta_i\right]\;\le\;\frac{2\tilde{c}_0c_2}{L}\Delta^2(f-f^*)+\frac{c_2}{L}\frac{c_3^2}{4\tilde{c}_0}\frac{s\log n\log T_n}{n}.$$

We then obtain the result by taking expectations and setting

$$\ell(f)\;=\;4\tilde{c}_0c_2\Delta^2(f-f^*).$$

Given that

$$\begin{aligned}\mathbb{E}\left(u_{ij}(f)|\delta_i\right)\;=\;&\int_{[0,1]^d}\left(\int_0^{f(x)-f^*(x)}\left(\mathbb{P}\left(y_{ij}\;\le\;f^*(x_{ij})+u\mid\delta_i,x_{ij}=x\right)\right.\right.\\&\left.\left.-\mathbb{P}\left(y_{ij}\;\le\;f^*(x_{ij})+u\mid x_{ij}=x\right)\right)du\right)dx.\end{aligned}\tag{232}$$

Thus, we modify the function $\Pi(x,u)$ in (8) as follows:

$$\Pi(x,u) \coloneqq \frac{1}{|\mathcal{J}_{e,l}|}\sum_{i\in\mathcal{J}_{e,l}} \left[\mathbb{P}(y_{i1} \leq f^*(x_{i1})+u \mid x_{i1}=x) - \mathbb{P}(y_{i1} \leq f^*(x_{i1})+u \mid \delta_i, x_{i1}=x)\right]. \tag{233}$$

Similarly, as (104), we then have that

$$\begin{aligned}
&\left|\int_{[0,1]^d}\int_0^{f(x)-f^*(x)} \Pi(x,u)\,du\,dx\right| \\
&\leq \int_{[0,1]^d} |f(x)-f^*(x)| \cdot \max_{x\in[0,1]^d}\max_{u\in[-1,1]} |\Pi(x,u)| \cdot \mathbb{I}_{[|f(x)-f^*(x)|\leq 1]}\,dx \\
&\quad + \int_{[0,1]^d} |f(x)-f^*(x)| \cdot \max_{x\in[0,1]^d}\max_{u\in[-|f(x)-f^*(x)|,|f(x)-f^*(x)|]} |\Pi(x,u)| \\
&\quad \cdot \mathbb{I}_{[|f(x)-f^*(x)|>1]}\,dx
\end{aligned}$$

Now, for the $\max_{x\in[0,1]^d}\max_{u\in[-|f(x)-f^*(x)|,|f(x)-f^*(x)|]}|\Pi(x,u)|$ term above, we have

$$\begin{aligned}
&\max_{x\in[0,1]^d}\max_{u}\left|\frac{1}{|\mathcal{J}_{e,l}|}\sum_{i\in\mathcal{J}_{e,l}}\Bigg[\mathbb{P}(y_{i1} \leq f^*(x_{i1})+u \mid x_{i1}=x)\right. \\
&\qquad \left. - \mathbb{P}(y_{i1} \leq f^*(x_{i1})+u \mid \delta_i, x_{i1}=x)\Bigg]\right|.
\end{aligned}$$

Let $T_n > 0$ be a constant that depends on n, then we have

$$\begin{aligned}
&\max_{u>T_n}\max_{x\in[0,1]^d}\left|\frac{1}{|\mathcal{J}_{e,l}|}\sum_{i\in\mathcal{J}_{e,l}}\left[\mathbb{P}(\delta_i(x_{i1})+e_{i1} \leq u \mid \delta_i, x_{i1}=x) - \mathbb{P}(\delta_i(x_{i1})+e_{i1} \leq u \mid x_{i1}=x)\right]\right| \\
&\overset{(a)}{\leq} \tilde{c}_0 \text{ with high probability.}
\end{aligned}$$

The (a) holds by Assumption 4.

Similarly, we have

$$\begin{aligned}
&\max_{u<-T_n}\max_{x\in[0,1]^d}\left|\frac{1}{|\mathcal{J}_{e,l}|}\sum_{i\in\mathcal{J}_{e,l}}\left[\mathbb{P}(\delta_i(x_{i1})+e_{i1} \leq u \mid \delta_i, x_{i1}=x) - \mathbb{P}(\delta_i(x_{i1})+e_{i1} \leq u \mid x_{i1}=x)\right]\right| \\
&\overset{(a)}{\leq} \tilde{c}_0 \text{ with high probability.}
\end{aligned}$$

The (a) holds by Assumption 4.

Next, for $u \in [-T_n, T_n]$, we can follow the proof for (105) to get,

$$\max_{x_S\in[0,1]^s}\max_{u\in[-T_n,T_n]} |\Pi(x,u)| \leq c_3\sqrt{\frac{s\log n\log T_n}{n}} = O_p\left(\sqrt{\frac{s\log n\log T_n}{n}}\right).$$

where the c_3 is a constant that depends on c_1 and C_β.

Returning to (231), we obtain:

$$
\begin{aligned}
&\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}|\mathbb{E}[u_{ij}(f)|\delta_i]| \\
&\leq \frac{c_2}{L}\Bigg|\frac{1}{|\mathcal{J}_{e,l}|}\sum_{i\in\mathcal{J}_{e,l}}\int_{[0,1]^d}\int_0^{f(x)-f^*(x)}\Big[\mathbb{P}(y_{i1}\leq f^*(x_{i1})+u\mid\delta_i,x_{i1}=x) \\
&\qquad\qquad -\mathbb{P}(y_{i1}\leq f^*(x_{i1})+u\mid x_{i1}=x)\Big]du\,dx\Bigg| \\
&= \frac{c_2}{L}\Bigg|\frac{1}{|\mathcal{J}_{e,l}|}\sum_{i\in\mathcal{J}_{e,l}}\int_{[0,1]^d}\int_0^{f(x)-f^*(x)}\Big[\mathbb{P}(y_{i1}\leq f^*(x_{i1})+u\mid x_{i1}=x) \\
&\qquad\qquad -\mathbb{P}(y_{i1}\leq f^*(x_{i1})+u\mid\delta_i,x_{i1}=x)\Big]du\,dx\Bigg| \\
&\leq \frac{2\tilde{c}_0c_2}{L}\Delta^2(f-f^*)+\frac{c_2}{L}\frac{c_3^2}{4\tilde{c}_0}\frac{s\log n\log T_n}{n}.
\end{aligned}
$$

Thus, similar to (106), we get

$$
\begin{aligned}
&\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*,s}\left[\left(\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\leq\tilde{t}^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*,s}\left[u_{ij}(f)|\delta_i\right]\right)\cdot\mathbf{1}_{\Omega_\beta}\right] \\
&\leq \frac{2\tilde{c}_0c_2}{L}\Delta^2(f-f^*)+\frac{c_2}{L}\frac{c_3^2}{4\tilde{c}_0}\frac{s\log n\log T_n}{n}.
\end{aligned}
$$

By letting $\ell(f)=4\tilde{c}_0\Delta^2(f-f^*)$, we have

$$
\begin{aligned}
&E_6:=\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\left(\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\leq\tilde{t}^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\mathbb{E}_{x,e|\delta_i}\left[u_{ij}(f)\right]-\frac{\ell(f)}{2L}\right)\cdot\mathbf{1}_{\Omega_\beta}\right] \\
&\leq \frac{c_2}{L}\frac{c_3^2}{4\tilde{c}_0}\frac{s\log n\log T_n}{n}.
\end{aligned}
$$

Step 4: Back to (228), we have

$$
\begin{aligned}
&\tilde{E}_1:=\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\left(\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\leq\tilde{t}^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\tilde{Z}_{ij}(f)-Z_{ij}(f)-\frac{\ell(f)}{2L}\right)\cdot\mathbf{1}_{\Omega_\beta}\right] \\
&\leq 2\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\left(\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\leq\tilde{t}^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}s_{ij}u_{ij}(f)\right)\cdot\mathbf{1}_{\Omega_\beta}\right]+\frac{c_4}{L}\frac{s\log n\log T_n}{n} \\
&\overset{(a)}{\leq} 2\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\left(\sup_{\substack{f\in\mathrm{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\leq\tilde{t}^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}s_{ij}f(x_{ij}^*)\right)\cdot\mathbf{1}_{\Omega_\beta}\right]+\frac{c_4}{L}\frac{s\log n\log T_n}{n}
\end{aligned}
$$

$$\overset{\text{(b)}}{\leq} \frac{c_5}{L} R_{\mathcal{J}_{e,l}}(\text{Star}(\mathcal{F}, f^*) \cap \{c_0 \Delta^2(f - f^*) \leq \tilde{t}^2\}) + \frac{c_4}{L} \frac{s \log n \log T_n}{n}.$$

Where (a) follows the Lipschitz condition of $u_{ij}(f)$ in Lemma 19, and then apply the Ledoux-Talagrand contraction principle in Theorem 62, and apply the Lemma 25. Finally, we get the conclusion. ■

Corollary 59 (Bound of $\tilde{E}_3$, $\tilde{F}_3$) *Let Ω_β be defined in (76). Let $\bar{Z}(f)$, $Z(f)$ been defined in (69) and (67), let $\{s_{i,j}\}_{i\in\mathcal{J}_{e,l}, j\in[m_i]}$ be independent Rademacher variables, independent of the data $\{(x_{ij}, e_{ij}, \delta_i)\}$ and their coupled copies $\{(x^*_{ij}, e^*_{ij}, \delta^*_i)\}$, let $\tilde{E}_3$ be the quantity introduced in expression (222), then we have*

$$\begin{aligned}
\tilde{E}_3 \;&=\; \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\left(\sup_{\substack{f\in Star(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\leq\tilde{t}^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\left\{Z_{ij}(f) - \bar{Z}_{ij}(f)\right\}\right)\cdot \mathbf{1}_{\Omega_\beta}\right] \\
&\leq 2\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\left(\sup_{\substack{f\in Star(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\leq\tilde{t}^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i} s_{ij}Z_{ij}(f)\right)\cdot \mathbf{1}_{\Omega_\beta}\right] \\
&= \frac{c_5}{L} R_{\mathcal{J}_{e,l}}(Star(\mathcal{F}, f^*) \cap \{c_0\Delta^2(f-f^*) \leq \tilde{t}^2\}).
\end{aligned}$$

Similarly, for $\tilde{F}_3$ in (223) we have

$$\tilde{F}_3 \;\leq\; \frac{c_5}{L} R_{\mathcal{J}_{o,l}}(Star(\mathcal{F}, f^*) \cap \{c_0\Delta^2(f-f^*) \leq \tilde{t}^2\}).$$

Proof

Let $\{x^*_{ij}\}_{i\in\mathcal{J}_{e,l}, j\in[m_i]}$ be an independent and identically distributed copy of $\{x_{ij}\}_{i\in\mathcal{J}_{e,l}, j\in[m_i]}$, then we define the independent and identically distributed copy of $Z_{ij}(f)$ as

$$Z^*_{ij}(f) \;:=\; \mathbb{E}\left[\rho_\tau(f^*(x^*_{ij}) + \delta_i(x^*_{ij}) + e_{ij} - f(x^*_{ij})) - \rho_\tau(\delta_i(x^*_{ij}) + e_{ij}) \mid x^*_{ij}\right].$$

Now we control the $\mathbb{E}\left[Z_{ij}(f) - Z^*_{ij}(f)\right]$, the expectation takes on x^*_{ij}. So we have

$$\begin{aligned}
&\sup_{\substack{f\in \text{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\leq\tilde{t}^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\left\{\mathbb{E}\left[Z_{ij}(f)\right] - \mathbb{E}\left[Z^*_{ij}(f)\right]\right\} \\
&\overset{\text{(a)}}{\leq} \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*,s}\left[\left(\sup_{\substack{f\in \text{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\leq\tilde{t}^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\left\{Z_{ij}(f) - Z^*_{ij}(f)\right\}\right)\cdot \mathbf{1}_{\Omega_\beta}\right].
\end{aligned} \tag{234}$$

The equality (a) is held by Jensen's inequality. Then, the left-hand side of the above, can be further written as

$$\begin{aligned}
&\sup_{\substack{f\in \text{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\leq\tilde{t}^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\left\{\mathbb{E}\left[Z_{ij}(f)\right] - \mathbb{E}\left[Z^*_{ij}(f)\right]\right\} \\
&\overset{\text{(a)}}{=} \sup_{\substack{f\in \text{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\leq\tilde{t}^2}} \frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\left\{Z_{ij}(f) - \bar{Z}_{ij}(f)\right\}.
\end{aligned} \tag{235}$$

The equality (a) holds since $\bar{Z}_{ij}(f)$ and $\mathbb{E}\left[Z^*_{ij}(f)\right]$ are the same quantity.

Combining (234) and (235), taking expectation of x_{ij} we have

$$
\begin{aligned}
&\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\left(\sup_{\substack{f\in\text{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\left\{Z_{ij}(f)-\bar{Z}_{ij}(f)\right\}\right)\cdot\mathbf{1}_{\Omega_\beta}\right]\\
&\le \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*,s}\left[\left(\sup_{\substack{f\in\text{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\left\{Z_{ij}(f)-Z^*_{ij}(f)\right\}\right)\cdot\mathbf{1}_{\Omega_\beta}\right]\\
&\overset{\text{(a)}}{=} \mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\left(\sup_{\substack{f\in\text{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}s_{ij}\left\{Z_{ij}(f)-Z^*_{ij}(f)\right\}\right)\cdot\mathbf{1}_{\Omega_\beta}\right]\\
&\overset{\text{(b)}}{\le} 2\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*}\left[\left(\sup_{\substack{f\in\text{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}s_{ij}Z_{ij}(f)\right)\cdot\mathbf{1}_{\Omega_\beta}\right].
\end{aligned}
$$

The (a) holds since the $\left\{Z_{ij}(f)-Z^*_{ij}(f)\right\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$ is independently distributed and symmetric with mean zero and $\left\{s_{ij}(Z_{ij}(f)-Z^*_{ij}(f))\right\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$ and $\left\{Z_{ij}(f)-Z^*_{ij}(f)\right\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$ have the same distribution. The (b) follows because $\{-s_{ij}\}_{i\in\mathcal{J}_{e,l},j\in[m_i]}$ are also independent Rademacher variables.

We apply the Lipschitz condition of $\hat{Z}_{ij}(f)$, $Z_{ij}(f)$ and $r_{ij}(f)$ in Lemma 19, and then apply the Ledoux-Talagrand contraction principle in Theorem 62.

Then we have

$$
\begin{aligned}
&\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*,s}\left[\left(\sup_{\substack{f\in\text{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}\left\{\bar{Z}_{ij}(f)-Z_{ij}(f)\right\}\right)\cdot\mathbf{1}_{\Omega_\beta}\right]\\
&\le 2\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*,s}\left[\left(\sup_{\substack{f\in\text{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}s_{ij}Z_{ij}(f)\right)\cdot\mathbf{1}_{\Omega_\beta}\right]\\
&\le 2\mathbb{E}_{x,\delta,e,x^*,\delta^*,e^*,s}\left[\left(\sup_{\substack{f\in\text{Star}(\mathcal{F},f^*)\\ c_0\Delta^2(f-f^*)\le\tilde{t}^2}}\frac{1}{n}\sum_{i\in\mathcal{J}_{e,l}}\frac{1}{m_i}\sum_{j=1}^{m_i}s_{ij}f(x^*_{ij})\right)\cdot\mathbf{1}_{\Omega_\beta}\right]\\
&= \frac{c_5}{L}R_{\mathcal{J}_{e,l}}(\text{Star}(\mathcal{F},f^*)\cap\left\{c_0\Delta^2(f-f^*)\le\tilde{t}^2\right\}).
\end{aligned}
\tag{236}
$$

Where the last equality follows from Lemma 25. ∎

Lemma 60 (Bound of A_1, A_2, and A_3 by Rademacher width) *Let Ω_β be defined in (76). Let $\hat{f}$, belong to $\mathcal{F}$, and f^* not necessarily in $\mathcal{F}$, where $\mathcal{F}$ is not necessarily convex. Let the star-shaped set $Star(\mathcal{F}, f^*)$ be defined in (26). Suppose the distribution of $\{(x_{ij}, y_{i,j})\}_{i=1,j=1}^{n,m_i}$ obey Assumption 2, where c_0 satisfies (84). Let $R(\mathcal{F})$ be the Rademacher width defined in 12, let $\tilde{c}_0$ be a small constant given in Assumption 4. Let f_{Proj} of f^* in $\mathcal{F}$ such that*

$$\|f_{Proj} - f^*\|_\infty \le \phi_{nm}$$

Let

$$\overset{\approx}{t}{}^2 := t^2 + 2\left(C_0 + \frac{\bar{f}}{2}\right)\phi_{nm}^2, \quad \tilde{t}^2 = \overset{\approx}{t}{}^2 + \left(C_0 + \frac{\bar{f}}{2}\right)\phi_{nm}^2.$$

Then we have

$$\begin{aligned}
&\mathbb{P}\left(c_0\Delta^2(\hat{f} - f^*) > \tilde{t}^2\right) \\
&\le \frac{24c_5\tilde{t}^2}{\overset{\approx}{t}{}^2}\frac{1}{\tilde{t}^2}\left[\sum_{l=1}^{L}\left(R_{\mathcal{J}_{e,l}}\left(Star(\mathcal{F}, f^*) \cap \left\{f : \|f - f^*\|^2 \le c_0^{-1}\cdot\max\{B,1\}\cdot\tilde{t}^2\right\}\right)\right.\right. \\
&\left.\left.+R_{\mathcal{J}_{o,l}}\left(Star(\mathcal{F}, f^*) \cap \left\{f : \|f - f^*\|^2 \le c_0^{-1}\cdot\max\{B,1\}\cdot\tilde{t}^2\right\}\right)\right)\right] \\
&\quad + \frac{c_{12}}{\overset{\approx}{t}{}^2}\frac{s\log n\log T_n}{n}.
\end{aligned}$$

where $b(B) = \max\{B^{1/2}, 1\}$ is a constant that depends on B, and c is a positive constant that depends on $\underline{f}$.

Proof By (224), on event Ω, we have

$$\begin{aligned}
&\mathbb{P}\left(c_0\Delta^2(\hat{f} - f^*) > \tilde{t}^2\right) \\
&\le \sum_{l=1}^{L}\frac{4L}{\overset{\approx}{t}{}^2}\left(\tilde{A}_1 + \tilde{A}_2 + \tilde{A}_3 + \tilde{B}_1 + \tilde{B}_2 + \tilde{B}_3 + \tilde{E}_1 + \tilde{E}_2 + \tilde{E}_3 + \tilde{F}_1 + \tilde{F}_2 + \tilde{F}_3\right).
\end{aligned}$$

Then by Corollary 54, 55, 56, 57, 58, 59, we get the the following result.

$$\begin{aligned}
&\mathbb{P}\left(c_0\Delta^2(\hat{f} - f^*) > \tilde{t}^2\right) \\
&\le \frac{24c_5}{\overset{\approx}{t}{}^2}\sum_{l=1}^{L}\left(R_{\mathcal{J}_{e,l}}(\mathrm{Star}(\mathcal{F}, f^*) \cap \{c_0\Delta^2(f - f^*) \le \tilde{t}^2\})\right. \\
&\qquad \left.+ R_{\mathcal{J}_{o,l}}(\mathrm{Star}(\mathcal{F}, f^*) \cap \{c_0\Delta^2(f - f^*) \le \tilde{t}^2\})\right) \\
&\quad + \frac{16c_4}{\overset{\approx}{t}{}^2}\frac{s\log n\log T_n}{n}.
\end{aligned} \tag{237}$$

Based on the results in Lemma 60 and Lemma 16, we have on event $\Omega \cup \Omega^c$,

$$\begin{aligned}
&\mathbb{P}\left(c_0\Delta^2(\hat{f}-f^*) > \tilde{t}^2\right) \\
&\leq \frac{24c_5}{\overset{\approx}{t}^2}\sum_{l=1}^{L}\Big(R_{\mathcal{J}_{e,l}}(\mathrm{Star}(\mathcal{F},f^*)\cap\left\{c_0\Delta^2(f-f^*)\leq\tilde{t}^2\right\}) \\
&\quad + R_{\mathcal{J}_{o,l}}(\mathrm{Star}(\mathcal{F},f^*)\cap\left\{c_0\Delta^2(f-f^*)\leq\tilde{t}^2\right\})\Big) \\
&\quad + \frac{16c_4}{t^2}\frac{s\log n\log T_n}{n} + \frac{1}{\overset{\approx}{t}^2}\frac{3}{n}. \\
&\leq \frac{24c_5}{\overset{\approx}{t}^2}\sum_{l=1}^{L}\Big(R_{\mathcal{J}_{e,l}}(\mathrm{Star}(\mathcal{F},f^*)\cap\left\{c_0\Delta^2(f-f^*)\leq\tilde{t}^2\right\}) \\
&\quad + R_{\mathcal{J}_{o,l}}(\mathrm{Star}(\mathcal{F},f^*)\cap\left\{c_0\Delta^2(f-f^*)\leq\tilde{t}^2\right\})\Big) \\
&\quad + \frac{c_{12}}{\overset{\approx}{t}^2}\frac{s\log n\log T_n}{n}.
\end{aligned} \tag{238}$$

By Lemma 35, we have

$$\begin{aligned}
&R_{\mathcal{J}_{e,l}}\left(\mathrm{Star}(\mathcal{F},f^*)\cap\left\{c_0\Delta^2(f-f^*)\leq\tilde{t}^2\right\}\right) \\
&\leq R_{\mathcal{J}_{e,l}}\left(\mathrm{Star}(\mathcal{F},f^*)\cap\left\{f:\|f-f^*\|^2\leq c_0^{-1}\cdot\max\{B,1\}\cdot\tilde{t}^2\right\}\right).
\end{aligned}$$

and we have

$$\begin{aligned}
&R_{\mathcal{J}_{o,l}}\left(\mathrm{Star}(\mathcal{F},f^*)\cap\left\{c_0\Delta^2(f-f^*)\leq\tilde{t}^2\right\}\right) \\
&\leq R_{\mathcal{J}_{o,l}}\left(\mathrm{Star}(\mathcal{F},f^*)\cap\left\{f:\|f-f^*\|^2\leq c_0^{-1}\cdot\max\{B,1\}\cdot\tilde{t}^2\right\}\right).
\end{aligned}$$

Thus, we get

$$\begin{aligned}
&\mathbb{P}\left(c_0\Delta^2(\hat{f}-f^*) > \tilde{t}^2\right) \\
&\leq \frac{24c_5\tilde{t}^2}{\overset{\approx}{t}^2}\frac{1}{\tilde{t}^2}\left[\sum_{l=1}^{L}\left(R_{\mathcal{J}_{e,l}}\left(\mathrm{Star}(\mathcal{F},f^*)\cap\left\{f:\|f-f^*\|^2\leq c_0^{-1}\cdot\max\{B,1\}\cdot\tilde{t}^2\right\}\right)\right.\right. \\
&\left.\left.+R_{\mathcal{J}_{o,l}}\left(\mathrm{Star}(\mathcal{F},f^*)\cap\left\{f:\|f-f^*\|^2\leq c_0^{-1}\cdot\max\{B,1\}\cdot\tilde{t}^2\right\}\right)\right)\right] \\
&\quad + \frac{c_{12}}{\overset{\approx}{t}^2}\frac{s\log n\log T_n}{n}.
\end{aligned}$$

■

E.2 Proof of Theorem 7

Proof : From Lemma 60, we get

$$\mathbb{P}\left(c_0\Delta^2(\hat{f}-f^*) > \tilde{t}^2\right)$$

$$
\begin{aligned}
&\leq \frac{24c_5\tilde{t}^2}{\underset{t}{\approx}^2}\frac{1}{\tilde{t}^2}\left[\sum_{l=1}^{L}\left(R_{\mathcal{J}_{e,l}}\left(\mathrm{Star}(\mathcal{F},f^*)\cap\left\{f:\|f-f^*\|^2\leq c_0^{-1}\cdot\max\{B,1\}\cdot\tilde{t}^2\right\}\right)\right.\right.\\
&\left.\left.+R_{\mathcal{J}_{o,l}}\left(\mathrm{Star}(\mathcal{F},f^*)\cap\left\{f:\|f-f^*\|^2\leq c_0^{-1}\cdot\max\{B,1\}\cdot\tilde{t}^2\right\}\right)\right)\right]\\
&\quad+\frac{c_{12}}{\underset{t}{\approx}^2}\frac{s\log n\log T_n}{n}.
\end{aligned}
$$

Then by Corollary 38, we have

$$
\Delta^2(\hat{f}-f^*)\;=\;O_p\left(\tilde{t}^2\right), \tag{239}
$$

where

$$
\begin{aligned}
\tilde{t}^2\asymp\sum_{l=1}^{L}&\left(R_{\mathcal{J}_{e,l}}\left(\mathrm{Star}(\mathcal{F},f^*)\cap B_2\left(c_0^{-\frac{1}{2}}\cdot b(B)\cdot\tilde{t}\right)\right)\right.\\
&\left.+R_{\mathcal{J}_{o,l}}\left(\mathrm{Star}(\mathcal{F},f^*)\cap B_2\left(c_0^{-\frac{1}{2}}\cdot b(B)\cdot\tilde{t}\right)\right)\right)\\
&+2\left(C_0+\frac{\bar{f}}{2}\right)\phi_{nm}^2+\frac{s\log^{\mu+1}(n)\log T_n}{n},
\end{aligned}
$$

for some $\mu>0$.

With such choice of $\tilde{t}^2$, suppose

$$
\begin{aligned}
&\frac{1}{\tilde{t}^2}\left[\sum_{l=1}^{L}\left(R_{\mathcal{J}_{e,l}}\left(\mathrm{Star}(\mathcal{F},f^*)\cap\left\{f:\|f-f^*\|^2\leq c_0^{-1}\cdot\max\{B,1\}\cdot\tilde{t}^2\right\}\right)\right.\right.\\
&\left.\left.+R_{\mathcal{J}_{o,l}}\left(\mathrm{Star}(\mathcal{F},f^*)\cap\left\{f:\|f-f^*\|^2\leq c_0^{-1}\cdot\max\{B,1\}\cdot\tilde{t}^2\right\}\right)\right)\right]=O(\gamma_n),
\end{aligned}
$$

with γ_n is a sequence goes to zero with such $\tilde{t}^2$. Then we have

$$
\begin{aligned}
&\frac{24c_5\tilde{t}^2}{\underset{t}{\approx}^2}\frac{1}{\tilde{t}^2}\left[\sum_{l=1}^{L}\left(R_{\mathcal{J}_{e,l}}\left(\mathrm{Star}(\mathcal{F},f^*)\cap\left\{f:\|f-f^*\|^2\leq c_0^{-1}\cdot\max\{B,1\}\cdot\tilde{t}^2\right\}\right)\right.\right.\\
&\left.\left.+R_{\mathcal{J}_{o,l}}\left(\mathrm{Star}(\mathcal{F},f^*)\cap\left\{f:\|f-f^*\|^2\leq c_0^{-1}\cdot\max\{B,1\}\cdot\tilde{t}^2\right\}\right)\right)\right]\\
&\quad+\frac{c_{12}}{\underset{t}{\approx}^2}\frac{s\log n\log T_n}{n}=O\left(\gamma_n+\frac{\left(C_0+\frac{\bar{f}}{2}\right)\phi_{nm}^2}{\underset{t}{\approx}^2}\gamma_n+\gamma_n^{\mu}\right)\;=\;O\left(\gamma_n+\gamma_n^{\mu}\right).
\end{aligned}
$$

Thus, it holds that

$$
\begin{aligned}
\Delta^2(\hat{f}-f^*)\;=\;O_p&\left(\sum_{l=1}^{L}\left(R_{\mathcal{J}_{e,l}}\left(\mathrm{Star}(\mathcal{F},f^*)\cap\left\{f:\|f-f^*\|\;\leq\;c_0^{-\frac{1}{2}}\cdot b(B)\cdot\tilde{t}\right\}\right)\right.\right.\\
&+R_{\mathcal{J}_{o,l}}\left(\mathrm{Star}(\mathcal{F},f^*)\cap\left\{f:\|f-f^*\|\;\leq\;c_0^{-\frac{1}{2}}\cdot b(B)\cdot\tilde{t}\right\}\right)\\
&\left.\left.+2\left(C_0+\frac{\bar{f}}{2}\right)\phi_{nm}^2+\frac{s\cdot\mathrm{poly}(\log n)\log(T_n)}{n}\right)\right),\;\text{ and}
\end{aligned}
$$

$$\left\|\hat{f} - f^*\right\|_2^2 = O_p\left(b^2(B) \cdot \sum_{l=1}^{L}\left(R_{\mathcal{J}_{e,l}}\left(\mathrm{Star}(\mathcal{F}, f^*) \cap \left\{f : \|f - f^*\| \leq c_0^{-\frac{1}{2}} \cdot b(B) \cdot \tilde{t}\right\}\right)\right.\right.$$
$$+R_{\mathcal{J}_{o,l}}\left(\mathrm{Star}(\mathcal{F}, f^*) \cap \left\{f : \|f - f^*\| \leq c_0^{-\frac{1}{2}} \cdot b(B) \cdot \tilde{t}\right\}\right)$$
$$\left.\left.+b^2(B)2\left(C_0 + \frac{\bar{f}}{2}\right)\phi_{nm}^2 + \frac{b^2(B) \cdot s \cdot \mathrm{poly}(\log n)\log(T_n)}{n}\right)\right).$$

Appendix F. Proof of Theorem 10

Corollary 61 (Neural Network When $f^* \notin \mathcal{F}$ and $\mathcal{F}$ is not Convex) *, Let $F(\mathcal{L}, r)$ defined as in (2) where $\sigma : \mathbb{R} \to \mathbb{R}$ with $\sigma(x) = \max\{x, 0\}$, let Star$(F(\mathcal{L}, r), f^*)$ follow the definition in (26) where f^* not necessarily in $F(\mathcal{L}, r)$, let Let f_{Proj} in $\mathcal{F}$ such that*

$$\|f_{Proj} - f^*\|_\infty \leq \phi_{nm}$$

Let

$$\tilde{t}^2 := t^2 + 2\left(C_0 + \frac{\bar{f}}{2}\right)\phi_{nm}^2,$$

then we have

$$\Delta^2(\hat{f} - f^*) = O_p\Bigg(\frac{\max\{B, 1\}\log^3(n) \cdot \log(\mathcal{L} \cdot r^2) \cdot \mathcal{L}^2 \cdot r^2}{nm} + 2\left(C_0 + \frac{\bar{f}}{2}\right)\phi_{nm}^2 + \frac{s \cdot \mathit{poly}(\log n)\log(T_n)}{n}\Bigg). \tag{240}$$

Proof

We analyze

$$\frac{R_{\mathcal{J}_{e,l}}\left(\mathcal{F} \cap \left\{c_0\Delta^2(f - f^*) \leq \tilde{t}^2\right\}\right) + R_{\mathcal{J}_{o,l}}\left(\mathcal{F} \cap \left\{c_0\Delta^2(f - f^*) \leq \tilde{t}^2\right\}\right)}{\tilde{\tilde{t}}^2},$$

Also, let $\tilde{b} = c^{-1/2}\max\left\{B^{1/2}, 1\right\} \cdot \tilde{t}$.

For any function class $\mathcal{F} \cap B_2(\tilde{b})$, we have

$$R_{\mathcal{J}_{e,l}}\left(\mathcal{F} \cap B_2(\tilde{b})\right) = \mathbb{E}\left[\mathbb{E}\Big[\sup_{g \in \mathcal{F} \cap B_2(b)} \frac{1}{\sum_{i \in \mathcal{J}_{e,l}} m_i} \sum_{i \in \mathcal{J}_{e,l}} \sum_{j=1}^{m_i} s_{ij} g(x_{ij}) | \{x_{ij}\}\Big]\right]$$

Denote

$$R_{\mathcal{J}_{elmn}}\left(\mathcal{F} \cap B_2(\tilde{b})\right) = \mathbb{E}\left[\sup_{g \in \mathcal{F} \cap B_2(\tilde{b})} \frac{1}{\sum_{i \in \mathcal{J}_{e,l}} m_i} \sum_{i \in \mathcal{J}_{e,l}} \sum_{j=1}^{m_i} s_{ij} g(x_{ij}) | \{x_{ij}\}\right] \tag{241}$$

Also denote by $\|g\|_{mn} = \left(\frac{1}{\sum_{i \in \mathcal{J}_{e,l}} m_i} \sum_{i \in \mathcal{J}_{e,l}} \sum_{j=1}^{m_i} g(x_{ij})^2\right)^{1/2}$

Corollary 2.2 of Bartlett et al. (2005) gives

$$\mathcal{F} \cap B_2(\tilde{b}) \subseteq \mathcal{F} \cap B_{mn}(\sqrt{2}\tilde{b}),$$

with probability at least $1 - \frac{1}{\sum_{i \in \mathcal{J}_{e,l}} m_i} \geq 1 - \frac{1}{C_\beta} \frac{2 \log n}{cc_1 nm}$, where c appears in Assumption 1.9. Define this good event as $\mathcal{E}$. On $\mathcal{E}$ using Dudley's entropy integral (see Dudley (1967) or Lemma 64), Equation (75), and the assumption $m_i \asymp m$ specified in the statement of the Lemma, with c defined in Assumption 1.9, we bound the empirical Rademacher complexity defined in Equation (241) as follows

The Rademacher complexity defined in Equation (241) is bounded by

$$\begin{aligned}
R_{\mathcal{J}_{elmn}}(\mathcal{F} \cap B_2(\tilde{b})) \;&\leq\; R_{\mathcal{J}_{e1mn}}\left(\left\{g \in F : ||g||^2_{mn} \;\leq\; 2\tilde{b}^2\right\}\right) \\
&\leq\; 12 \inf_{0<\alpha<\sqrt{2}\tilde{b}} \left[\alpha + \frac{1}{\sqrt{\sum_{i \in \mathcal{J}_{e,1} m_i}}} \int_\alpha^\delta \sqrt{\log N(t, || \cdot ||_{mn}, \mathcal{F})}\, dt\right] \\
&\leq\; 12 \inf_{0<\alpha<\sqrt{2}\tilde{b}} \left[\alpha + \frac{1}{\sqrt{C_\beta cc_1 nm}} \int_\alpha^\delta \sqrt{\log N(t, || \cdot ||_{mn}, \mathcal{F})}\, dt\right].
\end{aligned}$$

Under the event $\mathcal{E}^c$, given that $F \subseteq B_\infty(B)$, we have that $R_{\mathcal{J}_{e1mn}}(\mathcal{F} \cap B_2(b)) \leq B$. Therefore, splitting the expectation over $\mathcal{E}$ and $\mathcal{E}^c$,

Set that $\mathcal{F} \equiv \mathrm{Star}(F(\mathcal{L}, r), f^*)$, from Lemma 65, we obtain that.

$$\log N(\epsilon, \|\cdot\|_{nm}, \mathrm{Star}(F(\mathcal{L}, r), f^*)) \;\leq\; \log\left(\frac{2B}{\epsilon}\right) + \mathcal{L}^2 r^2 \log(\mathcal{L}r^2) \log(B\epsilon^{-1}).$$

Denote $\frac{1}{\sqrt{C_\beta cc_1}}$ as c_8.

$$\begin{aligned}
&R_{\mathcal{J}_{elmn}}(\mathrm{Star}(F(\mathcal{L}, r), f^*) \cap B_{nm}(\sqrt{2}\tilde{b})) \\
&\leq\; 12 \inf_{0<\alpha<\sqrt{2}\tilde{b}} \left[\alpha + \frac{c_8}{\sqrt{nm}} \int_\alpha^{\sqrt{2}\tilde{b}} \sqrt{\log N(\epsilon, || \cdot ||_{mn}, \mathrm{Star}(F(\mathcal{L}, r))}\, d\epsilon\right] \\
&\overset{(a)}{\leq}\; 12 \inf_{0<\alpha<\sqrt{2}\tilde{b}} \left[\alpha + \frac{c_8}{\sqrt{nm}} \int_\alpha^{\sqrt{2}\tilde{b}} \sqrt{\mathcal{L}^2 r^2 \log(\mathcal{L}r^2) \log(B\epsilon^{-1})} d\epsilon + \frac{c_8}{\sqrt{nm}} \int_\alpha^{\sqrt{2}\tilde{b}} \sqrt{c_1 \log\left(\frac{2B}{\epsilon}\right)} d\epsilon\right] \\
&\overset{(b)}{\leq}\; \frac{\tilde{b}^2}{mn} + \frac{12c_8}{\sqrt{mn}} \int_{\frac{\tilde{b}^2}{12\log(nm)}}^{\sqrt{2}\tilde{b}} \sqrt{\mathcal{L}^2 r^2 \log(\mathcal{L}r^2) \log(\epsilon^{-1})} d\epsilon + \frac{12c_8}{\sqrt{mn}} \int_{\frac{\tilde{b}^2}{12\log(nm)}}^{\sqrt{2}\tilde{b}} \sqrt{c_1 \log\left(\frac{2B}{\epsilon}\right)} d\epsilon \\
&\leq\; \frac{\tilde{b}^2}{\log(mn)} + \frac{12\sqrt{2}\tilde{b}c_8}{\sqrt{mn}} \sqrt{\mathcal{L}^2 r^2 \log(\mathcal{L}r^2) \log\left(\frac{12B\log(mn)}{\tilde{b}^2}\right)} + \frac{12\sqrt{2}\tilde{b}c_8}{\sqrt{mn}} \sqrt{c_1 \log\left(\frac{24B\log(mn)}{\tilde{b}^2}\right)} \\
&=\; \frac{\tilde{b}^2}{\log(mn)} + \frac{12\sqrt{2}\tilde{b}c_8\mathcal{L}r\sqrt{\log(\mathcal{L}r^2)\log\left(\frac{12B\log(mn)}{\tilde{b}^2}\right)}}{\sqrt{mn}} + \frac{12\sqrt{2}\tilde{b}c_8\sqrt{c_1\log\left(\frac{24B\log(mn)}{\tilde{b}^2}\right)}}{\sqrt{mn}} \\
&\leq\; \frac{\tilde{b}^2}{\log(mn)} + \frac{12\sqrt{2}\tilde{b}c_8c_9\mathcal{L}r(\log(\mathcal{L}r^2))^{1/2}\sqrt{\log\left(\frac{12B\log(mn)}{\tilde{b}^2}\right)}}{\sqrt{mn}}.
\end{aligned}$$

Where inequality (a) follows the Corollary 67 and (b) holds by choosing $\alpha = \frac{\tilde{b}^2}{12\log(mn)}$.

The Rademacher complexity defined in Equation (120) is bounded by

$$
\begin{aligned}
& R_{\mathcal{J}_{e,l}}(\mathrm{Star}(F(\mathcal{L},r),f^*)\cap B_2(b)) \\
& = \mathbb{E}[R_{\mathcal{J}_{e1mn}}(\mathrm{Star}(F(\mathcal{L},r),f^*)\cap B_2(b))] \\
& \leq \frac{\tilde{b}^2}{\log(nm)} + \frac{12\sqrt{2}\tilde{b}c_8c_9\mathcal{L}r(\log(\mathcal{L}r^2))^{1/2}\sqrt{\log\left(\frac{24B\log(mn)}{\tilde{b}^2}\right)}}{\sqrt{mn}} + \frac{2Bc_8^2\log(n)}{nm} \\
& = \frac{\tilde{b}^2}{\log(nm)} + \frac{c_8c_9\tilde{b}\mathcal{L}r(\log(\mathcal{L}r^2))^{1/2}\sqrt{\log\left(\frac{24B\log(nm)}{\tilde{b}^2}\right)}}{\sqrt{mn}} + \frac{2Bc_8^2\log(n)}{nm}
\end{aligned}
$$

Now, we see that

$$
\begin{aligned}
& \frac{R_{\mathcal{J}_{e,1}}\left(\mathcal{F}\cap\left\{c_0\Delta^2(f-f^*)\leq\tilde{t}^2\right\}\right)+R_{\mathcal{J}_{o,1}}\left(\mathcal{F}\cap\left\{c_0\Delta^2(f-f^*)\leq\tilde{t}^2\right\}\right)}{\overset{\approx}{t}^2} \\
& \leq \frac{\tilde{b}^2}{\overset{\approx}{t}^2\log(nm)} + \frac{c_8c_9\tilde{b}\mathcal{L}r(\log(\mathcal{L}r^2))^{1/2}\sqrt{\log\left(\frac{24B\log(nm)}{\tilde{b}^2}\right)}}{\overset{\approx}{t}^2\sqrt{nm}} + \frac{2Bc_8^2\log(n)}{\overset{\approx}{t}^2 nm} \\
& = \frac{\tilde{t}^2c_0^{-1}\max\{B,1\}}{\overset{\approx}{t}^2\log(nm)} + \frac{\tilde{t}^2}{\overset{\approx}{t}^2}\frac{c_8c_9c_0^{-1}\max\{B,1\}\,\tilde{b}\mathcal{L}r(\log(\mathcal{L}r^2))^{1/2}\sqrt{\log\left(\frac{24B\log(nm)}{\tilde{b}^2}\right)}}{c_0^{-1}\max\{B,1\}\,\tilde{t}^2\sqrt{nm}} + \frac{2Bc_8^2\log(n)}{\overset{\approx}{t}^2 nm} \\
& = \frac{\tilde{t}^2c_0^{-1}\max\{B,1\}}{\overset{\approx}{t}^2\log(nm)} + \frac{\tilde{t}^2}{\overset{\approx}{t}^2}\frac{c_8c_9c_0^{-1}\max\{B,1\}\,\tilde{b}\mathcal{L}r(\log(\mathcal{L}r^2))^{1/2}\sqrt{\log\left(\frac{24B\log(nm)}{\tilde{b}^2}\right)}}{\tilde{b}^2\sqrt{nm}} + \frac{2Bc_8^2\log(n)}{\overset{\approx}{t}^2 nm} \\
& = \frac{\tilde{t}^2c_0^{-1}\max\{B,1\}}{\overset{\approx}{t}^2\log(nm)} + \frac{\tilde{t}^2}{\overset{\approx}{t}^2}\frac{c_{10}\max\{B,1\}\,\mathcal{L}r(\log(\mathcal{L}r^2))^{1/2}\sqrt{\log\left(\frac{24B\log(nm)}{\tilde{b}^2}\right)}}{\tilde{b}\sqrt{nm}} + \frac{2Bc_8^2\log(n)}{\overset{\approx}{t}^2 nm}
\end{aligned}
\tag{242}
$$

If $B \geq 1$, from Inequality (242), we get

$$
\begin{aligned}
& \frac{R_{\mathcal{J}_{e,1}}\left(\mathcal{F}\cap\left\{c_0\Delta^2(f-f^*)\leq\tilde{t}^2\right\}\right)+R_{\mathcal{J}_{o,1}}\left(\mathcal{F}\cap\left\{c_0\Delta^2(f-f^*)\leq\tilde{t}^2\right\}\right)}{\overset{\approx}{t}^2} \\
& \leq \frac{\tilde{t}^2c_0^{-1}\max\{B,1\}}{\overset{\approx}{t}^2\log(nm)} + \frac{\tilde{t}^2}{\overset{\approx}{t}^2}\frac{c_{10}\max\{B,1\}\,\mathcal{L}r(\log(\mathcal{L}r^2))^{1/2}\sqrt{\log\left(\frac{24B\log(nm)}{\tilde{b}^2}\right)}}{\tilde{b}\sqrt{nm}} + \frac{2Bc_8^2\log(n)}{\overset{\approx}{t}^2 nm}.
\end{aligned}
\tag{243}
$$

Then we let

$$
\tilde{b}^2 = \frac{c_{11}B^2\log^2(n)\cdot\log(\mathcal{L}\cdot r^2)\cdot\mathcal{L}^2\cdot r^2}{nm}
$$

The (243) is $O\left(\frac{\tilde{t}^2}{\overset{\approx}{t}^2}\frac{1}{\log(n)}\right)$.

Furthermore, we know $\tilde{b}^2 = c_0^{-1}\max\{B,1\}\,\tilde{t}^2 = c_0^{-1}\cdot B\cdot\tilde{t}^2$ when $B>1$, so we get

$$\tilde{t}^2 = \frac{c_{12}B\log^2(n)\cdot\log(\mathcal{L}\cdot r^2)\cdot\mathcal{L}^2\cdot r^2}{nm}.$$

If $B<1$, from Inequality (242),

$$\begin{aligned}
&\frac{R_{\mathcal{J}_{e,1}}\left(\mathcal{F}\cap\left\{c_0\Delta^2(f-f^*)\le\tilde{t}^2\right\}\right)+R_{\mathcal{J}_{o,1}}\left(\mathcal{F}\cap\left\{c_0\Delta^2(f-f^*)\le\tilde{t}^2\right\}\right)}{\overset{\approx}{t}^2}\\
&\le\frac{\tilde{t}^2c_0^{-1}\max\{B,1\}}{\overset{\approx}{t}^2\log(nm)}+\frac{\tilde{t}^2}{\overset{\approx}{t}^2}\frac{c_{10}\max\{B,1\}\,\mathcal{L}r(\log(\mathcal{L}r^2))^{1/2}\sqrt{\log\left(\frac{24B\log(nm)}{\tilde{b}^2}\right)}}{\tilde{b}\sqrt{nm}}+\frac{2Bc_8^2\log(n)}{\overset{\approx}{t}^2nm}.
\end{aligned}\tag{244}$$

then we let

$$\tilde{b}^2 = \frac{c_{11}\log^2(n)\cdot\log(\mathcal{L}\cdot r^2)\cdot\mathcal{L}^2\cdot r^2}{nm}$$

The (244) is $O\left(\frac{\tilde{t}^2}{\overset{\approx}{t}^2}\frac{1}{\log(n)}\right)$.

Furthermore, we know $\tilde{b}^2 = c_0^{-1}\max\{B,1\}\,\tilde{t}^2 = c_0^{-1}\cdot\tilde{t}^2$ when $B\le 1$, so we get

$$\tilde{t}^2 = \frac{c_{13}\log^2(n)\cdot\log(\mathcal{L}\cdot r^2)\cdot\mathcal{L}^2\cdot r^2}{nm},\ \text{when } B\le 1.$$

So we can let

$$\tilde{t}^2 = \frac{c_{14}\max\{B,1\}\log^2(n)\cdot\log(\mathcal{L}\cdot r^2)\cdot\mathcal{L}^2\cdot r^2}{nm}$$

to get

$$\frac{R_{\mathcal{J}_{e,1}}\left(\mathcal{F}\cap\left\{c_0\Delta^2(f-f^*)\le\tilde{t}^2\right\}\right)+R_{\mathcal{J}_{o,1}}\left(\mathcal{F}\cap\left\{c_0\Delta^2(f-f^*)\le\tilde{t}^2\right\}\right)}{\overset{\approx}{t}^2}=O\left(\frac{\tilde{t}^2}{\overset{\approx}{t}^2}\frac{1}{\log(n)}\right).\tag{245}$$

Then by Theorem 7, we know

$$\begin{aligned}
&\mathbb{P}\left(c_0\Delta^2(\hat{f}-f^*)>\tilde{t}^2\right)\\
&\le\frac{24c_5}{\overset{\approx}{t}^2}\sum_{l=1}^{L}\Big(R_{\mathcal{J}_{e,l}}(\mathrm{Star}(\mathcal{F},f^*)\cap\left\{c_0\Delta^2(f-f^*)\le\tilde{t}^2\right\})\\
&\quad+R_{\mathcal{J}_{o,l}}(\mathrm{Star}(\mathcal{F},f^*)\cap\left\{c_0\Delta^2(f-f^*)\le\tilde{t}^2\right\})\Big)\\
&\quad+\frac{c_{12}}{\overset{\approx}{t}^2}\frac{s\log n\log T_n}{n},
\end{aligned}\tag{246}$$

and then by (245) and the choice of $L=\left\lceil\frac{1}{C_\beta}2\log(n)\right\rceil$ in (73), We can let

$$\tilde{t}^2 = \frac{c_{15} \max\{B, 1\} \log^3(n) \cdot \log(\mathcal{L} \cdot r^2) \cdot \mathcal{L}^2 \cdot r^2}{nm}$$

to make

$$\begin{aligned} &\frac{24c_5}{\approx^2_t} \sum_{l=1}^{L} \Big(R_{\mathcal{J}_{e,l}}(\mathrm{Star}(\mathcal{F}, f^*) \cap \{c_0 \Delta^2(f - f^*) \le \tilde{t}^2\}) \\ &\quad + R_{\mathcal{J}_{o,l}}(\mathrm{Star}(\mathcal{F}, f^*) \cap \{c_0 \Delta^2(f - f^*) \le \tilde{t}^2\}) \Big) \\ &= O\left(\frac{\tilde{t}^2}{\approx^2_t} \frac{1}{\log(n)} \right). \end{aligned}$$

Furthermore, we know

$$\tilde{t}^2 = \overset{\approx}{t}^2 + \left(C_0 + \frac{\bar{f}}{2} \right) \phi^2_{nm},$$

so we can let

$$\overset{\approx}{t}^2 = \frac{c_{15} \max\{B, 1\} \log^3(n) \cdot \log(\mathcal{L} \cdot r^2) \cdot \mathcal{L}^2 \cdot r^2}{nm} + \left(C_0 + \frac{\bar{f}}{2} \right) \phi^2_{nm}.$$

to make

$$\begin{aligned} &\frac{24c_5}{\approx^2_t} \sum_{l=1}^{L} \Big(R_{\mathcal{J}_{e,l}}(\mathrm{Star}(\mathcal{F}, f^*) \cap \{c_0 \Delta^2(f - f^*) \le \tilde{t}^2\}) \\ &\quad + R_{\mathcal{J}_{o,l}}(\mathrm{Star}(\mathcal{F}, f^*) \cap \{c_0 \Delta^2(f - f^*) \le \tilde{t}^2\}) \Big) \\ &= O\left(\frac{1}{\log(n)} + \frac{\left(C_0 + \frac{\bar{f}}{2}\right) \phi^2_{nm}}{\approx^2_t} \frac{1}{\log(n)} \right) = O\left(\frac{1}{\log(n)} \right). \end{aligned}$$

Overall, we can let

$$\tilde{t}^2 = \frac{c_{15} \max\{B, 1\} \log^3(n) \cdot \log(\mathcal{L} \cdot r^2) \cdot \mathcal{L}^2 \cdot r^2}{nm} + 2 \left(C_0 + \frac{\bar{f}}{2} \right) \phi^2_{nm} + \frac{s \log^{1+\mu}(n) \log(T_n)}{n},$$

for some $\mu > 0$, to have the right hand side of (246) to be

$$O\left(\frac{1}{\log(n)} + \frac{1}{\log^{\mu}(n)} \right).$$

In the end, we have

$$\Delta^2(\hat{f} - f^*) = O_p\left(r_n^2\right),$$

by setting that

$$r_n^2 \asymp \frac{\max\{B, 1\} \log^3(n) \cdot \log(\mathcal{L} \cdot r^2) \cdot \mathcal{L}^2 \cdot r^2}{nm} + 2 \left(C_0 + \frac{\bar{f}}{2} \right) \phi^2_{nm} + \frac{s \log^{1+\mu}(n) \log(T_n)}{n}.$$

Thus, we get

$$\Delta^2(\hat{f}-f^*) \;=\; O_p\Bigg(\frac{\max\{B,1\}\log^3(n)\cdot\log(\mathcal{L}\cdot r^2)\cdot\mathcal{L}^2\cdot r^2}{nm}$$
$$+2\left(C_0+\frac{\bar{f}}{2}\right)\phi_{nm}^2+\frac{s\cdot\operatorname{poly}(\log n)\log(T_n)}{n}\Bigg), \quad \text{and}$$

$$\left\|\hat{f}-f^*\right\|_2^2 \;=\; O_p\Bigg(\frac{\max\left\{B^2,B\right\}\log^3(n)\cdot\log(\mathcal{L}\cdot r^2)\cdot\mathcal{L}^2\cdot r^2}{nm}$$
$$+2\max\{B,1\}\left(C_0+\frac{\bar{f}}{2}\right)\phi_{nm}^2+\max\{B,1\}\,\frac{s\cdot\operatorname{poly}(\log n)\log(T_n)}{n}\Bigg).$$

Then using the same notations as in Theorem 1 of Kohler and Langer (2019), let

$$L = L_{nm} \asymp \log\left(\sum_{i=1}^n m_i\right) \overset{\text{(a)}}{\asymp} \log(nm), \tag{247}$$

$$r \;=\; r_{nm} \asymp \max_{(p,K)\in\mathcal{P}}\left(\sum_{i=1}^n m_i\right)^{\frac{K}{2(2p+K)}} \asymp \max_{(p,K)\in\mathcal{P}}(nm)^{\frac{K}{2(2p+K)}}, \tag{248}$$

where the (a) holds by the assumption $m_i \asymp m$, or

$$L \;=\; L_{nm} \asymp \log(nm)\left(\max_{(p,K)\in\mathcal{P}}(nm)^{\frac{K}{2(2p+K)}}\right), \tag{249}$$

$$r \;=\; r_{nm} \asymp 1. \tag{250}$$

It follows from Theorem 2 of Kohler and Langer (2019) that

$$\inf_{f\in\mathcal{F}(L,r)}\|f-f^*\|_\infty \;\le\; \phi_{nm}, \tag{251}$$

where

$$\phi_{nm}^2 \;=\; \max_{(p,K)\in\mathcal{P}}\left(\sum_{i=1}^n m_i\right)^{\frac{-2p}{2p+K}} \asymp \max_{(p,K)\in\mathcal{P}}(nm)^{\frac{-2p}{2p+K}}, \tag{252}$$

up to a logarithmic factor depending on n and $\{m_i\}_{i=1}^n$. Then we have

Thus, we get

$$\Delta^2(\hat{f}-f^*) \;=\; O_p\Bigg(\frac{\max\{B,1\}\log^3(n)\cdot\log^2(nm)\cdot\max_{(p,K)\in\mathcal{P}}(nm)^{\frac{2K}{2(2p+K)}}}{nm}$$
$$+2\left(C_0+\frac{\bar{f}}{2}\right)\max_{(p,K)\in\mathcal{P}}(nm)^{\frac{-2p}{2p+K}}+\frac{s\cdot\operatorname{poly}(\log n)\log(T_n)}{n}\Bigg)$$
$$= O_p\left(\left(\max\{B,1\}\log^3(n)\cdot\log^2(nm)+2C_0+\bar{f}\right)\max_{(p,K)\in\mathcal{P}}(nm)^{\frac{-2p}{2p+K}}+\frac{s\cdot\operatorname{poly}(\log n)\log(T_n)}{n}\right), \quad \text{and}$$

$$\left\|\hat{f}-f^*\right\|_2^2 \;=\; O_p\Bigg(\max\left\{B^2,B\right\}\cdot\log^3(n)\cdot\log^2(nm)\cdot\max_{(p,K)\in\mathcal{P}}(nm)^{\frac{-2p}{2p+K}}$$

$$+\max\{B,1\}\cdot\left(2C_0+\bar{f}\right)\cdot\max_{(p,K)\in\mathcal{P}}(nm)^{\frac{-2p}{2p+K}}+\max\{B,1\}\,\frac{s\cdot\mathrm{poly}(\log n)\log(T_n)}{n}\Bigg).$$

■

Appendix G. Auxiliary Lemmas

Lemma 62 (Ledoux-Talagrand contraction, see Theorem7 in Duchi (2009).) *Let $f:\mathbb{R}_+\to\mathbb{R}$ be convex and increasing. Let $\phi:\mathbb{R}\to\mathbb{R}$ satisfy $\phi(0)=0$ and be Lipschitz with constant L, i.e., $|\phi(a)-\phi(b)|\le L|a-b|$. Let σ_i be independent Rademacher random variables. For any $T\subseteq\mathbb{R}^n$,*

$$\mathbb{E}f\left(\frac{1}{2}\sup_{t\in T}\sum_{i=1}^n\sigma_i\phi(t_i)\right)\le\mathbb{E}f\left(L\cdot\sup_{t\in T}\sum_{i=1}^n\sigma_i t_i\right).$$

Lemma 63 (See Lemma 8 in Liang et al. (2015)) *For any scale $\epsilon>0$, the covering number of $\mathcal{F}\subset B_2(1)$ and that of star($\mathcal{F}$) are bounded in the sense*

$$\log N(\mathcal{F},\|\cdot\|_{nm},2\epsilon)\le\log N(star(\mathcal{F}),\|\cdot\|_{nm},2\epsilon)\le\log\frac{2}{\epsilon}+\log N(\mathcal{F},\|\cdot\|_{nm},\epsilon).$$

Lemma 64 (Dudley's Chaining, see Lemma 3 in Farrell et al. (2021)) *Let $N(\delta,\mathcal{F},\|\cdot\|_\infty)$ denote the metric entropy for class $\mathcal{F}$ (with covering radius δ and metric $\|\cdot\|_\infty$), then*

$$\mathbb{E}R_{nm}\{f:f\in\mathcal{F},\|f\|_{nm}\le r\}\le\inf_{0<\alpha<r}\left\{4\alpha+\frac{12}{\sqrt{n}}\int_\alpha^r\sqrt{\log N(\delta,\mathcal{F},\|\cdot\|_{nm})}d\delta\right\}.$$

Corollary 65 *For any scale $\epsilon>0$, the covering number of $\mathcal{F}\subset B_\infty(B)$ and then*

$$\log N(\epsilon,\|\cdot\|_{nm},star(\mathcal{F}))\le\log\left(\frac{2B}{\epsilon}\right)+\log N\left(\epsilon/B,\|\cdot\|_{nm},\mathcal{F}/B\right).$$

Proof We know

$$\mathcal{F}/B\subset B_\infty(1)\subset B_{nm}(1).$$

Then based on Lemma 63, we have

$$\begin{aligned}&\log N(\epsilon,\|\cdot\|_{nm},\mathrm{star}(\mathcal{F}))\\&\le\log N\left(\epsilon/B,\|\cdot\|_{nm},\mathrm{star}(\mathcal{F}/B)\right)\\&\le\log\left(\frac{2B}{\epsilon}\right)+\log N\left(\epsilon/B,\|\cdot\|_{nm},\mathcal{F}/B\right).\end{aligned}$$

■

Lemma 66 *Let $F(\mathcal{L},r)$ defined as in (2) where $\sigma:\mathbb{R}\to\mathbb{R}$ with $\sigma(x)=\max\{x,0\}$. Let $\mathcal{L},r\in\mathbb{N}$. Then*

$$\log(N(\epsilon,\|\cdot\|_{nm},F(\mathcal{L},r)))\le\mathcal{L}^2r^2\log(\mathcal{L}r^2)\log(B\epsilon^{-1}).$$

Proof It follows from Theorem 6 of Bartlett et al. (2019) (see also the proof of Lemma 19 on page 78 of Kohler and Langer (2019)) that the VC-dimension of $F(\mathcal{L}, r)$ satisfies

$$\mathrm{VC}(F(\mathcal{L}, r)) \le L^2 r^2 \log(Lr^2).$$

By a slight variant of Lemma 19 of Kohler and Langer (2019), where L_1 norm is replaced by L_2 norm (which is implied by Lemma 9.2 and Theorem 9.4 of Györfi et al. (2002)), it follows that

$$\log N(\epsilon, F(\mathcal{L}, r), \|\cdot\|_{nm}) \lesssim \mathcal{L}^2 r^2 \log(\mathcal{L}r^2) \left\{\log(B^2\epsilon^{-2}) + \log\log(B^2\epsilon^{-2})\right\} \lesssim \mathcal{L}^2 r^2 \log(\mathcal{L}r^2) \log(B\epsilon^{-1}).$$

■

Corollary 67 *Let $F(\mathcal{L}, r)$ defined as in (2) where $\sigma : \mathbb{R} \to \mathbb{R}$ with $\sigma(x) = \max\{x, 0\}$, let Star$(F(\mathcal{L}, r), f^*)$ follow the definition in (26) where f^* not necessarily in $F(\mathcal{L}, r)$, For any scale $\epsilon > 0$, we have*

$$\log N(\epsilon, \|\cdot\|_{nm}, \mathit{Star}(F(\mathcal{L}, r), f^*)) \le \log\left(\frac{2B}{\epsilon}\right) + \mathcal{L}^2 r^2 \log(\mathcal{L}r^2) \log(B\epsilon^{-1}).$$

Proof We apply the Lemma 63 and Lemma 66 to obtain the conclusion with the following derivations.

$$\begin{aligned}
&\log N(\epsilon, \|\cdot\|_{nm}, \mathrm{Star}(F(\mathcal{L}, r), f^*)) \\
&\le \log\left(\frac{2B}{\epsilon}\right) + \log N\left(\epsilon/B, \|\cdot\|_{nm}, F(\mathcal{L}, r)/B\right) \\
&\le \log\left(\frac{2B}{\epsilon}\right) + \mathcal{L}^2 r^2 \log(\mathcal{L}r^2) \log(B\epsilon^{-1}).
\end{aligned}$$

■

Appendix H. β-mixing coefficients

In this section, we discuss some results on mixing. For additional proofs and a more comprehensive discussion, the reader is directed to Doukhan (2012). Consider a random variable X in $\mathbb{R}^d$. The σ-algebra generated by X is denoted as $\sigma(X)$. Furthermore, for a collection of random variables $\{X_t\}, t \in I \subset \mathbb{R}$, the associated σ-algebra is denoted by $\sigma(X_t, t \in I)$, which is generated by the random variables $X_t, t \in I$. A sequence of random variables $\{X_t, t \in \mathbb{Z}\} \subset \mathbb{R}^d$ is defined to be β-mixing if

$$\beta_l = \sup_{t \in \mathbb{Z}} \beta(\sigma(X_s, s \le t), \sigma(X_s, s \ge t + l)) \to 0, \quad l \to +\infty,$$

where the β coefficients are defined as,

$$\beta(\sigma(X_s, s \le t), \sigma(X_s, s \ge t + l)) = \mathbb{E} \sup_{C \in (\sigma(X_s, s \ge t+l))} |P(C) - P(C|\sigma(X_s, s \le t))|.$$

Note, that the right-hand side is the worst-case scenario of how much the past could influence the future. Hence, intuitively, larger β-mixing coefficients indicate a higher degree of dependence, while smaller values suggest more independence.

Theorem 68 (Doukhan (2012) Theorem 1) . *Let E and F be two polish spaces and (X, Y) some $E \times F$-valued random variables. A random variable Y^* can be defined with the same probability distribution as Y, independent of X and such that*

$$P(Y \neq Y^*) \;=\; \beta(\sigma(X), \sigma(Y)).$$

For some measurable function f on $E \times F \times [0,1]$, and some uniform random variable Δ on the interval $[0,1]$, Y^* takes the form $Y^* = f(X, Y, \Delta)$.

Note that Theorem 68 says that Y is originally dependent on X, and Y^* is constructed such that retains Y's distribution but is independent of X, by introducing the Δ. Let's divide the sequence $[1 \dots n]$ into K groups of size L. For any $L \in \{1, \dots, n\}$ and $K = \lfloor n/L \rfloor$, denote

$$\mathcal{I}_s = \begin{cases} \{(s-1)L, sL\} & \text{when } 1 \;\le\; s \;\le\; K, \\ \{(KL, n]\} & \text{when } s = K+1. \end{cases}$$

So $\mathcal{I}_s$ is a set that contains the start and end indices of the s-th group in a sequence of data points. Specifically, the s-th group starts at the index $(s-1)L$ and ends at index sL. When $s = K+1$, the set $\mathcal{I}_s$ is defined to contain the indices from (KL) to n,

Denote for $1 \;\le\; s \;\le\; L$,

$$\mathcal{J}_{e,l} = \{i \in \mathcal{I}_s \text{ for } s\%2 = 0 \;\text{ and }\; i\%L = l\} \;\text{ and }\; \mathcal{J}_{o,l} = \{i \in \mathcal{I}_s \text{ for } s\%2 = 1 \;\text{ and }\; i\%L = l\}.$$

Note that the $\mathcal{J}_{e,l}$ and $\mathcal{J}_{o,l}$ represent the set of the l indices for K groups. Note that for $i \in \mathcal{I}_s$, we have $i = (s-1)L + l$. Then

$$\frac{K-1}{2} \;\le\; |\mathcal{J}_{e,l}| \;\le\; \frac{K+2}{2} \qquad \text{and} \qquad \frac{K-1}{2} \;\le\; |\mathcal{J}_{o,l}| \;\le\; \frac{K+2}{2},$$

and that

$$\bigcup_{l=1}^{L} \mathcal{J}_{e,l} \cup \bigcup_{l=1}^{L} \mathcal{J}_{o,l} = \{1, \dots, n\}.$$

Lemma 69 *Let $\{x_{ij}\}_{i=1}^{n,m_i} \in [0,1]^d$ are identically distributed random variables such that $\{\sigma(i)\}_{i=1}^{n}$ are β-mixing, where $\sigma(i) \;=\; \sigma(x_{ij}, j \in [m_i])$. Moreover, suppose that for any $i \in [n]$ fixed, $\{x_{ij}\}_{j=1}^{m_i}$ are independent. For any $l \in [1, L]$, there exists a collection of independent identically distributed random variables $\{x^*_{ij}\}_{j \le m_i, i \in \mathcal{J}_{e,l}}$, with same distribution as x_{11}, such that for any $i \in \mathcal{J}_{e,l}$,*

$$P(x^*_{ij} \neq x_{ij} \textit{ for some } j \in [1, \dots, m_i]) \;\le\; \beta(L).$$

The exact same result is attained for $\mathcal{J}_{o,l}$.

and denote,

$$\mathcal{J}_{o,l,K} = \{l, 2L + l, 4L + l, \dots, (2K)L + l\} \cap [1, n],$$

which implies $\bigcup_{k \ge 0} \mathcal{J}_{o,l,K} \;=\; \mathcal{J}_{o,l}$. Therefore, the result is followed by replicating the analysis performed above.

Proof By Theorem 5, for any $i \in \{1, \dots, n\}$, there exists $\{x^*_{i,j}\}_{j=1}^{m_i}$ such that

$$P(x^*_{i,j} \neq x_{i,j} \text{ for some } j \in \{0, \dots, m_i\}) \;=\; \beta(L), \tag{253}$$

where $\{x^*_{i,j}\}_{j=1}^{m_i}$ and $\{x_{i,j}\}_{j=1}^{m_i}$ have the same marginal distribution, and that $\{x^*_{i,j}\}_{j=1}^{m_i}$ is independent of $\sigma(\{x_{t,s}\}_{t=1,s=1}^{n,m_t\backslash\{i,j\}})$. In addition, there exists a uniform random variable Δ_i, independent of σ_z, such that $\{x^*_{i,j}\}_{j=1}^{m_i}$ is measurable with respect to $\sigma(\{x_{i,j}\}_{j=1}^{m_i},\Delta)$. Note that $\{x_{i,j}\}_{j\in I_r}$ are identically distributed because $\{x^*_{i,j}\}_{j\in I_r}$ have the same marginal density. Therefore it suffices to justify the independence.

Note that $J_{l,1}$ is defined as

$$J_{l,1} \;=\; \{L+l, 3L+l, 5L+l, \ldots\} \cap [1,n]. \tag{254}$$

Denote

$$J_{l,K} \;=\; \{L+l, 3L+l, 5L+l, \ldots, (2K+1)L+l\} \cap [1,n], \tag{255}$$

and so $\bigcup_{k=0}^{K} J_{l,k} \;=\; J_{l,1}$. By induction, suppose $\{x^*_{i,j}\}_{i\in J_{l,K}, 1\,\leq\, j\,\leq\, m_i}$ are jointly independent. Let $\{B_{i,j}\}_{i\in J_{l,1}, 1\,\leq\, j\,\leq\, m_i}$ be any collection of Borel sets in $[0,1]^d$. Then

$$\begin{aligned}\mathbb{P}\left(x^*_{i,j}\in B_{i,j}, i\in J_{l,K+1}, 1\,\leq\, j\,\leq\, m_i\right) \;=\; &\mathbb{P}\left(x^*_{i,j}\in B_{i,j}, i\in J_{l,K}, 1\,\leq\, j\,\leq\, m_i\right)\times\\ &\mathbb{P}\left(x^*_{i,j}\in B_{i,j}, i\in J_{l,K+1}\setminus J_{l,K}, 1\,\leq\, j\,\leq\, m_i\right),\end{aligned} \tag{256}$$

where the equation follows because $\{x^*_{i,j}\}_{i\in J_{l,K+1}\backslash J_{l,K}, 1\,\leq\, j\,\leq\, m_i}$ are independent of $\{x^*_{i,j}\}_{i\in J_{l,K}, 1\,\leq\, j\,\leq\, m_i}$. By induction,

$$\mathbb{P}\left(x^*_{i,j}\in B_{i,j}, i\in J_{l,K}, 1\,\leq\, j\,\leq\, m_i\right) = \prod_{i\in J_{l,K}, 1\,\leq\, j\,\leq\, m_i} \mathbb{P}\left(x^*_{i,j}\in B_{i,j}\right). \tag{257}$$

Since for any $i\in\{1,\ldots,n\}$, $\{x^*_{i,j}\}_{j=1}^{m_i}$ and $\{x_{i,j}\}_{j=1}^{m_i}$ have the same marginal distribution,

$$\mathbb{P}\left(\bigcap_{j=1}^{m_i}\left\{x^*_{i,j}\in B_{(2k+3)L+j}\right\}\right) \;=\; \prod_{j=1}^{m_i}\mathbb{P}\left(x^*_{i,j}\in B_{(2k+3)L+j}\right). \tag{258}$$

So

$$\mathbb{P}\left(\bigcap_{i\in J_{l,K+1}, 1\,\leq\, j\,\leq\, m_i}\left\{x^*_{i,j}\in B_{i,j}\right\}\right) \;=\; \prod_{i\in J_{l,K+1}, 1\,\leq\, j\,\leq\, m_i}\mathbb{P}\left(x^*_{i,j}\in B_{i,j}\right). \tag{259}$$

This implies that $\{x^*_{i,j}\}_{i\in J_{l,K+1}, 1\,\leq\, j\,\leq\, m_i}$ are jointly independent. To analyze the case where the indices are considered in $J_{0,l}$, we follow the same line of arguments, but now we note that

$$J_{0,l} \;=\; \{l, 2L+l, 4L+l, \ldots\} \cap [1,n], \tag{260}$$

and denote,

$$J_{0,l,K} \;=\; \{l, 2L+l, 4L+l, \ldots, (2K)L+l\} \cap [1,n], \tag{261}$$

which implies $\bigcup_{k\,\geq\,0} J_{0,l,K} \;=\; J_{0,l}$. Therefore, the result is followed by replicating the analysis performed above. ■

Lemma 70 *Let $\{\varepsilon_{ij}\}_{i=1}^{n,m_i}$ in $\mathbb{R}$ are identically distributed random variables such that $\{\sigma(i)\}_{i=1}^{n}$ are β-mixing, where $\sigma(i) = \sigma(\varepsilon_{ij}, j \in [m_i])$. Moreover, suppose that for any $i \in [n]$ fixed, $\{\varepsilon_{ij}\}_{j=1}^{m_i}$ are independent. For any $l \in [1, L]$, there exists a collection of independent identically distributed random variables $\{\varepsilon_{ij}^*\}_{j \le m_i, i \in \mathcal{J}_{e,l}}$, with same distribution as ε_{11}, such that for any $i \in \mathcal{J}_{e,l}$,*

$$P(\varepsilon_{ij}^* \neq \varepsilon_{ij} \textit{ for some } j \in [1, \ldots, m_i]) \le \beta(L).$$

The exact same result is attained for $\mathcal{J}_{o,l}$.

Proof The arguments used in this proof parallel to those employed in the proof of Lemma 69. The key distinction lies in our utilization of Borel sets within $\mathbb{R}$ for the present context. ■

Lemma 71 *Let $\{\theta_{i,j}\}_{i=1,j=1}^{n,m_i}$ be identically distributed random functions in $[0,1]^d$, such that $\{\sigma(\theta_{i,j})\}_{j=1}^{m_i}$ are σ-mixing, where $\sigma(\delta_i) = \sigma(\delta)$. For any $l \in [0, \ldots, L-1]$, there exists a collection $\{\theta_{i,j}^*\}_{j \in J_l}$ of independent identically distributed random functions in $[0,1]^d$, with the same distribution as δ_i, such that for any $i \in J_l$,*

$$P(\theta_i^* \neq \theta_i) \le \beta(L). \tag{262}$$

The exact same result is attained for $J_{0,l}$.

Proof By Theorem 5, for any $i \in [1, \ldots, n]$, there exists θ_i^* such that

$$P(\theta_i^* \neq \theta_i) = \beta(L), \tag{263}$$

where equation (263) follows because θ_i^* is independent of $\sigma(\{\theta_t\}_{t \in \mathcal{J}_{e,l}})$. By induction,

$$P(\theta_i^*(t) \in B_i, i \in J_{l,K+1}, 1 \le t \le m_i) = \prod_{i \in J_{l,K+1}, 1 \le t \le m_i} P(\theta_i^*(t) \in B_i). \tag{264}$$

This implies that $\{\theta_i^*\}_{i \in J_{l,K+1}}$ are jointly independent. To analyze the case where the indices are considered in $J_{0,l}$, we note that

$$J_{0,l} = \{l, 2L+l, 4L+l, \ldots\} \cap [1, n], \tag{265}$$

$$J_{0,l,K} = \{l, 2L+l, 4L+l, \ldots, (2K)L+l\} \cap [1, n], \tag{266}$$

which implies $\bigcup_{k \ge 0} J_{0,l,K} = J_{0,l}$. Therefore, the result is followed by replicating the analysis performed above. ■

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